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(12) **United States Patent**
Borriello(10) **Patent No.:** **US 12,403,186 B2**(45) **Date of Patent:** ***Sep. 2, 2025**(54) **ALLOGENEIC TUMOR CELL VACCINE**(71) Applicant: **ALLOPLEX BIOTHERAPEUTICS, INC.**, Woburn, MA (US)(72) Inventor: **Frank Borriello**, Winchester, MA (US)(73) Assignee: **Alloplex Biotherapeutics, Inc.**, Woburn, MA (US)

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See application file for complete search history.

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The described invention provides a tumor cell vaccine comprising genetically modified tumor cell line of a particular tumor type that stably expresses high levels of two or more immunomodulators. According to some embodiments, an immunogenic amount of the tumor cell line variants may be selected for concomitant expression of two or more of recombinant membrane expressed IgG1, CD40L, TNF- α , as well as membrane and soluble forms of GM-CSF, and Flt-3L peptides that are effective to elicit an anti-tumor immune response compared to the parent unmodified tumor cell line as measured in vitro by a one-way mixed lymphocyte tumor reaction assay using human peripheral blood mononuclear cells and the genetically modified allogeneic cell vaccine candidate. According to some embodiments, the tumor cell vaccine candidate will induce an immune response in the recipient cancer patient that cross reacts with the patient's own (autologous) tumor cells, the effects of which will be sufficient to result in enhanced anti-tumor immunity contributing to the increased survival of a vaccinated patient cohort compared to a matched unvaccinated patient cohort.

5 Claims, 45 Drawing Sheets**Specification includes a Sequence Listing.**

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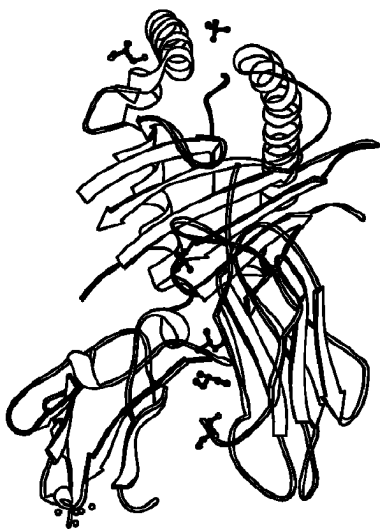
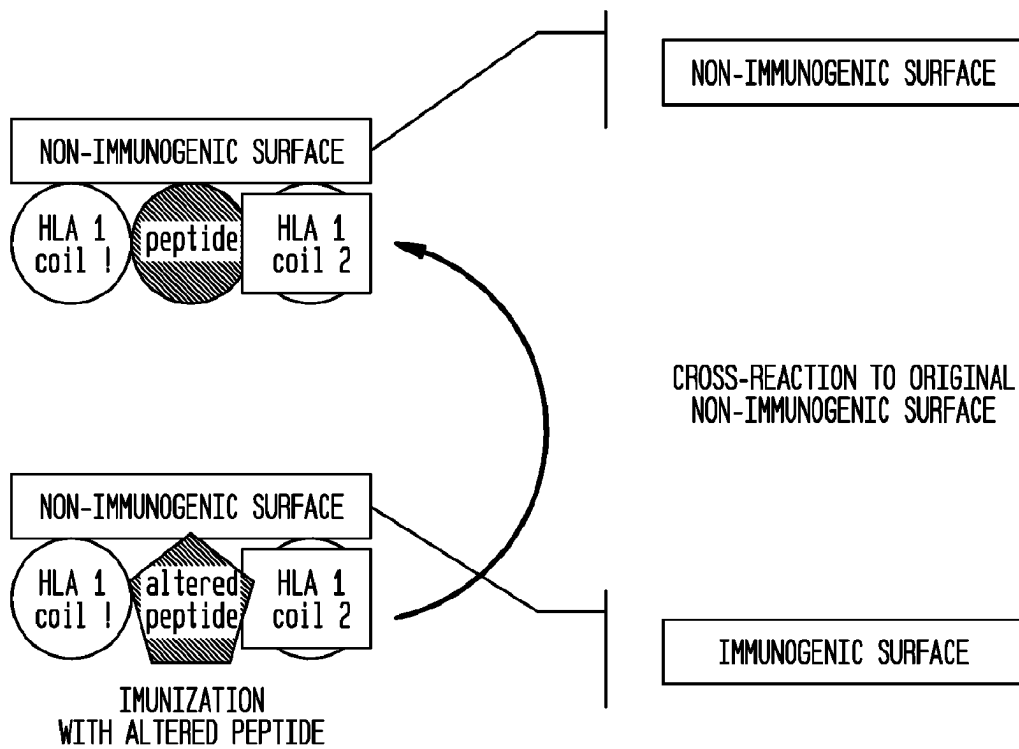
FIG. 1RIBBON STRUCTURE OF AN HLA
CLASS I SURFACE ANTIGENRIBBON STRUCTURE OF AN HLA
CLASS I SURFACE ANTIGEN

FIG. 2A

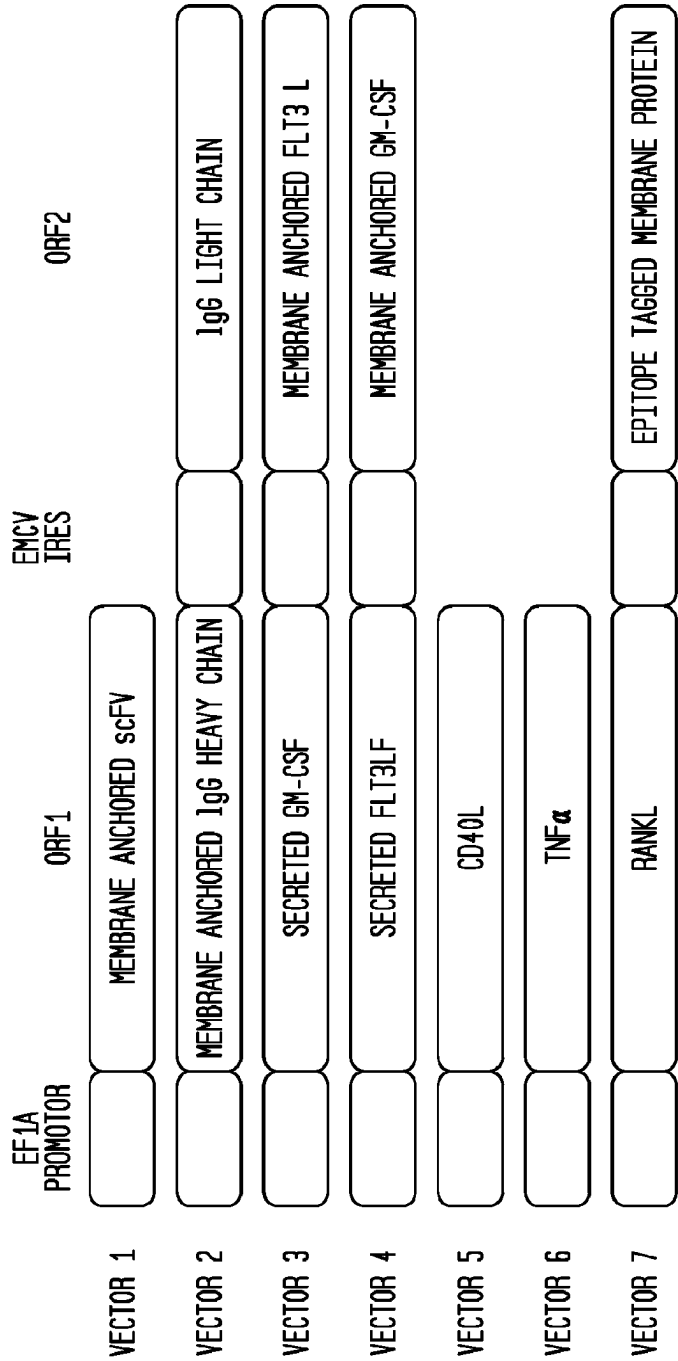


FIG. 2B

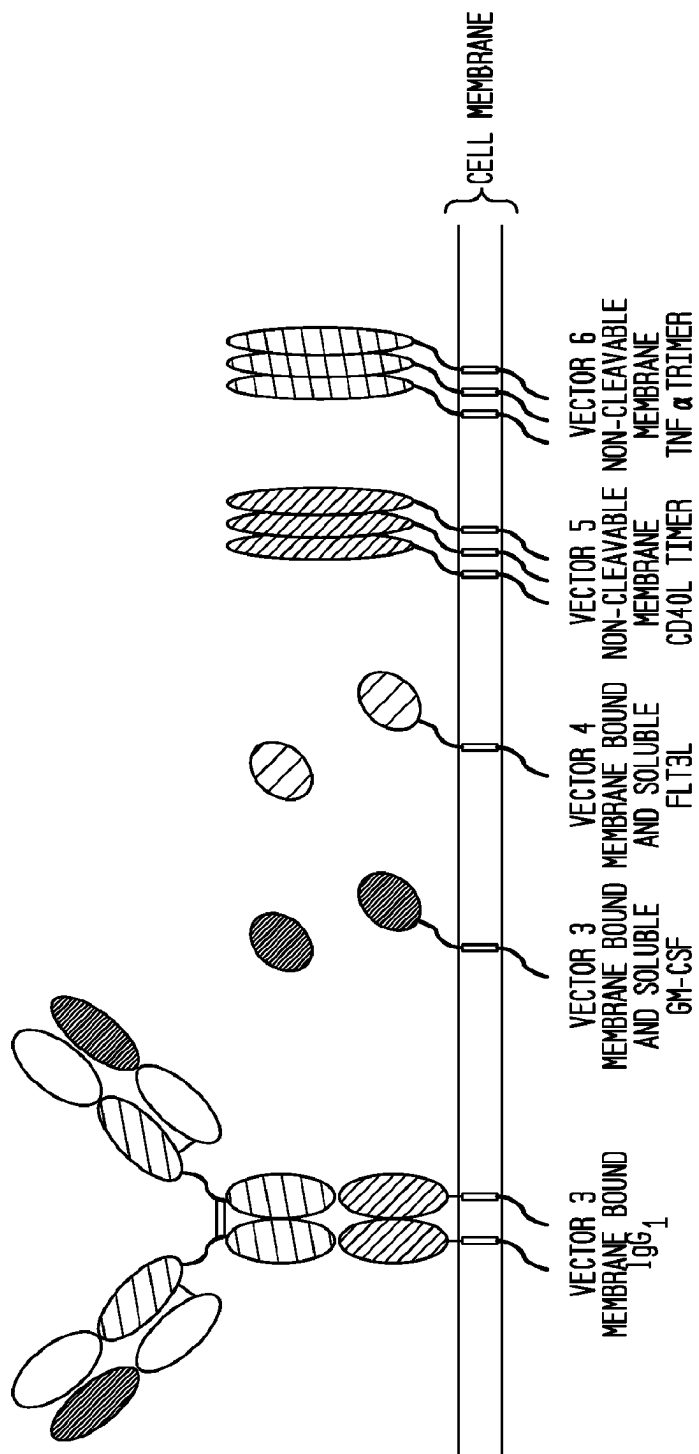


FIG. 3A

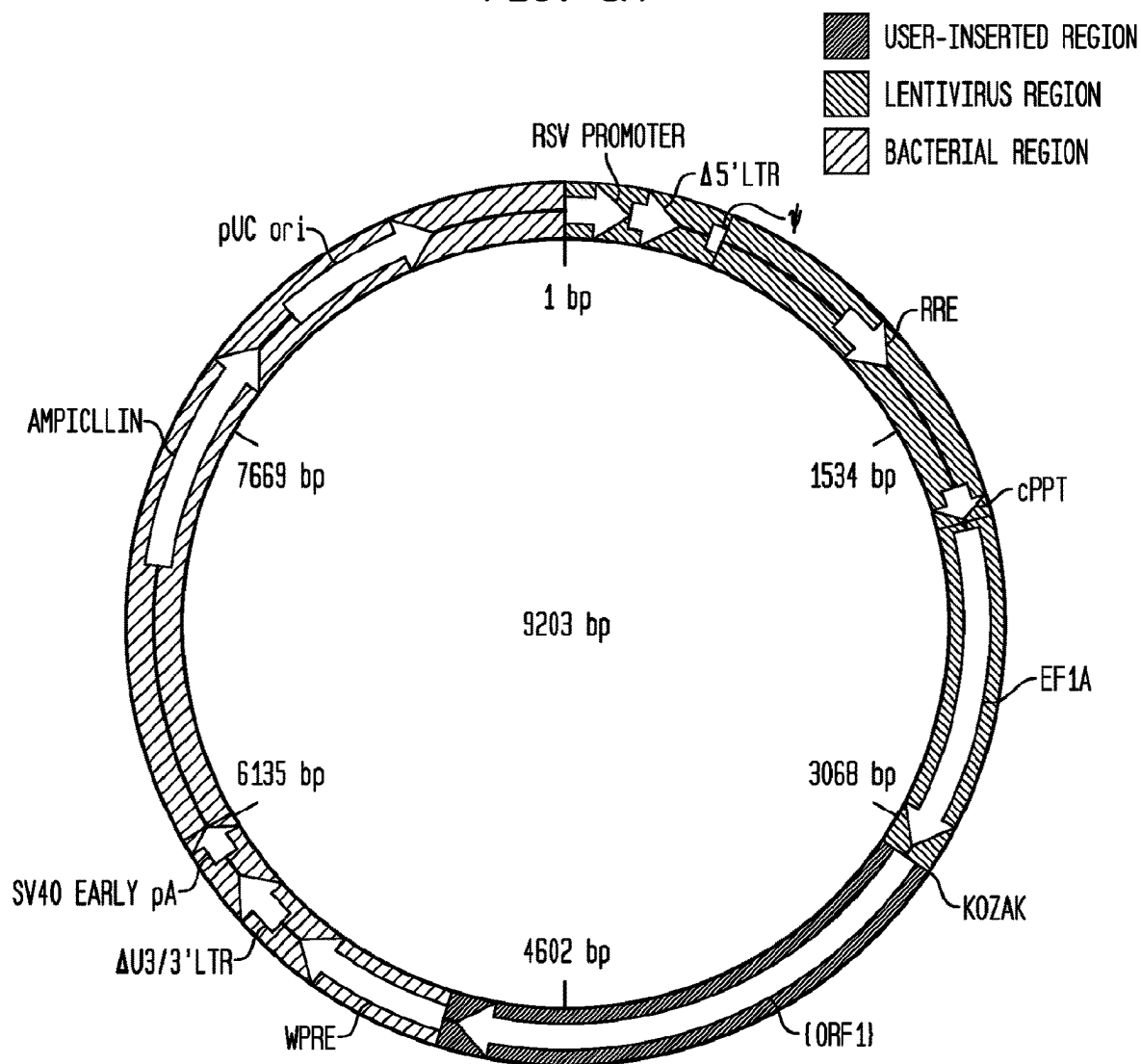


FIG. 3B

SEQ ID NO. 49

1	AATGTAGTCT	TATGCAATAC	TCTTGTAGTC	TTGCAACATG	GTAACGATGA	GTTAGCAACA
61	TGCCTTACAA	GGAGAGAAAA	AGCACCGTGC	ATGCCGATTG	GTGGAAGTAA	GGTGGTACGA
121	TCGTGCCCTTA	TTAGGAAGGC	AACAGACGGG	TCTGACATGG	ATTGGACGAA	CCACTGAATT
181	GCCGCATTGC	AGAGATATTG	TATTTAAGTG	CCTAGCTCGA	TACATAAACG	GGTCTCTCTG
241	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC
301	TCAATAAAGC	TTGCCCTTGAG	TGCTTCAAGT	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG
361	TAACTAGAGA	TCCCTCAGAC	CCTTTTAGTC	AGTGTGGAAG	ATCTCTAGCA	GTGGCGCCCG
421	AACAGGGACT	TGAAAGCGAA	AGGGAACCA	GAGGAGCTCT	CTCGACGCAG	GACTCGGCTT
481	GCTGAAGCGC	GCACGGCAAG	AGGCGAGGGG	CGGCGACTGG	TGAGTACGCC	AAAAATTTTG
541	ACTAGCGGAG	GCTAGAAGGA	GAGAGATGGG	TGCGAGAGCG	TCAGTATTAA	GCGGGGGAGA
601	ATTAGATCGC	GATGGGAAAA	AATTCGGTTA	AGGCCAGGGG	GAAAGAAAAA	ATATAAATTA
661	AAACATATAG	TATGGGCAAG	CAGGGAGCTA	GAACGATTCT	CAGTTAATCC	TGGCCTGTTA
721	GAAACATCAG	AAGGCTGTAG	ACAAATACTG	GGACAGCTAC	AACCATCCCT	TCAGACAGGA
781	TCAGAAGAAC	TTAGATCATT	ATATAATACA	GTAGCAACCC	TCTATTGTGT	GCATCAAAGG
841	ATAGAGATAA	AAGACACCAA	GGAAAGCTTTA	GACAAGATAG	AGGAAGAGCA	AAACAAAAGT
901	AAGACCACCG	CACAGCAAGC	GGCCGCTGAT	CTTCAGACCT	GGAGGAGGAG	ATATGAGGGA
961	CAATTGGAGA	AGTGAATTAT	ATAAATATAA	AGTAGTAAAA	ATTGAACCAT	TAGGAGTAGC
1021	ACCCACCAAG	GCAAAGAGAA	GAGTGGTGCA	GAGAGAAAAA	AGAGCAGTGG	GAATAGGAGC
1081	TTTGTTCCTT	GGGTTCTTGG	GAGCAGCAGG	AAGCACTATG	GGCGCAGCGT	CAATGACGCT
1141	GACGGTACAG	GCCAGACAAT	TATTGTCTGG	TATAGTGCAG	CAGCAGAACA	ATTTGCTGAG
1201	GGCTATTGAG	GCGCAACAGC	ATCTGTTGCA	ACTCACAGTC	TGGGGCATCA	AGCAGCTCCA
1261	GGCAAGAATC	CTGGCTGTGG	AAAGATACCT	AAAGGATCAA	CAGCTCCTGG	GGATTTGGGG
1321	TTGCTCTGGA	AAACTCATTT	GCACCACTGC	TGTGCCCTTG	AATGCTAGTT	GGAGTAATAA
1381	ATCTCTGGAA	CAGATTGGA	ATCACACGAC	CTGGATGGAG	TGGGACAGAG	AAATTAACAA
1441	TTACACAAGC	TTAATACACT	CCTTAATTGA	AGAATCGCAA	AACCAGCAAG	AAAAGAATGA
1501	ACAAGAATTA	TTGGAATTAG	ATAAATGGGC	AAGTTTGTGG	AATTGGTTTA	ACATAACAAA
1561	TTGGCTGTGG	TATATAAAAT	TATTCATAAT	GATAGTAGGA	GGCTTGGTAG	GTTTAAGAAT
1621	AGTTTTTGCT	GTACTTTCTA	TAGTGAATAG	AGTTAGGCAG	GGATATTCAC	CATTATCGTT
1681	TCAGACCCAC	CTCCCAACCC	CGAGGGGACC	CGACAGGCCC	GAAGGAATAG	AAGAAGAAGG
1741	TGGAGAGAGA	GACAGAGACA	GATCCATTCT	ATTAGTGAAC	GGATCTCGAC	GGTATCGCTA
1801	GCTTTTAAAA	GAAAAGGGGG	GATTGGGGGG	TACAGTGCAG	GGGAAAGAAT	AGTAGACATA
1861	ATAGCAACAG	ACATACAAAC	TAAAGAATTA	CAAAAACAAA	TTACAAAAT	TCAAAATTTT
1921	ACTAGTGATT	ATCGGATCAA	CTTTGTATAG	AAAAGTTGGG	CTCCGGTGCC	CGTCAGTGGG
1981	CAGAGCGCAC	ATCGCCACAA	GTCCCGGAGA	AGTTGGGGGG	AGGGGTGGCG	AATTGAACCG
2041	GTGCCTAGAG	AAGGTGGCGC	GGGGTAAACT	GGGAAAGTGA	TGTCGTGTAC	TGGCTCCGCC
2101	TTTTTCCCGA	GGGTGGGGGA	GAACCGTATA	TAAGTGCAGT	AGTCGCCGTG	AACGTTCTTT
2161	TTCGCAACGG	GTTTGCCGCC	AGAACACAGG	TAAGTGCCGT	GTGTGTTTCC	CGCGGGCCTG
2221	GCCTCTTTAC	GGGTTATGGC	CCTTGCGTGC	CTTGAATTAC	TTCCACCTGG	CTGCAGTACG
2281	TGATTCTTGA	TCCCGAGCTT	CGGGTTGGAA	GTGGGTGGGA	GAGTTCGAGG	CCTTGCCTTT
2341	AAGGAGCCCC	TTCCGCTCGT	GCTTGAGTTG	AGGCCTGGCC	TGGCGCTGG	GGCCGCCCGC
2401	TGCGAATCTG	GTGGCACCTT	CGCGCTGTCT	TCGCTGCTTT	CGATAAGTCT	CTAGCCATTT
2461	AAAATTTTGG	ATGACCTGCT	GCGACGCTTT	TTTTCTGGCA	AGATAGTCTT	GTAATGCGG
2521	GCCAAGATCT	GCACACTGGT	ATTTGCGTTT	TTGGGGCCGC	GGGCGCGAC	GGGGCCCGTG
2581	CGTCCCAGCG	CACATGTTCT	GCGAGGCGGG	GCCTGCGAGC	GCGGCCACCG	AGAATCGGAC
2641	GGGGGTAGTC	TCAAGCTGGC	CGGCTGTCTC	TGGTGCCTGG	TCTGCGCCCG	CCGTGTATCG
2701	CCCCGCCCTG	GGCGGCAAGG	CTGCCCCGGT	CGGCACCACT	TGCGTGAGCG	GAAAGATGGC
2761	CGCTTCCCGG	CCCTGCTGCA	GGGAGCTCAA	AATGGAGGAC	GCGGCGCTCG	GGAGAGCGGG
2821	CGGGTGAGTC	ACCCACACAA	AGGAAAAGGG	CCTTTCCTGC	CTCAGCCGTC	GCTTCATGTG
2881	ACTCCACGGA	GTACCGGGCG	CCGTCCAGGC	ACCTCGATTA	GTTCTCGAGC	TTTTGGAGTA
2941	CGTCGTCTTT	AGGTTGGGGG	GAGGGGTTTT	ATGCGATGGA	GTTTCCCCAC	ACTGAGTGGG
3001	TGGAGACTGA	AGTTAGGCCA	GCTTGGCACT	TGATGTAATT	CTCCTTGGAA	TTTGCCCTTT

FIG. 3B-1

3061	TTGAGTTTGG	ATCTTGGTTC	ATTCTCAAGC	CTCAGACAGT	GGTTCAAAGT	TTTTTTCTTC
3121	CATTTTCAGGT	GTCTGTACAA	GTTTGTACAA	AAAAGCAGGC	TGCCACCATG	GAGTTCGGCC
3181	TGAGCTGGGT	GTTCTTGGTG	GCCCTGTTCA	GAGGCGTGCA	GTGCCACCTG	AAGCTGCAGG
3241	AGAGCGGCCC	CGGCTTGGTG	GCCCCAGCC	AGAGCCTGAG	CATCACCTGC	ACCGTGAGCG
3301	GCTTCAGCCT	GACCGCCTAC	GGCGTGGACT	GGGTGAGACA	GCCCCCGGC	AAGTGCCCTGG
3361	AGTGGCTGGG	CGTGATCTGG	GGCGGCGGCA	GAACCAACTA	CAACAGCGGC	CTGATGAGCA
3421	GACTGAGCAT	CAGAAAGGAC	AACAGCAAGA	GCCAGGTGTT	CCTGACCATG	AACAGCCTGC
3481	AGACCGACGA	CACCGCCAAG	TACTACTGCG	TGAAGCACAC	CAACTGGGAG	GGCGGCTTCG
3541	CCTACTGGGG	CCAGGGCACC	ACCGTGACCG	TGAGCAGCGG	CGCGGGCGGC	AGCGGCGGCG
3601	GCGGCAGCGG	CGCGGGCGGC	AGCGGCAGCC	CCGGCCAGAG	CGTGAGCATC	AGCTGCAGCG
3661	GCAGCAGCAG	CAACATCGGC	AACAACACG	TGTACTGGTA	CCAGACCTTG	CCCGGCACCG
3721	CCCCCAAGCT	GCTCATCTAC	AGCGACACCA	AGAGACCCAG	CGGCCTGCC	GACAGAATCA
3781	GCGGCAGCAA	GAGCGGCACC	AGCGCCAGCC	TGGCCATCAG	CGGCCTGCAG	AGCGAGGACG
3841	AGGCCGACTA	CTACTGCGCC	AGCTGGGACG	ACAGCCTGGA	CGGCCCGGTG	TTCGGCTGCG
3901	GCACCAAGCT	GACCGTGCTG	CTGAAGACCC	CCCTGGGCGA	CACCAACCAC	ACCTGCCCCA
3961	GATGCCCCGA	GCCCAAGAGC	TGCGACACCC	CCCCCCCCTG	CCCCAGATGC	CCCGAGCCCA
4021	AGAGCTGCGA	CACCCCCCCC	CCCTGCCCCA	GATGCCCCGA	GCCCCAAGAGC	TGCGACACCC
4081	CCCCCCCCTG	CCCCAGATGC	CCCGCCCCCG	AGCTGCTGGG	CGGCCCCAGC	GTGTTCTCTG
4141	TCCCCCCCAA	GCCCAAGGAC	ACCCTGATGA	TCAGCAGAGC	CCCCGAGGTG	ACCTGCGTGG
4201	TGGTGGACGT	GAGCCACGAG	GACCCCGAGG	TGAAGTTCAA	CTGGTACGTG	GACGGCGTGG
4261	AGGTGCACAA	CGCCAAGACCC	AAGCCAGAG	AGGAGCAGTA	CAACAGCACC	TACAGAGTGG
4321	TGAGCGTGCT	GACCGTGCTG	CACCAGGACT	GGCTGAACGG	CAAGGAGTAC	AAGTGCAAGG
4381	TGAGCAACAA	GGCCCTGCC	GCCCCCATCG	AGAAGACCAT	CAGCAAGGCC	AAGGGCCAGC
4441	CCAGAGAGCC	CCAGGTGTAC	ACCCTGCCCC	CCAGCAGAGA	CGAGCTGACC	AAGAACCAGG
4501	TGAGCCTGAC	CTGCCTGGTG	AAGGGCTTCT	ACCCAGCGGA	CATCGCCGTG	GAGTGGGAGA
4561	GCAACGGCCA	GCCCGAGAAC	AACTACAAGA	CCACCCCCCC	CGTGCTGGAC	AGCGACGGCA
4621	GCTTCTTCCT	GTACAGCAAG	CTGACCGTGG	ACAAGAGCAG	ATGGCAGCAG	GGCAACGTGT
4681	TCAGCTGCAG	CGTGATGCAC	GAGGCCCTGC	ACAACCACTA	CACCCAGAAG	AGCCTGAGCC
4741	TGAGCCCCGA	GCTGCAGCTG	GAGGAGAGCT	GGCCGAGGGC	CCAGGACGGC	GAGCTGGACG
4801	GCCTGTGGAC	CACCATCACC	ATCTTCATCA	CCCTGTTCTT	GCTGAGCGTG	TGCTACAGCG
4861	CCACCGTGAC	CTTCTTCAAG	GTGAAGTGGG	TCTTCAGCAG	CGTGGTGGAC	CTGAAGCAGA
4921	CCATCATCCC	CGACTACAGA	AACATGATCG	GCCAGGGCGC	CTAAACCCAG	CTTCTTGTA
4981	CAAAGTGGTG	ATAATCGAAT	TCTAAACCCA	GCTTCTTGT	ACAAAGTGGT	GATAATCGAA
5041	TTCCGATAAT	CAACCTCTGG	ATTACAAAT	TTGTGAAAGA	TTGACTGGTA	TTCTTAACCTA
5101	TGTTGCTCCT	TTTACGCTAT	GTGGATACGC	TGCTTTAATG	CCTTTGTATC	ATGCTATTGC
5161	TTCCCGTATG	GCTTTCATTT	TCTCTCCTT	GTATAAATCC	TGTTGCTGT	CTCTTTATGA
5221	GGAGTTGTGG	CCCGTTGTCA	GGCAACGTGG	CGTGGTGTGC	ACTGTGTTTG	CTGACGCAAC
5281	CCCCACTGGT	TGGGGCATTG	CCACCACCTG	TCAGCTCCTT	TCCGGGACTT	TCGCTTTCCC
5341	CCTCCCTATT	GCCACGGCGG	AACTCATCGC	CGCCTGCCTT	GCCCCTGCT	GGACAGGGGC
5401	TCGGCTGTTG	GGCACTGACA	ATTCCGTGGT	GTTGTCGGGG	AAGCTGACGT	CCTTTCATG
5461	GCTGCTCGCC	TGTGTTGCCA	CCTGGATTCT	GGCGGGGACG	TCCTTCTGCT	ACGTCCCTTC
5521	GGCCCTCAAT	CCAGCGGACC	TTCCTTCCCG	CGGCTGCTG	CCGGCTCTGC	GGCCTCTTCC
5581	GCGTCTTCGC	CTTCGCCCTC	AGACGAGTCG	GATCTCCCTT	TGGGCCGCGT	CCCCGCATCG
5641	GGAATTCCTG	CGGTTGCTTT	TAAGACCAAT	GACTTACAAG	GCAGCTGTAG	ATCTTAGCCA
5701	CTTTTTAAAA	GAAAAGGGGG	GACTGGAAGG	GCTAATTAC	TCCCAACGAA	GACAAGATCT
5761	GCTTTTTGCT	TGTAAGGGT	CTCTCTGGTT	AGACCAGATC	TGAGCCTGGG	AGCTCTCTGG
5821	CTAACTAGGG	AACCCACTGC	TTAAGCCTCA	ATAAAGCTTG	CCTTGAGTGC	TTCAAGTAGT
5881	GTGTGCCCCG	CTGTTGTGTG	ACTCTGGTAA	CTAGAGATCC	CTCAGACCTT	TTTAGTCAGT
5941	GTGGAAAATC	TCTAGCAGTA	GTAGTTCAAT	TCATCTTATT	ATTGAGTATT	TATAACTTGC
6001	AAAGAAATGA	ATATCAGAGA	GTGAGAGGAA	CTTGTTTATT	GCAGCTTATA	ATGCTTACAA
6061	ATAAGCAAT	AGCATCACAA	ATTCACAAA	TAAAGCATTT	TTTTCACTGC	ATTCTAGTTG
6121	TGGTTTGTCC	AAACTCATCA	ATGTATCTTA	TCATGTCTGG	CTCTAGCTAT	CCCGCCCTTA
6181	ACTCCGCCCA	TCCCGCCCTT	AACTCCGCCC	AGTTCCGCCC	ATTCTCCGCC	CCATGGCTGA
6241	CTAATTTTTT	TTATTTATGC	AGAGGCCGAG	GCCGCTCGG	CCTCTGAGCT	ATTCCAGAAG
6301	TAGTGAGGAG	GCTTTTTTGG	AGGCCTAGGG	ACGTACCCAA	TTCGCCCTAT	AGTGAGTCGT
6361	ATTACGCGCG	CTCACTGGCC		AAAGGGGGGA	CTGGGAAAC	TAATTCACTC

FIG. 3B-2

6421	CCCAACTTAA	TCGCCTTGCA	GCACATCCCC	CTTTCGCCAG	CTGGCGTAAT	AGCGAAGAGG
6481	CCCGCACC GA	TCGCCCTTCC	CAACAGTTGC	GCAGCCTGAA	TGGCGAATGG	GACGCGCCCT
6541	GTAGCGGCGC	ATTAAGCGCG	GCGGGTGTGG	TGGTTACGCG	CAGCGTGACC	GCTACACTTG
6601	CCAGCGCCCT	AGCGCCCGCT	CTTTTCGCTT	TCTTCCCTTC	CTTTCTCGCC	ACGTTTCGCCG
6661	GCTTTCGCCG	TCAAGCTCTA	AATCGGGGGC	TCCCTTTAGG	GTTCCGATTT	AGTGCTTTAC
6721	GGCACCTCGA	CCCCAAAAAA	CTTGATTAGG	GTGATGGTTC	ACGTAGTGGG	CCATCGCCCT
6781	GATAGACGGT	TTTTCGCCCT	TTGACGTTGG	AGTCCACGTT	CTTTAATAGT	GGACTCTTGT
6841	TCCAAACTGG	AACAACACTC	AACCCTATCT	CGGTCTATTC	TTTTGATTTA	TAAGGGATTT
6901	TGCCGATTTC	GGCCTATTGG	TAAAAAATG	AGCTGATTTA	ACAAAAATTT	AACGCGAATT
6961	TTAACAAAAT	ATTAACGCTT	ACAATTTAGG	TGGCACTTTT	CGGGGAAATG	TGCGCGGAAC
7021	CCCTATTTGT	TTATTTTTCT	AAATACATTC	AAATATGTAT	CCGCTCATGA	GACAATAACC
7081	CTGATAAATG	CTTCAATAAT	ATTGAAAAAG	GAAGAGTATG	AGTATTCAAC	ATTTCCGTGT
7141	CGCCCTTATT	CCCTTTTTTG	CGGCATTTTG	CCTTCCTGTT	TTTGCTCACC	CAGAAACGCT
7201	GGTGAAAGTA	AAAGATGCTG	AAGATCAGTT	GGGTGCACGA	GTGGGTTACA	TCGAACTGGA
7261	TCTCAACAGC	GGTAAGATCC	TTGAGAGTTT	TCGCCCCGAA	GAACGTTTTT	CAATGATGAG
7321	CACTTTTTAA	GTTCTGCTAT	GTGGCGCGGT	ATTATCCCGT	ATTGACGCCG	GGCAAGAGCA
7381	ACTCGGTCCG	CGCATACACT	ATTCTCAGAA	TGACTTG GTT	GAGTACTCAC	CAGTCACAGA
7441	AAAGCATCTT	ACGGATGGCA	TGACAGTAAG	AGAATTATGC	AGTGCTGCCA	TAACCATGAG
7501	TGATAACACT	GCGGCCAACT	TACTTCTGAC	AACGATCGGA	GGACCGAAGG	AGCTAACCGC
7561	TTTTTTGCAC	AACATGGGGG	ATCATGTAAC	TCGCCTTGAT	CGTTGGGAAC	CGGAGCTGAA
7621	TGAAGCCATA	CCAAACGACG	AACTACTTAC	CACGATGCCT	GTAGCAATGG	CAACAACGTT
7681	GCGCAAACTA	TTAAGTGGCG	AACTACTTAC	TCTAGCTTCC	CGGCAACAAT	TAATAGACTG
7741	GATGGAGGCG	GATAAAGTTG	CAGGACCACT	TCTGCGCTCG	GCCCTTCCGG	CTGGCTGGTT
7801	TATTGCTGAT	AAATCTGGAG	CCGGTGAGCG	TGGGTCTCGC	GGTATCATTG	CAGCACTGGG
7861	GCCAGATGGT	AAGCCCTCCC	GTATCGTAGT	TATCTACACG	ACGGGGAGTC	AGGCAACTAT
7921	GGATGAACGA	AATAGACAGA	TCGTGAGAT	AGGTGCCTCA	CTGATTAAAG	ATTGGTAACT
7981	GTAGACCCAA	GTTTACTCAT	ATATACTTTA	GATTGATTTA	AAACTTCATT	TTTAATTTAA
8041	AAGGATCTAG	GTGAAGATCC	TTTTTGATAA	TCTCATGACC	AAAATCCCTT	AACGTGAGTT
8101	TTCGTTCCAC	TGAGCGTCAG	ACCCCGTAGA	AAAGATCAAA	GGATCTTCTT	GAGATCCTTT
8161	TTTTCTGCGC	GTAATCTGCT	GCTTGCAAAC	AAAAAAACCA	CCGCTACCAG	CGGTGGTTTG
8221	TTTGCCGGAT	CAAGAGCTAC	CAACTCTTTT	TCCGAAGGTA	ACTGGCTTCA	GCAGAGCGCA
8281	GATACCAAAT	ACTGTTCTTC	TAGTGTAGCC	GTAGTTAGGC	CACCACTTCA	AGAACTCTGT
8341	AGCACCGCCT	ACATACCTCG	CTCTGCTAAT	CCTGTTACCA	GTGGCTGCTG	CCAGTGGCGA
8401	TAAGTCGTGT	CTTACCGGGT	TGGACTCAAG	ACGATAGTTA	CCGGATAAGG	CGCAGCGGTC
8461	GGGCTGAACG	GGGGGTTCTG	GCACACAGCC	CAGCTTGAGG	CGAACGACCT	ACACCGAACT
8521	GAGATACCTA	CAGCGTGAGC	TATGAGAAAG	CGCCACGCTT	CCCGAAGAGA	GAAGGGCGGA
8581	CAGGTATCCG	GTAAGCGGCA	GGGTCGGAAC	AGGAGAGCGC	ACGAGGGAGC	TTCCAGGGGG
8641	AAACGCCCTGG	TATCTTTATA	GTCTGTCTGG	GTTTCGCCAC	CTCTGACTTG	AGCGTCGATT
8701	TTTGTGATGC	TCGTCAGGGG	GGCGGAGCCT	ATGGAAAAAC	GCCAGCAACG	CGGCCTTTTT
8761	ACGGTTGGTG	GCCTTTTGCT	GGCCTTTTGC	TCACATGTTT	TTTCTGCGT	TATCCCTGTA
8821	TTCTGTGGAT	AACCGTATTA	CCGCCTTTGA	GTGAGCTGAT	ACCGCTCGCC	GCAGCCGAAC
8881	GACCGAGCGC	AGCGAGTCAG	TGAGCGAGGA	AGCGGAAGAG	CGCCAATAC	GCAAACCGCC
8941	TCTCCCCGCG	CGTTGGCCGA	TTCATTAAATG	CAGCTGGCAC	GACAGGTTTC	CCGACTGGAA
9001	AGCGGGCAGT	GAGCGCAACG	CAATTAATGT	GAGTTAGCTC	ACTCATTAGG	CACCCAGGCG
9061	TTTACACTTT	ATGCTTCCGG	CTCGTATGTT	GTGTGGAATT	GTGAGCGGAT	AACAATTTCA
9121	CACAGGAAAC	AGCTATGACC	ATGATTACGC	CAAGCGCGCA	ATTAACCTCT	ACTAAAGGGA
9181	ACAAAAGCTG	GAGCTGCAAG	CTT			

FIG. 4A

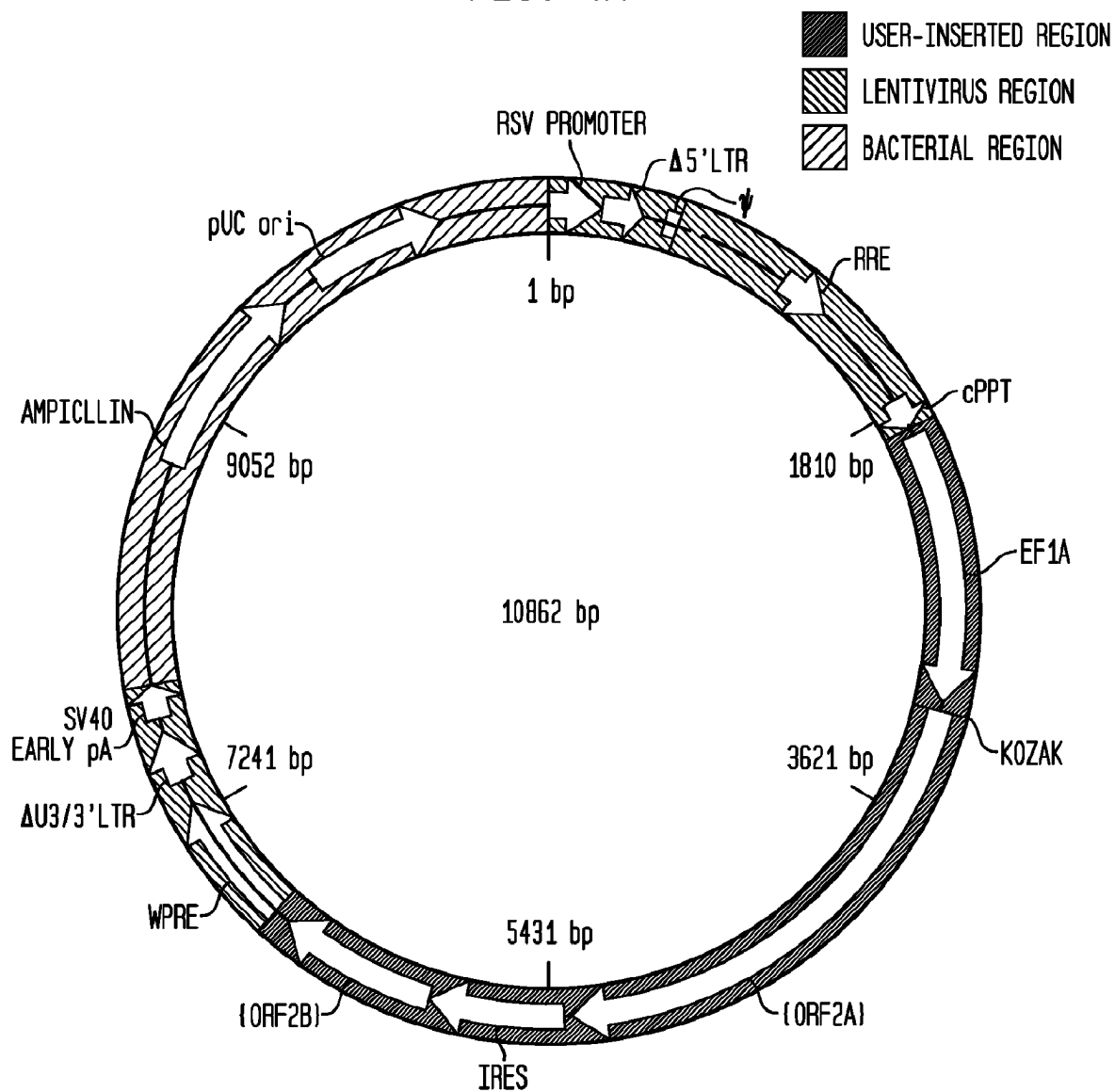


FIG. 4B

SEQ ID NO. 48

1	AATGTAGTCT	TATGCAATAC	TCTTGTAGTC	TTGCAACATG	GTAACGATGA	GTTAGCAACA
61	TGCCTTACAA	GGAGAGAAAA	AGCACCGTGC	ATGCCGATTG	GTGGAAGTAA	GGTGGTACGA
121	TCGTGCCCTTA	TTAGGAAGGC	AACAGACGGG	TCTGACATGG	ATTGGACGAA	CCACTGAATT
181	GCCGCATTGC	AGAGATATTG	TATTTAAGTG	CCTAGCTCGA	TACATAAACG	GGTCTCTCTG
241	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC
301	TCAATAAAGC	TTGCCCTTGAG	TGCTTCAAGT	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG
361	TAACTAGAGA	TCCCTCAGAC	CTTTTAGTC	AGTGTGAAA	ATCTCTAGCA	ATATAAATTA
421	AACAGGGACT	TGAAAGCGAA	AGGGAACCA	GAGGAGCTCT	CTCGACGCAG	GACTCGGCTT
481	GCTGAAGCGC	GCACGGCAAG	AGCGGAGGGG	CGGCGACTGG	TGAGTACGCC	AAAAATTTTG
541	ACTAGCGGAG	GCTAGAAGGA	GAGAGATGGG	TGCGAGAGCG	TCAGTATTA	GCGGGGAGA
601	ATTAGATCGC	GATGGGAAAA	AATTCGGTTA	AGGCCAGGGG	GAAGAAAAA	ATATAAATTA
661	AAACATATAG	TATGGGCAAG	CAGGGAGCTA	GAACGATTCT	CAGTTAATCC	TGGCCTGTTA
721	GAACATCAG	AAGGCTGTAG	ACAAATACTG	GGACAGCTAC	AACCATCCCT	TCAGACAGGA
781	TCAGAAGAAC	TTAGATCATT	ATATAATACA	GTAGCAACCC	TCTATTGTGT	GCATCAAAGG
841	ATAGAGATAA	AAGACACCAA	GGAGCTTTA	GACAAGATAG	AGGAAGAGCA	AAACAAAGT
901	AAGACCACCG	CACAGCAAGC	GGCCGCTGAT	CTTCAGACCT	GGAGGAGGAG	ATATGAGGGA
961	CAATTGGAGA	AGTGAATTAT	ATAAATATAA	AGTAGTAAAA	ATTGAACCAT	TAGGAGTAGC
1021	ACCCACCAAG	GCAAAGAGAA	GAGTGGTGCA	GAGAGAAAAA	AGAGCAGTGG	ATATGAGGGA
1081	TTTGTTCCTT	GGGTTCTTGG	GAGCAGCAGG	AAGCACTATG	GGCGCAGCGT	CAATGACGCT
1141	GACGGTACAG	GCCAGACAAT	TATTGTCTGG	TATAGTGCAG	CAGCAGAACA	ATTTGCTGAG
1201	GGCTATTGAG	GCGCAACAGC	ATCTGTTGCA	ACTCACAGTC	TGGGGCATCA	AGCAGCTCCA
1261	GGCAAGAATC	CTGGCTGTGG	AAAGATACCT	AAAGGATCAA	CAGCTCCTGG	GGATTTGGGG
1321	TTGCTCTGGA	AAACTCATTT	GCACCACTGC	TGTGCCCTTG	AATGCTAGTT	GGAGTAATAA
1381	ATCTCTGGAA	CAGATTGGGA	ATCACACGAC	CTGGATGGAG	TGGGACAGAG	AAATTAACAA
1441	TTACACAAGC	TTAATACACT	CCTTAATTGA	AGAATCGCAA	AACCAGCAAG	AAAAGAATGA
1501	ACAAGAATTA	TTGGAATTAG	ATAAATGGGC	AAGTTTGTGG	AATTGGTTTA	ACATAACAAA
1561	TTGGCTGTGG	TATATAAAAT	TATTCATAAT	GATAGTAGGA	GGCTTGGTAG	GTTTAAGAAT
1621	AGTTTTTGCT	GTACTTTCTA	TAGTGAATAG	AGTTAGGCAG	GGATATTCAC	CATTATCGTT
1681	TCAGACCCAC	CTCCCAACCC	CGAGGGGACC	CGACAGGCCC	GAAGGAATAG	AAGAAGAAGG
1741	TGGAGAGAGA	GACAGAGACA	GATCCATTCT	ATTAGTGAAC	GGATCTCGAC	GGTATCGCTA
1801	GCTTTTAAAA	GAAAAGGGGG	GATTGGGGGG	TACAGTGCAG	GGGAAAGAAT	AGTAGACATA
1861	ATAGCAACAG	ACATACAAAC	TAAAGAATTA	CAAAAACAAA	TTACAAAAAT	TCAAAATTTT
1921	ACTAGTATCA	ACTTTGTATA	GAAAAGTTGG	GCTCCGGTGC	CCGTCACTGG	GCAGAGCGCA
1981	CATCGCCAC	CATCGCCAC	AAGTTGGGGG	GAGGGGTCGG	CAATTGAACC	GGTGCCTAGA
2041	GAAGGTGGCG	CGGGGTAAAC	TGGGAAAGTG	ATGTCGTGTA	CTGGCTCCGC	CTTTTTCCCG
2101	AGGGTGGGGG	AGAACCGTAT	ATAAGTGCAG	TAGTCGCCGT	GAACGTTCTT	TTTCGCAACG
2161	GGTTTGCCGC	CAGAACACAG	GTAAGTGCCG	TGTGTGGTTC	CCGCGGGCCT	GGCCTCTTTA
2221	CGGGTTATGG	CCCTTCGCTG	CCTTGAATTA	CTTCCACCTG	GCTGCAGTAC	GTGATTCTTG
2281	ATCCCGAGCT	TCGGGTTGGA	AGTGGGTGGG	AGAGTTCGAG	GCCTTGCGCT	TAAGGAGCCC
2341	CTTCGCCCTCG	TGCTTGAGTT	GAGGCCTGGC	CTGGGCGCTG	GGGCCGCCGC	CTGCGAATCT
2401	GGTGGCACCT	TCGCGCCTGT	CTCGCTGCTT	TCGATAAGTC	TCTAGCCATT	TAAATTTTTT
2461	GATGACCTGC	TGCGACGCTT	TTTTTCTGGC	AAGATAGTCT	TGTAATGCG	GGCCAAGATC
2521	TGCACACTGG	TATTTCTGTT	TTTGGGGCCG	CGGGCGCCGA	CGGGGCCCGT	GCGTCCAGC
2581	GCACATGTTC	GGCGAGGCGG	GGCTGCGAG	CGCGGCCACC	GAGAATCGGA	CGGGGGTAGT
2641	CTCAAGCTGG	CCGGCCTGCT	CTGGTGCCTG	GTCTCGCGCC	GCCGTGTATC	GCCCCGCCCT
2701	GGCGGGCAAG	GCTGGCCCCG	TCGGCACCAG	TTGCGTGAGC	GGAAAGATGG	CCGCTTCCCG
2761	GCCCTGCTGC	AGGGAGCTCA	AAATGGAGGA	CGCGGCGCTC	GGGAGAGCGG	GCGGGTAGT

FIG. 4B-1

2821	CACCCACACA	AAGGAAAAGG	GCCTTTCGGT	CCTCAGCCGT	CGCTTCATGT	GACTCCACGG
2881	AGTACCGGGC	GCCGTCCAGG	CACCTCGATT	AGTTCTCGAG	CTTTTGGAGT	ACGTCGTCTT
2941	TAGGTTGGGG	GGAGGGGTTT	TATGCGATGG	AGTTTCCCCA	CACTGAGTGG	GTGGAGACTG
3001	AAGTTAGGCC	AGCTTGGCAC	TTGATGTAAT	TCTCCTTGGG	ATTTGCCCTT	TTTGAGTTTG
3061	GATCTTGTTT	CATTCTCAAG	CCTCAGACAC	TGGTTCAAAG	TTTTTTTCTT	CCATTTCAGG
3121	TGTCGTGACA	AGTTTGTACA	AAAAAGCAGG	CTGCCACCAT	GGAGTTCGGC	CTGAGCTGGG
3181	TGTTCTCGGT	GGCCCTGTTC	AGAGGCGTGC	AGTGCCAGGT	GAAGCTGCAG	GAGAGCGGGC
3241	CCGGCCTGGT	GGCCCCCAGC	CAGAGCCTGA	GCATCACCTG	CACCGTGAGC	GGCTTCAGCC
3301	TGACCGCCTA	CGGCGTGGAC	TGGGTGAGAC	AGCCCCCGG	CAAGGGCCTG	GAGTGGCTGG
3361	GCGTGATCTG	GGGCGGCGGC	AGAACCAACT	ACAACAGCGG	CCTGATGAGC	AGACTGAGCA
3421	TCAGAAAGGA	CAACAGCAAG	AGCCAGGTGT	TCCTGACCAT	GAACAGCCTG	CAGACCGACG
3481	ACACCGCCAA	GTACTACTGC	GTGAAGCACA	CCAAC TGGA	CGGCGGCTTC	GCCTACTGGC
3541	GCCAGGGCAC	CACCGTGACC	GTGAGCAGCC	CCAGCGTGTT	CCCCCTGGCC	CCCAGCAGCA
3601	AGAGCACCAG	CGGCGGCACC	GCGGCCCTGG	GCTGCCTGGT	GAAGGACTAC	TTCCCCGAGC
3661	CCGTGACCGT	GAGCTGGAAC	AGCGGCGCC	TGACCAGCGG	CGTGACACAC	TTCCCCGCCG
3721	TGCTGCAGAG	CAGCGCCTG	TACAGCCTGA	GCAGGTTGGT	GACCGTGCCC	AGGAGCAGCC
3781	TGGGCACCCA	GACCTACATC	TGCAACGTGA	ACCACAAGCC	CAGCAACACC	AAGGTGGACA
3841	AGAAGGTGGA	GCTGAAGACC	CCCCTGGGCG	ACACCACCCA	CACCTGCCCC	AGATGCCCCG
3901	AGCCCAAGAG	CTGCGACACC	CCCCCCCCCT	GCCCCAGATG	CCCCGAGCCC	AAGAGCTGCG
3961	ACACCCCCC	CCCCTGCCCC	AGATGCCCCG	AGCCCAAGAG	CTGCGACACC	CCCCCCCCCT
4021	GCCCCAGATG	CCCCGCCCCC	GAGTGCTGG	GCGGCCCCAG	CGTGTTCTTG	TTCCCCCCCCA
4081	AGCCCAAGGA	CACCCTGATG	ATCAGCAGAG	CCCCCGAGGT	GACCTGCGTG	GTGGTGGACG
4141	TGAGCCACGA	GGACCCCGAG	GTGAAGTTCA	ACTGGTACGT	GGACGCGGTG	GAGGTGCACA
4201	ACGCCAAGAC	CAAGCCCAGA	GAGGAGCAGT	ACAACAGCAC	CTACAGAGTG	GTGAGCGTGC
4261	TGACCGTGCT	GCACCAGGAC	TGGCTGAACG	GCAAGGACTA	CAAGTGCAAG	GTGAGCAACA
4321	AGGCCCTGCC	CGCCCCCATC	GAGAAGACCA	TCAGCAAGGC	CAAGGGCCAG	CCCAGAGAGC
4381	CCCAGGTGTA	CACCCTGCCC	CCCAGCAGAG	ACGAGCTGAC	CAAGAACCAG	GTGAGCCTGA
4441	CCTGCCTGGT	GAAGGGCTTC	TACCCAGCG	ACATCGCGGT	GGAGTGGGAG	AGCAACGGCC
4501	AGCCCGAGAA	CAACTACAAG	GACAAGAGCA	GATGGCAGCA	GGGCAACGTG	TTGAGTGGCA
4561	GCGTGATGCA	CGAGGCCCTG	CACAACCACT	ACACCAGAA	GAGCCTGAGC	CTGAGCCCCG
4621	AGCTGCAGCT	GGAGGAGAGC	TGCGCCGAGG	CCCAGGACGG	CGAGCTGGAC	GGCCTGTGGA
4681	CCACCATCAC	CATCTTCATC	ACCCTGTTCC	TGCTGAGCGT	GTGCTACAGC	GCCACCGTGA
4741	CCTTCTTCAA	GGTGAAGTGG	ATCTTCAGCA	GCGTGGTGGA	CCTGAAGCAG	ACCATCATCC
4801	CCGACTACAG	AAACATGATC	GGCCAGGGCG	CCTAAAACAA	CAACAATTGC	ATTCATTTTA
4861	TGTTTCAGGT	TCAGGGGGAG	GTGTGGGAGG	TTTTTTAAAG	CAAGTAAAC	CTCTACAAAT
4921	GTGGTACGCG	TTAACAACAA	CAATTGCATT	CATTTTATGT	TTGAGTTTCA	GGGGGAGGTG
4981	TGGGAGGTTT	TTTAAACGAA	GTAAACCTC	TACAAATGTG	GTACGCGTTA	CCCAGCTTTC
5041	TTGTACAAAG	TGGTAAATAG	ATAGAACAAC	AACAATTGCA	TTTATTTTGT	ATTTGAGGTT
5101	CAGGGGGAGG	TGTGGGAGGT	TTTTTAAAGC	AAGTAAACC	TCTACACTGA	CGGTACGCCG
5161	TAACAACAAC	AATTGCATTG	ATTTGTAGTT	TCAGGTTTCA	GGGGAGGTGT	GGGAGGTTTT
5221	TTAAAGCAAG	TTAAACCTCT	AAAATAGTGG	TACGCGTTAC	CCAGCTTTCT	TGTACAAAGT
5281	GGACCCAGCT	TTCTTGTACA	AAGTGGGCCC	CTCTCCCTCC	CCCCCCCCCTA	ACGTTACTGG
5341	CCGAAGCCGC	TTGGAATAAG	GCCGGTGTGC	GTTTGTCTAT	ATGTTATTTT	CCACCATATT
5401	GCCGTCTTTT	GGCAATGTGA	GGGCCCGGAA	ACCTGGCCCT	GTCTTCTTGA	ACGTTACTGG
5461	TAGGGGTCTT	TCCCTCTCTG	CCAAAGGAAT	GCAAGGTCTG	TTGAATGTCT	TGAAGGAAGC
5521	AGTTCCTCTG	GAAGCTTCTT	GAAGACAAAC	AACGTCTGTA	GCGACCTTTT	GCAGGCAGCG
5581	GAACCCCCCA	CCTGGCGACA	GGTGCTCTG	CGGCCAAAAG	CCAGCTGTAT	AAGATACACC
5641	TGCAAAAGGCG	GCACAACCCC	AGTGCCACGT	TGTGAGTTGG	ATAGTTGTGG	AAAGAGTCAA
5701	ATGGCTCTCC	TCAAGCGTAT	TCAACAAGGG	GCTGAAGGAT	GCCCAAGAGG	TACCCCATTG
5761	TATGGGATCT	GATCTGGGGC	CTCGGTGCAC	ATGCTTTACA	TGTGTTTAGT	CGAGGTAAAA

FIG. 4B-2

5881	AAAACGTCTA	GGCCCCCGA	ACCACGGGGA	CGTGGTTTTC	CTTTGAAAAA	CACGATGATA
5941	ATATGGCCAC	AACCATGGCC	ACCGACATGA	GAGTGGCCGC	CCAGCTGCTG	GGCCTGCTGC
6001	TGCTGTGGCT	GAGCGGCGCC	AGATGCGGCA	GCCCCGGCCA	GAGCGTGAGC	ATCAGCTGCA
6061	GCGGCAGCAG	CAGCAACATC	GGCAACAAC	ACGTGTACTG	GTACCAGCAC	CTGCCCGGCA
6121	CCGGGCCCAA	GCTGTGATC	TACAGCGACA	CCAAGAGACC	CAGCGGCGTG	CCCGACAGAA
6181	TCAGCGGCAG	CAAGAGCGGC	ACCAGCGCCA	GCCTGGCCAT	CAGCGGCGTG	CAGAGCGAGG
6241	ACGAGGCCGA	CTACTACTGC	GCCAGCTGGG	ACGACAGGCT	GGACGGCCCC	GTGTTGCGGG
6301	GCGGCACCAA	GCTGACCCTG	CTGGGCCAGC	CCAAGGCCAA	CCCCACCCTG	ACCTGTTC
6361	CCCCCAGCAG	CGAGGAGCTG	CAGGCCAACA	AGGCCACCCT	GGTGTGCCTG	ATCAGCGACT
6421	TCTACCCCGG	CGCCGTGACC	GTGGCCTGGA	AGGCCGACGG	CAGCCCCGTG	AAGGCCGGCG
6481	TGGAGACCAC	CAAGCCAGC	AAGCAGAGCA	ACAACAAGTA	CGCCGCCAGC	AGCTACCTGA
6541	GCCTGACCCC	CGAGCAGTGG	AAGAGCCACA	GAAGCTACAG	CTGCCAGGTG	ACCCACGAGG
6601	GCAGCACCGT	GGAGAAGACC	GTGGCCCCCA	CCGAGTGCA	CTAACAACTT	TATTATACAT
6661	AGTTGATCAA	TTCCAACCTT	ATTATACATA	GTTGATCAAT	TCCGATAATC	AACCTCTGGA
6721	TTACAAAATT	TGTGAAAGAT	TGACTGGTAT	TCTTAACTAT	GTTGCTCCTT	TTACGCTATG
6781	TGGATACGCT	GCTTTAATGC	CITTTGATCA	TGCTATTGCT	TCCCGTATGG	CTTTCATTTT
6841	CTCCTCCTTG	TATAAATCCT	GTTGTGCTGC	TCTTTATGAG	GAGTTGTGGC	CCGTTGTGAG
6901	GCAACGTGGC	GTGGTGTGCA	CTGTGTTTGC	TGACGCAACC	CCCACTGGTT	CGGGCATTGC
6961	CACCACCTGT	CAGCTCCTTT	CCGGGACTTT	CGCTTTCCTC	CTCCCTATTG	CCACGGCGGA
7021	ACTCATCGCG	GCCTGCGTTG	CCCGCTGCTG	GACAGGGCCT	CGGCTGTTGG	GCACTGACAA
7081	TTCCGTGGTG	TTGTGCGGGA	AGCTGACGTC	CTTTCATG6	CTGCTCGCCT	GTGTTGCCAC
7141	CTGGATTCTG	CGCGGGACGT	CCTTCTGCTA	CCTCCCTTC6	GCCCTCAATC	GCCCTCAATC
7201	TCCTTCCCGC	CGGCTCTGCG	GCCTCTTCGG	GCCTCTTC6	CGTCTTCGCC	TTCGCCCTCA
7261	GACGACTCGG	ATCTCCCTTT	GGGCCGCTC	CCCGCATCG6	GAATTCCCGC	GGTTCGCTTT
7321	AAGACCAATG	ACTTACAAGG	CAGCTGTAGA	TCTTAGCCAC	TTTTTAAAG	AAAAGGGGGG
7381	ACTGGAAGGG	CTAATCACT	CCCAACGAAG	ACAAGATCT6	CTTTTGTGTT	GTAAGGGGTC
7441	TCTCTGGTTA	GACCAGATCT	GAGCCTGGGA	GCTCTCTGGC	TAAGTAGGGA	ACCCACTGCT
7501	TAAGCCTCAA	TAAAGCTTGC	CTTGAGTGCT	TCAAGTAGT6	TGTGCCCCGT	TGTTGTGTGA
7561	CTCTG6TAAC	TAGAGATCCC	TCAGACCCTT	TTAGTCACT6	TGAAAATCT	CTAGCAGTAG
7621	TAGTTATGTT	CATCTTATTA	TTAGTATTTT	ATAACTTGCA	AAGAAATGAA	TATCAGAGAG
7681	TGAGAGGAAC	TTGTTTATTG	CAGCTTATAA	TGGTTACAAA	TAAAGCAATA	GCATCACAAA
7741	TTTACAAAAT	AAAGCATTTT	TTTCACTGCA	TTCTAGTTGT	GGTTTGTCCA	AACTCATCAA
7801	TGTATCTTAT	CATGTGTGGC	TCTAGCTATC	CCGCCCTTAA	CTCCGCCCAT	CCCGCCCTTA
7861	ACTCCGCCCA	GTTCGCCCCA	TTCTCGCCCC	CATGGCTGAC	TATTTTTTTT	TATTTATGCA
7921	GAGGCCGAGG	CCGCCCTGGC	CTCTGAGCTA	TTCCAGAAAG	ACTGAGGAGG	CTTTTTTGCA
7981	GGCCTAGGGA	CGTACCCAAT	TCGCCCTATA	GTGAGTCGTA	TTACGCGCGC	TCACTGGCCG
8041	TCGTTTTACA	ACGTCGTGAC	TGGGAAAACC	CTGGCGTTAC	CCAACTTAAT	CGCCTTGCA
8101	CACATCCCCC	TTTCGCCAGC	TGGCGTAATA	GCGAAGAGGC	CCGCACCGAT	CGCCCTTCCC
8161	AACAGTTGCG	CAGCCTGAAT	GGCGAATGGG	ACGCGCCCT6	TAGCGGCGCA	TTAAGCGCGG
8221	CGGCTGTGGT	GGTTACGCGC	AGCGTGACCG	CTACACTTGC	CAGCGCCCTA	CGCGCCGCTC
8281	CTTTCGCTTT	CTTCCCTTCC	TTTCTCGCCA	CGTTCCGCCG	CTTTCCTCCG	CAAGCTCTAA
8341	ATCGGGGGCT	CCCTTTAGGG	TTCCGATTTA	GTGCTTTACG	GCACCTCGAC	CCCAAAAAAC
8401	TTGATTAGGG	TGATGTTTCA	CGTAGTGGGC	CATCGCCCT6	ATAGACGGTT	TTTCGCCCTT
8461	TGACGTTGGA	GTCCACGTTT	TTTAATAGTG	GACTCTTGTT	CCAACTGGA	ACAACACTCA
8521	ACCCTATCTC	GGTCTATTCT	TTTGATTTAT	AAGGGATTTT	GCCGATTTTC	GCCTATTGGT
8581	TAAAAAATGA	GCTGATTTAA	CAAAAAATTTA	ACGCGAATTT	TAACAAAATA	TTAACGCTTA
8641	CAATTTAGGT	GGCACTTTTC	GGGGAATGT	GCGCGGAACC	CCTATTTGTT	TATTTTCTTA
8701	AATACATTCA	AATATGTATC	CGCTCATGAG	ACAATAACCC	TGATAAATGC	TTCAATAATA
8761	TTGAAAAAGG	AAGAGTATGA	GTATTCAACA	TTTCCGTGTC	GCCCTTATTC	CCTTTTTTGC
8821	GGCATTTTGC	CTTCTGTTT	TTGCTCACCC	AGAAACGCT6	GTGAAAGTAA	AAGATGCTGA
8881	AGATCAGTTG	GGTGACGAG	TGGTTACAT	CGAACTGGAT	CTCAACAGCG	GTAAGATCCT

FIG. 4B-3

8941	TGAGAGTTTT	CGCCCCGAAG	AACGTTTTCC	AATGATGAGC	ACTTTTAAAG	TTCTGCTATG
9001	TGGCGCGGTA	TTATCCCGTA	TTGACGCCGG	GCAAGAGCAA	CTCGGTCGCC	GCATACACTA
9061	TTCTCAGAAT	GACTTGTTG	AGTACTCACC	AGTCACAGAA	AAGCATCTTA	CGGATGGCAT
9121	GACAGTAAGA	GAATTATGCA	GTGCTGCCAT	AACCATGAGT	GATAAACTG	CGGCCAACTT
9181	ACTTCTGACA	ACGATCGGAG	GACCGAAGGA	GCTAACCCCT	TTTTTGACA	ACATGGGGGA
9241	TCATGTAACT	CGCCTTGATC	GTTGGGAACC	GGAGCTGAAT	GAAGCCATAC	CAAACGACGA
9301	GCGTGACACC	ACGATGCCTG	TAGCAATGGC	AACAACGTTG	CGCAAATAT	TAAGTGGCGA
9361	ACTACTTACT	CTAGCTTCCC	GGCAACAATT	AATAGACTGG	ATGGAGGCGG	ATAAAGTTGC
9421	AGGACCACTT	CTGCGCTCGG	CCCTTCGGGC	TGGCTGGTTT	ATTCGTGATA	AATCTGGAGC
9481	CGGTGAGCGT	GGGTCTCGCG	GTATCATTGC	AGCACTGGGG	CCAGATGGTA	AGCCCTCCCG
9541	TATCGTAGTT	ATCTACACGA	CGGGGAGTCA	GGCAACTATG	GATGAACGAA	ATAGACAGAT
9601	CGCTGAGATA	GGTGCCCTCAC	TGATTAAGCA	TTGGTAACTG	TCAGACCAAG	TTTACTCATA
9661	TATACTTTAG	ATTGATTTAA	AAC TTCATT	TTAATTTAAA	AGGATCTAGG	TGAAGATCCT
9721	TTTTGATAAT	CTCATGACCA	AAATCCCTTA	ACGTGAGTTT	TCGTTCCACT	GAGCGTCAGA
9781	CCCCGTAGAA	AAGATCAAAG	GATCTTCTTG	AGATCCTTTT	TTTCTGCGCG	TAATCTGCTG
9841	CTTCCAACA	AAAAAACAC	CGTACCAGC	GGTGGTTTGT	TTGCCGGATC	AAGAGCTACC
9901	AAC TCTTTT	CCGAAGGTAA	CTGGCTTCAG	CAGAGCGCAG	ATACCAAATA	CTGTTCTTCT
9961	AGTGTAGCCG	TAGTTAGGCC	ACCACTTCAA	GAAC TCTGTA	GCACCGCCTA	CATACCTCGC
10021	TCTGCTAATC	CTGTTACCA	TGGCTGCTGC	CAGTGCCGAT	AAGTCGTGTC	TTACCGGGTT
10081	GGACTCAAGA	CGATAGTTAC	CGGATAAGGC	GCAGCGGTCG	GGCTGAACGG	GGGGTTCGTG
10141	CACACAGCCC	AGCTTGAGC	GAACGACCTA	CACCGAACTG	AGATACCTAC	AGCGTGAGCT
10201	ATGAGAAAGC	GCCACGCTTC	CCGAAGAGAG	AAAGGCGGAC	AGGTATCCGG	TAAGCGGCAG
10261	GGTCGGAACA	GGAGAGCGCA	CGAGGGAGCT	TCCAGGGGGA	AACGCTTGGT	ATCTTTATAG
10321	TCCTGTGCGG	TTTCGCCACC	TCTGACTTGA	GCCTCGATTT	TTGTGATGCT	CGTCAGGGGG
10381	GCGGAGCCTA	TGGAAAAACG	CCAGCAACGC	GGCCTTTTTA	CGGTTCTTGG	CCTTTTGCTG
10441	GCCTTTTGCT	CACATGTTCT	TTCTGCGTT	ATCCCCTGAT	TCGTGGGATA	ACCGTATTAC
10501	CGCCTTTGAG	TGAGCTGATA	CCGCTCGCCG	CAGCCGAACG	ACCGAGCGCA	GCGAGTCAGT
10561	GAGCGAGGAA	GCGGAAGAGC	GCCCAATACG	CAAACCGCCT	CTCCCCGCGC	GTTGGCCGAT
10621	TCATTAATGC	AGCTGGCAGC	ACAGGTTTCC	CGACTGGAAA	GCGGGCAGTG	AGCGCAACGC
10681	AATTAATGTG	AGTTAGCTCA	CTCATTAGGC	ACCCAGGGCT	TTACACTTTA	TGCTTCGGC
10741	TCGTATGTTG	TGTGGAATTG	TGAGCGGATA	ACAATTTAC	ACAGGAAACA	GCTATGACCA
10801	TGATTACGCC	AAGCGCGCAA	TTAACCCTCA	CTAAAGGGAA	CAAAAGCTGG	AGCTGCAAGC
10861	TT					

FIG. 5A

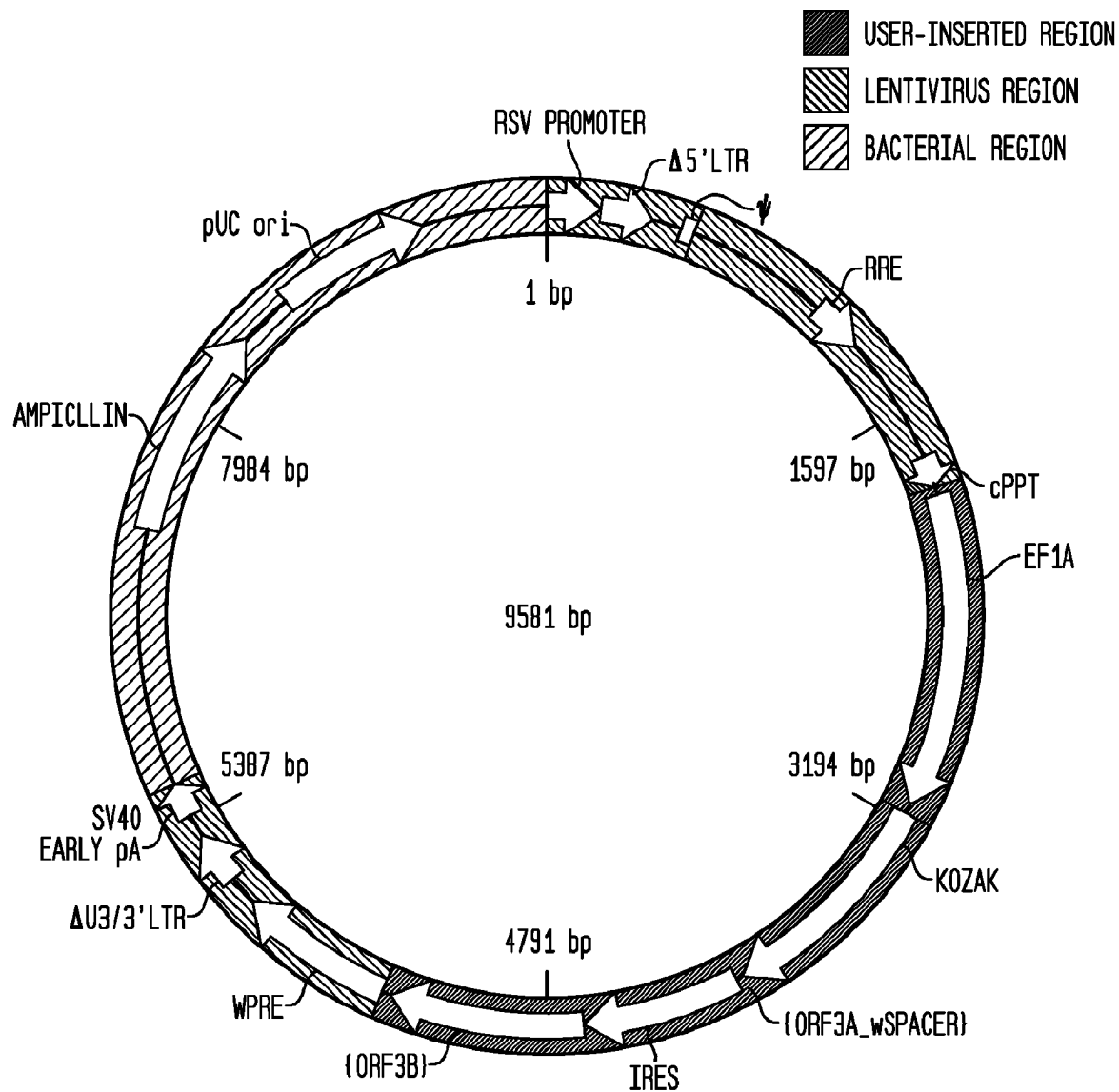


FIG. 5B

SEQ ID NO. 49

1	AATGTAGTCT	TATGCAATAC	TCTTGTAGTC	TTGCAACATG	GTAACGATGA	GTTAGCAACA
61	TGCCTTACAA	GGAGAGAAAA	AGCACCGTGC	ATGCCGATTG	GTGGAAGTAA	GGTGTACGA
121	TCGTGCCCTTA	TTAGGAAGGC	AACAGACGGG	TCTGACATGG	ATTGGACGAA	CCACTGAATT
181	GCCGCATTGC	AGAGATATTG	TATTTAAGTG	CCTAGCTCGA	TACATAAACG	GGTCTCTCTG
241	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC
301	TCAATAAAGC	TTGCCTTGAG	TGCTTCAAGT	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG
361	TAACTAGAGA	TCCCTCAGAC	CCTTTTAGTC	AGTGTGGA	ATCTCTAGCA	GTGGCGCCCG
421	AACAGGGACT	TGAAAGCGAA	AGGGAACCA	GAGGAGCTCT	CTCGACGCA	GACTCGGCTT
481	GCTGAAGCGC	GCACGGCAAG	AGGCGAGGGG	CGGCGACTGG	TGAGTACGCC	AAAAATTTTG
541	ACTAGCGGAG	GCTAGAAGGA	GAGAGATGGG	TGCGAGAGCG	TCAGTATTAA	GCGGGGAGAA
601	ATTAGATCGC	GATGGGAAAA	AATTCGGTTA	AGGCCAGGGG	GAAAGAAAAA	ATATAAATTA
661	AAACATATAG	TATGGGCAAG	CAGGGAGCTA	GAACGATTGG	CAGTTAATCC	TGGCCTGTTA
721	GAAACATCAG	AAGGCTGTAG	ACAAATACTG	GGACAGCTAC	AACCATCCCT	TCAGACAGGA
781	TCAGAAGAAC	TTAGATCATT	ATATAATACA	GTAGCAACCC	TCTATTGTGT	GCATCAAAGG
841	ATAGAGATAA	AAGACACCAA	GGAAAGCTTTA	GACAAGATAG	AGGAAGAGCA	AAACAAAAGT
901	AAGACCACCG	CACAGCAAGC	GGCCGCTGAT	CTTCAGACCT	GGAGGAGGAG	ATATGAGGGA
961	CAATTGGAGA	AGTGAATTAT	ATAAATATAA	AGTAGTAAAA	ATTGAACCAT	TAGGAGTAGC
1021	ACCCACCAAG	GCAAAGAGAA	GAGTGGTGCA	GAGAGAAAAA	AGAGCAGTGG	GAATAGGAGC
1081	TTTGTTCCCT	GGGTTCTTGG	GAGCAGCAGG	AAGCACTATG	GGCGCAGCGT	CAATGACGCT
1141	GACGGTACAG	GCCAGACAAT	TATTGTCTGG	TATAGTGCAG	CAGCAGAACA	ATTTGCTGAG
1201	GGCTATTGAG	GCGCAACAGC	ATCTGTTGCA	ACTCACAGTC	TGGGGCATCA	AGCAGCTCCA
1261	GGCAAGAATC	CTGGCTGTGG	AAAGATACCT	AAAGGATCAA	CAGCTCCTGG	GGATTTGGGG
1321	TTGCTCTGGA	AAACTCATTT	GCACCACTGC	TGTGCCTTGG	AATGCTAGTT	GGAGTAATAA
1381	ATCTCTGGAA	CAGATTTGGA	ATCACACGAC	CTGGATGGAG	TGGGACAGAG	AAATTAACAA
1441	TTACACAAGC	TTAATACACT	CCTTAATTGA	AGAATCGCAA	AACCAGCAAG	AAAAGAATGA
1501	ACAAGAATTA	TTGGAATTAG	ATAAATGGGC	AAGTTTGTGG	AATTGGTTTA	ACATAACAAA
1561	TTGGCTGTGG	TATATAAAAT	TATTCATAAT	GATAGTAGGA	GGCTTGGTAG	GTTTAAGAAT
1621	AGTTTTTGCT	GTACTTTCTA	TAGTGAATAG	AGTTAGGCAG	GGATATTCAC	CATTATCGTT
1681	TCAGACCCAC	CTCCCAACCC	CGAGGGGACC	CGACAGGCC	GAAGGAATAG	AAGAAGAAGG
1741	TGGAGAGAGA	GACAGAGACA	GATCCATTGG	ATTAGTGAAC	GGATCTCGAC	GGTATCGCTA
1801	GCTTTTAAAA	GAAAAGGGGG	GATTGGGGGG	TACAGTGCAG	GGGAAAGAAT	AGTAGACATA
1861	ATAGCAACAG	ACATACAAC	TAAAGAATTA	CAAAAACAAA	TTACAAAAAT	TCAAAATTTT
1921	ACTAGTATCA	ACTTTGTATA	GAAAAGTTGG	GCTCCGGTGC	CCGTCAGTGG	GCAGAGCGCA
1981	CATCGCCAC	AGTCCCGAG	AAGTTGGGGG	GAGGGGTGCG	CAATTGAACC	GGTGCTTAGA
2041	GAAGGTGGCG	CGGGGTAAC	TGGGAAAGTG	ATGTCGTGTA	CTGGCTCCGC	CTTTTCCCG
2101	AGGGTGGGGG	AGAACCGTAT	ATAAGTGCA	TAGTCGCCGT	GAACGTTCTT	TTTCGCAACG
2161	G6TTTGCCGC	CAGAACACAG	GTAAGTGCCG	TGTGTG6TTC	CCGCGGGCCT	GGCCTCTTTA
2221	CGGGTTATGG	CCCTTGCGTG	CCTTGAATTA	CTCCACCTG	GCTGCAGTAC	GTGATTCTTG
2281	ATCCCGAGCT	TCGGGTTGGA	AGTGGGTGGG	AGAGTTGAG	GCCTTGCGCT	TAAGGAGCCC
2341	CTTCGCTCG	TGCTTGAGTT	GAGGCTGGC	CTGGGCGCTG	GGGCCCGCGC	GTGCGAATCT
2401	GGTGGCACCT	TGCGGCTGT	CTCGCTGCTT	TCGATAAGTC	TCTAGCCATT	TAAAAATTTT
2461	GATGACCTGC	TGCGACGCTT	TTTTCTGGC	AAGATAGTCT	TGTAAATGCG	GGCCAAGATC
2521	TGCACACTGG	TATTTGCGTT	TTTGGGGCCG	CGGGCGGCGA	CGGGGCGCGT	GCCTCCACGC
2581	GCACATGTT	GGCGAGGCGG	GGCCTGCGAG	CGCGGCCACC	GAGAAATCGGA	CGGGGGTAGT
2641	CTCAAGCTGG	CCGGCTGCT	CTGGTGCTG	GTCTCGCGCC	GCCGTGTATC	GCCCCGCCCT
2701	GGGCGGCAAG	GCTGGCCCGG	TCGGCACCAG	TTGCGTGAGC	GGAAAGATGG	CCGCTTCCCG
2761	GCCCTGCTGC	AGGGAGCTCA	AAATGGAGGA	CGCGGCGCTC	GGGAGAGCGG	GCGGGTGAGT
2821	CACCCACACA	AAGGAAAAGG	GCCTTTCGCT	CCTCAGCCGT	CGCTTCATGT	GACTCCACGG
2881	AGTACCGGGC	GCCGTCCAGG	CACCTCGATT	AGTTCTCGAG	CTTTTGGAGT	ACGTCGTCTT

FIG. 5B-1

2941	TAGGTTGGGG	GGAGGGGTTT	TATGCGATGG	AGTTTCCCA	CACTGAGTGG	GTGGAGACTG
3001	AAGTTAGGCC	AGCTTGGCAC	TTGATGTAAT	TCTCCTTGA	ATTGCCCCTT	TTTGAGTTTG
3061	GATCTTGGTT	CATTCTCAAG	CCTCAGACAG	TGGTTCAAAG	TTTTTTTCTT	CCATTTCAGG
3121	TGTCGTGACA	AGTTTGTACA	AAAAAGCAGG	CTGCCACCAT	GTGGCTGCAG	AGCCTGCTGC
3181	TGCTGGGCAC	CGTGGCCTGC	AGCATCAGCG	CCCCCGCCAG	AAGCCCCAGC	CCAGCAGCCC
3241	AGCCCTGGGA	GCACGTGAAC	GCCATCCAGG	AGGCCAGAAG	ACTGCTGAAC	CTGAGCAGAG
3301	ACACCGCCGC	CGAGATGAAC	GAGACCGTGG	AGGTGATCAG	CGAGATGTTT	GACCTGCAGG
3361	AGCCACCTG	CCTGCAGACC	AGACTGGAGC	TGTACAAGCA	GGCCTGAGA	GGCAGCCTGA
3421	CCAAGCTGAA	GGGCCCCCTG	ACCATGATGG	CCAGCCACTA	CAAGCAGCAC	TGCCCCCCA
3481	CCCCGAGAC	CAGCTGCGCC	ACCCAGATCA	TCACCTTCGA	GAGCTTCAAG	GAGAACCTGA
3541	AGGACTTCCT	GCTGGTGATC	CCCTTCGACT	GCTGGGAGCC	CGTGCAGGAG	TAAAACAACA
3601	ACAATTGCAT	TCATTTTATG	TTTCAGGTTT	AGGGGGAGGT	GTGGGAGGTT	TTTTAAAGCA
3661	AGTAAACCT	CTACAAATGT	GGTACGCGTT	AACAACAACA	ATTGCATTCA	TTTTATGTTT
3721	CAGGTTCAGG	GGGAGGTGTG	GGAGGTTTTT	TAAAGCAAGT	AAAACCTCTA	CAAAATGTGGT
3781	ACGCGTTACC	CAGCTTTCTT	GTACAAAGTG	GTAAATAGAT	AGAACAACAA	CAATTGCATT
3841	CATTTTGTAT	TTCAGGTTCA	GGGGGAGGTG	TGGGAGGTTT	TTTAAAGCAA	GTAAACCTC
3901	TACACTGACG	GTACGCGTTA	ACAACAACAA	TTGCATTGAT	TTGTAGTTTC	AGGTTCAGGG
3961	GGAGGTGTGG	GAGGTTTTTT	AAAGCAAGTT	AAACCTCTAA	AATAGTGGTA	CGCGTTACCC
4021	AGCTTCTTG	TACAAAGTGG	ACCCAGCTTT	CTTGTACAAA	GTGGGCCCTT	CTCCCTCCCC
4081	CCCCCTAAC	GTTACTGACC	GAAGCCGCTT	GGATAAGGC	CGGTGTGCGT	TTGTCTATAT
4141	GTTATTTTCC	ACCATATTGC	CGTCTTTTGG	CAATGTGAGG	GGCCGGAAC	CTGGCCCTGT
4201	CTTCTTGACG	AGCATTCTTA	GGGGTCTTTC	CCCTCTCGCC	AAAGGAATGC	AAGGTCGTGT
4261	GAATGTCGTG	AAGGAAGCAG	TTCTCTGGA	AGCTTCTTGA	AGACAAACAA	CGTCTGTAGC
4321	AGCCCTTTGC	AGGCAGCGGA	ACCCCCACCC	TGGCGACAGG	TGCCCTCTCG	GCCAAAAGCC
4381	ACGTGTATAA	GATACACCTG	CAAAGGCGGC	ACAACCCAG	TGCCACGTTG	TGAGTTGGAT
4441	AGTTGTGGAA	AGAGTCAAAT	GGCTCTCCTC	AAGCGTATTC	AACAAGGGGC	TGAAGGATGC
4501	CCAGAAGGTA	CCCCATTGTA	TGGGATCTGA	TCTGGGGCCT	CGGTGCACAT	GCTTTACATG
4561	TGTTTAGTCG	AGGTTAAAAA	AACGTCTAGG	CCCCCGAAC	CACGGGGACG	TGGTTTTCTT
4621	TTGAAAAACA	CGATGATAAT	ATGGCCACAA	CCATGGCCAC	CGTGCTGGCC	CCCGCTGGA
4681	GCCCCACCAC	CTACCTGCTG	CTGCTGCTGC	TGCTGAGCAG	CGGCTTGAGC	GGCACCAGG
4741	ACTGCAGCTT	CCAGCACAGC	CCATCAGCA	GCGACTTCGC	CGTGAAGATC	AGAGAGCTGA
4801	GCGACTACCT	GCTGCAGGAC	TACCCGCTGA	CCGTGGCCAG	CAACCTGCAG	GACGAGGAGC
4861	TGTGCGCGGG	CCTGTGGAGA	CTGGTGCTGG	CCCAGAGATG	GATGGAGAGA	CTGAAGACCG
4921	TGGCGGGCAG	CAAGATGCAG	GGCTGCTGG	AGAGAGTGAA	CACCGAGATC	CACTTCGTGA
4981	CCAAGTGCGC	CTTCCAGCCC	CCCCCAGCT	GCCTGAGATT	CGTGCAGACC	AACATCAGCA
5041	GACTGCTGCA	GGAGACCAGC	GAGCAGCTGG	TGGCCCTGAA	GCCTGGATC	ACCAGACAGA
5101	ACTTCAGCAG	ATGCCTGGAG	CTGCAGTGCC	AGCCCGACAG	CAGCACCTTG	CCCCCCCCCT
5161	GGAGCCCCAG	ACCCCTGGAG	GCCACCGCCC	CCACCGCCCC	CCAGCCCCCT	CTGCTGCTGC
5221	TGCTGCTGCT	GCCCGTGGGC	CTGCTGCTGC	TGGCCGCGCG	CTGGTGCCTG	CACTGGCAGA
5281	GAACCAGAAG	AAGAACCCCC	AGACCGGGCG	AGCAGGTGCC	CCCCGTGCC	AGCCCCCAGG
5341	ACCTGCTGCT	GGTGGAGCAC	TAACAACCTT	ATTATACATA	GTTGATCAAT	TCCAACCTTA
5401	TTATACATAG	TTGATCAATT	CCGATAATCA	ACCTCTGGAT	TACAAAATTT	GTGAAAGATT
5461	GACTGGTATT	CTTAACATAG	TTGCTCCTTT	TACGCTATGT	GGATACGCTG	CTTTAATGCC
5521	TTTGTATCAT	GCTATTGCTT	CCCGTATGGC	TTTCATTTTC	TCCTCCTTGT	ATAAATCCTG
5581	GTTGCTGTCT	CTTTATGAGG	AGTTGTGGCC	CGTTGTGAGG	CAACGTGGCG	TGGTGTGCAC
5641	TGTGTTTGCT	GACGCAACCC	CCACTGCTTG	GGGCATTGCC	ACCACCTGTC	AGCTCCTTTC
5701	CGGGACTTTT	GCTTTCCCCC	TCCCTATTGC	CACGGCGGAA	CTCATCGCCG	CCTGCCCTTG
5761	CCGCTGCTGG	ACAGGGGCTC	GGCTGTTGGG	CAGTGACAAT	TCCGTGGTGT	TGTCGGGGAA
5821	GCTGACGTCC	TTTCCATGGC	TGCTGCGCTG	TGTTGCCACC	TGGATTCTGC	GCGGGACGTC
5881	CTTCTGCTAC	GTCCCTTCGG	CCCTCAATCC	AGCGGACCTT	CCTTCCCGCG	GCCTGCTGCC
5941	GGCTCTGCGG	CCTCTTCGGC	GTCTTCGCCT	TGCGCCTCAG	ACGAGTCGGA	TCTCCCTTTG
6001	GGCCGCTCC	CCGCATCGGG	AATTCGCGCG	GTTGCTTTTA	AGACCAATGA	CTTACAAGGC
6061	AGCTGTAGAT	CTTAGCCACT	TTTTAAAGA	AAAGGGGGGA	CTGGAAGGGC	TAATTCACTC

FIG. 5B-2

6121	CCAACGAAGA	CAAGATCTGC	TTTTTGCTTG	TACTGGGTCT	CTCTGGTTAG	ACCAGATCTG
6181	AGCCTGGGAG	CTCTCTGGCT	AACTAGGGAA	CCCACTGCTT	AAGCCTCAAT	AAAGCTTGCC
6241	TTGAGTGCTT	CAAGTAGTGT	GTGCCCCTCT	GTTGTGTGAC	TCTGGTAAC	AGAGATCCCT
6301	CAGACCCTTT	TAGTCAGTGT	GGAAATCTC	TAGCAGTAGT	AGTTCATGTC	ATCTTATTAT
6361	TCAGTATTTA	TAAGTTGCAA	AGAAATGAAT	ATCAGAGAGT	GAGAGGAAC	TGTTTATTGC
6421	AGCTTATAAT	GGTTACAAAT	AAAGCAATAG	CATCACAAAT	TTCACAAATA	AAGCATTTTT
6481	TTCAGTGCAT	TCTAGTTGTG	GTTTGTCCTA	ACTCATCAAT	GTATCTTATC	ATGTCTGGCT
6541	CTAGCTATCC	CGCCCCTAAC	TCCGCCCATC	CCGCCCTAA	CTCCGCCAG	TCCGCCCAT
6601	TCTCCGCCCC	ATGGCTGACT	AATTTTTTTT	ATTTATGCAG	AGGCCGAGGC	CGCCTCGGCC
6661	TCTGAGCTAT	TCCAGAAGTA	GTGAGGAGGC	TTTTTTGSAG	GCCTAGGGAC	GTACCCAAAT
6721	CGCCCTATAG	TGAGTCGTAT	TACGCGCGCT	CACTGGCCGT	CGTTTTACAA	CGTCGTGACT
6781	GGGAAAACCC	TGGCGTTACC	CAACTTAATC	GCCTTGACGC	ACATCCCCCT	TTCGCCAGCT
6841	GGCGTAATAG	CGAAGAGGCC	CGCACCGATC	GCCCTTCCCA	ACAGTTGCGC	AGCCTGAATG
6901	GCGAATGGGA	CGCGCCCTGT	AGCGGCGCAT	TAAGCGCGGC	GGGTGTGGTG	GTTACGCGCA
6961	GCGTGACCGC	TACACTTGCC	AGCGCCCTAG	CGCCCGCTCC	TTTCGCTTTC	TTCCCTTCCT
7021	TTCTCGCCAC	GTTTCGCCGC	TTTCCCGTCT	AAGCTCTAAA	TCGGGGGCTC	CCTTTAGGGT
7081	TCCGATTTAG	TGCTTTACGG	CACCTCGACC	CCAAAAACT	TGATTAGGGT	GATGGTTCAC
7141	GTAGTGGGCC	ATCGCCCTGA	TAGACGGTTT	TTCGCCCTTT	GACGTTGGAG	TCCACGTTCT
7201	TTAATAGTGG	ACTCTTGTTT	CAAACTGGAA	CAAACTCAA	CCCTATCTCG	GTCTATTCTT
7261	TTGATTTATA	AGGGATTTTG	CCGATTTCCG	CCTATTGGTT	AAAAAATGAG	CTGATTTAAC
7321	AAAAATTTAA	CGCGAATTTT	AACAAAATAT	TAACGCTTAC	AATTTAGGTG	GCATTTTCG
7381	GGGAAATGTG	CGCGGAACCC	CTATTTGTTT	ATTTTCTAA	ATACATTCAA	ATATGTATCC
7441	GCTCATGAGA	CAATAACCTT	GATAAATGCT	TCAATAATAT	TGAAAAAGGA	AGAGTATGAG
7501	TATTCAACAT	TTCCGTGTCT	CCCTTATTCC	CTTTTTTGCG	GCATTTTGCC	TTCTGTTTT
7561	TGCTCACCCA	GAAACGCTGG	TGAAAGTAAA	AGATGCTGAA	GATCAGTTGG	GTGCACGAGT
7621	GGGTTACATC	GAACTGGATC	TCAACAGCGG	TAAGATCCTT	GAGAGTTTTC	GCCCCGAAGA
7681	ACGTTTTTCCA	ATGATGAGCA	CTTTTAAAGT	TCTGCTATGT	GGCGCGGTAT	TATCCCGTAT
7741	TGACGCCGGG	CAAGAGCAAC	TCGGTCGCCG	CATACACTAT	TCTCAGAATG	ACTTGGTTGA
7801	GTACTACCCA	GTACACAGAA	AGCATCTTAC	GGATGGCATG	ACAGTAAGAG	AATTATGCAG
7861	TGCTGCCATA	ACCATGAGTG	ATAACACTGC	GGCCAACTTA	CTTCTGACAA	CGATCGGAGG
7921	ACCGAAGGAG	CTAACCGCTT	TTTTGCACAA	CATGGGGGAT	CATGTAAC	GCCTTGATCG
7981	TTGGGAACCG	GAGCTGAATG	AAGCCATACC	AAACGACGAG	CGTGACACCA	CGATGCCTGT
8041	AGCAATGGCA	ACAACGTTGC	GCAAACTATT	AACTGGCGAA	CTACTTACTC	TAGCTTCCCG
8101	GCAACAATTA	ATAGACTGGA	TGGAGGCGGA	TAAAGTTGCA	GGACCACTTC	TGCCTCGGCG
8161	CCTTCCGGCT	GGCTGGTTTA	TTGCTGATAA	ATCTGGAGCC	GGTGAGCGTG	GGTCTCGCGG
8221	TATCATTGCA	GCACTGGGGC	CAGATGGTAA	GCCCTCCCGT	ATCGTAGTTA	TCTACACGAC
8281	GGGGAGTCAG	GCAACTATGG	ATGAACGAAA	TAGACAGATC	GCTGAGATAG	GTGCCTCACT
8341	GATTAAGCAT	TGGTAAGTGT	CAGACCAAGT	TTACTCATAT	ATACTTTAGA	TTGATTTAAA
8401	ACTTCATTTT	TAATTTAAAA	GGATCTAGGT	GAAGATCCTT	TTTGATAATC	TCATGACCAA
8461	AATCCCTTAA	CGTGAGTTTT	CGTTCCACTG	AGCGTCAGAC	CCCGTAGAAA	AGATCAAAGG
8521	ATCTTCTTGA	GATCCTTTTT	TTCTGCCTGT	AATCTGCTGC	TTGCAACAA	AAAAACCACC
8581	GCTACGAGCG	GTGGTTTGT	TGCCGGATCA	AGAGCTACCA	ACTCTTTTTT	CGAAGGTAAC
8641	TGGCTTCAGC	AGAGCGCAGA	TACCAAAATC	TGTTCTTCTA	GTGTAGCCGT	AGTTAGGCCA
8701	CCACTTCAAG	AACTCTGTAG	CACCGCCTAC	ATACCTCGCT	CTGCTAATCC	TGTTACCACT
8761	GGCTGCTGCC	AGTGGCGATA	AGTCGTGTCT	TACCGGGTTG	GACTCAAGAC	GATAGTTACC
8821	GGATAAGGCG	CAGCGGTCGG	GCTGAACGGG	GGGTTTCTGC	ACACAGCCCA	GCTTGGAGCG
8881	AACGACCTAC	ACCGAACTGA	GATACCTACA	GCGTGAGCTA	TGAGAAAGCG	CCACGCTTCC
8941	CGAAGAGAGA	AAGGCGGACA	GGTATCCGGT	AAGCGGCAGG	GTGGAACAG	GAGAGCGCAC
9001	GAGGGAGCTT	CCAGGGGGAA	ACGCTTGGTA	TCTTTATAGT	CCTGTGCGGT	TTCGCCACCT
9061	CTGACTTGAG	CGTCGATTTT	TGTGATGCTC	GTGAGGGGGG	CGGAGCCTAT	GGAAAAACGC
9121	CAGCAACGCG	GCCTTTTTAC	GGTTCCTGGC	CTTTTGCTGG	CCTTTTGCTC	ACATGTTCTT
9181	TCCTGCGTTA	TCCCTGATT	CTGTGGATAA	CCGTATTACC	GCCTTTGAGT	GAGCTGATAC
9241	CGCTCGCCGC	AGCCGAACGA	CCGAGCGCAG	CGAGTCAGTG	AGCGAGGAAG	CGGAAGAGCG

FIG. 5B-3

9301	CCCAATACGC	AAACCGCCTC	TCCCCGCGCG	TTGGCCGATT	CATTAATGCA	GCTGGCACGA
9361	CAGGTTTCCC	GACTGGAAAAG	CGGGCAGTGA	GCGCAACGCA	ATTAATGTGA	GTAGCTCAC
9421	TCATTAGGCA	CCCCAGGCTT	TACACTTTAT	GCTTCCGGCT	CGTATGTTGT	GTGGAATTGT
9481	GAGCGGATAA	CAATTTCAACA	CAGGAAACAG	CTATGACCAT	GATTACGCCA	AGCGCGCAAT
9541	TAACCCTCAC	TAAAGGGAAC	AAAAGCTGGA	GCTGCAAGCT	T	

FIG. 6A

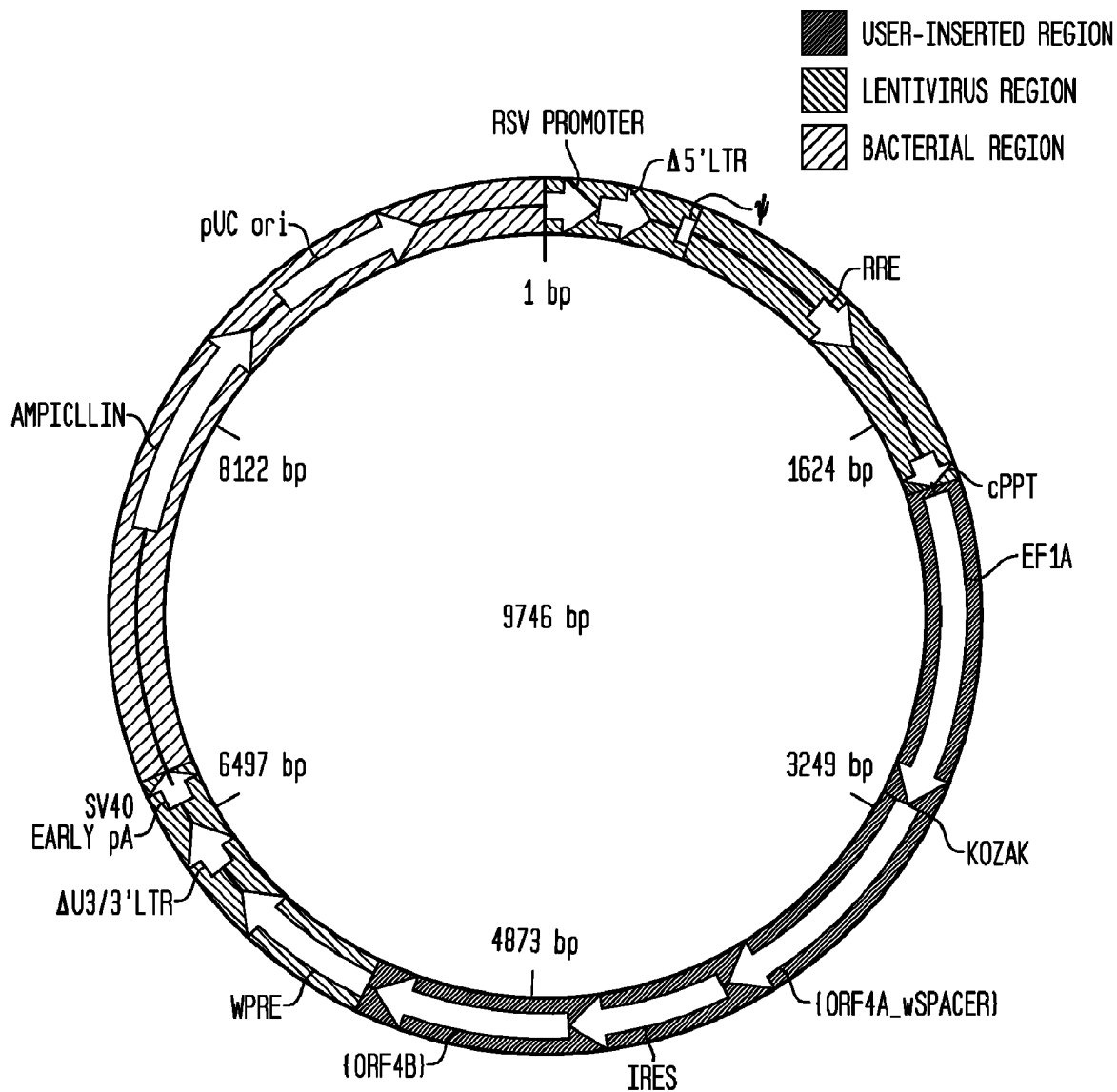


FIG. 6B

SEQ ID NO. 50

1	AATGTAGTCT	TATGCAATAC	TCTTGTAGTC	TTGCAACATG	GTAACGATGA	GTTAGCAACA
61	TGCCTTACAA	GGAGAGAAAA	AGCACCGTGC	ATGCCGATTG	GTGGAAGTAA	GGTGGTACGA
121	TCGTGCCCTTA	TTAGGAAGGC	AACAGACGGG	TCTGACATGG	ATTGGACGAA	CCACTGAATT
181	GCCGCATTGC	AGAGATATTG	TATTTAAGTG	CCTAGCTCGA	TACATAAACG	GGTCTCTCTG
241	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC
301	TCAATAAAGC	TTGCCCTTGAG	TGCTTCAAGT	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG
361	TAACTAGAGA	TCCCTCAGAC	CTTTTAGTC	AGTGTGAAA	ATCTCTAGCA	GTGGCGCCCG
421	AACAGGGACT	TGAAAGCGAA	AGGGAACCA	GAGGAGCTCT	CTCGACGCAG	GACTCGGCTT
481	GCTGAAGCGC	GCACGGCAAG	AGCGGAGGGG	CGGCGACTGG	TGAGTACGCC	AAAAATTTTG
541	ACTAGCGGAG	GCTAGAAGGA	GAGAGATGGG	TGCGAGAGCG	TCAGTATTAA	GCGGGGAGA
601	ATTAGATCGC	GATGGGAAAA	AATTCGGTTA	AGGCCAGGGG	GAAGAAAAAA	ATATAAATTA
661	AAACATATAG	TATGGGCAAG	CAGGGAGCTA	GAACGATTCT	CAGTTAATCC	TGGCCTGTTA
721	GAACATCAG	AAGGCTGTAG	ACAAATACTG	GGACAGCTAC	AACCATCCCT	TCAGACAGGA
781	TCAGAAGAAC	TTAGATCATT	ATATAATACA	GTAGCAACCC	TCTATTGTGT	GCATCAAAGG
841	ATAGAGATAA	AAGACACCAA	GGAAAGCTTTA	GACAAGATAG	AGGAAGAGCA	AAACAAAAGT
901	AAGACCACCG	CACAGCAAGC	GGCCGCTGAT	CTTCAGACCT	GGAGGAGGAG	ATATGAGGGA
961	CAATTGGAGA	AGTGAATTAT	ATAAATATAA	AGTAGTAAAA	AGAGCAGTGG	GAATAGGAGC
1021	ACCCACCAAG	GCAAGAGAA	GAGTGGTGCA	GAGAGAAAAA	AGAGCAGTGG	GAATAGGAGC
1081	TTTGTTCCTT	GGGTTCTTGG	GAGCAGCAGG	AAGCACTATG	GGCGCAGCGT	CAATGACGCT
1141	GACGGTACAG	GCCAGACAAT	TATTGTCTGG	TATAGTGCAG	CAGCAGAACA	ATTTGCTGAG
1201	GGCTATTGAG	GCGCAACAGC	ATCTGTTGCA	ACTCACAGTC	TGGGCGATCA	AGCAGCTCCA
1261	GGCAAGAATC	CTGGCTGTGG	AAAGATACCT	AAAGGATCAA	CAGCTCCTGG	GGATTTGGGG
1321	TTGCTCTGGA	AAACTCATTT	GCACCACTGC	TGTGCCCTTG	AATGCTAGTT	GGAGTAATAA
1381	ATCTCTGGAA	CAGATTGGGA	ATCACACGAC	CTGGATGGAG	TGGGACAGAG	AAATTAACAA
1441	TTACACAAGC	TTAATACACT	CCTTAATTGA	AGAATCGCAA	AACCAGCAAG	AAAAGAATGA
1501	ACAAGAATTA	TTGGAATTAG	ATAAATGGGC	AAGTTTGTGG	AATTGGTTTA	ACATAACAAA
1561	TTGGCTGTGG	TATATAAAAT	TATTCATAAT	GATAGTAGGA	GGCTTGGTAG	GTTTAAGAAT
1621	AGTTTTTGCT	GTACTTTCTA	TAGTGAATAG	AGTTAGGCAG	GGATATTCAC	CATTATCGTT
1681	TCAGACCCAC	CTCCCAACCC	CGAGGGGACC	CGACAGGCC	GAAGGAATAG	AAGAAGAAGG
1741	TGGAGAGAGA	GACAGAGACA	GATCCATTCT	ATTAGTGAAC	GGATCTCGAC	GGTATCGCTA
1801	GCTTTTAAAA	GAAAAGGGGG	GATTGGGGGG	TACAGTGCAG	GGGAAAGAAT	AGTAGACATA
1861	ATAGCAACAG	ACATACAAAC	TAAAGAATTA	CAAAAACAAA	TTACAAAAT	TCAAAATTTT
1921	ACTAGTATCA	ACTTTGTATA	GAAAAGTTGG	GCTCCGGTGC	CCGTCACTGG	GCAGAGCGCA
1981	CATCGCCAC	AGTCCCGGAG	AAGTTGGGGG	GAGGGGTCTG	CAATTGAACC	GGTGCCTAGA
2041	GAAGGTGGCG	CGGGGTAAAC	TGGGAAAGTG	ATGTCGTGTA	CTGGCTCCGC	CTTTTTCCCG
2101	AGGGTGGGGG	AGAACCGTAT	ATAAGTGCAG	TAGTCGCCGT	GAACGTTCTT	TTTCGCAACG
2161	GGTTTGCCGC	CAGAACACAG	GTAAGTGCCG	TGTGTGGTTC	CCGCGGGCCT	GGCCTCTTTA
2221	CGGGTTATGG	CCCTTGCGTG	CCTTGAATTA	CTTCCACCTG	GCTGCAGTAC	GTGATTCTTG
2281	ATCCCGAGCT	TCGGGTGGA	AGTGGGTGGG	AGAGTTCGAG	GCCTTGCGCT	TAAGGAGCCC
2341	CTTCGCCCTCG	TGCTTGAGTT	GAGGCCTGGC	CTGGCGCTG	GGGCGCCGC	GTGCGAATCT
2401	GGTGGCACCT	TCGCGCCTGT	CTCGCTGCTT	TCGATAAGTC	TCTAGCCATT	TAAATTTTTT
2461	GATGACCTGC	TGCGACGCTT	TTTTTCTGGC	AAGATAGTCT	TGTAATGCG	GGCCAAGATC
2521	TGCACACTGG	TATTTCTGTT	TTTGGGGCCG	CGGGCGCCGA	CGGGGCCCGT	GCGTCCAGC
2581	GCACATGTTT	GGCGAGGCGG	GGCTGCGAG	CGCGGCCACC	GAGAATCGGA	CGGGGTAGT
2641	CTCAAGCTGG	CCGGCTGCT	CTGGTGCCTG	GTCTCGCGCC	GCCGTGTATC	GCCCCGCCCT
2701	GGCGGGCAAG	GCTGGCCCGG	TCGGCACCAG	TTGCGTGAGC	GGAAAGATGG	CCGCTTCCCG
2761	GCCCTGCTGC	AGGGAGCTCA	AAATGGAGGA	CGCGGCGCTC	GGGAGAGCGG	GCGGGTGAAT
2821	CACCCACACA	AAGGAAAAGG	GCCTTTCCTG	CCTCAGCCGT	CGCTTCATGT	GACTCCACGG
2881	AGTACCGGGC	GCCGTCCAGG	CACCTCGATT	AGTTCTCGAG	CTTTTGAGT	ACGTCGTCTT

FIG. 6B-1

2941	TAGGTTGGGG	GGAGGGGTTT	TATGCGATGG	AGTTTCCCA	CACTGAGTGG	GTGGAGACTG
3001	AAGTTAGGCC	AGCTTGGCAC	TTGATGTAAT	TCTCCTTGGA	ATTTGCCCTT	TTTGAGTTTG
3061	GATCTTGGTT	CATTCTCAAG	CCTCAGACAG	TGGTTCAAAG	TTTTTTTCTT	CCATTTCAGG
2121	TGTCGTGACA	AGTTTGTACA	AAAAAGCAGG	CTGCCACCAT	GACCGTGCTG	GCCCCGCCT
3181	GGAGCCCCAC	CACCTACCTG	CTGCTGCTGC	TGCTGCTGAG	CAGCGGCTG	AGCGGCACCC
3241	AGGACTGCAG	CTTCCAGCAC	AGCCCCATCA	GCAGCGACTT	CGCCGTGAAG	ATCAGAGAGC
3301	TGAGCGACTA	CCTGCTGCAG	GACTACCCCG	TGACCGTGGC	CAGCAACCTG	CAGGACGAGG
3361	AGCTGTGCGG	CGGCCTGTGG	AGACTGGTGC	TGGCCAGAG	ATGGATGGAG	AGACTGAAGA
3421	CCGTGGCCGG	CAGCAAGATG	CAGGGCCTGC	TGGAGAGAGT	GAACACCGAG	ATCCACTTCG
3481	TGACCAAGTG	CGCCTTCCAG	CCCCCCCCCA	GCTGCCTGAG	ATTCGTGCAG	ACCAACATCA
3541	GCAGACTGCT	GCAGGAGACC	AGCGAGCAGC	TGGTGGCCCT	GAAGCCCTGG	ATCACCAGAC
3601	AGAACTTCAG	CAGATGCCTG	GAGCTGCAGT	GCCAGCCCGA	CAGCAGCACC	CTGCCCCCCC
3661	CCTGGAGCCC	CAGACCCCTG	GAGCTGCAGT	CCCCCACC GC	CCCCCAGTAA	AACAACAACA
3721	ATTGCATTCA	TTTTATGTTT	CAGGTTCAAG	GGGAGGTGTG	GGAGGTTTTT	TAAAGCAAGT
3781	AAAACCTCTA	CAAATGTGGT	ACGCGTTAAC	AACAACAATT	GCATTCATTT	TAAAGCAAGT
3841	GTTCAGGGGG	AGGTGTGGGA	GGTTTTTTAA	AGCAAGTAAA	ACCTCTACAA	ATGTGGTACG
3901	CGTTACCCAG	CTTTCTTGTA	CAAAGTGGTA	AATAGATAGA	ACAACAACAA	TTGCATTTCAT
3961	TTTTGATTTT	AGGTTCAAGG	GGAGGTGTGG	GAGGTTTTTT	AAAGCAAGTA	AAACCTCTAC
4021	ACTGACGGTA	CGCGTTAACA	ACAACAATTG	CATTCAATTG	TAGTTTCAGG	TTCAGGGGGA
4081	GGTGTGGGAG	GTTTTTTAAA	GCAAGTAAA	CCTCTAAAAT	AGTGGTACGC	GTTACCCAGC
4141	TTTCTGTAC	AAAGTGGACC	CAGCTTCTTT	GTACAAAAGT	GGCCCCCTCT	CCTCCCCCCC
4201	CCCTAACGTT	ACTGCCCCAA	GCCGCTTGGA	ATAAGGCCGG	TGTGCGTTTG	TCTATATGTT
4261	ATTTTCCACC	ATATTGCCGT	CTTTTGCCAA	TGTGAGGGCC	CGGAAACCTG	GCCCTGTCTT
4321	CTTGACGAGC	ATTCTTAGGG	GTCTTTCCCC	TCTCGCCAAA	GGAAATGCAAG	GTCTGTTGAA
4381	TGTCGTGAAG	GAAGCAGTTC	CTCTGGAAGC	TTCTTGAAGA	CAAACAACGT	CTGTAGCGAC
4441	CCTTTGCAGG	CAGCGGAACC	CCCCACCTGG	CGACAGGTGC	CTCTGCGGCC	CCCCCCAGAA
4501	TGTATAAGAT	ACACCTGCAA	AGCGGGCACA	TGGCCACCGT	GCTGGCCCCC	GCCTGGAGCC
4561	TGTGGAAAGA	GTCAAATGGC	TCTCTCAAG	CGTATTCAAC	AAGGGGCTGA	AGGATGCCCA
4621	GAAGGTACCC	CATTGTATGG	GATCTGATCT	GGGGCCTCGG	TGCACATGCT	TTACATGTGT
4681	TTAGTCGAGG	TTAAAAAAC	GTCTAGGCC	CCCGAACCAC	GGGGACGTGG	TTTTCTTTTG
4741	AAAAACACGA	TGATAATATG	GCCACAACCA	TGGCCACCGT	GCTGGCCCCC	GCCTGGAGCC
4801	CCACCACCTA	CCTGCTGCTG	CTGCTGCTGC	TGAGCAGCGG	CCTGAGCGCC	CCCCCCAGAA
4861	GCCCCAGCCC	CAGCACCCAG	CCCTGGGAGC	ACGTGAACGC	CATCCAGGAG	GCCAGAAGAC
4921	TGCTGAACCT	GAGCAGAGAC	ACCGCCGCCG	AGATGAACGA	GACCGTGGAG	GCCAGAAGAC
4981	AGATGTTCGA	CCTGCAGGAG	CCCACCTGCC	TGCAGACCAG	ACTGGAGCTG	TACAAGCAGG
5041	GCCTGAGAGG	CAGCCTGACC	AAGCTGAAGG	GCCCCCTGAC	CATGATGGCC	AGCCACTACA
5101	AGCAGCACTG	CCCCCCCACC	CCCGAGACCA	GCTGCGCCAC	CCAGATCATC	ACCTTCGAGA
5161	GCTTCAAGGA	GAACCTGAAG	GACTTCTGTC	TGGTGATCCC	CTTCGACTGC	TGGGAGCCCC
5221	TGCAGGAGCC	CACCACCACC	CCCGCCCCCA	GACCCCCCAC	CCCCGCCCCC	ACCATCGCCA
5281	GCCAGCCCTT	GAGCCTGAGA	CCCGAGGCCT	GCAGACCCGC	CGCCGGCGGC	GCCGTGCACA
5341	CCAGAGGCCT	GGACTTCGCC	TGCGACATCT	ACATCTGGGC	CCCCCTGGCC	GGCACCTGCG
5401	GCGTGCTGCT	GCTGAGCCTG	GTGATCACCC	TGTA CTGCAA	CCACAGAAAC	AGAAGAAGAG
5461	TGTGCAAGTG	CCCCAGACCC	GTGGTGAAGA	GCGGCACAAA	GCCCAGCCTG	AGCGCCAGAT
5521	ACGTGTAACA	ACTTTATTAT	ACATAGTTGA	TCAATTCCAA	CTTTATTATA	CATAGTTGAT
5581	CAATTCCGAT	AATCAACCTC	TGGATTACAA	AATTTGTGAA	AGATTGACTG	GTATTCTTAA
5641	CTATGTTGCT	CCTTTTACGC	TATGTGGATA	CGCTGCTTTA	ATGCCCTTGT	ATCATGCTAT
5701	TGCTTCCCGT	ATGGCTTTCA	TTTTCTCCTC	CTTGATATAA	TCCTGGTTGC	TGTCTCTTTA
5761	TGAGGAGTTG	TGGCCCGTTG	TCAGGCAACG	TGGCGTGGTG	TGCACTGTGT	TTGCTGACGC
5821	AACCCCCACT	GGTTGGGGCA	TTGCCACCAC	CTGTCAAGCTC	CTTTCCGGGA	CTTTGCTTTT
5881	CCCCCTCCCT	ATTGCCACGG	CGGAACCTCAT	CGCCGCTGCT	CTTGCCCGCT	GCTGGACAGG
5941	GGCTCGGCTG	TTGGGCACTG	ACAATTCCGT	GGTGTGTGCT	CTTGCCCGCT	GCTGGACAGG
6001	ATGGCTGCTC	GCCTGTGTTG	CCACCTGGAT	TCTGCGCGGG	ACGTCTTCT	GCTACGTCCC
6061	TTCGGCCCTC	AATCCAGCGG	ACCTTCCTTC	CCGCGGCTG	CTGCCGGCTC	TGCGGCTCT

FIG. 6B-2

6121	TCCGCGTCTT	CGCCTTCGCC	CTCAGACGAG	TGGATCTCC	CTTTGGGCCG	CCTCCCGCA
6181	TGGGAATTC	CCGCGGTTCC	CTTTAAGACC	AATGACTTAC	AAGGCAGCTG	TAGATCTTAG
6241	CCACTTTTTA	AAAGAAAAGG	GGGACTGGA	AGGGCTAATT	CACTCCCAAC	GAAGACAAGA
6301	TCTGCTTTTT	GCTTGTAAG	GGTCTCTCTG	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC
6361	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC	TCAATAAAGC	TTGCCCTTGAG	TGCTTCAAGT
6421	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG	TAAGTAGAGA	TCCCTCAGAC	CCTTTTAGTC
6481	AGTGTGGAAA	ATCTCTAGCA	GTAGTAGTTC	ATGTCATCTT	ATTATTCAGT	ATTTATAACT
6541	TGCAAAGAAA	TGAATATCAG	AGAGTGAGAG	GAACCTGTTT	ATTGCAGCTT	ATAATGGTTA
6601	CAAATAAAGC	AATAGCATCA	CAAATTTTAC	AAATAAAGCA	TTTTTTTTCAC	TGCATTCTAG
6661	TTGTGGTTTG	TCCAAACTCA	TCAATGTATC	TTATCATGTC	TGGCTCTAGC	TATCCCGCCC
6721	CTAACTCCGC	CCATCCCGCC	CCTAACTCCG	CCCAGTTCCG	CCCATTCTCC	GCCCATGCG
6781	TGACTAATTT	TTTTTATTTA	TGCAGAGGCC	GAGGCCGCCT	CGGCCTCTGA	GCTATTCCAG
6841	AAGTAGTGAG	GAGGCTTTTT	TGGAGGCCTA	GGGACGTACC	CAATTCGCCC	TATAGTGAGT
6901	CGTATTACGC	GCGCTCACTG	GCCGTCGTTT	TACAACGTCG	TGACTGGGAA	AACCTGGCG
6961	TTACCCAACT	TAATCGCCTT	GCAGCACATC	CCCTTTTCGC	CAGCTGGCGT	AATAGCGAAC
7021	AGGCCCGCAC	CGATCGCCCT	TCCCAACAGT	TGCGCAGCCT	GAATGGCGAA	TGGACGCGC
7081	CCTGTAGCGG	CGCATTAAAG	GCGGCGGGTG	TGGTGGTTAC	GCGCAGCGTG	ACCGCTACAC
7141	TTGCCAGCGC	CCTAGCGCCC	GCTCCTTTCC	CTTTCTTCCC	TTCTTTTCTC	GCCAGCTTCG
7201	CCGGCTTTCC	CCGTCAAAGT	CTAAATCGGG	GGCTCCCTTT	AGGGTTCCGA	TTTAGTGCTT
7261	TACGGCACCT	CGACCCCAAA	AAACTTGATT	AGGGTGATGG	TTACAGTAGT	GGGCCATCGC
7321	CCTGATAGAC	GGTTTTTCGC	CCTTTGACGT	TGGAGTCCAC	GTCTTTTAAT	AGTGGACTCT
7381	TGTTCCAAAC	TGGAACAACA	CTCAACCCTA	TCTCGTCTA	TTCTTTTGAT	TTATAAGGGA
7441	TTTTGCCGAT	TTCGGCCTAT	TGGTTAAAAA	ATGAGCTGAT	TTAACAACAA	TTTAACGCGA
7501	ATTTTAACAA	AATATTAACG	CTTACAATTT	AGGTGGCACT	TTTCGGGGAA	ATGTGCGCGG
7561	AACCCCTATT	TGTTTATTTT	TCTAAATACA	TTCAAATATG	TATCCGCTCA	TGAGACAATA
7621	ACCCTGATAA	ATGCTTCAAT	AATATTGAAA	AAGGAAGAGT	ATGAGTATTC	AACATTTCCG
7681	TGTCGCCCTT	ATTCCCTTTT	TTGCGGCATT	TTGCCCTTCT	GTTTTTGCTC	ACCCAGAAAC
7741	GCTGGTGAAA	GTAAAGATG	CTGAAGATCA	GTTGGGTGCA	CGAGTGGGTT	ACATCGAACT
7801	GGATCTCAAC	AGCGGTAAGA	TCCTTGAGAG	TTTTCGCCCC	GAAGAAGCTT	TTCCAATGAT
7861	GAGCACTTTT	AAAGTTCTGC	TATGTGGCGC	GGTATTATCC	CGTATTGACG	CCGGCAAGA
7921	GCAACTCGGT	CGCCGCATAC	ACTATTCTCA	GAATGACTTG	GTTGAGTACT	CACCACTCAC
7981	AGAAAAGCAT	CTTACGGATG	GCATGACAGT	AAGAGAATTA	TGCAGTGCTG	CCATAACCAT
8041	GAGTGATAAC	ACTCGGCCA	ACTTACTTCT	GACAACGATC	GGAGGACCGA	AGGAGCTAAC
8101	CGCTTTTTTG	CACAACATGG	GGGATCATGT	AACTCGCCTT	GATCGTTGGG	AACCGGAGCT
8161	GAATGAAGCC	ATACCAAACG	ACGAGCGTGA	CACCACGATG	CCTGTAGCAA	TGGCAACAAC
8221	GTTGCGCAAA	CTATTAAGTG	GAGCCGGTGA	GCGTGGGTCT	CGCGGTATCA	TTGCAGCACT
8281	CTGGATGGAG	GCGGATAAAG	TTGCAGGACC	ACTTCTGCGC	TCGGCCCTTC	CGGCTGGCTG
8341	GTTTATTGCT	GATAAATCTG	GAGCCGGTGA	GCGTGGGTCT	CGCGGTATCA	TTGCAGCACT
8401	GGGGCCAGAT	GGTAAGCCCT	CCCGTATCGT	AGTTATCTAC	ACGACGGGGA	GTCAGGCAAC
8461	TATGGATGAA	CGAAATAGAC	AGATCGCTGA	GATAGGTGCC	TCACTGATTA	AGCATTGGTA
8521	ACTGTCAGAC	CAAGTTTACT	CATATATACT	TTAGATTGAT	TTAAAACCTC	ATTTTTTAAT
8581	TAAAAGGATC	TAGGTGAAGA	TCCTTTTTGA	TAATCTCATG	ACCAAAATCC	CTTAACGTGA
8641	GTTTTCTGTC	CACTGAGCGT	CAGACCCCGT	AGAAAAGATC	AAAGGATCTT	CTTGAGATCC
8701	TTTTTTCTG	CGCGTAATCT	GCTGCTTGCA	AACAAAAAAA	CCACCGCTAC	CAGCGGTGGT
8761	TTGTTTGCCG	GATCAAGAGC	TACCAACTCT	TTTTCCGAAG	GTAAGTGGCT	TCAGCAGAGC
8821	GCAGATACCA	AATACTGTTT	TTCTAGTGTA	GCCGTAGTTA	GGCCACCACT	TCAAGAACTC
8881	TGTAGCACCG	CCTACATACC	TCGCTCTGCT	AATCCTGTTA	CCAGTGGCTG	CTGCCAGTGG
8941	CGATAAGTCG	TGCTTACCAG	GGTTGGACTC	AAGACGATAG	TTACCGGATA	AGGCGCAGCG
9001	GTCGGGCTGA	ACGGGGGGTT	CGTGCACACA	GCCCAGCTTG	GAGCGAACGA	CCTACACCGA
9061	ACTGAGATAC	CTACAGCGTG	AGCTATGAGA	AAGCGCCACG	CTTCCCGAAG	AGAGAAAGGC
9121	GGACAGGTAT	CCGGTAAGCG	GCAGGGTCGG	AACAGGAGAG	CGCACGAGGG	AGCTTCCAGG
9181	GGGAAACGCC	TGGTATCTTT	ATAGTCTGT	CGGGTTTCGC	CACCTCTGAC	TTGAGCGTCG
9241	ATTTTTGTGA	TGCTCGTCAG	GGGGCGGAG	CCTATGGAAA	AACGCCAGCA	ACGCGGCCTT

FIG. 6B-3

9301	TTTACGGTTC	CTGGCCTTTT	GCTGGCCTTT	TGCTCACATG	TTCTTTCCTG	CGTTATCCCC
9361	TGATTCTGTG	GATAACCGTA	TTACCGCCTT	TGAGTGAGCT	GATACCGCTC	GCCGCAGCCG
9421	AACGACCGAG	CGCAGCGAGT	CAGTGAGCGA	GGAAGCGGAA	GAGCGCCCAA	TACGCAAACC
9481	GCCTCTCCCC	GCGCGTTGGC	CGATTCATTA	ATGCAGCTGG	CACGACAGGT	TTCCCGACTG
9541	GAAAGCGGGC	AGTGAGCGCA	ACGCAATTAA	TGTGAGTTAG	CTCACTCATT	AGGCACCCCA
9601	GGCTTTACAC	TTTATGCTTC	CGGCTCGTAT	GTTGTGTGGA	ATTGTGAGCG	GATAAGAATT
9661	TCACACAGGA	AACAGCTATG	ACCATGATTA	CGCCAAGCGC	GCAATTAACC	CTCACTAAAG
9721	GGAACAAAAG	CTGGAGCTGC	AAGCTT			

FIG. 7A

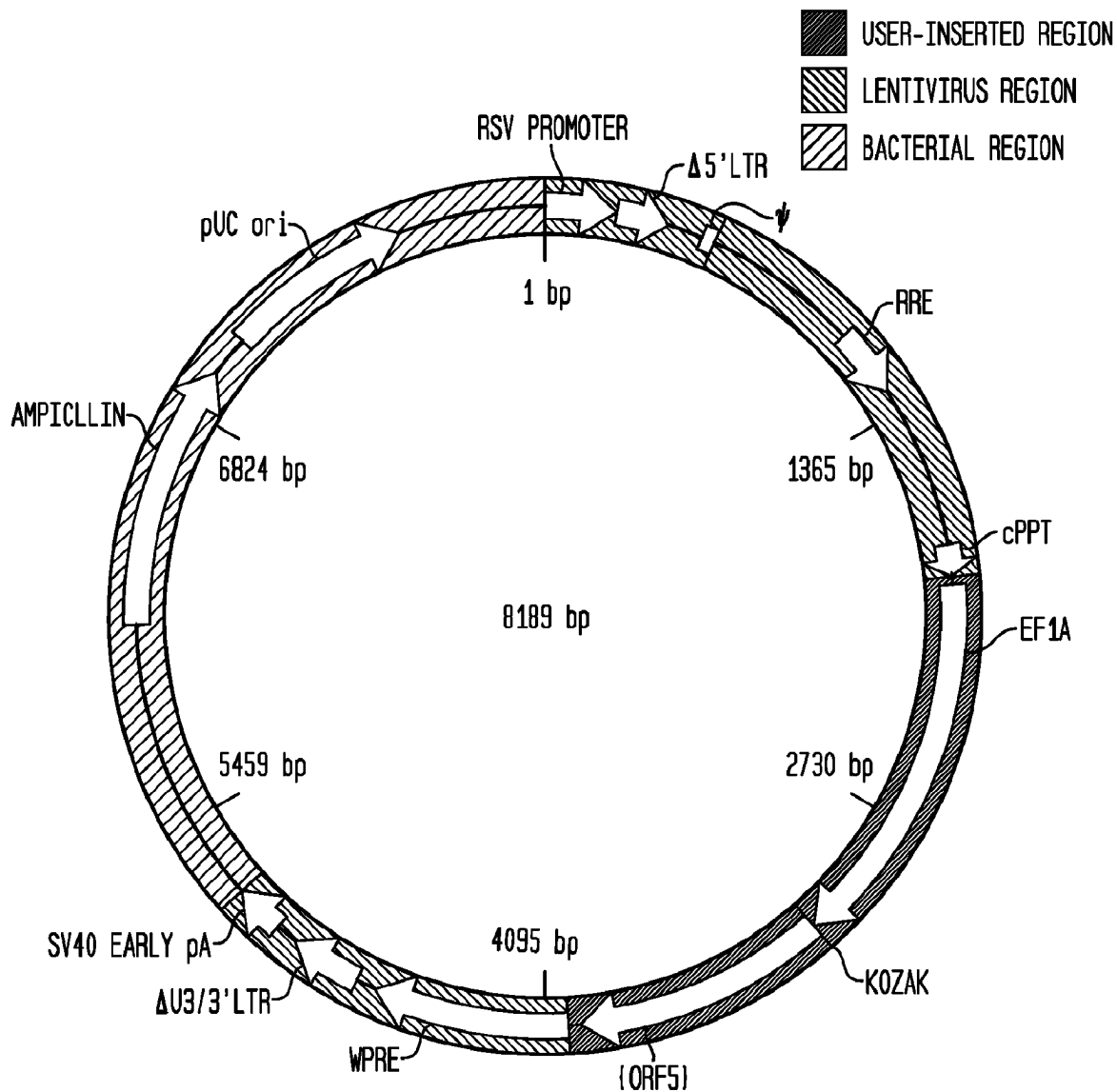


FIG. 7B

SEQ ID NO. 51

1	AATGTAGTCT	TATGCAATAC	TCTTGTAGTC	TTGCAACATG	GTAACGATGA	GTTAGCAACA
61	TGCCTTACAA	GGAGAGAAAA	AGCACCGTGC	ATGCCGATTG	GTGGAAGTAA	GGTGGTACGA
121	TCGTGCCCTTA	TTAGGAAGGC	AACAGACGGG	TCTGACATGG	ATTGGACGAA	CCACTGAATT
181	GCCGCATTGC	AGAGATATTG	TATTTAAGTG	CCTAGCTCGA	TACATAAACG	GGTCTCTCTG
241	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC
301	TCAATAAAGC	TTGCCCTTGAG	TGCTTCAAGT	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG
361	TAACTAGAGA	TCCCTCAGAC	CCTTTTAGTC	AGTGTGGAAG	ATCTCTAGCA	GTGGCGCCCG
421	AACAGGGACT	TGAAAGCGAA	AGGGAACCA	GAGGAGCTCT	CTCGACGCAG	GACTCGGCTT
481	GCTGAAGCGC	GCACGGCAAG	AGCGGAGGGG	CGGCGACTGG	TGAGTACGCC	AAAAATTTTG
541	ACTAGCGGAG	GCTAGAAGGA	GAGAGATGGG	TGCGAGAGCG	TCAGTATTAA	GCGGGGGAGA
601	ATTAGATCGC	GATGGGAAAA	ATTGCGTTA	AGGCCAGGGG	GAAAGAAAAA	ATATAAATTA
661	AAACATATAG	TATGGGCAAG	CAGGGAGCTA	GAACGATTCT	CAGTTAATCC	TGGCCTGTTA
721	GAAACATCAG	AAGGCTGTAG	ACAAATACTG	GGACAGCTAC	AACCATCCCT	TCAGACAGGA
781	TCAGAAGAAC	TTAGATCATT	ATATAATACA	GTAGCAACCC	TCTATTGTGT	GCATCAAAAG
841	ATAGAGATAA	AAGACACCAA	GGAAAGCTTTA	GACAAGATAG	AGGAAGAGCA	AAACAAAAGT
901	AAGACCACCG	CACAGCAAGC	GGCCGCTGAT	CTTCAGACCT	GGAGGAGGAG	ATATGAGGGA
961	CAATTGGAGA	AGTGAATTAT	ATAAATATAA	AGTAGTAAAA	ATTGAACCAT	TAGGAGTAGC
1021	ACCCACCAAG	GCAAAGAGAA	GAGTGGTGCA	GAGAGAAAAA	AGAGCAGTGG	GAATAGGAGC
1081	TTTGTTCCTT	GGGTTCTTGG	GAGCAGCAGG	AAGCACTATG	GGCGCAGCGT	CAATGACGCT
1141	GACGGTACAG	GCCAGACAAT	TATTGTCTGG	TATAGTGCAG	CAGCAGAACA	ATTTGCTGAG
1201	GGCTATTGAG	GCGCAACAGC	ATCTGTTGCA	ACTCACAGTC	TGGGCGATCA	AGCAGCTCCA
1261	CGCAAGAATC	CTGGCTGTGG	AAAGATACCT	AAAGGATCAA	CAGCTCCTGG	GGATTTGGGG
1321	TTGCTCTGGA	AAACTCATTT	GCACCACTGC	TGTGCCCTTG	AATGCTAGTT	GGAGTAATAA
1381	ATCTCTGGAA	CAGATTGGA	ATCACACGAC	CTGGATGGAG	TGGGACAGAG	AAATTAACAA
1441	TTACACAAGC	TTAATACACT	CCTTAATTGA	AGAATCGCAA	AACCAGCAAG	AAAAGAATGA
1501	ACAAGAATTA	TTGGAATTAG	ATAAATGGGC	AAGTTTGTGG	AATTGGTTTA	ACATAACAAA
1561	TTGGCTGTGG	TATATAAAAT	TATTCATAAT	GATAGTAGGA	GGCTTGGTAG	GTTTAAGAAT
1621	AGTTTTTGCT	GTACTTTCTA	TAGTGAATAG	AGTTAGGCAG	GGATATTCAC	CATTATCGTT
1681	TCAGACCCAC	CTCCCAACCC	CGAGGGGACC	CGACAGGCC	GAAGGAATAG	AAGAAGAAGG
1741	TGGAGAGAGA	GACAGAGACA	GATCCATTCT	ATTAGTGAAC	GGATCTCGAC	GGTATCGCTA
1801	GCTTTTAAAA	GAAAAGGGGG	GATTGGGGGG	TACAGTGCAG	GGGAAAGAAT	AGTAGACATA
1861	ATAGCAACAG	ACATACAAAC	TAAAGAATTA	CAAAAACAAA	TTACAAAAT	TCAAAATTTT
1921	ACTAGTGATT	ATCGGATCAA	CITTGTATAG	AAAAGTTGGG	CTCCGGTGCC	CGTCAGTGGG
1981	CAGAGCGCAC	ATCGCCACAA	GTCCCGGAGA	AGTTGGGGGG	AGGGGTCCGC	AATTGAACCG
2041	GTGCCTAGAG	AAGGTGGCGC	GGGTAAACT	GGGAAAGTGA	GTGCGTGATC	TGGCTCCGCC
2101	TTTTTCCCGA	GGGTGGGGGA	GAACCGTATA	TAAGTGCAGT	AGTCGCCGTG	AACGTTCTTT
2161	TTGCAACCGG	GTTTGCCGCC	AGAACACAGG	TAAGTGCCGT	GTGTGGTTCC	CGCGGGCCTG
2221	GCCTCTTTAC	GGGTTATGGC	CCTTGCCTGC	CTTGAATTAC	TTCCACCTGG	CTGCAGTACG
2281	TGATTCTTGA	TCCCGAGCTT	CGGTTGGGAA	GTGGGTGGGA	GAGTTCGAGG	CCTTGCGCTT
2341	AAGGAGCCCC	TTGCGCTCGT	GCTTGAATTG	AGGCCTGGCC	TGGCGCTGG	GGCCGCGCGG
2401	TGCGAATCTG	GTGGCACCTT	CGCGCTGTCT	TCGCTGCTTT	CGATAAGTCT	CTAGCCATTT
2461	AAAATTTTGT	ATGACCTGCT	GCGACGCTTT	TTTTCTGGCA	AGATAGTCTT	GTAAATGCGG
2521	GCCAAGATCT	GCACACTGGT	ATTTCGTTTT	TTGGGGCCGC	GGGCGGCGAC	GGGCCCCGTG
2581	CGTCCCAGCG	CACATGTTCT	GCGAGGCGGG	GCCTGCGAGC	GCGGCCACCG	AGAATCGGAC
2641	GGGGGTAGTC	TCAAGCTGGC	CGCCCTGCTC	TGGTGCCTGG	TCTGCGCCCG	CCGTGTATCG
2701	CCCCGCCCTG	GGCGGCAAGG	CTGCCCCGGT	CGGCACCACT	TGCGTGAGCG	GAAAGATGGC
2761	CGCTTCCCGG	CCCTGCTGCA	GGGAGCTCAA	AATGGAGGAC	GCGGCGCTCG	GGAGAGCGGG
2821	CGGGTGAGTC	ACCCACACAA	AGGAAAAGGG	CCTTTCGCTC	CTCAGCCGTC	GCTTCATGTG
2881	ACTCCACGGA	GTACCGGGCG	CCGTCCAGGC	ACCTCGATTA	GTTCTCGAGC	TTTTGGAGTA

FIG. 7B-1

2941	CGTCGTCTTT	AGGTTGGGGG	GAGGGGTTTT	ATGCGATGGA	GTTTCCCCAC	ACTGAGTGGG
3001	TGGAGACTGA	AGTTAGGCCA	GCTTGGCACT	TGATGTAATT	CTCCTTGGAA	TTTGGCCCTT
3061	TTGAGTTTGG	ATCTTGGTTC	ATTCTCAAGC	CTCAGACAGT	GGTTCAAAGT	TTTTTCTTTC
3121	CATTTCAAGT	GTCGTGACAA	GTTTGTACAA	AAAAGCAGGC	TGCCACCATG	AAGATCTTCA
3181	ACAACCAGCA	CAGCCCCAGA	AGCGCCGCCA	CCGGCCTGCC	CATCAGCATG	AAGATCTTCA
3241	TGTACCTGCT	GACCGTGTTT	CTGATCACCC	AGATGATCGG	CAGCGCCCTG	TTCGCCGTGT
3301	ACCTGCACAG	AAGACTGGAC	AAGATCGAGG	ACGAGAGAAA	CCTGCACGAG	GACTTCGTGT
3361	TCATGAAGAC	CATCCAGAGA	TGCAACACCG	GCCAGAGAA	CCTGAGCCTG	CTGAACCTGG
3421	AGGAGATCAA	GAGCCAGTTC	GAGGGCTTCG	TGAAGGACAT	CATGCTGAAC	AAGGAGGAGA
3481	CCAAGAAGGA	GAACAGCTTC	GAGATGCCCA	GAGGCGAGGA	GGACAGCCAG	ATCGCCGCCC
3541	ACGTGATCAG	CGAGGCCAGC	AGCAAGACCA	CCAGCGTGCT	GCAGTGGGCC	GAGAAGGGCT
3601	ACTACACCAT	GAGCAACAAC	CTGGTGACCC	TGGAGAACGG	CAAGCAGCTG	ACCGTGAAGA
3661	GACAGGGCCT	GTAATACATC	TACGCCCAGG	TGACCTTCTG	CAGCAACAGA	GAGGCCAGCA
3721	GCCAGGCCCC	CTTCATCGCC	CACAGCAGCG	CCAAGCCCTG	CGGCCAGCAG	AGCATCCACC
3781	TGCTGAGAGC	CGCCAACACC	CACAGCAGCG	CCAAGCCCTG	CGGCCAGCAG	AGCATCCACC
3841	TGGGCGGCGT	GTTGAGCTG	CAGCCCGGCG	CCAGCGTGTT	CGTGAACGTG	ACCGACCCCA
3901	GCCAGGTGAG	CCACGGCACC	GGCTTCACCA	GCTTCGGCCT	GCTGAAGCTG	TAAACCAGC
3961	TTTCTTGATC	AAAGTGGTGA	TAATCGAATT	CACCCAGCTT	TCTTGACAAA	AGTGGTGATA
4021	ATCGAATTCC	GATAATCAAC	CTCTGGATTA	CAAAATTTGT	GAAAGATTGA	CTGGTATTCT
4081	TAACTATGTT	GCTCCTTTTA	CGCTATGTGG	ATACGCTGCT	TTAATGCCTT	TGTATCATGC
4141	TATTCGTTCC	CGTATGGCTT	TCATTTTCTC	CTCCTTGAT	AAATCCTGGT	TGCTGTCTCT
4201	TTATGAGGAG	TTGTGGCCCG	TTGTGAGGCA	ACGTGGCGTG	GTGTGCACTG	TGTTTGCTGA
4261	CGCAACCCCC	ACTGGTTGGG	GCATTGCCAC	CACCTGTCA	CTCCTTTCCG	GGACTTTCGC
4321	TTTCCCCCTC	CCTATTGCCA	CGCGGGA	CATCGCCGCC	TGCCCTTGCC	GCTGTGGAC
4381	AGGGGCTCGG	CTGTTGGGCA	CTGACAATTC	CGTGGTGTTG	TGCGGGAAGC	TGACGTCTCT
4441	TCCATGGCTG	CTCGCCTGTG	TTGCCACCTG	GATTCTGCGC	GGGACGTCTT	TCTGTACGT
4501	CCCTTCGGCC	CTCAATCCAG	CGGACCTTCC	TTCCCGCGGC	CTGCTGCCGG	CTCTGCCGCC
4561	TCTTCGGCGT	CTTCGCCCTT	GCCCTCAGAC	GAGTCGGATC	TCCCTTTGGG	CCGCCTCCCC
4621	GCATCGGGAA	TTCCCGCGGT	TCGCTTTAAG	ACCAATGACT	TACAAGGCAG	CTGTAGATCT
4681	TAGCCACTTT	TTAAAGAAA	AGGGGGGACT	GGAGGGCTA	ATTCACTCCC	AACGAAGACA
4741	AGATCTGCTT	TTTGCTTGTA	CTGGGTCTCT	CTGGTTAGAC	CAGATCTGAG	CCTGGGAGCT
4801	CTCTGGCTAA	CTAGGGAACC	CACTGCTTAA	GCCTCAATAA	AGCTTGCTTT	GAGTGCTTCA
4861	AGTAGTGTGT	GCCCCGTCTG	TGTGTGACTC	TGGTAAGTAC	AGATCCCTCA	GACCCCTTTA
4921	GTCAGTGTGG	AAAATCTCTA	GCAAGTAGTAG	TTCATGTCAT	CTTATTATTC	AGTATTATTA
4981	ACTTGCAAAG	AAATGAATAT	CAGAGAGTGA	GAGGAACTTG	TTTATTGCAG	CTTATAATGG
5041	TTACAAATAA	AGCAATAGCA	TCACAAATTT	CACAAATAAA	GCATTTTTTT	CACTGCATTG
5101	TAGTTGTGGT	TTGTCCAAAC	TCATCAATGT	ATCTTATCAT	GTCTGGCTCT	AGCTATCCCG
5161	CCCCTAATC	CGCCCATCCC	GCCCTAATC	CGCCCAAGTT	CGCCCAATTC	TCCGCCCCAT
5221	GGCTGACTAA	TTTTTTTTAT	TTATGCAGAG	GCCGAGGCCG	CCTCGGCCCT	TGAGCTATTG
5281	CAGAAAGTAG	GAGGAGGCTT	TTTTGGAGGC	CTAGGGACGT	ACCCAATTCC	CCCTATAGTG
5341	AGTCGTATTA	CGCGCGCTCA	CTGCGCGTCG	TTTTACAACG	TCGTGACTGG	GAAAACCCTG
5401	GCGTTACCCA	ACTTAATCGC	CTTGACAGC	ATCCCCCTTT	CGCCAGCTGG	CGTAATAGCG
5461	AAGAGGCCCG	CACCGATCGC	CCTTCCCAAC	AGTTGCGCAG	CCTGAATGGC	GAATGGGACG
5521	CGCCCTGTAG	CGGCGCATT	AGCGCGGCGG	GTGTGGTGGT	TACGCGCAGC	GTGACCGCTA
5581	CACTTGCCAG	CGCCCTAGCG	CCGCTCTCTT	TCGCTTTCTT	CCCTTCTTTT	CTCGCCACGT
5641	TCGCCGGCTT	TCCCCGTCAA	GCTCTAAATC	GGGGGCTCCC	TTTAGGGTTC	CGATTTAGTG
5701	CTTTACGGCA	CCTCGACCCC	AAAAAACTTG	ATTAGGGTGA	TGGTTCACGT	AGTGGGCCAT
5761	CGCCCTGATA	GACGGTTTTT	CGCCCTTTGA	CGTTGGAGTC	CACGTTCTTT	AATAGTGAC
5821	TCTTGTTCCA	AACTGGAACA	AACTCAAC	CTATCTCGGT	CTATTCTTTT	GATTTATAAG
5881	GGATTTTGCC	GATTTGCGCC	TATTGGTTAA	AAAATGAGCT	GATTTAACAA	AAATTTAACG
5941	CGAATTTTAA	CAAAATATTA	ACGCTTACAA	TTTAGGTGGC	ACTTTTCGGG	GAAATGTGCG
6001	CGGAACCCCT	ATTTGTTTAT	TTTTCTAAAT	ACATTCAAAT	ATGTATCCGC	TCATGAGACA
6061	ATAACCCTGA	TAAATGCTTC	AATAATATTG	AAAAAGGAAG	AGTATGAGTA	TTCAACATTT

FIG. 7B-2

6121	CCGTGTCGCC	CTTATTCCT	TTTTTGCGGC	ATTTTGCCCT	CCTGTTTTTG	CTCACCAG
6181	AACGCTGGTG	AAAGTAAAAG	ATGCTGAAGA	TCAGTTGGGT	GCACGAGTGG	GTTACATCGA
6241	ACTGGATCTC	AACAGCGGTA	AGATCCTTGA	GAGTTTTCGC	CCCGAAGAAC	GTTTTCCAAT
6301	GATGAGCACT	TTTAAAGTTC	TGCTATGTGG	CGCGGTATTA	TCCCGTATTG	ACGCCGGGCA
6361	AGAGCAACTC	GGTCGCCGCA	TACACTATTC	TCAGAATGAC	TTGGTTGAGT	ACTCACCAGT
6421	CACAGAAAAG	CATCTTACGG	ATGGCATGAC	AGTAAGAGAA	TTATGCAGTG	CTGCCATAAC
6481	CATGAGTGAT	AACACTGCGG	CCAACCTACT	TCTGACAACG	ATCGGAGGAC	CGAAGGAGCT
6541	AACCGCTTTT	TTGCACAACA	TGGGGGATCA	TGTAACCTCGC	CTTGATCGTT	GGGAACCGGA
6601	GCTGAATGAA	GCCATACCAA	ACGACGAGCG	TGACACCACG	ATGCCTGTAG	CAATGGCAAC
6661	AACGTTGCGC	AAACTATTAA	CTGGCGAACT	ACTTACTCTA	GCTTCCCGGC	AACAATTAAT
6721	AGACTGGATG	GAGGCGGATA	AAGTTGCAGG	ACCACTTCTG	CGCTCGGCC	TTCCGGCTGG
6781	CTGGTTTATT	GCTGATAAAT	CTGGAGCCGG	TGAGCGTG66	TCTCGCGGTA	TCATTGCAGC
6841	ACTGGGGCCA	GATGGTAAGC	CCTCCCGTAT	CGTAGTTATC	TACACGACGG	GGAGTCAGGC
6901	AACTATGGAT	GAACGAAATA	GACGATCGC	TGAGATAGGT	GCCTCACTGA	TTAAGCATTG
6961	GTAAGTGTCA	GACCAAGTTT	ACTCATATAT	ACTTTAGATT	GATTTAAAC	TTCATTTTAA
7021	ATTTAAAGG	ATCTAGGTGA	AGATCCTTTT	TGATAATCTC	ATGACCAAAA	TCCCTTAACG
7081	TGAGTTTTCG	TTCCACTGAG	CGTCAGACCC	CGTAGAAAA6	ATCAAAGGAT	CTTCTTGAGA
7141	TCCTTTTTTT	CTGCGGTAA	TCTGCTGCTT	GCAAAACAAA	AAACCACCGC	TACCAGCGGT
7201	GGTTTGTTTG	CCGGATCAAG	AGTACCAAC	TCTTTTCCG	AAGGTAAC TG	GCTTCAGCAG
7261	AGCGCAGATA	CCAAATACTG	TTCTTCTAGT	GTAGCCGTAG	TTAGGCCACC	ACTTCAAGAA
7321	CTCTGTAGCA	CCGCCTACAT	ACCTCGCTCT	GCTAATCCTG	TTACCAAGTG	CTGCTGCCAG
7381	TGGCGATAAG	TCGTGTCTTA	CCGGGTTGGA	CTCAAGACGA	TAGTTACCGG	ATAAGGCCCA
7441	GCGGTCGGGC	TGAACGGGGG	GTTCTGTGAC	ACAGCCCAGC	TTGGAGCGAA	CGACCTACAC
7501	CGAACTGAGA	TACCTACAGC	GTGAGCTATG	AGAAAGCGCC	ACGCTTCCCG	AAGAGAGAAA
7561	GGCGGACAGG	TATCCGTAA	GCGGCAGGGT	CGGAACAGGA	GAGCGCACGA	GGGAGCTTCC
7621	AGGGGGAAC	GCCTGGTATC	TTTATAGTCC	TGTCGGGTTT	CGCCACCTCT	GACTTGAGCG
7681	TCGATTTTTG	TGATGCTCGT	CAGGGGGGCG	GAGCCTATGG	AAAAACGCCA	GCAACGCGGC
7741	CTTTTTACGG	TTCTTGCCCT	TTTGCTGGCC	TTTTGCTCAC	ATGTTCTTTC	CTGCGTTATC
7801	CCCTGATTCT	GTGGATAACC	GTATTACCGC	CTTTGAGTGA	GCTGATACCG	CTCGCCGCAG
7861	CCGAACGACC	GAGCGCAGCG	AGTCAGTGAG	CGAGGAAGCG	GAAGAGCGCC	CAATACGCAA
7921	ACCGCCTCTC	CCCGCGCGTT	GGCCGATTCA	TTAATGCAGC	TGGCACGACA	GGTTTCCCGA
7981	ACCGCCTCTC	GGCAGTGAGC	GCAACGCAAT	TAATGTGAGT	TAGCTCACTC	ATTAGGCACC
8041	CCAGGCTTTA	CACTTTATGC	TTCCGGCTCG	TATGTTGTGT	GGAAATTGTA	GCGGATAACA
8101	ATTTACACA	GGAAACAGCT	ATGACCATGA	TTACGCCAAG	AGAGCAATTA	ACCCTCACTA
8161	AAGGGAACAA	AAGCTGGAGC	TGCAAGCTT			

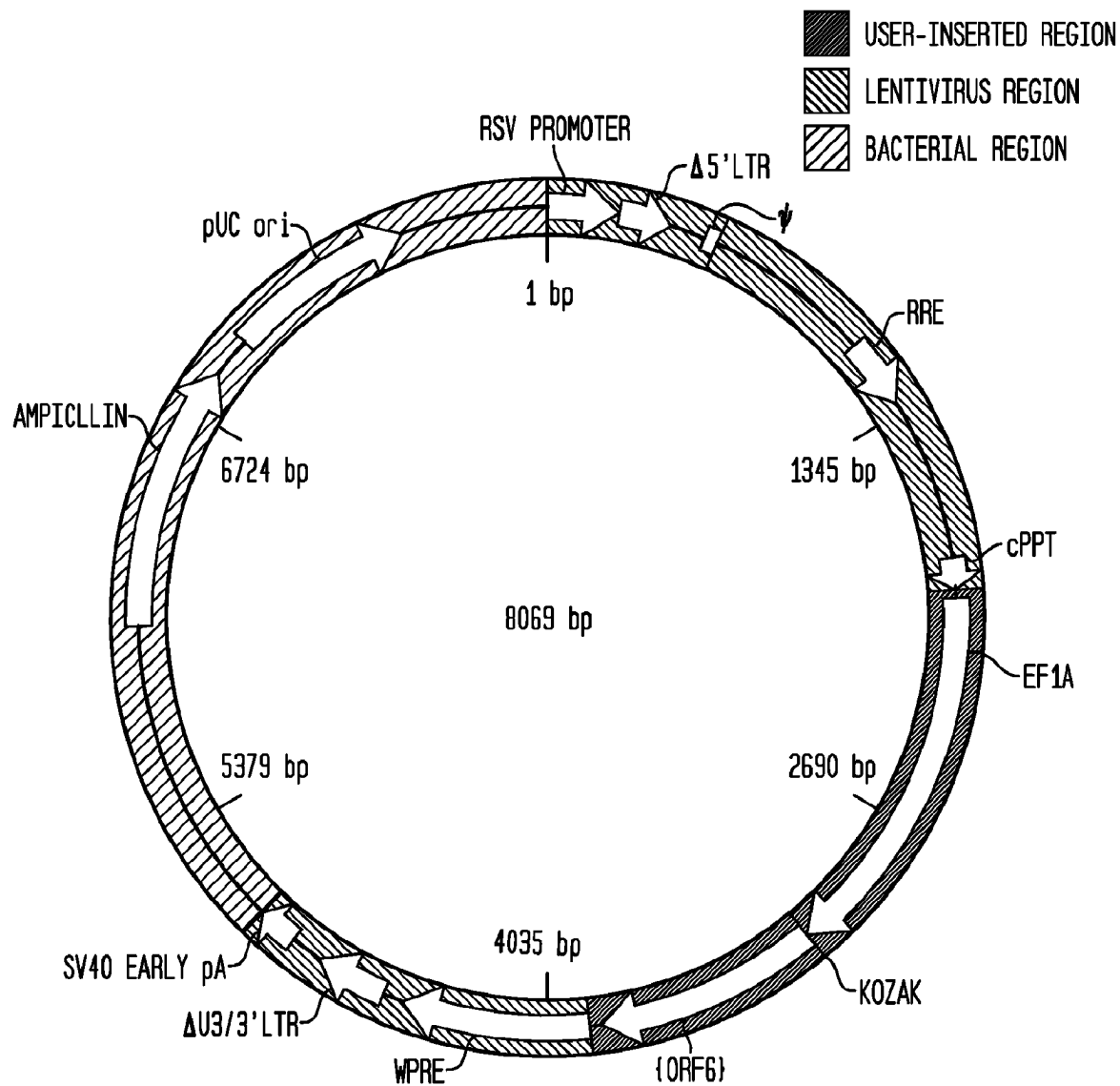
FIG. 8A

FIG. 8B

SEQ ID NO. 52

1	AATGTAGTCT	TATGCAATAC	TCTTGTAGTC	TTGCAACATG	GTAACGATGA	GTTAGCAACA
61	TGCCTTACAA	GGAGAGAAAA	AGCACCCTGC	ATGCCGATTG	GTGGAAGTAA	GGTGGTACGA
121	TCGTGCCTTA	TTAGGAAGGC	AACAGACGGG	TCTGACATGG	ATTGGACGAA	CCACTGAATT
181	GCCGCATTGC	AGAGATATTG	TATTTAAGTG	CCTAGCTCGA	TACATAAACG	GGTCTCTCTG
241	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC
301	TCAATAAAGC	TTGCCCTTGAG	TGCTTCAAGT	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG
361	TAACTAGAGA	TCCCTCAGAC	CCTTTTAGTC	AGTGTGGAAA	ATCTCTAGCA	GTGGCGCCCG
421	AACAGGGACT	TGAAAGCGAA	AGGGAACCA	GAGGAGCTCT	CTCGACGCAG	GACTCGGCTT
481	GCTGAAGCGC	GCACGGCAAG	AGCGAGGGG	CGGCGACTGG	TGAGTACGCC	AAAAATTTTG
541	ACTAGCGGAG	GCTAGAAGGA	GAGAGATGGG	TGCGAGAGCG	TCAGTATTAA	GCGGGGAGAG
601	ATTAGATCGC	GATGGGAAAA	AATTCGGTTA	AGGCCAGGGG	GAAGAAAAAA	ATATAAATTA
661	AAACATATAG	TATGGGCAAG	CAGGGAGCTA	GAACGATTCT	CAGTTAATCC	TGGCCTGTTA
721	GAACATCAG	AAGGCTGTAG	ACAAATACTG	GGACAGCTAC	AACCATCCCT	TCAGACAGGA
781	TCAGAAGAAC	TTAGATCATT	ATATAATACA	GTAGCAACCC	TCTATTGTGT	GCATCAAAAG
841	ATAGAGATAA	AAGACACCAA	GGAACTTTTA	GACAAGATAG	AGGAAGAGCA	AAACAAAAGT
901	AAGACCACCG	CACAGCAAGC	GGCCGCTGAT	CTTCAGACCT	GGAGGAGGAG	ATATGAGGGA
961	CAATTGGAGA	AGTGAATTAT	ATAAATATAA	AGTAGTAAAA	ATTGAACCAT	TAGGAGTAGC
1021	ACCCACCAAG	GCAAGAGAA	GAGTGGTGCA	GAGAGAAAAA	AGAGCAGTGG	GAATAGGAGC
1081	TTTGTTCCTT	GGGTTCCTTG	GAGCAGCAGG	AAGCACTATG	GGCGCAGCGT	CAATGACGCT
1141	GACGGTACAG	GCCAGACAAT	TATTGTCTGG	TATAGTGCAG	CAGCAGAACA	ATTTGCTGAG
1201	GGCTATTGAG	GCGCAACAGC	ATCTGTTGCA	ACTCACAGTC	TGGGGCATCA	AGCAGCTCCA
1261	GGCAAGAATC	CTGGCTGTGG	AAAGATACCT	AAAGGATCAA	CAGCTCCTGG	GGATTTGGGG
1321	TTGCTCTGGA	AAACTCATTT	GCACCACTGC	TGTGCCTTGG	AATGCTAGTT	GGAGTAATAA
1381	ATCTCTGGAA	CAGATTTGGA	ATCACACGAC	CTGGATGGAG	TGGGACAGAG	AAATTAACAA
1441	TTACACAAGC	TTAATACACT	CCTTAATTGA	AGAATCGCAA	AACCAGCAAG	AAAAGAATGA
1501	ACAAGAATTA	TTGGAATTAG	ATAAATGGGC	AAGTTTGTGG	AATTGGTTTA	ACATAACAAA
1561	TTGGCTGTGG	TATATAAAAT	TATTCATAAT	GATAGTAGGA	GGCTTGTTAG	GTTTAAGAAT
1621	AGTTTTTGCT	GTACTTTCTA	TAGTGAATAG	AGTTAGGCAG	GGATATTCAC	CATTATCGTT
1681	TCAGACCCAC	CTCCCAACCC	CGAGGGGACC	CGACAGGCC	GAAGGAATAG	AAGAAGAAGG
1741	TGGAGAGAGA	GACAGAGACA	GATCCATTCT	ATTAGTGAAC	GGATCTCGAC	GGTATCGCTA
1801	GCTTTTAAAA	GAAAAGGGGG	GATTGGGGGG	TACAGTGCAG	GGGAAAGAAT	AGTAGACATA
1861	ATAGCAACAG	ACATACAAAC	TAAAGAATTA	CAAAAACAAA	TTACAAAAAT	TCAAAATTTT
1921	ACTAGTGATT	ATCGGATCAA	CTTTGTATAG	AAAAGTTGGG	CTCCGGTGCC	CGTCAGTGGG
1981	CAGAGCGCAC	ATCGCCACAA	GTCCCGGAGA	AGTTGGGGGG	AGGGGTCCGC	AATTGAACCG
2041	GTGCCTAGAG	AAGGTGGCGC	GGGTAAACT	GGGAAAGTGA	TGTCGTGTAC	TGGCTCCGCC
2101	TTTTTCCCGA	GGGTGGGGGA	GAACCGTATA	TAAGTGCAGT	AGTCGCCGTG	AACGTTCTTT
2161	TTCGCAACGG	GTTTGGCCGC	AGAACACAGG	TAAGTGCCGT	GTGTGGTTCC	CGCGGGCCTG
2221	GCCTCTTTAC	GGGTATGGC	CCTTGCCTGC	CTTGAATTAC	TTCCACCTGG	CTGCAGTACG
2281	TGATTCTTGA	TCCCGAGCTT	CGGGTTGGAA	GTGGGTGGGA	GAGTTCGAGG	CCTTGCCTTT
2341	AAGGAGCCCC	TTCCGCTCGT	GCTTGAATTT	AGGCCTGGCC	TGGGCGCTGG	GGCCGCGCGG
2401	TGCGAATCTG	GTGGCACCTT	CGCGCTGTCT	TCGCTGCTTT	CGATAAGTCT	CTAGCCATTT
2461	AAAATTTTTG	ATGACCTGCT	GCGACGCTTT	TTTTCTGGCA	AGATAGTCTT	GTAAATGCGG
2521	GCCAAGATCT	GCACACTGGT	ATTTGCTTTT	TTGGGGCCGC	GGGCGGCGAC	GGGCCCCGTG
2581	CGTCCAGCG	CACATGTTCT	GCGAGGCGGG	GCCTGCGAGC	GCGGCCACCG	AGAATCGGAC
2641	GGGGGTAGTC	TCAAGCTGGC	CGCCCTGCTC	TGGTGCCTGG	TCTCGCGCCG	CCGTGTATCG
2701	CCCCGCCCTG	GGCGGCAAGG	CTGGCCCGGT	GCGCACCAAT	TGCGTGAGCG	GAAGATGGC
2761	CGCTTCCCGG	CCTGCTGCA	GGGAGCTCAA	AATGGAGGAC	GCGCGCTCG	GGAGAGCGGG
2821	CGGGTGAGTC	ACCCACACAA	AGGAAAAGGG	CCTTTCGGTC	CTCAGCCGTC	GCTTCATGTG

FIG. 8B-1

2881	ACTCCACGGA	GTACCGGGCG	CCGTCCAGGC	ACCTCGATTA	GTTCTCGAGC	TTTTGGAGTA
2941	CGTCGTCTTT	AGGTTGGGGG	GAGGGGTTTT	ATGCGATGGA	GTTTCCCCAC	ACTGAGTGGG
3001	TGGAGACTGA	AGTTAGGCCA	GCTTGGCACT	TGATGTAATT	CTCCTTGGAA	TTTGCCCTTT
3061	TTGAGTTTGG	ATCTTGTTTC	ATTCTCAAGC	CTCAGACAGT	G6TTCAAAGT	TTTTTTCTTC
3121	CATTTCAAGT	GTCGTGACAA	GTTTGACAA	AAAAGCAGGC	TGCCACCATG	AGCACCGAGA
3181	GCATGATCAG	AGACGTGGAG	CTGGCCGAGG	AGGCCCTGCC	CAAGAAGACC	GGCGGGCCCC
3241	AGGGCAGCAG	AAGATGCCTG	TTCTTGAGCC	TGTTCAAGCTT	CCTGATCGTG	GCCGGCGCCA
3301	CCACCCTGTT	CTGCCTGCTG	CACTTCGGCG	TGATCGGCC	CCAGAGAGAG	GAGTTCCTCA
3361	GAGACCTGAG	CCTGATCAGC	CCCCTGGCCC	AGGCCGTGGC	CCACGTGGTG	GCCAACCCCC
3421	AGGCCGAGGG	CCAGCTGCAG	TGGCTGAACA	GAAGAGCCAA	CGCCCTGCTG	GCCAACGGCG
3481	TGGAGCTGAG	AGACAACCAG	CTGGTGGTGC	CCAGCGAGGG	CCTGTACCTG	ATCTACAGCC
3541	AGGTGCTGTT	CAAGGGCCAG	GGCTGCCCCA	GCACCCACGT	GCTGCTGACC	CACACCATCA
3601	GCAGAATCGC	CGTGAGCTAC	CAGACCAAGG	TGAACCTGCT	GAGCGCCATC	AAGAGCCCTT
3661	GCCAGAGAGA	GACCCCGAG	GGCGCCGAGG	CCAAGCCCTG	GTACGAGCCC	ATCTACCTGG
3721	GGCGCGTGTT	CCAGCTGGAG	AAGGGCGACA	GACTGAGCGC	CGAGATCAAG	AGACCCGACT
3781	ACCTGGACTT	CGCCGAGAGC	GGCCAGGTGT	ACTTCGGCAT	CATCGCCCTG	TAAACCCAGC
3841	TTTCTTGATC	AAAGTGGTGA	TAATCGAATT	CACCCAGCTT	TCTGTACAA	AGTGGTGATA
3901	ATCGAATTCC	GATAATCAAC	CTCTGGATTA	CAAAATTTGT	GAAAGATTGA	CTGGTATTCT
3961	TAATATGTT	GCTCCTTTTA	CGCTATGTGG	ATACGCTGCT	TAAATGCCTT	TGTATCATGC
4021	TATTGCTTCC	CGTATGGCTT	TCATTTTCTC	CTCCTTGAT	AAATCCTGGT	TGCTGTCTCT
4081	TTATGAGGAG	TTGTGGCCCG	TTGTGAGGCA	ACGTGGCGTG	GTGTGCACTG	TGTTTGCTGA
4141	CGCAACCCCC	ACTGGTTGGG	GCATTGCCAC	CACCTGTCA	CTCCTTTCCG	GGACTTTTCGC
4201	TTTCCCCCTC	CCTATTGCCA	CGCGGGAAGT	CATCGCCGCC	TGCCCTTGCC	GCTGCTGGAC
4261	AGGGGCTCGG	CTGTTGGGCA	CTGACAATTC	CGTGGTGTG	TCGGGGAAGC	TGACGTCTCT
4321	TCCATGGCTG	CTCGCCTGTG	TTGCCACCTG	GATTCTGCGC	GGGACGTCTT	TCTGCTACGT
4381	CCCTTCGGCC	CTCAATCCAG	CGGACCTTCC	TTCCCGCGGC	CTGCTGCCGG	CTCTGGGGCC
4441	TCTTCGCGCT	CTTCGCTTTC	GCCCTCAGAC	GAGTCGGATC	TCCCTTTGGG	CCGCCTCCCC
4501	GCATCGGGAA	TTCCCGCGGT	TCGCTTTAAG	ACCAATGACT	TACAAGGCAG	CTGTAGATCT
4561	TAGCCACTTT	TTAAAGAA	AGGGGGGACT	GGAAAGGCTA	ATTCACTCCC	AACGAAGACA
4621	AGATCTGCTT	TTTGCTTGTA	CTGGTCTCT	CTGGTTAGAC	CAGATCTGAG	CCTGGGAGCT
4681	CTCTGGCTAA	CTAGGGAACC	CACTGCTTAA	GCCTCAATAA	AGCTTGCTTT	GAGTGCTTCA
4741	AGTAGTGTGT	GCCCCTCTGT	TGTGTGACTC	TGGTAAGTAC	AGATCCCTCA	GACCTTTTAA
4801	GTCAGTGTGG	AAAATCTCTA	GCAAGTAGAG	TTCATGTCA	CTTATTATTC	AGTATTTATA
4861	ACTTGCAAAG	AAATGAATAT	CAGAGAGTGA	GAGGAAGTTG	TTTATTGCAG	CTTATAATGG
4921	TTACAAATAA	AGCAATAGCA	TCACAAAT	CACAAATAAA	GCATTTTTTT	CACTGCATTTC
4981	TAGTTGTGGT	TTGTCCAAAC	TCATCAATGT	ATCTTATCAT	GTCTGGCTCT	AGCTATCCCC
5041	CCCCTAATC	CGCCCATCCC	GCCCTAATC	CGCCCAAGT	CGCCCATTC	TCCGCCCAT
5101	GGCTGACTAA	TTTTTTTTAT	TTATGCAGAG	GCCGAGGCCG	CCTCGGCCCTC	TGAGCTATTTC
5161	CAGAAAGTAGT	GAGGAGGCTT	TTTTGGAGGC	CTAGGGACGT	ACCCAATTCC	CCCTATAGTG
5221	AGTCGTATTA	CGCGCGCTCA	CTGGCCGTCTG	TTTTACAACG	TCGTGACTGG	GAAAACCCTG
5281	GCCTTACCCA	ACTTAATCGC	CTTGCAAGAC	ATCCCCCTTT	CGCCAGCTGG	CGTAATAGCG
5341	AAGAGGCCCG	CACCGATCGC	CCTTCCCAAC	AGTTGCGCAG	CCTGAATGGC	GAATGGGACG
5401	CGCCCTGTAG	CGGCGCATT	AGCGCGCGCG	GTGTGGTGGT	TACGCGCAGC	GTGACCGCTA
5461	CACTTGCCAG	CGCCCTAGCG	CCGCTTCTT	TCGCTTCTT	CCCTTCTTT	CTCGCCACGT
5521	TCGCCGGCTT	TCCCCGTCAA	GCTCTAAATC	GGGGGCTCCC	TTTAGGGTTC	CGATTTAGTG
5581	CTTTACGGCA	CCTCGACCCC	AAAAAACTTG	ATTAGGGTGA	TGGTTCACGT	AGTGGGCCAT
5641	CGCCCTGATA	GACGGTTTTT	CGCCCTTTGA	CGTTGGAGTC	CACGTTCTTT	AATAGTG6AC
5701	TCTTGTTCCA	AACTGGAACA	AACTCAACC	CTATCTCGGT	CTATTCTTTT	GATTTATAAG
5761	GGATTTTGCC	GATTTTGGCC	ATTTGGTTAA	AAAATGAGCT	GATTTAACA	AAATTTAAGC
5821	CGAATTTTAA	CAAAATATTA	ACGCTTACAA	TTTAGGTGGC	ACTTTTCCGG	GAAATGTGCG
5881	CGGAACCCCT	ATTTGTTTAT	TTTTCTAAAT	ACATTCAAT	ATGTATCCGC	TCATGAGACA
5941	ATAACCCCTGA	TAAATGCTTC	AATAATATTG	AAAAAGGAA	AGTATGAGTA	TTCAACATTT

FIG. 8B-2

6001	CCGTGTCGCC	CTTATTCCT	TTTTTGCGGC	ATTTTGCCCT	CCTGTTTTTG	CTCACCAG
6061	AACGCTGGTG	AAAGTAAAAG	ATGCTGAAGA	TCAGTTGGGT	GCACGAGTGG	GTTACATCGA
6121	ACTGGATCTC	AACAGCGGTA	AGATCCTTGA	GAGTTTTCGC	CCCGAAGAAC	GTTTTCCAAT
6181	GATGAGCACT	TTTAAAGTTC	TGCTATGTGG	CGCGGTATTA	TCCCGTATTG	ACGCCGGGCA
6241	AGAGCAACTC	GGTCGCCGCA	TACACTATTC	TCAGAATGAC	TTGGTTGAGT	ACTCACCAGT
6301	CACAGAAAAG	CATCTTACGG	ATGGCATGAC	AGTAAGAGAA	TTATGCAGTG	CTGCCATAAC
6361	CATGAGTGAT	AACACTGCGG	CCAACCTACT	TCTGACAACG	ATCGGAGGAC	CGAAGGAGCT
6421	AACCGCTTTT	TTGCACAACA	TGGGGGATCA	TGTAACCTCGC	CTTGATCGTT	GGGAACCGGA
6481	GCTGAATGAA	GCCATACCAA	ACGACGAGCG	TGACACCACG	ATGCCTGTAG	CAATGGCAAC
6541	AACGTTGCGC	AAACTATTAA	CTGGCGAACT	ACTTACTCTA	GCTTCCCGGC	AACAATTAAT
6601	AGACTGGATG	GAGGCGGATA	AAGTTGCAGG	ACCACTTCTG	CGCTCGGCC	TTCCGGCTGG
6661	CTGGTTTATT	GCTGATAAAT	CTGGAGCCGG	GTAGCGTGGG	TCTCGCGGTA	TCATTGCAGC
6721	ACTGGGGCCA	GATGGTAAGC	CCTCCCGTAT	CGTAGTTATC	TACACGACGG	GGAGTCAGGC
6781	AACTATGGAT	GAACGAAATA	GACAGATCGC	TGAGATAGGT	GCCTCACTGA	TTAAGCATTG
6841	GTAAGTGTCA	GACCAAGTTT	ACTCATATAT	ACTTTAGATT	GATTTAAAC	TTCATTTTAA
6901	TGAGTTTTCG	TTCCACTGAG	CGTCAGACCC	CGTAGAAAAG	ATCAAAGGAT	CTTCTTGAGA
6961	TGAGTTTTCG	TTCCACTGAG	CGTCAGACCC	CGTAGAAAAG	ATCAAAGGAT	CTTCTTGAGA
7021	TCCTTTTTTT	CTGCGCGTAA	TCTGCTGCTT	GCAAAACAAA	AAACCACCGC	TACCAGCGGT
7081	GGTTTGTTTG	CCGGATCAAG	AGTACCAAC	CTCTTTTCCG	AAGGTAAC TG	GCTTCAGCAG
7141	AGCGCAGATA	CCAAATACTG	TTCTTCTAGT	GTAGCCGTAG	TTAGGCCACC	ACTTCAAGAA
7201	CTCTGTAGCA	CCGCCTACAT	ACCTCGCTCT	GCTAATCCTG	TTACCACTGG	CTGCTGCCAG
7261	TGGCGATAAG	TCGTGTCTTA	CCGGGTGGA	CTCAAGACGA	TAGTTACCGG	ATAAGGCCCA
7321	GCGGTCGGGC	TGAACGGGGG	GTTCTGTGAC	ACAGCCCAGC	TTGGAGCGAA	CGACCTACAC
7381	CGAACTGAGA	TACCTACAGC	GTGAGCTATG	AGAAAGCGCC	ACGCTTCCCG	AAGAGAGAAA
7441	GCGGACAGG	TATCCGTAA	GCGGCAGGGT	CGGAACAGGA	GAGCGCACGA	GGGAGCTTCC
7501	AGGGGAAAC	GCCTGGTATC	TTTATAGTCC	TGTCGGGTTT	CGCCACCTCT	GACTTGAGCG
7561	TCGATTTTTG	TGATGCTCGT	CAGGGGGGCG	GAGCCTATGG	AAAAACGCCA	GCAACGCGGC
7621	CTTTTTACGG	TTCTTGCCCT	TTTCTGCGCC	TTTTGCTCAC	ATGTTCTTTC	CTGCGTTATC
7681	CCCTGATTCT	GTGGATAACC	GTATTACCGC	CTTTGAGTGA	GCTGATACCG	CTCGCCGCAG
7741	CCGAACGACC	GAGCGCAGCG	AGTCAGTGAG	CGAGGAAGCG	GAAGAGCGCC	CAATACGCAA
7801	ACCGCCTCTC	CCCGCGCGTT	GGCCGATTCA	TTAATGCAGC	TGGCACGACA	GGTTTCCCGA
7861	CTGGAAGCG	GGCAGTGAGC	GCAACGCAAT	TAATGTGAGT	TAGCTCACTC	ATTAGGCACC
7921	CCAGGCTTTA	CACTTTATGC	TTCCGGCTCG	TATGTTGTGT	GGAAATTGTA	GCGGATAACA
7981	ATTTACACA	GGAAACAGCT	ATGACCATGA	TTACGCCAAG	CGCGCAATTA	ACCCTCACTA
8041	AAGGGAACAA	AAGCTGGAGC	TGCAAGCTT			

FIG. 9A

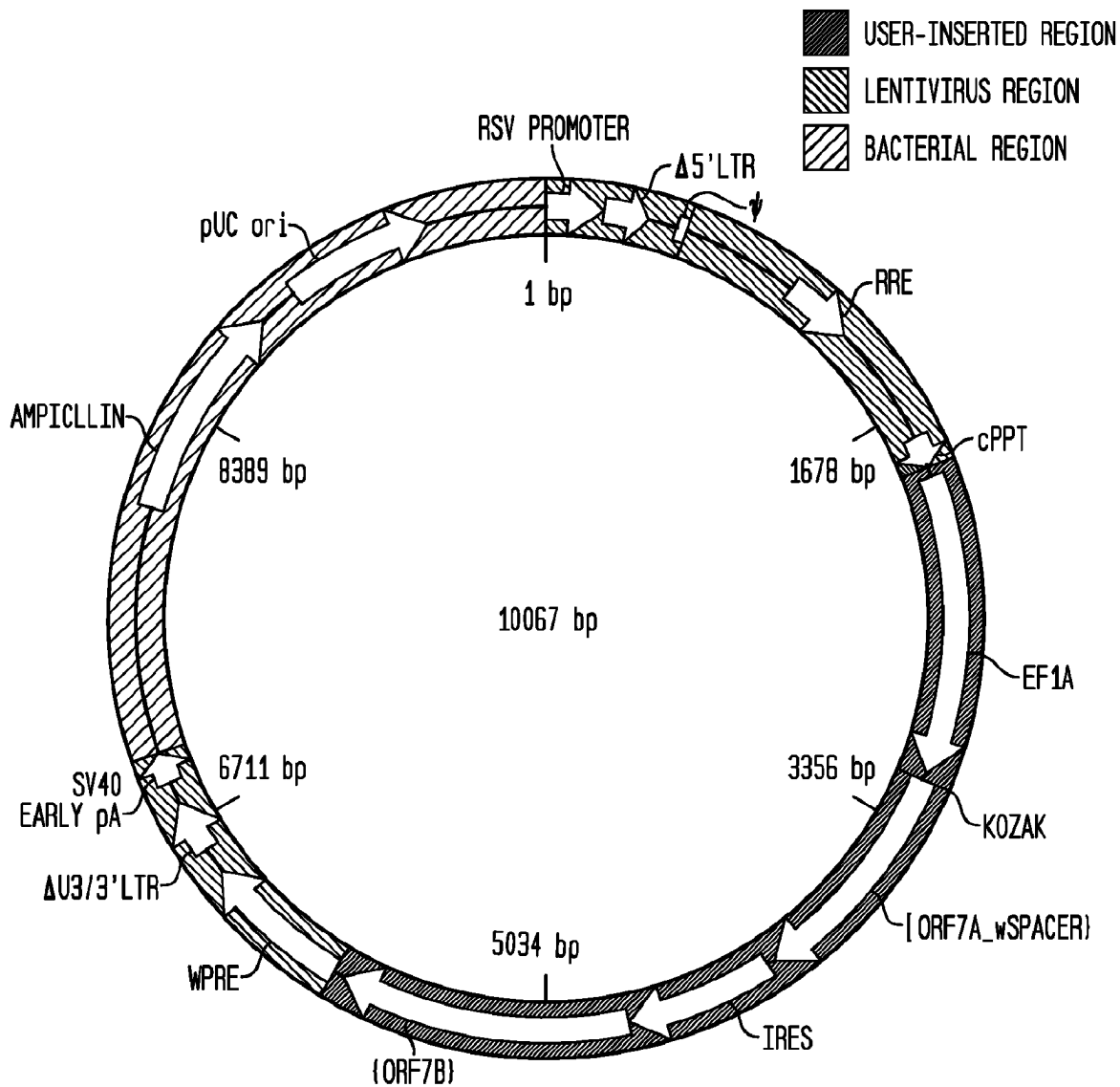


FIG. 9B

SEQ ID NO. 53

1	AATGTAGTCT	TATGCAATAC	TCTTGTAGTC	TTGCAACATG	GTAACGATGA	GTTAGCAACA
61	TGCCTTACAA	GGAGAGAAAA	AGCACCGTGC	ATGCCGATTG	GTGGAAGTAA	GGTGGTACGA
121	TCGTGCCTTA	TTAGGAAGGC	AACAGACGGG	TCTGACATGG	ATTGGACGAA	CCACTGAATT
181	GCCGCATTGC	AGAGATATTG	TATTTAAGTG	CCTAGCTCGA	TACATAAACG	GGTCTCTCTG
241	GTTAGACCAG	ATCTGAGCCT	GGGAGCTCTC	TGGCTAACTA	GGGAACCCAC	TGCTTAAGCC
301	TCAATAAAGC	TTGCCTTGAG	TGCTTCAAGT	AGTGTGTGCC	CGTCTGTTGT	GTGACTCTGG
361	TAACTAGAGA	TCCCTCAGAC	CCTTTTAGTC	AGTGTGGAAG	ATCTCTAGCA	GTGGCGCCCG
421	AACAGGGACT	TGAAAGCGAA	AGGGAACCA	GAGGAGCTCT	CTCGACGCAG	GACTCGGCTT
481	GCTGAAGCGC	GCACGGCAAG	AGGCGAGGGG	CGGCGACTGG	TGAGTACGCC	AAAAATTTTG
541	ACTAGCGGAG	GCTAGAAGGA	GAGAGATGGG	TGCGAGAGCG	TCAGTATTAA	GCGGGGGAGA
601	ATTAGATCGC	GATGGGAAAA	AATTCGTTA	AGGCCAGGGG	GAAAGAAAAA	ATATAAATTA
661	AAACATATAG	TATGGGCAAG	CAGGGAGCTA	GAACGATTCC	CAGTTAATCC	TGGCCTGTTA
721	GAAACATCAG	AAGGCTGTAG	ACAAATACTG	GGACAGCTAC	AACCATCCCT	TCAGACAGGA
781	TCAGAAGAAC	TTAGATCATT	ATATAATACA	GTAGCAACCC	TCTATTGTGT	GCATCAAAGG
841	ATAGAGATAA	AAGACACCAA	GGAAGCTTTA	GACAAGATAG	AGGAAGAGCA	AAACAAAAGT
901	AAGACCACCG	CACAGCAAGC	GGCCGCTGAT	CTTCAGACCT	GGAGGAGGAG	ATATGAGGGA
961	CAATTGGAGA	AGTGAATTAT	ATAAATATAA	AGTAGTAAAA	ATTGAACCAT	TAGGAGTAGC
1021	ACCCACCAAG	GCAAAGAGAA	GAGTGGTGCA	GAGAGAAAAA	AGAGCAGTGG	GAATAGGAGC
1081	TTTGTTCCTT	GGGTTCTTGG	GAGCAGCAGG	AAGCACTATG	GGCGCAGCGT	CAATGACGCT
1141	GACGGTACAG	GCCAGACAAT	TATTGTCTGG	TATAGTGCAG	CAGCAGAACA	ATTTGCTGAG
1201	GGCTATTGAG	GGCAACAGC	ATCTGTTGCA	ACTCACAGTC	TGGGGCATCA	AGCAGCTCCA
1261	GGCAAGAATC	CTGGCTGTGG	AAAGATACCT	AAAGGATCAA	CAGCTCCTGG	GGATTTGGGG
1321	TTGCTCTGGA	AAACTCATTT	GCACCACTGC	TGTGCCTTGG	AATGCTAGTT	GGAGTAATAA
1381	ATCTCTGGAA	CAGATTTGGA	ATCACACGAC	CTGGATGGAG	TGGGACAGAG	AAATTAACAA
1441	TTACACAAGC	TTAATACACT	CCTTAATTGA	AGAATCGCAA	AACCAGCAAG	AAAAGAATGA
1501	ACAAGAATTA	TTGGAATTAG	ATAAATGGGC	AAGTTTGTGG	AATTGGTTTA	ACATAACAAA
1561	TTGGCTGTGG	TATATAAAAT	TATTCATAAT	GATAGTAGGA	GGCTTGGTAG	GTTTAAGAAT
1621	AGTTTTTGCT	GTACTTTCTA	TAGTGAATAG	AGTTAGGCAG	GGATATTCAC	CATTATCGTT
1681	TCAGACCCAC	CTCCAACCC	CGAGGGGACC	CGACAGGCC	GAAGGAATAG	AAGAAGAAGG
1741	TGGAGAGAGA	GACAGAGACA	GATCCATTCC	ATTAGTGAAC	GGATCTCGAC	GGTATCGCTA
1801	GCTTTTAAAA	GAAAAGGGGG	GATTGGGGGG	TACAGTGCAG	GGGAAAGAAT	AGTAGACATA
1861	ATAGCAACAG	ACATACAAAC	TAAAGAATTA	CAAAAACAAA	TTACAAAAAT	TCAAAATTTT
1921	ACTAGTATCA	ACTTTGTATA	GAAAAGTTGG	GCTCCGGTGC	CCGTCAGTGG	GCAGAGCGCA
1981	CATCGCCAC	AGTCCCGGAG	AAGTTGGGGG	GAGGGGTCCG	CAATTGAACC	GGTGCCTAGA
2041	GAAGGTGGCG	CGGGGTAAAC	TGGGAAAGTG	ATGTCGTGTA	CTGGCTCCGC	CTTTTTCCCG
2101	AGGGTGGGGG	AGAACCGTAT	ATAAGTGCAG	TAGTCGCCGT	GAACGTTCTT	TTTCGCAACG
2161	GGTTTGCCGC	CAGAACACAG	GTAAGTGCCG	TGTGTGGTTC	CCGCGGGCCT	GGCCTCTTTA
2221	CGGGTTATGG	CCCTTGCGTG	CCTTGAATTA	CTTCCACCTG	GCTGCAGTAC	GTGATTCTTG
2281	ATCCCGAGCT	TCGGGTGGA	AGTGGGTGGG	AGAGTTCGAG	GCCTTGCGCT	TAAGGAGCCC
2341	CTTCGCTCG	TGCTTGAGTT	GAGGCCTGCC	CTGGGCGCTG	GGGCCGCCGC	GTGCGAATCT
2401	GGTGGCACCT	TCGCGCCTGT	CTCGCTGCTT	TCGATAAGTC	TCTAGCCATT	TAAAATTTTT
2461	GATGACCTGC	TGCGACGCTT	TTTTTCTGGC	AAGATAGTCT	TGTAAATGCG	GGCCAAGATC
2521	TGCACACTGG	TATTTGCGTT	TTTGGGGCCG	CGGGCGGCGA	CGGGGCCCGT	GGCTCCAGC
2581	GCACATGTTT	GGCGAGGCGG	GGCCTGCGAG	CGCGGCCACC	GAGAATCGGA	CGGGGGTAGT
2641	CTCAAGCTGG	CCGGCCTGCT	CTGGTGCTTG	GTCTCGCGCC	GCCGTGTATC	GGCCCGCCCT
2701	GGGCGGCAAG	GCTGGCCCGG	TCGGCACCAG	TTGCGTGAGC	GGAAAGATGG	CCGCTTCCCG

FIG. 9B-1

2761	GCCCTGCTGC	AGGGAGCTCA	AAATGGAGGA	CGCGGCGCTC	GGGAGAGCGG	GCGGGTGAGT
2821	CACCCACACA	AAGGAAAAGG	GCCTTCCGT	CCTCAGCCGT	CGCTTCATGT	GACTCCACGG
2881	AGTACCGGGC	GCCGTCCAGG	CACCTCGATT	AGTTCTCGAG	CTTTTGGAAT	ACGTCGTCTT
2941	TAGGTTGGGG	GGAGGGGTTT	TATGCGATGG	AGTTTCCCCA	CACCTAGTGG	GTGGAGACTG
3001	AAGTTAGGCC	AGCTTGGCAC	TTGATGTAAT	TCTCCTTGA	ATTTGCCCTT	TTTGAGTTTG
3061	GATCTTGTT	CATTCTCAAG	CCTCAGACAG	TGGTTCAAAG	TTTTTTTCTT	CCATTTACAGG
3121	TGTCGTGACA	AGTTTGTAACA	AAAAAGCAGG	CTGCCACCAT	GAGCACCGAG	AGCATGATCA
3181	GAGACGTGGA	GCTGGCCGAG	GAGGCCCTGC	CCAAGAAGAC	CGGCGGCCCC	CAGGGCAGCA
3241	GAAGATGCCT	GTTCTGTAGC	CTGTTCAAGT	TCCTGATCGT	GGCCGGCGCC	ACCACCCTGT
3301	TCTGCCTGCT	GCACCTCGGC	GTGATCGGCC	CCCAGAGAGA	GGAGTTCCCC	AGAGACCTGA
3361	GCCTGATCAG	CCCCCTGGCC	CAGGCCGTGG	CCCACGTGGT	GGCCAACCCC	CAGGCCGAGG
3421	GCCAGCTGCA	GTGGCTGAAC	AGAAGAGCCA	ACGCCCTGCT	GGCCAACGGC	GTGGAGCTGA
3481	GAGACAACCA	GCTGGTGTTG	CCCAGCGAGG	GCCTGTACCT	GATCTACAGC	CAGGTGCTGT
3541	TCAAGGGCCA	GGGCTGCCCC	AGCACCCACG	TGCTGCTGAC	CCACACCATC	AGCAGAATCG
3601	CCGTGAGCTA	CCAGACCAAG	GTGAACCTGC	TGAGCGCCAT	CAAGAGCCCC	TGCCAGAGAG
3661	AGACCCCGA	GGGCGCCGAG	GCCAAGCCCT	GGTACGAGCC	CATCTACCTG	GGCGGCGTGT
3721	TCCAGCTGGA	GAAGGGCGAC	AGACTGAGCG	CCGAGATCAA	CAGACCCGAC	TACCTGGACT
3781	TCGCCGAGAG	CGGCCAGGTG	TACTTCGGCA	TCATCGCCCT	GTAACCCAG	CTTCTTGTGA
3841	CAAAGTGTTG	ATAATCGAAT	TCTAAATAGA	TAGAACAACA	ACAATTGCAT	TCATTTTTGA
3901	TTTCAGGTTT	AGGGGGAGGT	GTGGGAGGTT	TTTTAAAGCA	AGTAAACCTT	CTACACTGAC
3961	GGTACGCGTT	AACAACAACA	ATTGCATTCA	TTTGTAGTTT	CAGGTTTCAGG	GGGAGGTGTG
4021	GGAGGTTTTT	TAAAGCAAGT	TAAACCTCTA	AAATAGTGGT	ACGCGTTACC	CAGCTTTCTT
4081	GTACAAAGTG	GACCCAGCTT	TCTTGTAACA	AGTGGGCCCC	TCTCCCTCCC	CCCCCCTAA
4141	CGTTACTGGC	CGAAGCCGCT	TGGAATAAGG	CCGGTGTGCG	TTTGCTCTATA	TGTTATTTTC
4201	CACCATATTG	CCGTCTTTTG	GCAATGTGAG	GGCCCGGAAA	CCTGGCCCTG	TCTTCTTGAC
4261	GAGCATTCTT	AGGGGTCTTT	CCCCTCTCGC	CAAAGGAATG	CAAGGTCTGT	TGAATGTCTG
4321	GAAGGAAGCA	GTTCTCTGCG	AAGCTTCTTG	AAGACAAACA	ACGTCTGTAG	CGACCCTTTG
4381	CAGGCAGCGG	AACCCCCAC	CTGGCGACAG	GTGCCTCTGC	GGCCAAAAGC	CACGTGTATA
4441	AGATACACCT	GCAAAGGCGG	CACAACCCCA	GTGCCACGTT	GTGAGTTGGA	TAGTTGTGGA
4501	AAGAGTCAAA	TGGCTCTCCT	CAAGCGTATT	CAACAAGGGG	CTGAAGGATG	CCCAGAAGGT
4561	ACCCCATTTG	ATGGGATCTG	ATCTGGGGCC	TCGGTGCACA	TGCTTTACAT	GTGTTTAGTC
4621	GAGGTTAAAA	AAACGTCTAG	GCCCCCGGAA	CCACGGGGAC	GTGGTTTTTC	TTTGAAAAAC
4681	ACGATGATAA	TATGGCCACA	ACCATGGCCA	CCGTGCTGGC	CCCCGCTTGG	AGCCCCACCA
4741	CCTACCTGCT	GCTGCTGCTG	CTGCTGAGCA	GCGGCCCTGAG	CGGCGGCGGC	GGCAGCGGCA
4801	AGCCCATCCC	CAACCCCTTG	CTGGGCTTGG	ACAGCACCGG	CGGCGGCGGC	AGCCAGGTGA
4861	AGCTGCAGGA	GAGCGGCCCC	GGCCTGTTGG	CCCCAGCCA	GAGCTGAGC	ATCACCTGCA
4921	CCGTGAGCGG	CTTCAGCCTG	ACCGCTACG	GCGTGGACTG	GGTGAGACAG	CCCCCGGCA
4981	AGTGCCTGGA	GTGGCTGGGC	GTGATCTGGG	GCGGCGGCA	AACCAACTAC	AACAGCGGCC
5041	TGATGAGCAG	ACTGAGCATC	AGAAAGGACA	ACAGCAAGAG	CCAGGTGTTT	CTGACCATGA
5101	ACAGCCTGCA	GACCGACGAC	ACCGCCAAGT	ACTACTGCGT	GAAGCACACC	AACTGGGACG
5161	GCGGCTTCGC	CTACTGGGGC	CAGGGCACCA	CCGTGACCGT	GAGCAGCGGC	GGCGGCGGCA
5221	GCGGCGGCGG	CGGCAGCGGC	GGCGGCGGCA	GCGGCAGCCC	CGGCAGAGC	GTGAGCATCA
5281	GCTGCAGCGG	CAGCAGCAGC	AACATCGGCA	ACAACACTGT	GACTGTTGAC	CAGCACCTGC
5341	CCGGCACCGC	CCCCAAGCTG	CTGATCTACA	GCGACACCAA	GAGACCCAGC	GGCGTGCCCG
5401	ACAGAATCAG	CGGCAGCAAG	AGCGGCACCA	GCGCCAGCCT	GGCCATCAGC	GGCCTGCAGA
5461	GCGAGGACGA	GGCCGACTAC	TACTGCGCCA	GCTGGGACGA	CAGCTGGAC	GGCCCCGTGT
5521	TCGGCTGCGG	CACCAAGCTG	ACCGTGCTGC	CCACCACCAC	CCCCGCCCCC	AGACCCCCCA
5581	CCCCCGCCCC	CACCATCGCC	AGCCAGCCCC	TGAGCCTGAG	ACCCGAGGCC	TGCAGACCCG
5641	CCGCGGCGGG	CGCCGTGCAC	ACCAGAGGCC	TGGACTTCGC	CTGCGACATC	TACATCTGGG
5701	CCCCCTGGC	CGGCACCTGC	GGCGTGCTGC	TGCTGAGCCT	GGTGATCACC	CGTGGTGAAG
5761	ACCACAGAAA	CAGAAGAAGA	GTGTGCAAGT	GCCCCAGACC	CGTGGTGAAG	AGCGGCGACA

FIG. 9B-2

5821	AGCCCAGCCT	GAGCGCCAGA	TACGTGTAAC	AACTTTATTA	TACATAGTTG	ATCAATTCCA
5881	ACTTTATTAT	ACATAGTTGA	TCAATTCGGA	TAATCAACCT	CTGGATTACA	AAATTTGTGA
5941	AAGATTGACT	GGTATTCTTA	ACTATGTTGC	TCCTTTTACG	CTATGTGGAT	ACGCTGCTTT
6001	AATGCCTTTG	TATCATGCTA	TTGCTTCCCG	TATGGCTTTC	ATTTTCTCCT	CCTTGATATA
6061	ATCCTGGTTG	CTGTCTCTTT	ATGAGGAGTT	GTGGCCCGTT	GTGAGGCAAC	GTGGCGTGGT
6121	GTGCACTGTG	TTTGCTGACG	CAACCCCCAC	TG6TTGGGGC	ATTGCCACCA	CCTGTCA6CT
6181	CCTTTCGGGG	ACTTTCGCTT	TCCCCCTCCC	TATTGCCACG	GCGGAACCTA	TGCGCGCCTG
6241	CCTTGCCCGC	TGCTGGACAG	GGGCTCGGCT	GTTGGGCACT	GACAATTCGG	TGGTGTGTGC
6301	GGGGAAGCTG	ACGTCCTTTC	CATGGCTGCT	CGCCTGTGTT	GCCACCTGGA	TTCTGCGCGG
6361	GACGTCTTTC	TGCTACGTCC	CTTCGGCCCT	CAATCCAGCG	GACCTTCCTT	CCGCGGCGCT
6421	GCTGCCGGCT	CTGCGGCCTC	TTCCGCGTCT	TCGCTTTCGC	CCTCAGACGA	GTGCGATCTC
6481	CCTTTGGGCC	GCCTCCCCGC	ATCGGGAATT	CCGCGGTTTC	GCTTTAAGAC	CAATGACTTA
6541	CAAGGCAGCT	GTAGATCTTA	GCCACTTTTT	AAAAGAAAAG	GGGGGACTGG	AAGGGCTAAT
6601	TCACTCCCAA	CGAAGACAAG	ATCTGCTTTT	TGCTTGTAAT	GGGTCTCTCT	GGTTAGACCA
6661	GATCTGAGCC	TGGGAGCTCT	CTGGCTAACT	AGGGAACCCA	CTGCTTAAGC	CTCAATAAAG
6721	CTTGCCTTGA	GTGCTTCAAG	TAGTGTGTGC	CCGTCTGTTG	TGTGACTCTG	GTAAGTAGAG
6781	ATCCCTCAGA	CCCTTTTAGT	CAGTGTGGAA	AATCTCTAGC	AGTAGTAGTT	CATGTCATCT
6841	TATTATTCAG	TATTTATAAC	TTGCAAAGAA	ATGAATATCA	GAGAGTGAGA	GGAACTTGTT
6901	TATTGCAGCT	TATAATGTTT	ACAAATAAAG	CAATAGCATC	ACAAATTTCA	CAAATAAAGC
6961	ATTTTTTTCA	CTGCATTCTA	GTTGTGGTTT	GTCCAAACTC	ATCAATGTAT	CTTATCATGT
7021	CTGGCTCTAG	CTATCCCGCC	CCTAACTCCG	CCCATCCCGC	CCCTAACTCC	GCCCAAGTTCC
7081	GCCCATCTCT	CGCCCATGGG	CTGACTAATT	TTTTTTATTT	ATGCAGAGGC	CGAGGCCGCC
7141	TCGGCTCTCT	AGCTATTCCA	GAAGTAGTGA	GGAGGCTTTT	TTGGAGGCCT	AGGGACGTAC
7201	CCAATTCGCC	CTATAGTGAG	TCGTATTACG	CGCGCTCACT	GCGCGTCTGT	TTACAACGTC
7261	GTGACTGGGA	AAACCCTGGC	GTTACCCAAC	TTAATCGCCT	TGCAGCACAT	CCCCCTTTCC
7321	CCAGCTGGCG	TAATAGCGAA	GAGGCCCGCA	CCGATCGCCC	TTCCCAACAG	TTGCGCAGCC
7381	TGAATGGCGA	ATGGGACGCG	CCCTGTAGCG	GC6CATTAA6	CGCGGCGGGT	GTGGTGGTTA
7441	CGCGCAGCGT	GACCGCTACA	CTTGCCAGCG	CCCTAGCGCC	CGCTCCTTTC	GCTTTCTTCC
7501	CTTCCTTTCT	CGCCACGTTT	GCCGGCTTTC	CCCGTCAAGC	TCTAAATCGG	GGGCTCCCTT
7561	TAGGGTTCCG	ATTTAGTGCT	TTACGGCACC	TCGACCCCAA	AAAACCTGAT	TAGGGTGATG
7621	GTTACAGTAG	TGGGCCATCG	CCCTGATAGA	CGGTTTTTCG	CCCTTTGACG	TTGGAGTCCA
7681	CGTTCCTTAA	TAGTGGACTC	TTGTTCCAAA	CTGGAACAAC	ACTCAACCCT	ATCTCGGTCT
7741	ATTCTTTTGA	TTTATAAGGG	ATTTTGCCGA	TTTCGGCCTA	TTGGTTAAAA	AATGAGCTGA
7801	TTTAACAAAA	ATTTAACGCG	AATTTTAACA	AAATATTAAC	GCTTACAATT	TAGGTGGCAC
7861	TTTTCGGGGA	AATGTGCGCG	GAACCCCTAT	TTGTTTATTT	TTCTAAATAC	ATTCAAATAT
7921	GTATCCGCTC	ATGAGACAAT	AACCCGTGATA	AATGCTTCAA	TAATATTGAA	AAAGGAAGAG
7981	TATGAGTATT	CAACATTTCC	GTGTCGCCCT	TATTCCTTTT	TTTGCGGCAT	TTTGCTTTCC
8041	TGTTTTTGCT	CACCCAGAAA	CGCTGGTGAA	AGTAAAAGAT	GCTGAAGATC	AGTTGGGTGC
8101	ACGAGTGGGT	TACATCGAAC	TGGATCTCAA	CAGCGGTAAG	ATCCTTGAGA	GTTCCTGCCC
8161	CGAAGAACGT	TTTCCAATGA	TGAGCACTTT	TAAAGTTCTG	CTATGTGGCG	CGGTATTATC
8221	CCGTATTGAC	GCCGGGCAAG	AGCAACTCGG	TCGCCGCATA	CACTATTCTC	AGAATGACTT
8281	GGTTGAGTAC	TCACCACTCA	CAGAAAAGCA	TCTTACGGAT	GGCATGACAG	TAAGAGAATT
8341	ATGCAGTGCT	GCCATAACCA	TGAGTGATAA	CACTGCGGCC	AACTTACTTC	TGACAACGAT
8401	CGGAGGACCG	AAGGAGCTAA	CCGCTTTTTT	GCACAACATG	GGGGATCATG	TAACTCGCCT
8461	TGATCGTTGG	GAACCGGAGC	TGAATGAAGC	CATACCAAAC	GACGAGCGTG	ACACCACGAT
8521	GCCTGTAGCA	ATGGCAACAA	CGTTGC6CAA	ACTATTAAC	GGCGAACTAC	TTACTCTAGC
8581	TTCCCGGCAA	CAATTAATAG	ACTGGATGGA	GGCGGATAAA	GTTGCAGGAC	CACCTCTGCG
8641	CTCGGCGCTT	CCGGCTGGCT	GGTTTATTGC	TGATAAATCT	GGAGCCGGTG	AGCGTGGGTC
8701	TCGCGGTATC	ATTGCAGCAC	TGGGGCCAGA	TGGTAAGCCC	TCCCGTATCG	TAGTTATCTA
8761	CACGACGGGG	AGTCAGGCAA	CTATGGATGA	ACGAAATAGA	CAGATCGCTG	AGATAGGTGC
8821	CTCACTGATT	AAGCATTGGT	AACTGTCAGA	CCAAGTTTAC	TCATATATAC	TTTAGATTGA

FIG. 9B-3

8881	TTTAAACTT	CATTTTAAAT	TTAAAGGAT	CTAGGTGAAG	ATCCTTTTTG	ATAATCTCAT
8941	GACCAAAATC	CCTTAACGTG	AGTTTTCGTT	CCACTGAGCG	TCAGACCCCG	TAGAAAAGAT
9001	CAAAGGATCT	TCITGAGATC	CTTTTTTCT	GCGCGTAATC	TGCTGCTTGC	AAACAAAAAA
9061	ACCACCGCTA	CCAGCGGTGG	TTTGTTTGGC	GGATCAAGAG	CTACCAACTC	TTTTTCCGAA
9121	GGTAACTGGC	TTCAGCAGAG	CGCAGATACC	AAATACTGTT	CTTCTAGTGT	AGCCGTAGTT
9181	AGGCCACCAC	TTCAAGAACT	CTGTAGCACC	GCCTACATAC	CTGGCTCTGC	TAATCTGTT
9241	ACCAGTGGCT	GCTGCCAGTG	GCGATAAGTC	GTGTCTTACC	GGGTTGGACT	CAAGACGATA
9301	GTTACCGGAT	AAGGCGCAGC	GGTCGGGCTG	AACGGGGGGT	TCGTGCACAC	AGCCGAGCTT
9361	GGAGCGAACG	ACCTACACCG	AACTGAGATA	CCTACAGCGT	GAGCTATGAG	AAAGCGCCAC
9421	GCTTCCCGAA	GAGAGAAAGG	CGGACAGGTA	TCCGGTAAGC	GGCAGGGTCG	GAACAGGAGA
9481	GCGCACGAGG	GAGCTTCCAG	GGGGAAACGC	CTGGTATCTT	TATAGTCCTG	TCGGGTTTCG
9541	CCACCTCTGA	CTTGAGCGTC	GATTTTTGTG	ATGCTCGTCA	GGGGGGCGGA	GCCTATGGAA
9601	AAACGCCAGC	AACGCGGCCT	TTTTACGGTT	CCTGGCCTTT	TGCTGGCCTT	TTGCTCACAT
9661	GTTCTTTCCT	GCGTTATCCC	CTGATTCTGT	GGATAACCGT	ATTACCGCCT	TTGAGTGAGC
9721	TGATACCGCT	CGCCGCAGCC	GAACGACCGA	GCGCAGCGAG	TCAGTGAGCG	AGGAAGCGGA
9781	AGAGCGCCCA	ATACGCAAAC	CGCCTCTCCC	CGCGCGTTGG	CCGATTCATT	AATGCAGCTG
9841	GCACGACAGC	TTTCCCGACT	GGAAAGCGGG	CAGTGAGCGC	AACGCAATTA	ATGTGAGTTA
9901	GCTCACTCAT	TAGGCACCCC	AGGCTTTACA	CTTTATGCTT	CCGGCTCGTA	TGTTGTGTGG
9961	AATTGTGAGC	GGATAACAAT	TTCACACAGG	AAACAGCTAT	GACCATGATT	ACGCCAAGCG
10021	CGCAATTAAC	CCTCACTAAA	GGGAACAAAA	GCTGGAGCTG	CAAGCTT	

FIG. 10

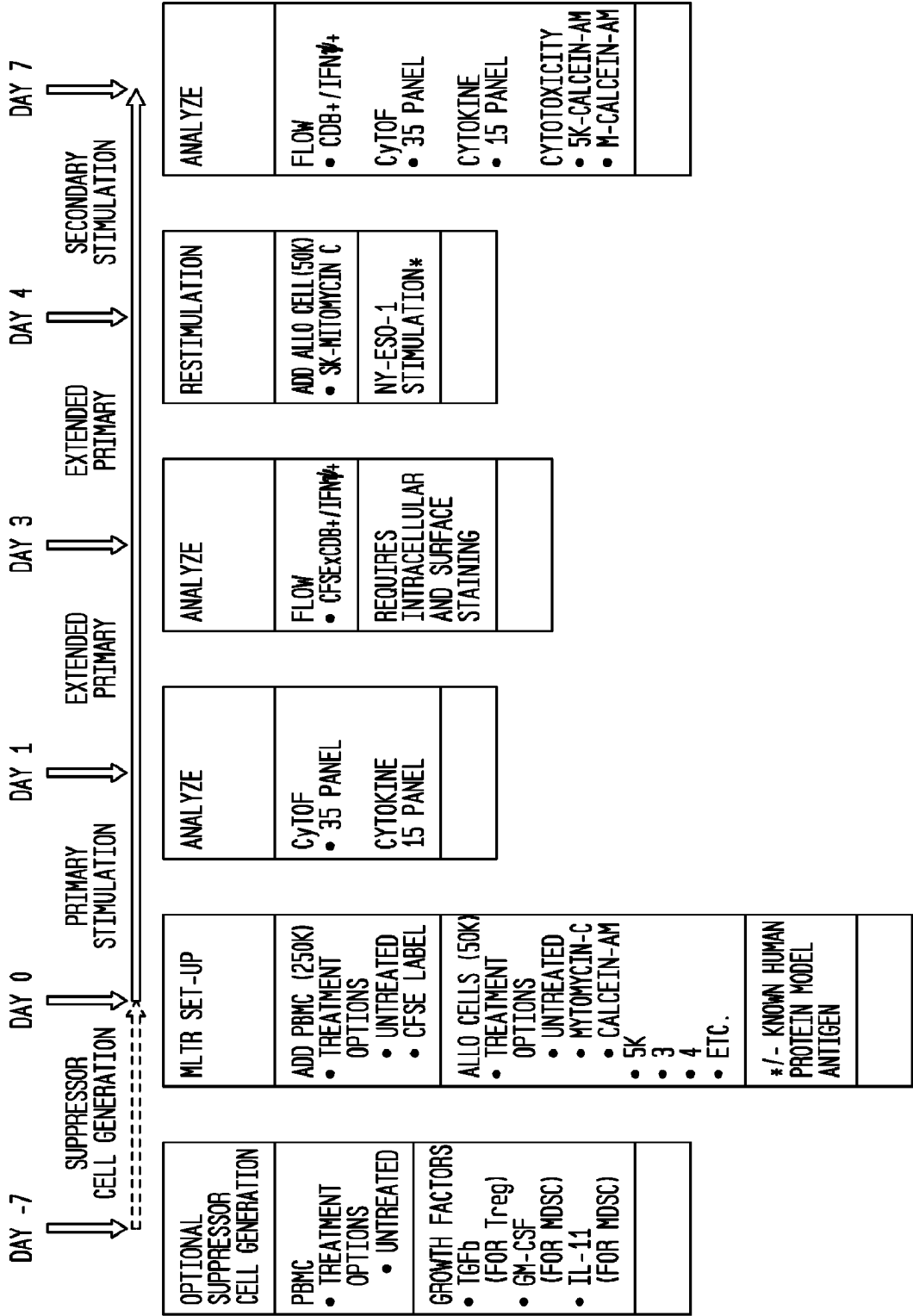
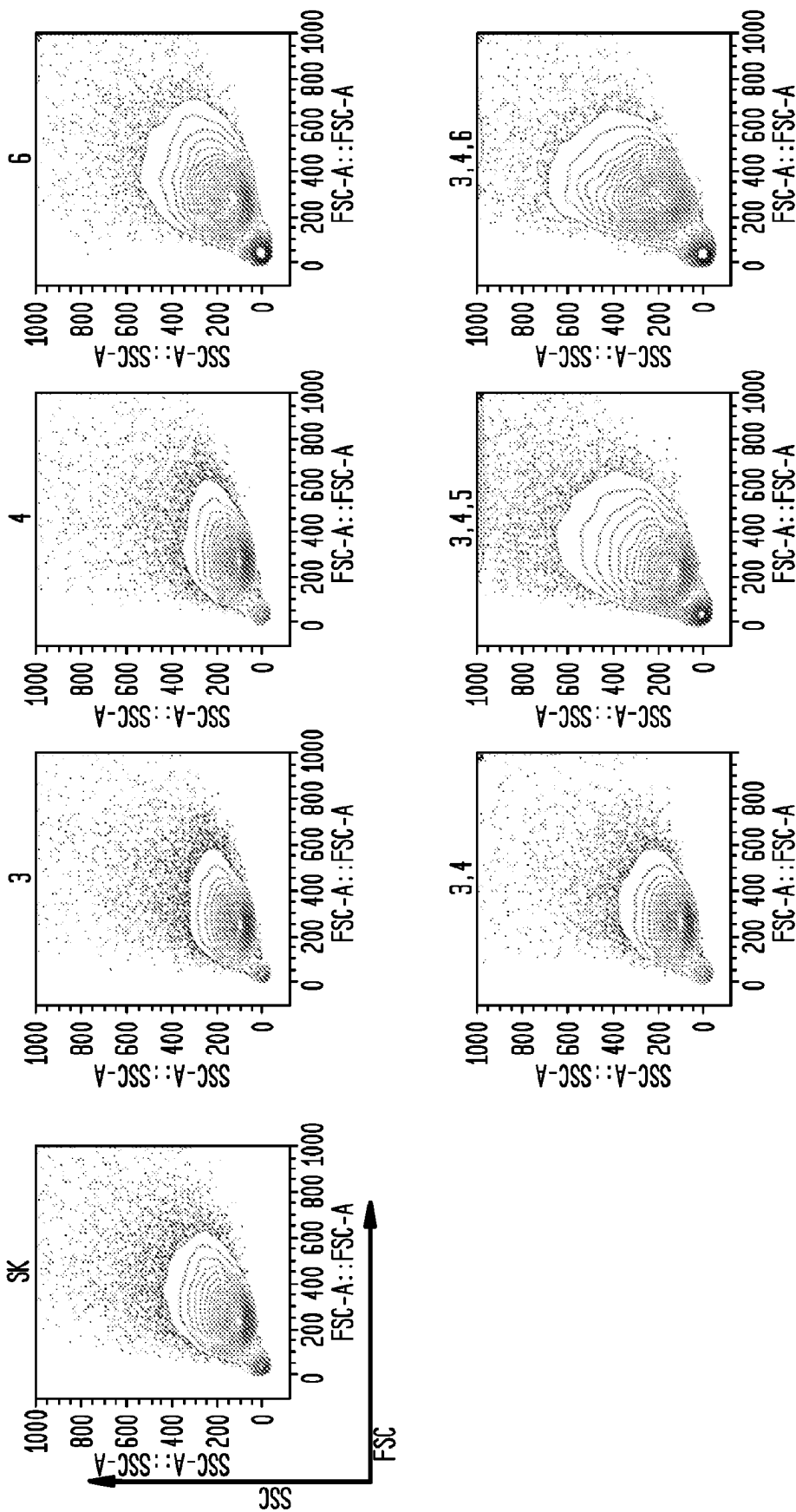


FIG. 11



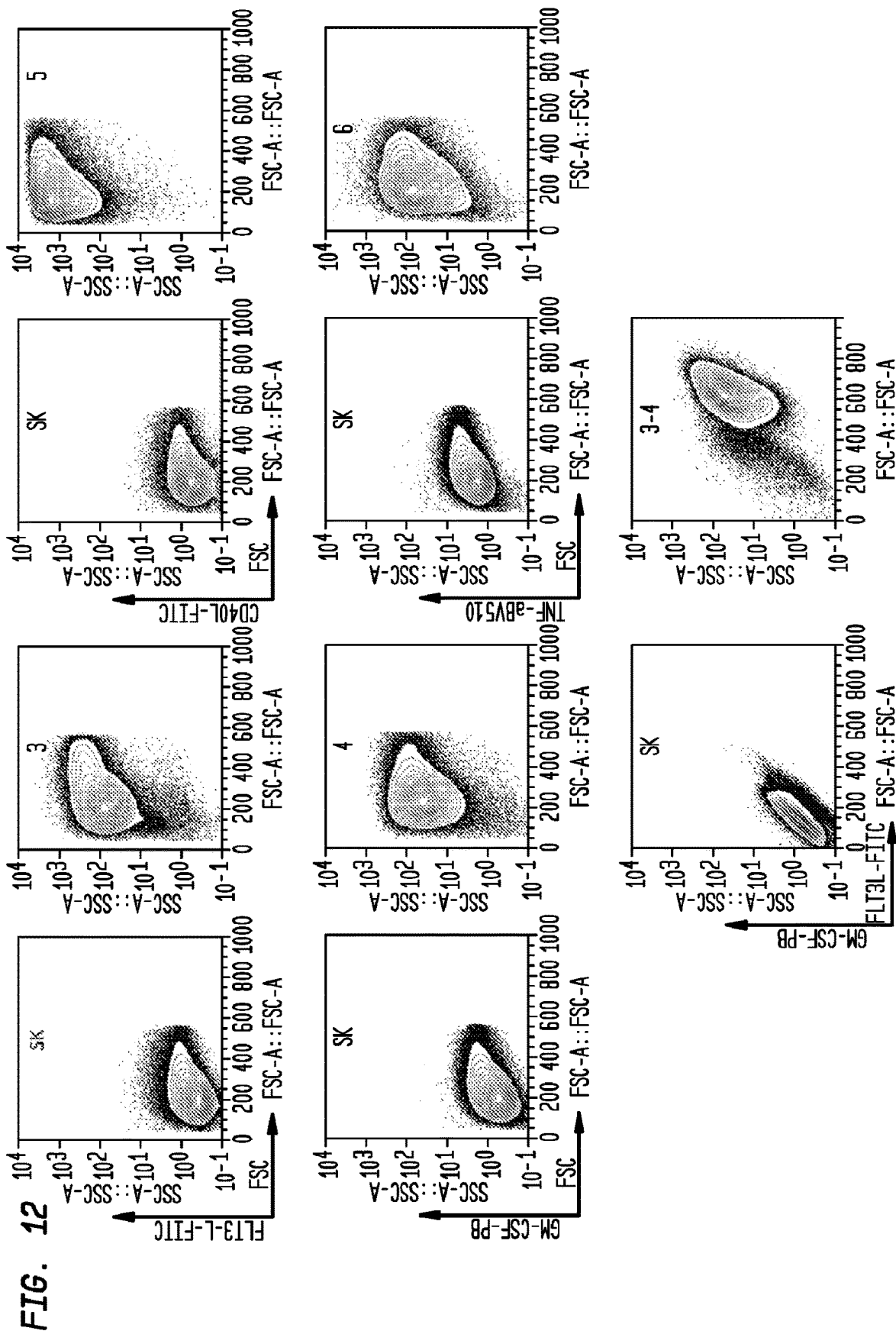


FIG. 13

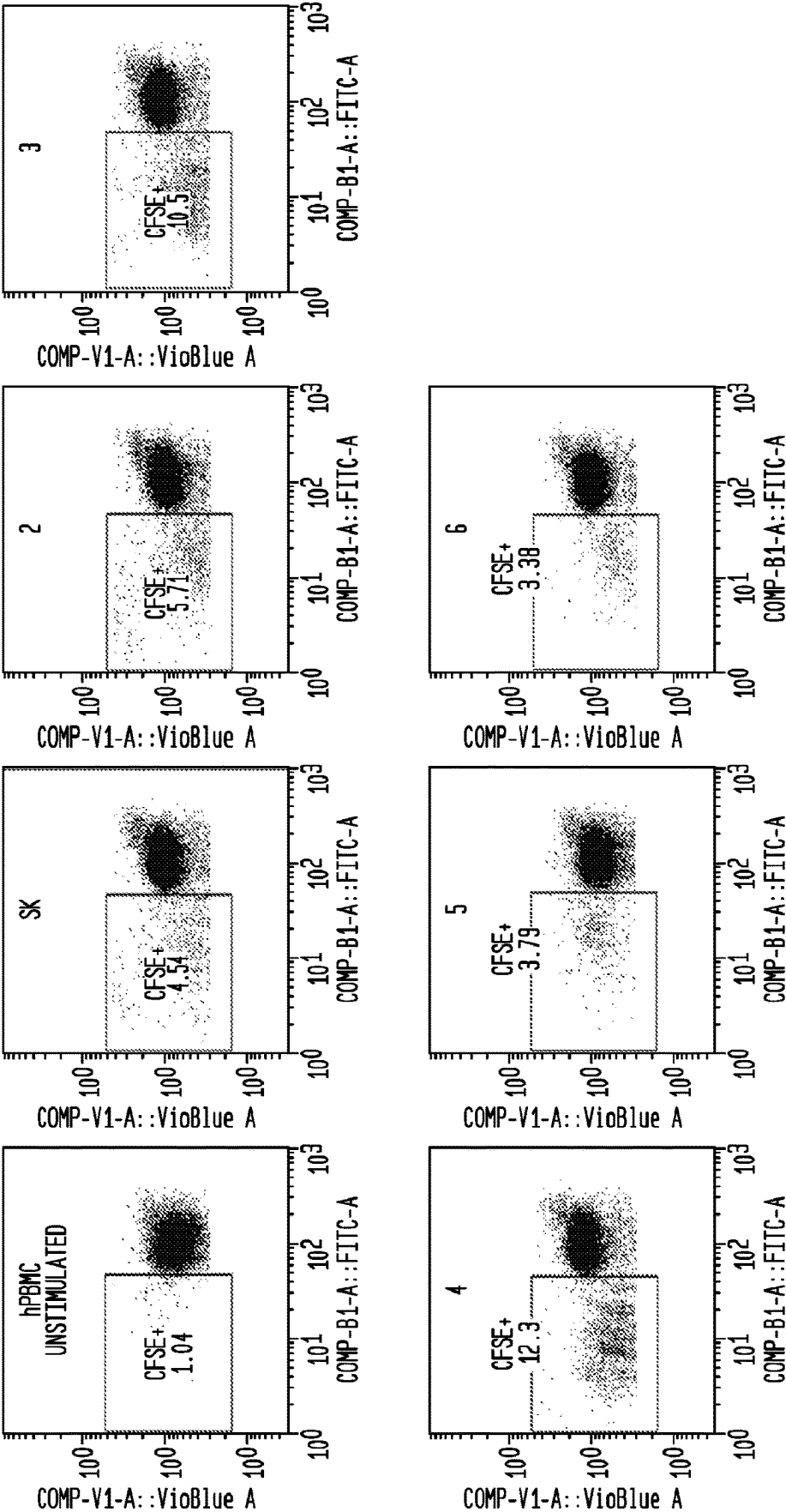
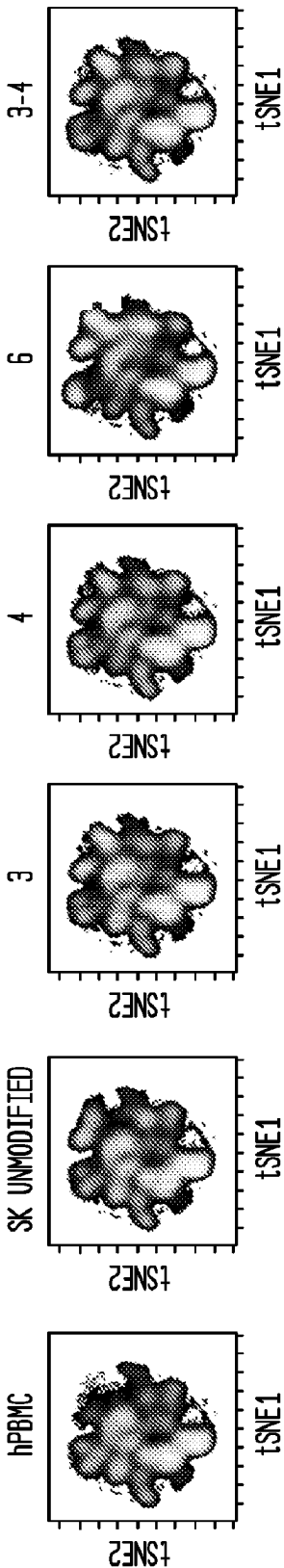
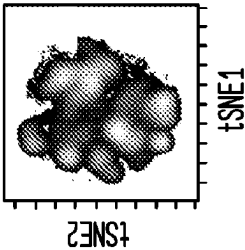


FIG. 14A



3-4-5



3-4-6

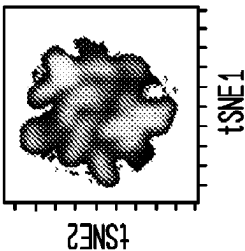


FIG. 14B

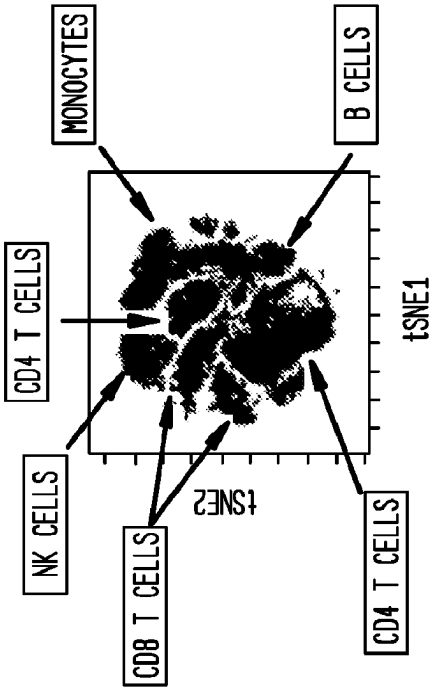


FIG. 15A

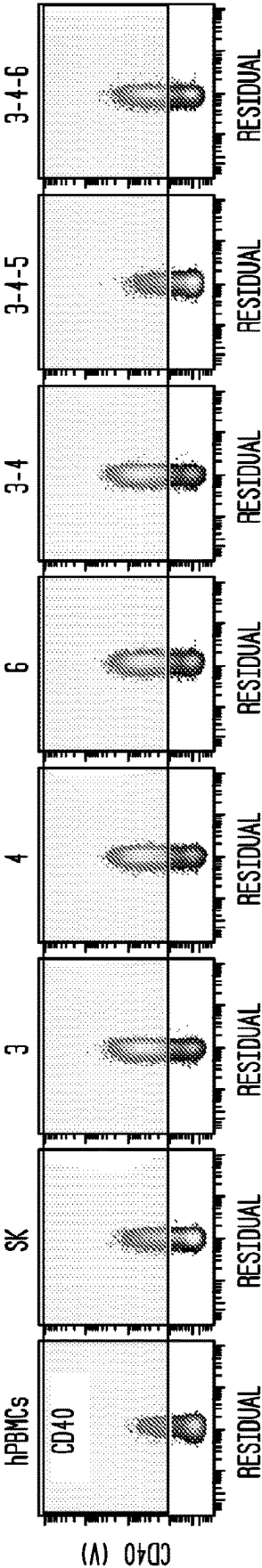


FIG. 15B

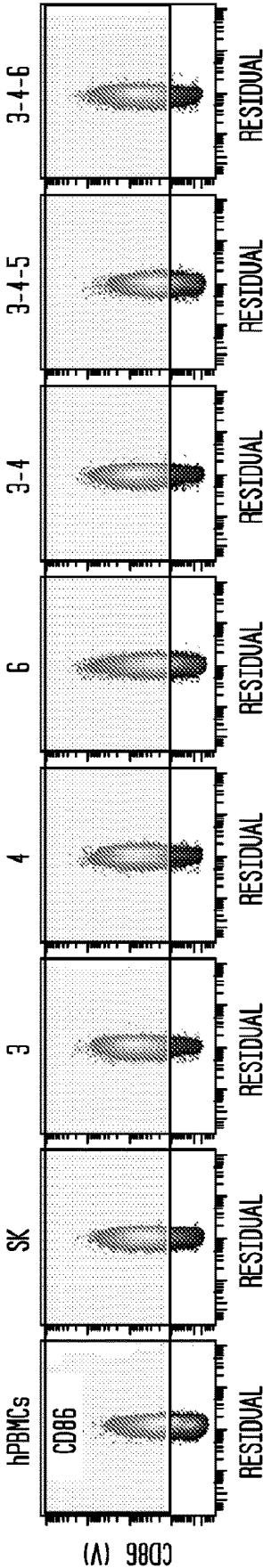


FIG. 15C

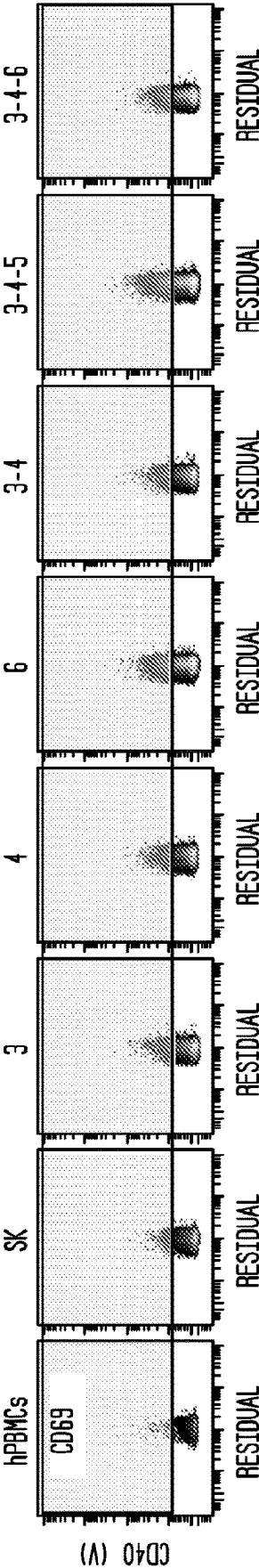


FIG. 15D

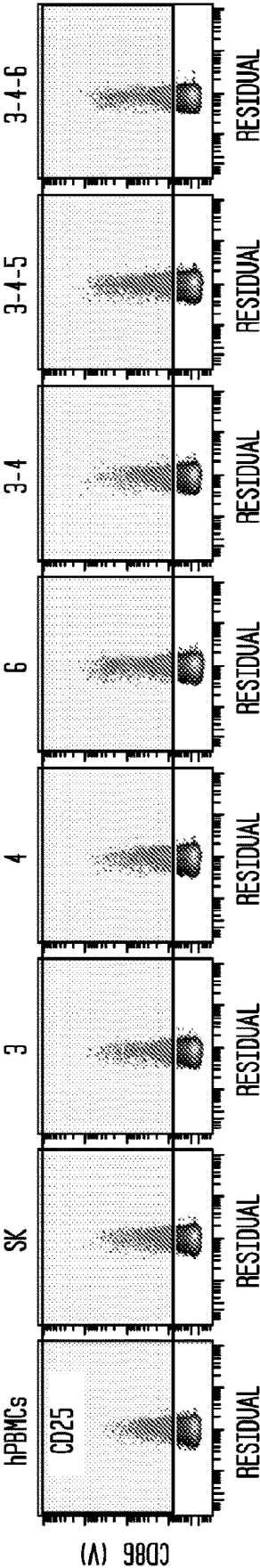


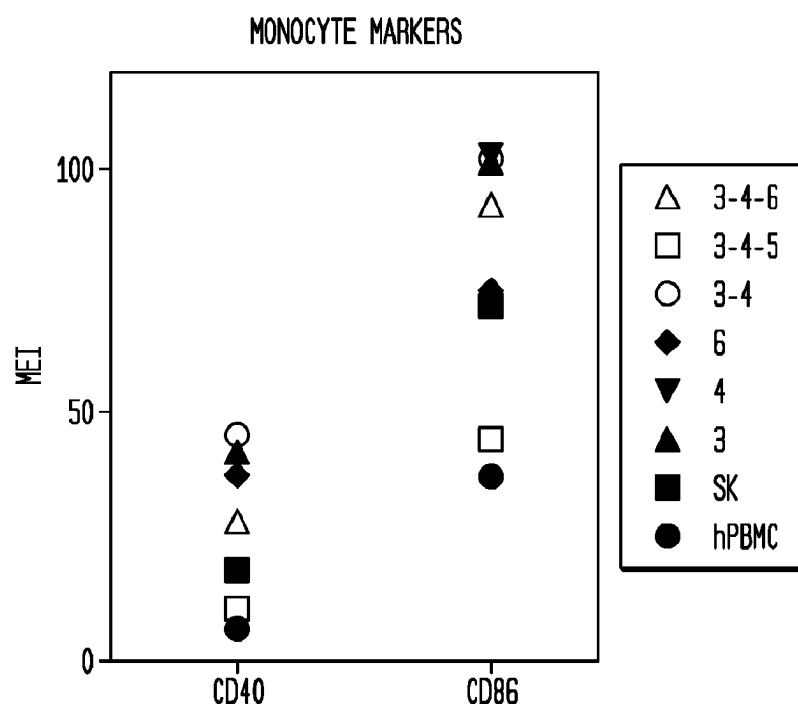
FIG. 15E

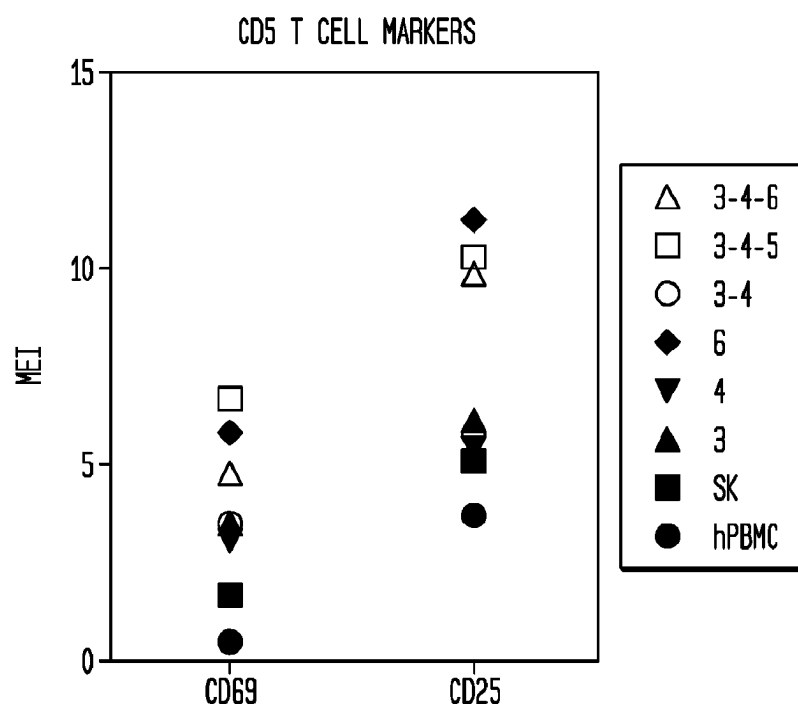
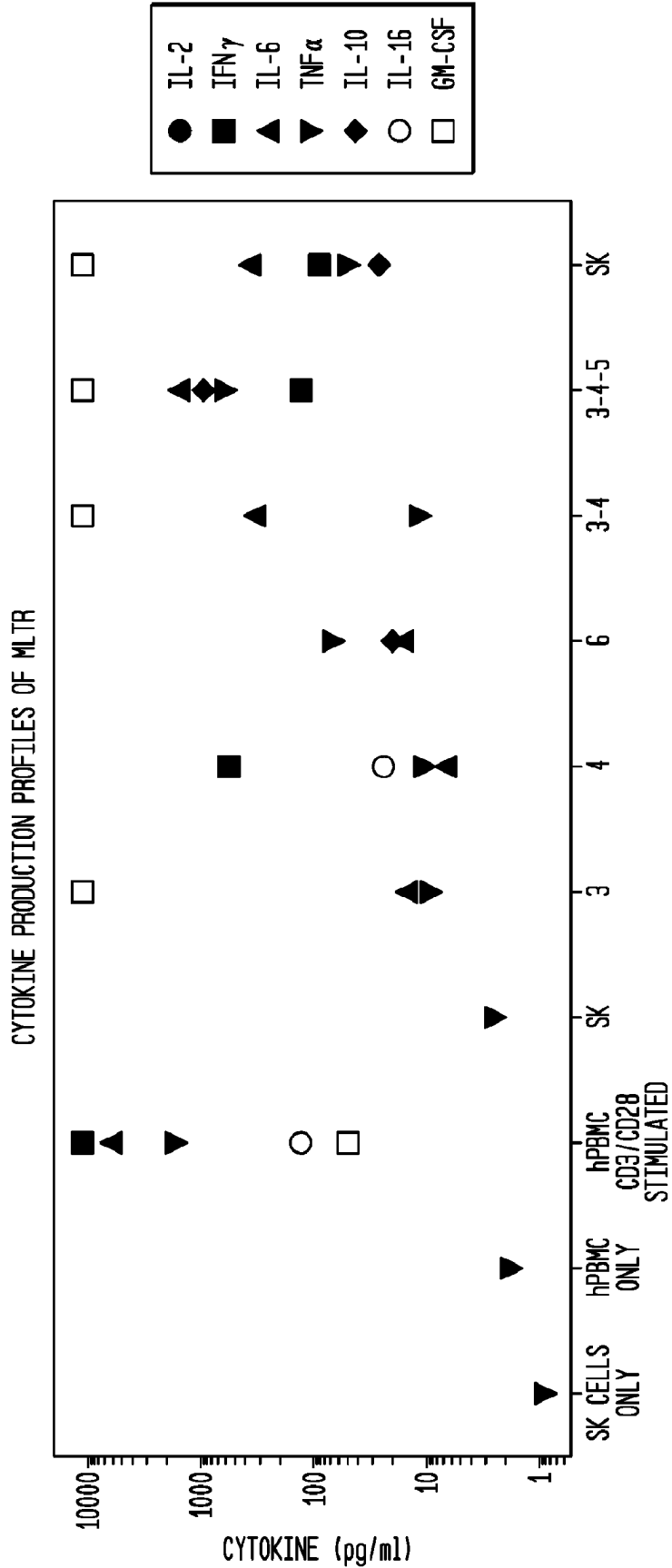
FIG. 15F

FIG. 16



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ALLOGENEIC TUMOR CELL VACCINE**RELATED APPLICATIONS**

This Application is a continuation of U.S. application Ser. No. 15/821,105, filed on Nov. 22, 2017, which claims priority to U.S. Provisional Application No. 62/425,424, filed on Nov. 22, 2016. The contents of each of these applications are incorporated herein by reference in their entirety.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format via EFS-Web in accordance with 37 C.F.R. §§ 1.821-1.825. Said ASCII copy, created on Apr. 15, 2021, is named "128663-00104SL" and is 247 KB in size.

FIELD OF THE INVENTION

The described invention relates generally to immunological approaches to the treatment of cancer, and more particularly to cancer vaccines comprising modified tumor cells.

BACKGROUND OF THE INVENTION**Immune Response**

Generally speaking, immune responses are initiated by an encounter between an individual and a foreign antigenic substance, e.g., an infectious microorganism. The infected individual rapidly responds with both a humoral immune response with the production of antibody molecules specific for the antigenic determinants/epitopes of the immunogen and a cell mediated immune response with the expansion and differentiation of antigen-specific regulatory and effector T-lymphocytes, including both cells that produce cytokines and killer T cells, capable of lysing infected cells. Primary immunization with a given microorganism evokes antibodies and T cells that are specific for the antigenic determinants/epitopes found on that microorganism, but that usually fail to recognize or recognize only poorly antigenic determinants expressed by unrelated microbes (Paul, W. E., "Chapter 1: The immune system: an introduction," *Fundamental Immunology*, 4th Edition, Ed. Paul, W. E., Lippincott-Raven Publishers, Philadelphia, (1999), at p. 102).

As a consequence of this initial response, the immunized individual develops a state of immunologic memory. If the same or a closely related microorganism is encountered again, a secondary response ensues. This secondary response generally consists of an antibody response that is more rapid, greater in magnitude and composed of antibodies that bind to the antigen with greater affinity and are more effective in clearing the microbe from the body, and a similarly enhanced and often more effective T-cell response. However, immune responses against infectious agents do not always lead to elimination of the pathogen. (Paul, W. E., "Chapter 1: The immune system: an introduction," *Fundamental Immunology*, 4th Edition, Ed. Paul, W. E., Lippincott-Raven Publishers, Philadelphia, (1999), at p. 102).

Immune Tolerance of Cancer

Cancer is characterized by genetic instability of particular cells but has also been described as a disorder of the immune system, based on the fact that the immune system fails, at least in certain segments of the afflicted human population, to respond optimally to cancerous cells that have taken on a distinctly non-self phenotype that should be recognized as

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foreign. Several reasons have been advanced to explain the basis of this observation. For example, first, cancer cells consist mainly of self-antigens, in striking contrast to the situation with infectious organisms. Some antigens that are classified as cancer antigens are actually normal antigens that are overexpressed, or normal antigens that have a mutation in only one or two amino acids in the polypeptide chain. Second, cancer cells down-regulate Major Histocompatibility Complex (MHC), and thus do not much present tumor cell-derived peptides by way of MHC. Third, cancer cells, and associated tumor-associated macrophages, express cytokines that dampen the immune response (see, e.g., Yu et al (2007) *Nature Rev. Immunol.* 7:41-51). This dampening is caused, for example, by the secretion of interleukin-10 (IL-10) by the cancer cells or by the associated macrophages. Fourth, unlike the situation with infections, cancer cells do not provide any immune adjuvant. Pathogens express a variety of naturally-occurring immune adjuvants, which take the form of toll-like receptor (TLR) agonists and NOD agonists (see, e.g., Kleinnijenhuis et al (2011) *Clin. Dev. Immunol.* 405310 (12 pages)). Generally, optimal activation of dendritic cells requires contact of an immune adjuvant with one or more toll-like receptors (TLRs) expressed by the dendritic cell. Without activation of the dendritic cell, contact between the dendritic cell and T cells (immune synapse) fails to result in optimal activation of the T cell.

Immune Surveillance and Immune Editing

Tumor immune editing is divided into three phases: an elimination phase, an equilibrium phase, and an escape phase. The elimination phase, also known as immune surveillance, is the process by which the immune system identifies cancerous or pre-cancerous cells and eliminates them before they grow out of control. This phase can be complete when all cancerous or precancerous cells are eliminated. If some tumor cells are not eliminated, a temporary state of equilibrium may be achieved between the immune system and tumor cell growth. In this equilibrium phase, tumors cells can either remain dormant or continue to evolve by accumulating further changes to genomic DNA that can modulate the antigens they present. During this process, the immune system exerts a selective pressure on evolving cells, whereby the tumor cells that are less able to be recognized have a survival advantage. Eventually the immune response is unable to recognize cells of the tumor, resulting in the transition to the escape phase wherein tumor cells progressively grow out of control.

Tumor Microenvironment

The tumor microenvironment provides a consistently effective barrier to immune cell function because tumors actively downregulate all phases of anti-tumor immune responses using a spectrum of different strategies and mechanisms. Many molecular mechanisms that cause dysfunction of immune cells in the tumor microenvironment have been identified, including those directly mediated by factors produced by tumors, and others resulting from alterations of normal tissue homeostasis in the presence of cancer. Most human tumors appear to be able to interfere with one or more stages of immune cell development, differentiation, migration, cytotoxicity and other effector functions (T L Whiteside, *The tumor microenvironment and its role in promoting tumor growth*, *Oncogene* (2008) 27, 5904-5912).

One such mechanism involves accumulation in tumors of T_{reg} ($CD4^+CD25^{bright}$ Foxp3⁺ T cells) and myeloid-derived cells ($CD34^+CD33^+CD13^+CD11b^+CD15^-$), which are common features of human tumors, and have been linked to poor prognosis in patients with cancer (T L Whiteside, *The tumor*

microenvironment and its role in promoting tumor growth, *Oncogene* (2008) 27, 5904-5912). Under normal conditions, T_{reg} cells are involved in the important role of preventing autoimmunity, but in cancer, they expand, migrate to tumors, downregulate autologous effector T-cell proliferation and suppress anti-tumor responses of both $CD4^+CD25^-$ and $CD8^+CD25^-$ T cells using distinct molecular pathways. The T_{reg} cells in the tumor are a heterogeneous population of regulatory $CD3^+CD4^+$ T cells, comprising natural T_{reg} , antigen-specific Tr1 cells, and other less well defined subsets of suppressor cells. Tr1 cells are induced in the tumor microenvironment, which is rich in IL-10, TGF- β , and prostaglandin E_2 (PGE_2), all of which have been shown to promote Tr1 generation (T L Whiteside, The tumor microenvironment and its role in promoting tumor growth, *Oncogene* (2008) 27, 5904-5912).

Myeloid suppressor cells (MSCs) also suppress T-cell responses in the tumor microenvironment, where they secrete TGF- β or induce TGF- β secretion. Immunosuppressive $CD34^+$ cell-derived myeloid cells have been identified in the peripheral blood of cancer patients. In tumor-bearing mice, MSCs accumulate in the spleen and peripheral circulation in very high amounts, exerting potent immunosuppression and favoring tumor growth. MSCs also control the availability of essential amino acids such as L-arginine and produce high levels of reactive oxygen species. The MSCs found in tumors also constitutively express iNOS and arginase 1, an enzyme involved in metabolism of L-arginine, which also synergizes with iNOS to increase superoxide and NO production, which have been found to interfere with lymphocyte responses. GM-CSF, which is also often secreted by tumor cells, recruits MSCs and induces dose-dependent *in vivo* immune suppression and tumor promotion, while at the same time, GM-CSF has been used as immune adjuvant in antitumor vaccines. GM-CSF was observed to increase a subset of TGF- β -producing MSCs in the circulation of patients with metastatic melanoma. The concurrent stimulatory and suppressive roles suggest that GM-CSF and MSCs are involved in maintaining immune homeostasis in normal tissue, but in the tumor microenvironment promote tumor cell escape (T L Whiteside, The tumor microenvironment and its role in promoting tumor growth, *Oncogene* (2008) 27, 5904-5912).

Tumor Immunotherapy

Cancer therapy is evolving rapidly as new molecular targets are being discovered. Despite the advent of biologics targeting specific pathways (e.g., HERCEPTIN® (Trastuzumab), ERBITUX® (Cetuximab)) and small molecules designed against specific targets (NOLVADEX® (Tamoxifen), GLEEVEC™ (Imatinib)), nonspecific modalities such as chemotherapy and radiation remain a standard of care.

Anti-cancer immunotherapy has been a goal for many years with a variety of approaches being tested. One difficulty of developing this immunotherapy is that target antigens are often tissue specific molecules found on both cancer cells and normal cells, and either do not elicit immunity or show non-specificity regarding cell killing (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). Furthermore, tumor cells have features that make immune recognition difficult, such as loss of expression of antigens that elicit immune response, lack of major histocompatibility (MHC) class II, and downregulation of MHC class I expression. These features can lead to non-recognition of tumor cells by both $CD4^+$ and $CD8^+$ T cells (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5,

67-121 (2007)). Tumors may also evade detection through active mechanisms, such as the production of immunosuppressive cytokines (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)).

DCs generated *ex vivo* by culturing hematopoietic progenitor cells or monocytes with cytokine combinations have been tested as therapeutic vaccines in cancer patients for more than a decade (Ueno H, et al., *Immunol. Rev.* (2010) 234: 199-212). For example, treatment of metastatic prostate cancer with sipuleucel-T (also known as APC 8015), which is a cellular product based on enriched blood APCs that are briefly cultured with a fusion protein of prostatic acid phosphatase (PAP) and granulocyte macrophage colony-stimulating factor (GM-CSF), resulted in an approximately 4-month-prolonged median survival in Phase III trials (Higano C S, et al., *Cancer* (2009) 115: 3670-3679; Kantoff P W, et al., *N. Engl. J. Med.* (2010) 363: 411-422). This study concluded that DC-based vaccines are safe and can induce the expansion of circulating $CD4^+$ T-cells and $CD8^+$ T-cells specific for tumor antigens. As a result of this and similar studies, sipuleucel-T has been approved by the US Food and Drug Administration (FDA) for the treatment of metastatic prostate cancer, thereby paving the clinical development and regulatory path for the next generation of cellular immunotherapy products (Palucka K and Banchereau J, *Nature Reviews Cancer* (April 2012) 12: 265-276).

Vaccination strategies involving DCs to induce tumor-specific effector T cells that can reduce the tumor mass specifically and that can induce immunological memory to control tumor relapse have been developed. For example, DCs can be provided with tumor-specific antigens by culturing DCs *ex vivo* with an adjuvant and a tumor-specific antigen, and then injecting these cells back into the patient. Tumor cells obtained from an excised tumor, needle biopsy, core biopsy, vacuum-assisted biopsy or peritoneal lavage have been used to generate immunogenic compositions comprising tumor-specific-antigen presenting dendritic cells.

Cancer Treatment Strategies

Antibody therapies such as HERCEPTIN™ (Trastuzumab) and ERBITUX™ (Cetuximab) are passive immunotherapies, but have yielded considerable improvement in clinical outcome, as measured by, e.g. the recurrence rate, progression free survival and overall survival. More recently, PD-1 and CTLA4 inhibitors have been reported to block discrete checkpoints in an active host immune response allowing an endogenous anti-cancer immune response to be sustained. The term "immune checkpoints" refers to the array of inhibitory pathways that are necessary for maintaining self-tolerance and modulating the duration and extent of immune responses to minimize damage to normal tissue. Immune checkpoint molecules such as PD-1, PD-L1, CTLA-4 are cell surface signaling receptors that play an important role in modulating the T-cell response in the tumor microenvironment. Tumor cells have been shown to utilize these checkpoints to their benefit by up regulating their expression and activity. With the tumor cell's ability to commandeer some immune checkpoint pathways as a mechanism of immune resistance, it has been hypothesized that checkpoint inhibitors that bind to molecules of immune cells to activate or inactivate them may relieve the inhibition of an immune response. Recent discoveries have identified immune checkpoints or targets, like PD-1, PD-L1, PD-L2, CTLA4, TIM3, LAG3, CCR4, OX40, OX40L, IDO, and A2AR, as proteins responsible for immune evasion. Specific immune checkpoint inhibitors, including antibodies against

CTLA-4, PD-1 receptor or its ligand PD-L1 have produced impressive results in the clinic in a range of cancers, leading to FDA approvals for YERVOY™ (Ipilimumab; CTLA-4 antagonist), OPDIVO™ (Nivolumab; PD-1 antagonist) and KEYTRUDA™ (Pembrolizumab; PD-1 antagonist) in multiple tumor indications and with ongoing registration trials in many more. This method of therapy, however, can only be successful if a pre-existing antitumor immune response is present within a patient (Pardoll, D., The blockade of immune checkpoints in cancer immunotherapy, *Nature Reviews: Cancer*, Vol. 12, April 2012, 253). Recent cellular therapies, such as chimeric antigen receptor T-cell therapy (CAR-T), attempt to use synthetic biology to redirect T-cells to specific cell surface tumor antigens. Genetic modification of T-cells is used to confer tumor antigen recognition by transgenic expression of chimeric antigen receptor (CAR). CARs are engineered molecules that can be introduced into T cells to enable them to target tumor antigens (Frey, N. V., Porter, D. L., The Promise of Chimeric Antigen Receptor T-Cell Therapy, *Oncology* (2016); 30(1)) pii 219281). CAR T cells have been shown to have some efficacy against hematologic malignancies and to a lesser extent solid tumors. CAR T therapy, however, has been shown to cause several types of toxicities, including cytokine release syndrome, neurological toxicity, non-tumor recognition, and anaphylaxis (Bonifant C L, et al., Toxicity and management in CAR T-cell therapy, *Molecular Therapy—Oncolytics* (2016) 3, 16011).

Cellular vaccines have also been used as a cancer treatment. GVAX™ (Ipilimumab), a prototypical example, is a GM-CSF gene transduced tumor vaccine within either an autologous or allogeneic population of tumor cells. It is believed that GM-CSF secretion of genetically modified tumor cells stimulates cytokine release at the vaccine site to activate antigen presenting cells to induce a tumor specific cellular immune response (Eager, R. & Nemunaitis, J., GM-CSF Gene-Transduced Tumor Vaccines, *Molecular Therapy*, Vol. 12, No. 1, 18 (July 2005)). However, GVAX™ yielded only limited clinical responses.

Dendritic cell (DC)-tumor cell fusions have been developed to generate hybrid cells that express the relevant tumor associated antigens derived from the parent tumor cells, and also have the ability to process and present such antigens to appropriate cells of the immune system. The DC-tumor cell fusions provide a greater variety of tumor antigens, but have been met with limited success in human trials, likely due to the autologous components required, the heterogeneity of the product caused by maturation of DC cells, and variations in antigen loading (Browning, M., Antigen presenting cell/tumor cell fusion vaccines for cancer, *Human Vaccines & Immunotherapeutics* 9:7, 1545-1548; July 2013; Butterfield, L., Dendritic Cells in Cancer Immunotherapy Clinical Trials: Are We Making Progress?, *Frontiers of Immunology*, 2013 4: 454).

Cells of the Immune System

There are a large number of cellular interactions that comprise the immune system. These interactions occur through specific receptor-ligand pairs that signal in both directions so that each cell receives instructions based on the temporal and spatial distribution of those signals.

Murine models have been highly useful in discovering immunomodulatory pathways, but clinical utility of these pathways does not always translate from an inbred mouse strain to an outbred human population, since an outbred human population may have individuals that rely to varying extents on individual immunomodulatory pathways.

Cells of the immune system include lymphocytes, monocytes/macrophages, dendritic cells, the closely related Langerhans cells, natural killer (NK) cells, mast cells, basophils, and other members of the myeloid lineage of cells. In addition, a series of specialized epithelial and stromal cells provide the anatomic environment in which immunity occurs, often by secreting critical factors that regulate growth and/or gene activation in cells of the immune system, which also play direct roles in the induction and effector phases of the response. (Paul, W. E., "Chapter 1: The immune system: an introduction," *Fundamental Immunology*, 4th Edition, Ed. Paul, W. E., Lippincott-Raven Publishers, Philadelphia, (1999), at p. 102).

The cells of the immune system are found in peripheral organized tissues, such as the spleen, lymph nodes, Peyer's patches of the intestine and tonsils. Lymphocytes also are found in the central lymphoid organs, the thymus, and bone marrow where they undergo developmental steps that equip them to mediate the myriad responses of the mature immune system. A substantial portion of lymphocytes and macrophages comprise a recirculating pool of cells found in the blood and lymph, providing the means to deliver immunocompetent cells to sites where they are needed and to allow immunity that is generated locally to become generalized. (Paul, W. E., "Chapter 1: The immune system: an introduction," *Fundamental Immunology*, 4th Edition, Ed. Paul, W. E., Lippincott-Raven Publishers, Philadelphia, (1999), at p. 102).

The term "lymphocyte" refers to a small white blood cell formed in lymphatic tissue throughout the body and in normal adults making up about 22-28% of the total number of leukocytes in the circulating blood that plays a large role in defending the body against disease. Individual lymphocytes are specialized in that they are committed to respond to a limited set of structurally related antigens through recombination of their genetic material (e.g. to create a T cell receptor and a B cell receptor). This commitment, which exists before the first contact of the immune system with a given antigen, is expressed by the presence of receptors specific for determinants (epitopes) on the antigen on the lymphocyte's surface membrane. Each lymphocyte possesses a unique population of receptors, all of which have identical combining sites. One set, or clone, of lymphocytes differs from another clone in the structure of the combining region of its receptors and thus differs in the epitopes that it can recognize. Lymphocytes differ from each other not only in the specificity of their receptors, but also in their functions. (Paul, W. E., "Chapter 1: The immune system: an introduction," *Fundamental Immunology*, 4th Edition, Ed. Paul, W. E., Lippincott-Raven Publishers, Philadelphia, (1999), at p. 102).

Two broad classes of lymphocytes are recognized: the B-lymphocytes (B-cells), which are precursors of antibody-secreting cells, and T-lymphocytes (T-cells).

B-Lymphocytes

B-lymphocytes are derived from hematopoietic cells of the bone marrow. A mature B-cell can be activated with an antigen that expresses epitopes that are recognized by its cell surface. The activation process may be direct, dependent on cross-linkage of membrane Ig molecules by the antigen (cross-linkage-dependent B-cell activation), or indirect, via interaction with a helper T-cell, in a process referred to as cognate help. In many physiological situations, receptor cross-linkage stimuli and cognate help synergize to yield more vigorous B-cell responses (Paul, W. E., "Chapter 1: The immune system: an introduction," *Fundamental Immu-*

nology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

Cross-linkage dependent B-cell activation requires that the antigen express multiple copies of the epitope complementary to the binding site of the cell surface receptors, because each B-cell expresses Ig molecules with identical variable regions. Such a requirement is fulfilled by other antigens with repetitive epitopes, such as capsular polysaccharides of microorganisms or viral envelope proteins. Cross-linkage-dependent B-cell activation is a major protective immune response mounted against these microbes (Paul, W. E., "Chapter 1: The immune system: an introduction", Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

Cognate help allows B-cells to mount responses against antigens that cannot cross-link receptors and, at the same time, provides costimulatory signals that rescue B cells from inactivation when they are stimulated by weak cross-linkage events. Cognate help is dependent on the binding of antigen by the B-cell's membrane immunoglobulin (Ig), the endocytosis of the antigen, and its fragmentation into peptides within the endosomal/lysosomal compartment of the cell. Some of the resultant peptides are loaded into a groove in a specialized set of cell surface proteins known as class II major histocompatibility complex (MHC) molecules. The resultant class II/peptide complexes are expressed on the cell surface and act as ligands for the antigen-specific receptors of a set of T-cells designated as CD4⁺ T-cells. The CD4⁺ T-cells bear receptors on their surface specific for the B-cell's class II/peptide complex. B-cell activation depends not only on the binding of the T cell through its T cell receptor (TCR), but this interaction also allows an activation ligand on the T-cell (CD40 ligand) to bind to its receptor on the B-cell (CD40) signaling B-cell activation. In addition, T helper cells secrete several cytokines that regulate the growth and differentiation of the stimulated B-cell by binding to cytokine receptors on the B cell (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

During cognate help for antibody production, the CD40 ligand is transiently expressed on activated CD4⁺ T helper cells, and it binds to CD40 on the antigen-specific B cells, thereby transducing a second costimulatory signal. The latter signal is essential for B cell growth and differentiation and for the generation of memory B cells by preventing apoptosis of germinal center B cells that have encountered antigen. Hyperexpression of the CD40 ligand in both B and T cells is implicated in pathogenic autoantibody production in human SLE patients (Desai-Mehta, A. et al., "Hyperexpression of CD40 ligand by B and T cells in human lupus and its role in pathogenic autoantibody production," J. Clin. Invest. Vol. 97(9), 2063-2073, (1996)).

T-Lymphocytes

T-lymphocytes derived from precursors in hematopoietic tissue, undergo differentiation in the thymus, and are then seeded to peripheral lymphoid tissue and to the recirculating pool of lymphocytes. T-lymphocytes or T cells mediate a wide range of immunologic functions. These include the capacity to help B cells develop into antibody-producing cells, the capacity to increase the microbicidal action of monocytes/macrophages, the inhibition of certain types of immune responses, direct killing of target cells, and mobilization of the inflammatory response. These effects depend on T cell expression of specific cell surface molecules and the secretion of cytokines (Paul, W. E., "Chapter 1: The

immune system: an introduction", Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

T cells differ from B cells in their mechanism of antigen recognition. Immunoglobulin, the B cell's receptor, binds to individual epitopes on soluble molecules or on particulate surfaces. B-cell receptors see epitopes expressed on the surface of native molecules. While antibody and B-cell receptors evolved to bind to and to protect against microorganisms in extracellular fluids, T cells recognize antigens on the surface of other cells and mediate their functions by interacting with, and altering, the behavior of these antigen-presenting cells (APCs). There are three main types of APCs in peripheral lymphoid organs that can activate T cells: dendritic cells, macrophages and B cells. The most potent of these are the dendritic cells, whose only function is to present foreign antigens to T cells. Immature dendritic cells are located in tissues throughout the body, including the skin, gut, and respiratory tract. When they encounter invading microbes at these sites, they endocytose the pathogens and their products, and carry them via the lymph to local lymph nodes or gut associated lymphoid organs. The encounter with a pathogen induces the dendritic cell to mature from an antigen-capturing cell to an APC that can activate T cells. APCs display three types of protein molecules on their surface that have a role in activating a T cell to become an effector cell: (1) MHC proteins, which present foreign antigen to the T cell receptor; (2) costimulatory proteins which bind to complementary receptors on the T cell surface; and (3) cell-cell adhesion molecules, which enable a T cell to bind to the APC for long enough to become activated ("Chapter 24: The adaptive immune system," Molecular Biology of the Cell, Alberts, B. et al., Garland Science, NY, (2002)).

T-cells are subdivided into two distinct classes based on the cell surface receptors they express. The majority of T cells express T cell receptors (TCR) consisting of α and β -chains. A small group of T cells express receptors made of γ and δ chains. Among the α/β T cells are two sub-lineages: those that express the coreceptor molecule CD4 (CD4⁺ T cells); and those that express CD8 (CD8⁺ T cells). These cells differ in how they recognize antigen and in their effector and regulatory functions.

CD4⁺ T cells are the major regulatory cells of the immune system. Their regulatory function depends both on the expression of their cell-surface molecules, such as CD40 ligand whose expression is induced when the T cells are activated, and the wide array of cytokines they secrete when activated.

T cells also mediate important effector functions, some of which are determined by the patterns of cytokines they secrete. The cytokines can be directly toxic to target cells and can mobilize potent inflammatory mechanisms.

In addition, T cells, particularly CD8⁺ T cells, can develop into cytotoxic T-lymphocytes (CTLs) capable of efficiently lysing target cells that express antigens recognized by the CTLs (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

T cell receptors (TCRs) recognize a complex consisting of a peptide derived by proteolysis of the antigen bound to a specialized groove of a class II or class I MHC protein. CD4⁺ T cells recognize only peptide/class II complexes while CD8⁺ T cells recognize peptide/class I complexes (Paul, W. E., "Chapter 1: The immune system: an introduc-

tion," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

The TCR's ligand (i.e., the peptide/MHC protein complex) is created within APCs. In general, class II MHC molecules bind peptides derived from proteins that have been taken up by the APC through an endocytic process. These peptide-loaded class II molecules are then expressed on the surface of the cell, where they are available to be bound by CD4⁺ T cells with TCRs capable of recognizing the expressed cell surface complex. Thus, CD4⁺ T cells are specialized to react with antigens derived from extracellular sources (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

In contrast, class I MHC molecules are mainly loaded with peptides derived from internally synthesized proteins, such as viral proteins. These peptides are produced from cytosolic proteins by proteolysis by the proteasome and are translocated into the rough endoplasmic reticulum. Such peptides, generally composed of nine amino acids in length, are bound into the class I MHC molecules and are brought to the cell surface, where they can be recognized by CD8⁺ T cells expressing appropriate receptors. This gives the T cell system, particularly CD8⁺ T cells, the ability to detect cells expressing proteins that are different from, or produced in much larger amounts than, those of cells of the remainder of the organism (e.g., viral antigens) or mutant antigens (such as active oncogene products), even if these proteins in their intact form are neither expressed on the cell surface nor secreted (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

T cells can also be classified based on their function as helper T cells; T cells involved in inducing cellular immunity; suppressor T cells; and cytotoxic T cells.

Helper T Cells

Helper T cells are T cells that stimulate B cells to make antibody responses to proteins and other T cell-dependent antigens. T cell-dependent antigens are immunogens in which individual epitopes appear only once or a limited number of times such that they are unable to cross-link the membrane immunoglobulin (Ig) of B cells or do so inefficiently. B cells bind the antigen through their membrane Ig, and the complex undergoes endocytosis. Within the endosomal and lysosomal compartments, the antigen is fragmented into peptides by proteolytic enzymes, and one or more of the generated peptides are loaded into class II MHC molecules, which traffic through this vesicular compartment. The resulting peptide/class II MHC complex is then exported to the B-cell surface membrane. T cells with receptors specific for the peptide/class II molecular complex recognize this complex on the B-cell surface. (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia (1999)).

B-cell activation depends both on the binding of the T cell through its TCR and on the interaction of the T-cell CD40 ligand (CD40L) with CD40 on the B cell. T cells do not constitutively express CD40L. Rather, CD40L expression is induced as a result of an interaction with an APC that expresses both a cognate antigen recognized by the TCR of the T cell and CD80 or CD86. CD80/CD86 is generally expressed by activated, but not resting, B cells so that the helper interaction involving an activated B cell and a T cell can lead to efficient antibody production. In many cases,

however, the initial induction of CD40L on T cells is dependent on their recognition of antigen on the surface of APCs that constitutively express CD80/86, such as dendritic cells. Such activated helper T cells can then efficiently interact with and help B cells. Cross-linkage of membrane Ig on the B cell, even if inefficient, may synergize with the CD40L/CD40 interaction to yield vigorous B-cell activation. The subsequent events in the B-cell response, including proliferation, Ig secretion, and class switching of the Ig class being expressed, either depend or are enhanced by the actions of T cell-derived cytokines (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

CD4⁺ T cells tend to differentiate into cells that principally secrete the cytokines IL-4, IL-5, IL-6, and IL-10 (T_H2 cells) or into cells that mainly produce IL-2, IFN- γ , and lymphotoxin (T_H1 cells). The T_H2 cells are very effective in helping B-cells develop into antibody-producing cells, whereas the T_H1 cells are effective inducers of cellular immune responses, involving enhancement of microbicidal activity of monocytes and macrophages, and consequent increased efficiency in lysing microorganisms in intracellular vesicular compartments. Although CD4⁺ T cells with the phenotype of T_H2 cells (i.e., IL-4, IL-5, IL-6 and IL-10) are efficient helper cells, T_H1 cells also have the capacity to be helpers (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

T Cell Involvement in Cellular Immunity Induction

T cells also may act to enhance the capacity of monocytes and macrophages to destroy intracellular microorganisms. In particular, interferon-gamma (IFN- γ) produced by helper T cells enhances several mechanisms through which mononuclear phagocytes destroy intracellular bacteria and parasitism including the generation of nitric oxide and induction of tumor necrosis factor (TNF) production. T_H1 cells are effective in enhancing the microbicidal action, because they produce IFN- γ . In contrast, two of the major cytokines produced by T_H2 cells, IL-4 and IL-10, block these activities (Paul, W. E., "Chapter 1: The immune system: an introduction," Fundamental Immunology, 4th Edition, Ed. Paul, W. E., Lippicott-Raven Publishers, Philadelphia, (1999)).

Regulatory T (Treg) Cells

Immune homeostasis is maintained by a controlled balance between initiation and downregulation of the immune response. The mechanisms of both apoptosis and T cell anergy (a tolerance mechanism in which the T cells are intrinsically functionally inactivated following an antigen encounter (Scwartz, R. H., "T cell anergy", Annu. Rev. Immunol., Vol. 21: 305-334 (2003)) contribute to the downregulation of the immune response. A third mechanism is provided by active suppression of activated T cells by suppressor or regulatory CD4⁺ T (Treg) cells (Reviewed in Kronenberg, M. et al., "Regulation of immunity by self-reactive T cells", Nature, Vol. 435: 598-604 (2005)). CD4⁺ Tregs that constitutively express the IL-2 receptor alpha (IL-2R α) chain (CD4⁺ CD25⁺) are a naturally occurring T cell subset that are anergic and suppressive (Taams, L. S. et al., "Human anergic/suppressive CD4⁺CD25⁺ T cells: a highly differentiated and apoptosis-prone population", Eur. J. Immunol. Vol. 31: 1122-1131 (2001)). Depletion of CD4⁺ CD25⁺ Tregs results in systemic autoimmune disease in mice. Furthermore, transfer of these Tregs prevents development of autoimmune disease. Human CD4⁺CD25⁺ Tregs, similar to their murine counterpart, are generated in the

thymus and are characterized by the ability to suppress proliferation of responder T cells through a cell-cell contact-dependent mechanism, the inability to produce IL-2, and the anergic phenotype in vitro. Human CD4⁺CD25⁺ T cells can be split into suppressive (CD25^{high}) and nonsuppressive (CD25^{low}) cells, according to the level of CD25 expression. A member of the forkhead family of transcription factors, FOXP3, has been shown to be expressed in murine and human CD4⁺CD25⁺ Tregs and appears to be a master gene controlling CD4⁺CD25⁺ Treg development (Battaglia, M. et al., "Rapamycin promotes expansion of functional CD4⁺CD25⁺FOXP3⁺ regulator T cells of both healthy subjects and type 1 diabetic patients", *J. Immunol.*, Vol. 177: 8338-8347, (2006)).

Cytotoxic T Lymphocytes

CD8⁺ T cells that recognize peptides from proteins produced within the target cell have cytotoxic properties in that they lead to lysis of the target cells. The mechanism of CTL-induced lysis involves the production by the CTL of perforin, a molecule that can insert into the membrane of target cells and promote the lysis of that cell. Perforin-mediated lysis is enhanced by granzymes, a series of enzymes produced by activated CTLs. Many active CTLs also express large amounts of fas ligand on their surface. The interaction of fas ligand on the surface of CTL with fas on the surface of the target cell initiates apoptosis in the target cell, leading to the death of these cells. CTL-mediated lysis appears to be a major mechanism for the destruction of virally infected cells.

Priming

The term "unprimed cells" (also referred to as virgin, naïve, or inexperienced cells) as used herein refers to T cells and B cells that have generated an antigen receptor (TCR for T cells, BCR for B cells) of a particular specificity, but have never encountered the antigen. The term "priming" as used herein refers to the process whereby T cells and B cell precursors encounter the antigen for which they are specific.

For example, before helper T cells and B cells can interact to produce specific antibody, the antigen-specific T cell precursors must be primed. Priming involves several steps: antigen uptake, processing, and cell surface expression bound to class II MHC molecules by an antigen presenting cell, recirculation and antigen-specific trapping of helper T cell precursors in lymphoid tissue, and T cell proliferation and differentiation (Janeway, C A, Jr., "The priming of helper T cells", *Semin. Immunol.*, Vol. 1(1): 13-20 (1989)). Helper T cells express CD4, but not all CD4 T cells are helper cells. Id. The signals required for clonal expansion of helper T cells differ from those required by other CD4 T cells. The critical antigen-presenting cell for helper T cell priming appears to be a macrophage; and the critical second signal for helper T cell growth is the macrophage product interleukin 1 (IL-1). Id. If the primed T cells and/or B cells receive a second, co-stimulatory signal, they become activated T cells or B cells.

Lymphocyte Activation

The term "activation" or "lymphocyte activation" refers to stimulation of lymphocytes by specific antigens, nonspecific mitogens, or allogeneic cells resulting in synthesis of RNA, protein and DNA and production of lymphokines; it is followed by proliferation and differentiation of various effector and memory cells. For example, a mature B cell can be activated by an encounter with an antigen that expresses epitopes that are recognized by its cell surface immunoglobulin Ig. The activation process may be a direct one, dependent on cross-linkage of membrane Ig molecules by the antigen (cross-linkage-dependent B cell activation) or an

indirect one, occurring most efficiently in the context of an intimate interaction with a helper T cell ("cognate help process"). T-cell activation is dependent on the interaction of the TCR/CD3 complex with its cognate ligand, a peptide bound in the groove of a class I or class II MHC molecule. The molecular events set in motion by receptor engagement are complex. Among the earliest steps appears to be the activation of tyrosine kinases leading to the tyrosine phosphorylation of a set of substrates that control several signaling pathways. These include a set of adapter proteins that link the TCR to the ras pathway, phospholipase C γ 1, the tyrosine phosphorylation of which increases its catalytic activity and engages the inositol phospholipid metabolic pathway, leading to elevation of intracellular free calcium concentration and activation of protein kinase C, and a series of other enzymes that control cellular growth and differentiation. Full responsiveness of a T cell requires, in addition to receptor engagement, an accessory cell-delivered costimulatory activity, e.g., engagement of CD28 on the T cell by CD80 and/or CD86 on the APC. The soluble product of an activated B lymphocyte is immunoglobulins (antibodies). The soluble product of an activated T lymphocyte is lymphokines.

Chemokines are chemotactic cytokines, which constitute a family of low molecular mass (8-11 kDa) structurally-related proteins with diverse immune and neural functions (Mackay C. R., "Chemokines: immunology's high impact factors", *Nat Immunol.*, Vol. 2: 95-101, (2001)); (Youn B. et al., "Chemokines, chemokine receptors and hematopoiesis", *Immunol Rev.*, Vol. 177: 150-174, (2000)) that can be categorized into four subfamilies (C, CC, CXC and CX3C) based on the relative positions of conserved cysteine residues (Rossi D. et al., "The biology of chemokines and their receptors", *Annu Rev Immunol.*, Vol. 18: 217-242, (2000)). Chemokines are essential molecules in directing leucocyte migration between blood, lymph nodes and tissues. They constitute a complex signaling network because they are not always restricted to one type of receptor (Loetscher P. et al., "The ligands of CXC chemokine receptor 3, I-TAC, Mig, and IP10, are natural antagonists for CCR3", *J. Biol. Chem.*, Vol. 276: 2986-2991, (2001)). Chemokines affect cells by activating surface receptors that are seven-transmembrane-domain G-protein-coupled receptors. Leukocyte responses to particular chemokines are determined by their expression of chemokine receptors. The binding of the chemokine to the receptor activates various signaling cascades, similar to the action of cytokines that culminate in the activation of a biological response. Secretion of the ligands for the CCR5 receptor, regulated upon activation normal T cell expressed and secreted (RANTES), macrophage inflammatory protein (MIP)-1 α and MIP-1 β (Schrum S. et al., "Synthesis of the CC-chemokines MIP-1 α , MIP-1 β , and RANTES is associated with a type 1 immune response", *J Immunol.*, Vol. 157: 3598-3604, (1996)) and the ligand for CXC chemokine receptor 3 (CXCR3), induced protein (IP)-10 (Taub D. D. et al., "Recombinant human interferon-inducible protein 10 is a chemoattractant for human monocytes and T lymphocytes and promotes T cell adhesion to endothelial cells", *J Exp Med.*, Vol. 177:1809-1814, (1993)) have been associated with unwanted heightened T_{H1} responses. Additionally, elevated damaging pro-inflammatory cytokine levels of IL-2 and IFN- γ correlate with type 1 diabetes (T1D) (Rabinovitch A. et al., "Roles of cytokines in the pathogenesis and therapy of type 1 diabetes", *Cell Biochem Biophys.*, Vol. 48(2-3): 159-63, (2007)). Chemokines have been observed in T_{H1} pancreatic infiltrates and other inflammatory lesions characterized by T cell infiltration (Bradley L. M. et al., "Islet-

specific Th1, but not Th2, cells secrete multiple chemokines and promote rapid induction of autoimmune diabetes”, *J Immunol*, Vol. 162:2511-2520, (1999)).

Pro-inflammatory cytokines like IL-1 β , IL-6, and TNF- α in the plasma have been primarily detected and involved in the insulin resistance and development of T2D which are kept in check and modulated by the anti-inflammatory and immune suppressive cytokines TGF- β 1 and IL-10 (Alexandaki K. et al., “Inflammatory process in type 2 diabetes: The role of cytokines”, *Annals of the New York Academy of Sciences*, 1084: 89-117, (2006); Kumar N. P. et al. 2015. *Eur J Immunol*. doi: 10.1002/eji.201545973 ahead of print). IL-17A is a well-known pro-inflammatory cytokine involved in several autoimmune diseases.

Immune Tolerance

The immune system is tolerant of self-antigens, i.e., it can discriminate between antigenic determinants expressed on foreign substances, and antigenic determinants expressed by tissues of the host. The capacity of the system to ignore host antigens, referred to as immune tolerance or immunological tolerance, is an active process involving the elimination or inactivation of cells that could recognize self-antigens through immunologic tolerance (Fundamental immunology, 4th Edn, William E. Paul, Ed. Lippincott-Raven Publishers, Philadelphia, (1999), at p. 2).

Immune tolerance is classified into 1) central tolerance or 2) peripheral tolerance depending on where the state is originally induced, i.e., whether it is in the thymus and bone marrow (central) or in other tissues and lymph nodes (peripheral). The biological mechanisms whereby these forms of tolerance are established are distinct, but the resulting effect is similar (Raker V. K. et al., “Tolerogenic Dendritic Cells for Regulatory T Cell Induction in Man”, *Front Immunol*, Vol., 6(569): 1-11, (2015)).

Central tolerance, the principal way in which the immune system is educated to discriminate self-molecules from non-self-molecules, is established by deleting autoreactive lymphocyte clones at a point before they mature into fully immunocompetent cells. It occurs during lymphocyte development in the thymus and bone marrow for T and B lymphocytes, respectively (Sprent J. et al., “The thymus and central tolerance”, *Philos Trans R Soc Lond B Biol Sci*, Vol. 356(1409): 609-616, (2001)). In these tissues, maturing lymphocytes are exposed to self-antigens presented by thymic epithelial cells and thymic dendritic cells, or bone marrow cells. Self-antigens are present due to endogenous expression, importation of antigen from peripheral sites via circulating blood, and in the case of thymic stromal cells, expression of proteins of other non-thymic tissues by the action of the transcription factor AIRE (Murphy, Kenneth. Janeway’s Immunobiology: 8th ed. Chapter 15: Garland Science. (2012), pp. 611-668; Klein L., “Aire gets company for immune tolerance”, *Cell*, Vol. 163(4):794-795, (2015)). Those lymphocytes that have receptors that bind strongly to self-antigens are removed by means of apoptosis of the autoreactive cells, or by induction of anergy (Id. at pp. 275-334). Weakly autoreactive B cells may also remain in a state of immunological inactivity where they do not respond to stimulation of their B cell receptor. Some weakly self-recognizing T cells are alternatively differentiated into natural regulatory T cells (nTreg cells), which act as sentinels in the periphery to lower potential instances of T cell autoreactivity (Id. at pp. 611-668).

The deletion threshold is more stringent for T cells than for B cells since T cells are the main populations of cells that can cause direct tissue damage. Furthermore, it is more advantageous for the organism to let its B cells recognize a

wider variety of antigens, so that they can elicit antibodies against a greater diversity of pathogens. Since B cells can only be fully activated after confirmation by more self-restricted T cells that recognize the same antigen, autoreactivity is held in great check (Murphy, Kenneth. Janeway’s Immunobiology: 8th ed. Chapter 8: Garland Sciences. pp. 275-334).

This process of negative selection ensures that T and B cells that potentially may initiate a potent immune response to the individual’s own tissues are destroyed while preserving the ability to recognize foreign antigens. This step in lymphocyte education is detrimental to preventing autoimmunity. Lymphocyte development and education is most active in fetal development, but continues throughout life as immature lymphocytes are generated, slowing as the thymus degenerates and the bone marrow shrinks in the adult life (Murphy, Kenneth. Janeway’s Immunobiology: 8th ed. Chapter 8: Garland Sciences. (2012), pp. 275-334; Jiang T. T., “Regulatory T cells: new keys for further unlocking the enigma of fetal tolerance and pregnancy complications”, *J Immunol*, Vol. 192(11): 4949-4956, (2014)).

Peripheral tolerance develops after T and B cells mature and enter the peripheral tissues and lymph nodes (Murphy, Kenneth. Janeway’s Immunobiology: 8th ed. Chapter 8: Garland Sciences. pp. 275-334). It is set forth by a number of overlapping mechanisms that predominantly involve control at the level of T cells, especially CD4⁺ helper T cells, which orchestrate immune responses and give B cells the confirmatory signals that the B cells need in order to progress to produce antibodies. Inappropriate reactivity toward a normal self-antigen that was not eliminated in the thymus can occur, since the T cells that leave the thymus are relatively, but not completely, safe. Some will have receptors (TCRs) that can respond to self-antigens that the T cell did not encounter in the thymus (Murphy, Kenneth. Janeway’s Immunobiology: 8th ed. Chapter 8: Garland Sciences. (2012), pp. 275-334). Those self-reactive T cells that escape intra-thymic negative selection in the thymus can inflict cell injury unless they are deleted in the peripheral tissue chiefly by nTreg cells.

Autoimmune regulator (Aire), usually expressed in thymic medullary epithelial cells, plays a role in immune tolerance by mediating ectopic expression of peripheral self-antigens and mediating the deletion of auto-reactive T cells. (Metzger T. C. et al., “Control of central and peripheral tolerance by Aire”, *Immunol. Rev*. 2011, Vol. 241: 89-103, (2011)).

Appropriate reactivity towards certain antigens can also be suppressed by induction of tolerance after repeated exposure. Naïve CD4⁺ helper T cells differentiate into induced Treg cells (iTreg cells) in the peripheral tissue, or accordingly, in nearby lymphoid tissue (lymph nodes, mucosal-associated lymphoid tissue, etc.). This differentiation is mediated by IL-2 produced upon T cell-activation, and TGF- β from any of a variety of sources, including tolerizing dendritic cells (DCs) or other antigen presenting cells (Curotto de Lafaille et al., “Effective recruitment and retention of older adults in physical activity research: PALS study”, *Immunity*, Vol. 30(6): 626-635, (2009)).

T-Memory Cells

Following the recognition and eradication of pathogens through adaptive immune responses, the vast majority (90-95%) of T cells undergo apoptosis with the remaining cells forming a pool of memory T cells, designated central memory T cells (TCM), effector memory T cells (TEM), and

resident memory T cells (TRM) (Clark, R. A., "Resident memory T cells in human health and disease", *Sci. Transl. Med.*, 7, 269rv1, (2015)).

Compared to standard T cells, these memory T cells are long-lived with distinct phenotypes such as expression of specific surface markers, rapid production of different cytokine profiles, capability of direct effector cell function, and unique homing distribution patterns. Memory T cells exhibit quick reactions upon re-exposure to their respective antigens in order to eliminate the reinfection of the offender and thereby restore balance of the immune system rapidly. Increasing evidence substantiates that autoimmune memory T cells hinder most attempts to treat or cure autoimmune diseases (Clark, R. A., "Resident memory T cells in human health and disease", *Sci. Transl. Med.*, Vol. 7, 269rv1, (2015)).

The Complement System

The complement system comprises over 30 different proteins that circulate in blood plasma. In the absence of an infection, the complement proteins circulate in an inactive form. In the presence of a pathogen, the complement proteins become activated to kill the pathogen either directly or by facilitating phagocytosis. There are three ways in which the complement system is activated.

Antibody-dependent cell mediated cytotoxicity (ADCC) is a mechanism by which effector cells of the immune system (e.g. natural killer cells) actively lyse target cells that have been bound by antibodies. The ADCC killing mechanism of an antibody-coated target cell by a cytotoxic effector cell is through a nonphagocytic process. This process involves the release of the content of cytotoxic granules or by the expression of cell death-inducing molecules. ADCC is triggered through interaction of target-bound antibodies (belonging to IgG or IgA or IgE classes) with certain Fc receptor glycoproteins present on the effector cell surface that bind the Fc region of immunoglobulins (Ig). Effector cells that mediate ADCC include natural killer (NK) cells, monocytes, macrophages, neutrophils, eosinophils and dendritic cells. ADCC is dependent on a number of parameters such as density and stability of the antigen on the surface of the target cell, antibody affinity, and FcR-binding affinity.

In contrast with ADCC, complement dependent cell cytotoxicity (CDCC) is a process of the immune system that kills pathogens by damaging target cell membrane without the involvement of antibodies. This alternative pathway is initiated by spontaneous hydrolysis and activation of the complement component C3, which binds directly to microbial surfaces. Alternatively the lectin pathway is initiated by soluble carbohydrate binding proteins that bind to specific carbohydrate molecules on microbial surfaces.

Each of the ADCC and CDCC mechanisms generates a C3 convertase that cleaves C3, leaving behind C3b bound to the pathogen's surface and releasing C3a. This results in recruitment of phagocytic cells to the site of an infection, phagocytosis of pathogens by immune cells, and/or formation of a membrane attack complex (MAC) that disrupts pathogen cell membrane and causes cell lysis.

Co-Stimulatory Molecules

Co-stimulatory molecules are the highly active immunomodulatory proteins that play a critical role in the development and maintenance of an immune response (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). The two signal hypothesis of T cell response involves the interaction between an antigen bound to an MHC molecule and the T cell receptor (TCR), and an interaction of a co-stimulatory molecule and its ligand. Specialized APCs, which are carriers of a co-

stimulatory second signal, are able to activate T cell responses following binding of the MHC molecule with TCR. By contrast, somatic tissues do not express the second signal and thereby induce T cell unresponsiveness (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). The two signal model explains the peripheral tolerance to self-antigens and why cancer cells can evade immune detection: tumor cells rarely express co-stimulatory molecules, and thereby lack the second signal critical to activating T cells.

Furthermore, many of the co-stimulatory molecules involved in the two-signal model can be blocked by co-inhibitory molecules that are expressed by normal tissue and by cancer cells (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). In fact, many types of interacting immunomodulatory molecules expressed on a wide variety of tissues may exert both stimulatory and inhibitory functions depending on the immunologic context (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)).

DNA motifs consisting of an unmethylated CpG dinucleotide flanked by two 5' purines (e.g. GpA) and two 3' pyrimidines (e.g. TpC or TpT) are capable of stimulating an innate immune response by mimicking bacterial DNA. CpG oligodeoxynucleotides can be used as immune adjuvants to improve the function of professional antigen-presenting cells and increase generation of humoral and cellular vaccine-specific immune responses. CpG DNA is able to directly activate dendritic cells and B cells, resulting in the induction of both innate and adaptive immune responses (Bode, C., CpG DNA as a vaccine adjuvant, *Expert Rev Vaccines*. 2011 April; 10(4): 499-511).

Cell-surface immunomodulatory molecules can be grouped according to structure into two large families of receptors/ligands: the B7/CD28 immunoglobulin family and the Tumor Necrosis Factor (TNF)-related family (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). Many members of these families have been characterized and evaluated for cancer immunotherapy.

CD28/CTLA-4/B7-1/B7-2 Family

B7-1 (CD80) and B7-2 (CD86) are expressed on activated APCs and bind to CD28 on T cells, providing the necessary co-stimulation for naïve T-cell activation, inducing IL-2 production, cell division, and the inhibition of activation induced cell death (AICD) (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). A homologue to CD28, Cytotoxic T lymphocyte-associated antigen 4 (CTLA-4, CD152), binds both B7-1 and B7-2 molecules and, in contrast to CD28, inhibits T-cell proliferation. B7 molecules therefore have two ligands, CD28 and CTLA-4, with opposing effects on T cells (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)).

Ligation of CTLA-4 in isolation may cause apoptosis of T cells, whereas CTLA-4 ligation in conjunction with signaling via the TCR and CD28 inhibits T-cell activation. Accordingly, CTLA-4 $-/-$ mice develop a fatal lymphoproliferative disorder (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)).

The differential expression of CD28 and CTLA-4 in time and location at the cell surface has implications for their respective roles in the generation of immune responses. While CD28 is uniformly distributed throughout the membrane and aggregates rapidly to the immunologic synapse

with T-cell activation, CTLA-4 is present in intracellular vesicles and is mobilized to the cell surface later (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)). Mobilization of CTLA-4 is tightly regulated by B7.1 expression on the APC, and by the strength of TCR stimulation. As a result, CTLA-4 may act to attenuate the T cell response, limiting the activity of high affinity T-cell clones (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

CTLA-4 has been implicated in many aspects of immune regulation. For example, it may be involved in causing T cell anergy, modulating memory T cell responses, shaping diversity of a polyclonal T cell response, and raising levels of inhibitory cytokines TGF-beta and IL10 (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)). CTLA-4 may also "back-signal" via B7 to down-regulate dendritic cell activation markers (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)). CTLA-4 may also play a role in regulatory T-cell (Treg) function, as it is expressed on Tregs and on cutaneous T cell lymphoma, which may arise from Tregs (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

The CD28/B7/CTLA-4 co-stimulatory mechanism has been investigated for cancer immunotherapy, such as by transfecting tumors with B7 molecules and use of anti-CTLA-4 antibodies.

In initial experiments involving transfection of B7.1 into poorly immunogenic melanoma cell lines, tumors grew but then regressed in a CD8+ T-cell-dependent process. Furthermore, animals treated with B7.1 melanoma cells became immune to further tumor challenge, demonstrating induction of immunologic memory, and inoculation with B7-expressing tumor cells caused regression of small pre-existing B7-negative tumors. Generally, larger tumors (greater than 2-3 mm) were not affected, and similar results were seen with B7.2-expressing tumors. Similar results have been shown in other tumor models including lymphoma and prostate cancer. The B7 surface molecule appears to be directly contacting and activating T cells, and B7-transfected tumor cells appear to function as APCs. Despite these promising results, human clinical trials of B7-containing vaccines have demonstrated increased immune response, but with only limited clinical benefit. (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

Anti-CTLA antibodies have been shown to be effective in murine colon carcinoma and fibrosarcoma, as well as in murine models of prostate cancer, breast cancer, and melanoma, but not in some models of poorly immunogenic tumors. Anti-CTLA-4 antibodies have also been combined with other modes of immunotherapy and conventional therapies (e.g. surgery, chemotherapy) in mouse models. The results of mouse models and human studies suggest that mechanisms by which CTLA-4 blockade enhances anti-tumor immunity are not due to regulatory T cell-mediated suppression but instead to enhanced proliferation of effector T cells through down-regulation of CTLA-4-mediated inhibition (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)). Human clinical trials of anti-CTLA-4 antibodies have demonstrated potential as an immune therapy, but because CTLA-4 plays a role in controlling T-cell responses, blocking its activity has the potential to lead to autoimmunity.

PD-1/PD-L1 (B7-H1), PD-L2 (B7-DC)

Programmed Death-1 (PD-1) is expressed by activated T cells, and is thought to be primarily an inhibitory modulator. Evidence from murine models suggests that expression of PD-L1 may protect tumors from the immune system. PD-L1 on tumors causes apoptosis in tumor-reactive T cells, and a myeloma cell line expressing PD-L1 fails to grow in PD-1 knock-out mice. In one model, PD-L1 blocking antibodies cured mice of squamous cell carcinoma. In another model, PD-L1 blocking antibodies restored responsiveness to immunologic therapy with a 4-1BB (CD137) agonist. Furthermore, PD-1 $-/-$ T cells have been shown to have enhanced anti-tumor characteristics. PD-L1 may also play an important role in the function of "suppressor" myeloid cells. It was reported that culturing dendritic cells in the presence of blocking antibody enhanced the development of T-cell responses against ovarian cancer. The mechanism through which PD-L1 may mediate immune suppression is through Interleukin-10 (IL-10) production. In contrast, the other PD-1 ligand, PD-L2, stimulated immunity in mice to the poorly immunogenic B16 melanoma (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

Many human cancers have been found to express PD-L1 including tumors of the breast, cervix, lung, ovary, colon, as well as melanoma, glioblastoma and primary T cell lymphomas, which is consistent with the role of the PD-L1 pathway in tumor immune evasion. Furthermore, a poor prognosis in esophageal cancer and renal cell cancer may be associated with expression of PD-L1. Similarly, PD-L2 is highly expressed in Hodgkin lymphoma cell lines and may also serve as a prognostic marker (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

CD27/CD70

Upon T cell activation, CD27 is transiently up-regulated and is also expressed on B cells and NK cells. The ligand of CD27, CD70, is expressed on activated lymphocytes and mature dendritic cells. A transition from central-memory to effector-memory phenotype is associated with loss of CD27 expression on CD8+ T cells, and CD27 $-/-$ mice show impaired memory T cell function along with decreased accumulation in peripheral tissues during viral infection. In contrast, mice with constitutive CD27 expression display accumulating increased T cell populations, and ultimately develop a paucity of B cells and eventually succumb to a lethal T-cell immunodeficiency, possibly due to an excessive shift in the T-cell population towards a terminally differentiated, non-reproducing memory phenotype (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

OX40/OX40L

OX40 (CD134) is expressed only on activated T-cells, and predominately CD4+ T cells. Its ligand, OX40L, is found on a wide variety of immune cells including activated B cells, T cells, dendritic cells, and vascular epithelial cells. Ligation of OX40 on T-cells promotes survival, expansion, and cytokine production, and studies in knock-out animals show that OX40 is critical for CD4, but not CD8 responses. OX40 is also important for the homeostasis and development of Tregs. In the context of immunotherapy, OX40 ligation may reverse T-cell anergy and render silent epitopes immunogenic (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

Various strategies of augmenting OX40 signaling in anti-tumor T cells have shown promise in developing tumor immunotherapy in mouse models. OX40 ligation has been found to increase tumor-free survival and cure some mice in

animal models of cancers such as melanoma, sarcoma, colon cancer, breast cancer, and glioma. Furthermore, treatment was effective in animal models of metastatic disease, where mice developed strong anti-tumor T-cell responses, in particular memory CD4⁺ T-cells, which protected them from further challenge with the same tumor. Furthermore, vaccines comprising cells transfected with OX40L and GM-CSF cure colon cancer in murine colon cancer models. OX40 ligation has also shown synergy with a combination of 4-1BB ligation and Interleukin 12 (IL-12). In total, evidence from murine studies suggests that ligation of OX40, combined with other immunotherapies, shows promise in the treatment of human cancers (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

4-1BB/4-1BBL

4-1BB (CD137) is expressed on activated T cells, NK cells, and dendritic cells, while 4-1BBL is expressed on activated antigen presenting cells (APCs). Studies have found that 4-1BB ligation particularly stimulates CD8⁺ T cells, and promotes their differentiation into effectors. It has been reported that 4-1BB signaling is able to reverse the anergy induced by soluble antigens and rescue CD28^{-/-} CD8⁺ T cells. Accumulation of such T cells occurs in the elderly, during chronic inflammation, and cancer. In contrast, 4-1BB ligation has been shown to suppress CD4⁺ T cells and B cells. Agonist anti-4-1BB antibody has been identified as being able to reverse autoimmunity in mice (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

Anti-4-1BB antibodies have been able to achieve eradication of established tumors in mouse models, and ligation of 4-1BB by systemically administered antibodies, as well as vaccination with 4-1BBL expressing tumor cells, have been shown to cause rejection of tumors. Furthermore, tumor cells transfected with single-chain Fv fragments specific for 4-1BB have also been found to be effective anti-tumor agents. CD8⁺ T cells are believed to primarily be effectors in 4-1BB mouse models, but tumor rejection has also been identified as being dependent on CD4⁺ T cells, NK cells, and on myeloid cells. Ligation of 4-1BB is ineffective, however, when CD28 is present and an immune response is already present. Thus, 4-1BB ligation has been used in combination with CD28 stimulation to target both pathways together (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

HVEM-LIGHT

Herpes Virus Entry Mediator (HVEM) is a biochemical switch regulating T cell activation in a costimulatory or co-inhibitory fashion. The stimulatory or inhibitory outcome depends on the specific ligand engaged (Cai, G., The CD160, BTLA, LIGHT/HVEM pathway: a bidirectional switch regulating T-cell activation, Immunol. Rev., May; 229(1):244-58 (2009)). HVEM binds to at least three ligands: lymphotoxin-like, exhibits inducible expression, and competes with herpes simplex virus glycoprotein D for HVEM, a receptor expressed by T lymphocytes (LIGHT), lymphotoxin alpha 3 (Lt α 3), and B- and T-lymphocyte attenuator (BTLA). LIGHT, Lt α 3, and BTLA are HVEM ligands. The binding of LIGHT or Lt α 3 to HVEM delivers a costimulatory signal, while binding of BTLA to HVEM delivers a co-inhibitory signal. The LIGHT receptor binds two receptors in addition to HVEM: LT β R and CdR3/TR6. HVEM is found on resting T cells, monocytes, and immature dendritic cells. LIGHT can be found on activated T cells, monocytes, and NK cells, and also on immature dendritic cells. LIGHT signaling causes proliferation of T cells stimu-

lated with CD3 or CD3/CD28, and can induce DC maturation, while over-expression of LIGHT can cause autoimmunity with increased T cell populations and inflammation of mucosal tissues. LIGHT deficiency causes CD8⁺ T-cell dysfunction. BTLA is expressed on activated T cells, B cells and dendritic cells, and its signals can suppress T-cell responses (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

LIGHT is believed to have an anti-tumor effect through apoptosis induction and immune activation, and it can kill tumors expressing HVEM via a death-domain pathway. Furthermore, transfection of tumor cells with LIGHT are capable of causing T-cell dependent tumor rejection, in some cases by inducing changes to tumor stromal cells facilitating entry of T cells into the tumor (Kaufman and Wolchok eds., General Principles of Tumor Immunotherapy, Chpt 5, 67-121 (2007)).

CpG

DNA motifs consisting of an unmethylated CpG dinucleotide flanked by two 5' purines (e.g. GpA) and two 3' pyrimidines (e.g. TpC or TpT) are capable of stimulating an innate immune response by mimicking bacterial DNA. CpG oligodeoxynucleotides can be used as immune adjuvants to improve the function of professional antigen-presenting cells and increase generation of humoral and cellular vaccine-specific immune responses. CpG DNA is able to directly activate dendritic cells and B cells, resulting in the induction of both innate and adaptive immune responses (Bode, C., CpG DNA as a vaccine adjuvant, Expert Rev Vaccines. 2011 April; 10(4): 499-511). The efficacy of oligodeoxynucleotides containing unmethylated CpG motifs as an immune therapy adjuvant is dependent on the spatial and temporal proximity between the CpG and an antigen. Studies have shown that physically attaching a CpG oligonucleotide to an antigen can increase immunity to that antigen by more than 100-fold relative to a CpG oligonucleotide diffusely mixed with an antigen. Furthermore, CpG conjugated increases dendritic cell uptake of cell based vaccines, increases co-stimulatory molecule expression, increases production of immunostimulatory cytokines, and causes expansion of cytotoxic T cells (Shirota, H., CpG-conjugated apoptotic tumor cells elicit potent tumor-specific immunity, Cancer Immunol Immunother (2011) 60:659-669 incorporated by reference herein in its entirety).

Immunogenic Potential of Vaccines

Vaccines against infectious agents are prime examples of specific receptor-ligand interactions being used to shape an immune response for the therapeutic goal of preventing or reducing infection (e.g. flu vaccine). Generally, an antigen is presented to the immune system in the context of an adjuvant (e.g., a synthetic small molecule immunomodulator).

The allogeneic tumor vaccines of the described invention are distinct from such vaccines in several key features. First, they are designed to be capable of treating existing tumors, although prevention of tumor formation is theoretically also possible. Second, their efficacy tends to be limited by the fact that while tumors express neoantigens (i.e. new, non-self elements) that are foreign and new to the individual, they are also undoubtedly human tumor cells and thus not always recognized as foreign (i.e. non-self) by the individual.

The aforementioned difficulties notwithstanding, evidence has now emerged that 1) endogenous antitumor responses exist, 2) that these immune responses can be modulated and 3) that this modulation can be measured in terms of overall survival in standard clinical trials.

According to aspects of the described invention, a series of immunomodulators that can be co-expressed either on a tumor cell line derived from a cancer patient, or on a multiply genetically modified allogeneic tumor cell line has been identified that, when used as a tumor vaccine, may serve 1) to efficiently load the broad array of tumor antigens into the endogenous antigen presenting cells, 2) to efficiently stimulate several cell types by enhancing the normal signals received during an immune response, 3) to impede the mechanisms by which T regulatory cells suppress the immune response, 4) to impede the signals by which immune responses are generally resolved, and 5) to result in enhanced overall survival of cancer patients vaccinated with such a formulation. Although in certain embodiments, the modified tumor cell line can be derived from the patient who receives the vaccine, the allogeneic tumor cell line vaccine approach is distinct from a personalized therapy approach, because the modified tumor cells are not necessarily derived from the individual who ultimately receives the vaccine. Instead, an allogeneic tumor cell vaccine aims to focus an immune response on the many elements that individual tumors of the same tumor type have in common.

One strategy for exploiting the large number of potential tumor antigens for each individual type of cancer is to vaccinate with whole tumor cells to avoid accidentally excluding potentially relevant antigens. The invention described herein provides, among other things, a vaccine with whole tumor cells possessing an array of antigens and modified to express two or more immune modulators.

BRIEF SUMMARY OF THE INVENTION

According to some aspects, a method of treating a cancer in a patient comprises the steps of (a) preparing an allogeneic tumor cell line variant transfected to express two or more immunomodulator peptides by: (1) providing an allogeneic parental tumor cell line; (2) transfecting or transducing recombinant DNA sequences coding for two or more of immunomodulator peptides selected from IgG1, CD40L, TNF-alpha, GM-CSF, and Flt-3L; (3) generating the tumor cell line variants by selecting for tumor cell clones that stably express an immunogenic amount of the two or more immune modulator peptides selected from IgG1, CD40L, TNF-alpha, GM-CSF, and Flt-3L; (4) selecting in a mixed lymphocyte tumor cell reaction clonally derived cell line variants by one or more of the following parameters selected from cellular proliferation, cellular subset differentiation, cytokine release profile, and tumor cell lysis; wherein the selected clonally derived cell line variant is effective to stimulate activation of one or more of T cells, B cells, and dendritic cells, and (b) administering to the patient that has cancer an immunostimulatory amount of the tumor cell line variant vaccine, wherein the immunostimulatory amount is effective to improve clinical outcome.

According to some embodiments, the immunomodulator peptides are selected from membrane expressed IgG1, CD40L, TNF-alpha, as well as membrane and soluble forms of GM-CSF, and Flt-3L.

According to some embodiments, the tumor cell line variant vaccine is effective to improve overall survival of cancer patients relative to placebo controls. According to some embodiments the parental tumor cell line is derived from a melanoma. According to some embodiments the parental tumor cell line is derived from a prostate cancer. According to some embodiments the parental tumor cell line is derived from a breast cancer.

According to some embodiments, the IgG1 immunomodulator peptide sequence is of at least 60% identity to SEQ ID NO: 45. According to some embodiments, the CD40L immune modulator peptides sequence is of at least 60% identity to SEQ ID NO: 7. According to some embodiments, the TNF-alpha immune modulator peptide sequence is of at least 60% identity to SEQ ID NO: 11. According to some embodiments, the GM-CSF immune modulator peptide sequence is of at least 60% identity to SEQ ID NO: 13 or SEQ ID NO: 5. According to some embodiments, the Flt-3L immune modulator peptide sequence is of at least 60% identity to SEQ ID NO: 14 or SEQ ID NO: 44.

According to some aspects, an allogeneic tumor cell vaccine comprises (1) a tumor cell line variant comprising (a) two or more stably expressed recombinant membrane bound immunomodulatory molecules selected from IgG1, CD40L, TNF-alpha, and Flt-3L peptides; and (b) stably expressed recombinant soluble GM-CSF peptides; and (2) a pharmaceutically acceptable carrier; wherein an immune stimulatory amount of the tumor cell line variant is effective to elicit an immune response that improves progression free survival, overall survival, or both relative to placebo controls.

According to some embodiments, the tumor cell line variant expresses two or more of (a) a membrane bound IgG1 peptide with at least 60% identity to SEQ ID NO: 45; (b) a membrane bound CD40L peptide with at least 60% identity to SEQ ID NO: 7; (c) a membrane bound form of TNF-alpha peptide with at least 60% identity to SEQ ID NO: 11; (d) a membrane bound form of Flt-3L peptide with at least 60% identity to SEQ ID NO: 14; and (e) a soluble GM-CSF peptide with at least 60% identity to SEQ ID NO: 13.

According to some embodiments, the tumor cell line variant comprises a membrane bound fusion protein of CD40L peptide and TNF-alpha peptide. According to some embodiments, the CD40L peptide is of at least 60% identity to SEQ ID NO: 9, and the TNF-alpha peptide is of at least 60% identity to SEQ ID NO: 10. According to some embodiments, the TNF-alpha peptide is of at least 60% identity to SEQ ID NO: 11. According to some embodiments, the tumor cell line variants comprise soluble GM-CSF and membrane bound IgG1, CD40L, TNF-alpha, and Flt-3L. According to some embodiments, the tumor cell line variant comprises a fusion of CD40L and TNFa peptides. According to some embodiments, the tumor cell line variant comprises an immune modulator peptide sequence of at least 60% identity to SEQ ID NO: 31. According to some embodiments, the tumor cell line variant comprises membrane and soluble forms of GM-CSF and membrane and soluble forms of Flt-3L. According to some embodiments, the tumor cell line variant comprises membrane bound forms of IgG, CD40L, and TNF-alpha.

These and other advantages of the invention will be apparent to those of ordinary skill in the art by reference to the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows one embodiment of a heteroclitic cross reaction between a peptide native to a tumor cell line and a peptide native to a tumor cell of a patient receiving immunotherapy.

FIG. 2A shows a schematic of the core vectors.

FIG. 2B shows a schematic of the proteins encoded by Vectors 2-6.

FIG. 3A shows a schematic of the organization of the scFv-anti-biotin-G3hinge-mIgG1 vector.

FIG. 3B shows the nucleotide sequence of vector 1 (SEQ ID NO. 47).

FIG. 4A shows a schematic of the organization of the full anti-biotin—G3hinge-mIgG1 vector.

FIG. 4B shows the nucleotide sequence of vector 2 (SEQ ID NO. 48).

FIG. 5A shows a schematic of the organization of the sGM-CSF/ires/mFLT3L vector.

FIG. 5B shows the nucleotide sequence of vector 3 (SEQ ID NO. 49).

FIG. 6A shows a schematic of the organization of the sFLT3L/ires/(FLT3 signal-GM-CSF-Tm) vector.

FIG. 6B shows the nucleotide sequence of vector 4 (SEQ ID NO. 50).

FIG. 7A shows a schematic of the organization of the mCD40L vector.

FIG. 7B shows the nucleotide sequence of vector 5 (SEQ ID NO. 51).

FIG. 8A shows a schematic of the organization of the mTNFa vector.

FIG. 8B shows the nucleotide sequence of vector 6 (SEQ ID NO. 52).

FIG. 9A shows a schematic of the organization of the mRANKL/ires/FLT3 signal-V5-scFV anti-biotin-Tm vector.

FIG. 9B shows the nucleotide sequence of vector 7 (SEQ ID NO. 53).

FIG. 10 is a schematic that shows the general experimental format.

FIG. 11 is a panel of graphs that show the results of flow cytometry experiments. Forward (FSC) and side scatter (SSC) plots for size and granularity. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT-3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF- α ; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3, 4 and 5; and 3-4-6 is a combination of 3, 4 and 6. Cell lines 6, 3-4-5 and 3-4-6 display a larger and more granular phenotype likely owing to the presence of receptors for TNF- α and CD40L on cells of epithelial origin.

FIG. 12 is a panel of graphs that show representative flow cytometry stains for CD4 cells in hPBMC in response to the indicated engineered cell lines with the indicated immunomodulators SK cell lines are represented by the following code; SK, unmodified parent line; 2, membrane expressed IgG1, 3, secreted GM-CSF and membrane expressed FLT-3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; and 6, a non-cleavable form of TNF.

FIG. 13 is a panel of graphs that show representative flow cytometry stains for the indicated engineered surface markers; GM-CSF, FLT3L, TNF- α and CD40L. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT-3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF- α ; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3, 4 and 5; and 3-4-6 is a combination of 3, 4 and 6.

FIG. 14A and FIG. 14B show the results of CyTOF mass cytometry single-cell phenotype analysis of hPBMC response to SK melanoma cells with modification by expression of immunomodulatory factors. FIG. 14A shows viSNE density contour plots of CyTOF staining data showing relative changes in immune cell subset abundance and

phenotype. FIG. 14B shows single-cell phenotype analysis. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT-3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF- α ; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3, 4 and 5; and 3-4-6 is a combination of 3, 4 and 6.

FIG. 15A-FIG. 15D shows CyTOF monocyte cluster analysis of hPBMC indicating changes in the activation markers CD40 (FIG. 15A), CD86 (FIG. 15B), CD69 (FIG. 15C) and CD25 (FIG. 15D) expression following 1 day stimulation with the indicated genetically modified SK lines at a 1:5 cell ratio. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT-3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF- α ; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3, 4 and 5; and 3-4-6 is a combination of 3, 4 and 6. FIG. 15E shows CyTOF monocyte cluster analysis of hPBMC indicating relative median expression levels of monocyte markers CD40 and CD86. FIG. 15F shows CyTOF monocyte cluster analysis of hPBMC indicating relative median expression levels of CD4 T cell markers CD69 and CD25.

FIG. 16 is a graph that shows the results of LUMINEX® multiplex cytokine profiling of human PBMC responses to SK parent line and genetically modified SK lines. Control cultures included SK cells alone, hPBMCs alone, and hPBMCs stimulated with a mixture of anti-CD3 and anti-CD28 antibodies (1 μ g/ml final concentration). Symbols indicate cytokine levels in pg/ml as estimated from a standard curve using recombinant cytokines. Absence of symbols indicates the cytokine was not detected. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT-3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF- α ; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3, 4 and 5; and 3-4-6 is a combination of 3, 4 and 6.

DETAILED DESCRIPTION

Definitions

The term “activation” or “lymphocyte activation” refers to stimulation of lymphocytes by specific antigens, nonspecific mitogens, or allogeneic cells resulting in synthesis of RNA, protein and DNA and production of lymphokines; it is followed by proliferation and differentiation of various effector and memory cells. For example, a mature B cell can be activated by an encounter with an antigen that expresses epitopes that are recognized by its cell surface immunoglobulin Ig). The activation process may be a direct one, dependent on cross-linkage of membrane Ig molecules by the antigen (cross-linkage-dependent B cell activation) or an indirect one, occurring most efficiently in the context of an intimate interaction with a helper T cell (“cognate help process”). T-cell activation is dependent on the interaction of the TCR/CD3 complex with its cognate ligand, a peptide bound in the groove of a class I or class II MHC molecule. The molecular events set in motion by receptor engagement are complex. Among the earliest steps appears to be the activation of tyrosine kinases leading to the tyrosine phosphorylation of a set of substrates that control several signaling pathways. These include a set of adapter proteins that

link the TCR to the ras pathway, phospholipase C γ 1, the tyrosine phosphorylation of which increases its catalytic activity and engages the inositol phospholipid metabolic pathway, leading to elevation of intracellular free calcium concentration and activation of protein kinase C, and a series of other enzymes that control cellular growth and differentiation. Full responsiveness of a T cell requires, in addition to receptor engagement, an accessory cell-delivered costimulatory activity, e.g., engagement of CD28 on the T cell by CD80 and/or CD86 on the antigen presenting cell (APC). The soluble product of an activated B lymphocyte is immunoglobulins (antibodies). The soluble product of an activated T lymphocyte is lymphokines.

As used herein, the term "administration" and its various grammatical forms as it applies to a mammal, cell, tissue, organ, or biological fluid, refers without limitation to contact of an exogenous ligand, reagent, placebo, small molecule, pharmaceutical agent, therapeutic agent, diagnostic agent, or composition to the subject, cell, tissue, organ, or biological fluid, and the like. "Administration" can refer, e.g., to therapeutic, pharmacokinetic, diagnostic, research, placebo, and experimental methods. "Administration" also encompasses in vitro and ex vivo treatments, e.g., of a cell, by a reagent, diagnostic, binding composition, or by another cell.

The term "allogeneic" as used herein means that the donor and the recipient (host) are of different genetic makeup, but of the same species. As used herein, an "allogeneic cell" refers to a cell that is not derived from the individual to which the cell is to be administered, that is, it has a different genetic constitution than the individual. An allogeneic cell is generally obtained from the same species as the individual to which the cell is to be administered. For example, the allogeneic cell can be a human cell, as disclosed herein, for administering to a human patient such as a cancer patient. As used herein, an "allogeneic tumor cell" refers to a tumor cell that is not derived from the individual to which the allogeneic cell is to be administered. Generally, the allogeneic tumor cell expresses one or more tumor antigens that can stimulate an immune response against a tumor in an individual to which the cell is to be administered. As used herein, an "allogeneic cancer cell," for example, a lung cancer cell, refers to a cancer cell that is not derived from the individual to which the allogeneic cell is to be administered.

The terms "amino acid residue" or "amino acid" or "residue" are used interchangeably to refer to an amino acid that is incorporated into a protein, a polypeptide, or a peptide, including, but not limited to, a naturally occurring amino acid and known analogs of natural amino acids that can function in a similar manner as naturally occurring amino acids. The amino acids may be L- or D-amino acids. An amino acid may be replaced by a synthetic amino acid, which is altered so as to increase the half-life of the peptide, increase the potency of the peptide, or increase the bioavailability of the peptide. The single letter designation for amino acids is used predominately herein. Such single letter designations are as follows: A is alanine; C is cysteine; D is aspartic acid; E is glutamic acid; F is phenylalanine; G is glycine; H is histidine; I is isoleucine; K is lysine; L is leucine; M is methionine; N is asparagine; P is proline; Q is glutamine; R is arginine; S is serine; T is threonine; V is valine; W is tryptophan; and Y is tyrosine. The following represents groups of amino acids that are conservative substitutions for one another: 1) Alanine (A), Serine (S), Threonine (T); 2) Aspartic Acid (D), Glutamic Acid (E); 3) Asparagine (N), Glutamine (Q); 4) Arginine (R), Lysine (K); 5) Isoleucine (I), Leucine (L), Methionine (M), Valine (V); and 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W).

The term "autologous" as used herein means derived from the same individual.

The term "cancer" as used herein refers to diseases in which abnormal cells divide without control and are able to invade other tissues. There are more than 100 different types of cancer. Most cancers are named for the organ or type of cell in which they start—for example, cancer that begins in the colon is called colon cancer; cancer that begins in melanocytes of the skin is called melanoma. Cancer types can be grouped into broader categories. The main categories of cancer include: carcinoma (meaning a cancer that begins in the skin or in tissues that line or cover internal organs, and its subtypes, including adenocarcinoma, basal cell carcinoma, squamous cell carcinoma, and transitional cell carcinoma); sarcoma (meaning a cancer that begins in bone, cartilage, fat, muscle, blood vessels, or other connective or supportive tissue); leukemia (meaning a cancer that starts in blood-forming tissue (e.g., bone marrow) and causes large numbers of abnormal blood cells to be produced and enter the blood; lymphoma and myeloma (meaning cancers that begin in the cells of the immune system); and Central nervous system cancers (meaning cancers that begin in the tissues of the brain and spinal cord). The term "myelodysplastic syndrome" refers to a type of cancer in which the bone marrow does not make enough healthy blood cells (white blood cells, red blood cells, and platelets) and there are abnormal cells in the blood and/or bone marrow. Myelodysplastic syndrome may become acute myeloid leukemia (AML).

The term "contact" and its various grammatical forms as used herein refers to a state or condition of touching or of immediate or local proximity. Contacting a composition to a target destination may occur by any means of administration known to the skilled artisan.

The term "costimulatory molecule" as used herein refers to one of two or more molecules that are displayed on the cell surface that have a role in activating a T cell to become an effector cell. For example MHC proteins, which present foreign antigen to the T cell receptor, also require costimulatory proteins which bind to complementary receptors on the T cell's surface to result in activation of the T cell.

The term "cytokine" as used herein refers to small soluble protein substances secreted by cells which have a variety of effects on other cells. Cytokines mediate many important physiological functions including growth, development, wound healing, and the immune response. They act by binding to their cell-specific receptors located in the cell membrane, which allows a distinct signal transduction cascade to start in the cell, which eventually will lead to biochemical and phenotypic changes in target cells. Cytokines can act both locally and distantly from a site of release. They include type I cytokines, which encompass many of the interleukins, as well as several hematopoietic growth factors; type II cytokines, including the interferons and interleukin-10; tumor necrosis factor ("TNF")-related molecules, including TNF α and lymphotoxin; immunoglobulin super-family members, including interleukin 1 ("IL-1"); and the chemokines, a family of molecules that play a critical role in a wide variety of immune and inflammatory functions. The same cytokine can have different effects on a cell depending on the state of the cell. Cytokines often regulate the expression of, and trigger cascades of other cytokines. Nonlimiting examples of cytokines include e.g., IL-1 α , IL- β , IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-11, IL-12/IL-23 P40, IL-13, IL-15, IL-17, IL-18, IL-21, IL-23, TGF- β , IFN- γ , GM-CSF, Gro. α , MCP-1 and TNF- α .

The term “derived from” as used herein encompasses any method for receiving, obtaining, or modifying something from a source of origin.

The term “derivative” or “variant” with respect to a peptide or DNA sequence (e.g. immune modulator peptide sequence) as used herein refers to a non-identical peptide or DNA sequence that is modified from its original sequence. The terms “derivative” or “variant” with respect to cells as used herein refers to a cell line that has been modified from its cell line of origin (e.g. modified to express recombinant DNA sequences).

The term “detectable marker” encompasses both selectable markers and assay markers. The term “selectable markers” refers to a variety of gene products to which cells transformed with an expression construct can be selected or screened, including drug-resistance markers, antigenic markers useful in fluorescence-activated cell sorting, adherence markers such as receptors for adherence ligands allowing selective adherence, and the like.

The term “detectable response” refers to any signal or response that may be detected in an assay, which may be performed with or without a detection reagent. Detectable responses include, but are not limited to, radioactive decay and energy (e.g., fluorescent, ultraviolet, infrared, visible) emission, absorption, polarization, fluorescence, phosphorescence, transmission, reflection or resonance transfer. Detectable responses also include chromatographic mobility, turbidity, electrophoretic mobility, mass spectrum, ultraviolet spectrum, infrared spectrum, nuclear magnetic resonance spectrum and x-ray diffraction. Alternatively, a detectable response may be the result of an assay to measure one or more properties of a biologic material, such as melting point, density, conductivity, surface acoustic waves, catalytic activity or elemental composition. A “detection reagent” is any molecule that generates a detectable response indicative of the presence or absence of a substance of interest. Detection reagents include any of a variety of molecules, such as antibodies, nucleic acid sequences and enzymes. To facilitate detection, a detection reagent may comprise a marker.

The term “dose” as used herein refers to the quantity of a therapeutic substance prescribed to be taken at one time.

The term “enrich” as used herein refers to increasing the proportion of a desired substance, for example, to increase the relative frequency of a subtype of cell compared to its natural frequency in a cell population. Positive selection, negative selection, or both are generally considered necessary to any enrichment scheme. Selection methods include, without limitation, magnetic separation and FACS. Regardless of the specific technology used for enrichment, the specific markers used in the selection process are critical, since developmental stages and activation-specific responses can change a cell’s antigenic profile.

As used herein, the term “expression” encompasses the biosynthesis of mRNA, polypeptide biosynthesis, polypeptide activation, e.g., by post-translational modification, or an activation of expression by changing the subcellular location or by recruitment to chromatin.

The term “expression vector” refers a DNA molecule comprising a gene that is expressed in a host cell. Typically, gene expression is placed under the control of certain regulatory elements including, but not limited to, promoters, tissue specific regulatory elements, and enhancers. Such a gene is said to be “operably linked to” the regulatory elements.

The term “flow cytometry” as used herein refers to a tool for interrogating the phenotype and characteristics of cells.

It senses cells or particles as they move in a liquid stream through a laser (light amplification by stimulated emission of radiation)/light beam past a sensing area. The relative light-scattering and color-discriminated fluorescence of the microscopic particles is measured. Flow Analysis and differentiation of the cells is based on size, granularity, and whether the cells is carrying fluorescent molecules in the form of either antibodies or dyes. As the cell passes through the laser beam, light is scattered in all directions, and the light scattered in the forward direction at low angles (0.5-10°) from the axis is proportional to the square of the radius of a sphere and so to the size of the cell or particle. Light may enter the cell; thus, the 90° light (right-angled, side) scatter may be labeled with fluorochrome-linked antibodies or stained with fluorescent membrane, cytoplasmic, or nuclear dyes. Thus, the differentiation of cell types, the presence of membrane receptors and antigens, membrane potential, pH, enzyme activity, and DNA content may be facilitated. Flow cytometers are multiparameter, recording several measurements on each cell; therefore, it is possible to identify a homogeneous subpopulation within a heterogeneous population (Marion G. Macey, Flow cytometry: principles and applications, Humana Press, 2007). Fluorescence-activated cell sorting (FACS), which allows isolation of distinct cell populations too similar in physical characteristics to be separated by size or density, uses fluorescent tags to detect surface proteins that are differentially expressed, allowing fine distinctions to be made among physically homogeneous populations of cells.

The term “functional equivalent” or “functionally equivalent” are used interchangeably herein to refer to substances, molecules, polynucleotides, proteins, peptides, or polypeptides having similar or identical effects or use.

The term “heteroclitic” is used herein to refer to peptides of higher biological potency than the original peptide. A “heteroclitic immunogen” is an immunogen that elicits an immune response that cross-reacts to an original poorly immunogenic antigen.

The terms “immune response” and “immune-mediated” are used interchangeably herein to refer to any functional expression of a subject’s immune system, against either foreign or self-antigens, whether the consequences of these reactions are beneficial or harmful to the subject.

The terms “immunomodulatory”, “immune modulator” and “immune modulatory” are used interchangeably herein to refer to a substance, agent, or cell that is capable of augmenting or diminishing immune responses directly or indirectly by expressing chemokines, cytokines and other mediators of immune responses.

As used herein the term “immunostimulatory amount” of the disclosed compositions refers to an amount of an immunogenic composition that is effective to stimulate an immune response, for example, as measured by ELISPOT assay (cellular immune response), ICS (intracellular cytokine staining assay) and major histocompatibility complex (MHC) tetramer assay to detect and quantify antigen-specific T cells, quantifying the blood population of antigen-specific CD4+ T cells, or quantifying the blood population of antigen specific CD8+ T cells by a measurable amount, or where the increase is by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, at least 100%, when compared to a suitable control (e.g., a control composition where dendritic cells are not loaded with tumor-specific cells, or not loaded with peptide derived from tumor-specific cells).

The term “integrate into the genome” as used herein refers to a recombinant DNA sequence being concomitantly joined to the genomic DNA comprising a host cell’s genome.

The term “Kaplan Meier plot” or “Kaplan Meier survival curve” as used herein refers to the plot of probability of clinical study subjects surviving in a given length of time while considering time in many small intervals. The Kaplan Meier plot assumes that: (i) at any time subjects who are censored (i.e., lost) have the same survival prospects as subjects who continue to be followed; (ii) the survival probabilities are the same for subjects recruited early and late in the study; and (iii) the event (e.g., death) happens at the time specified. Probabilities of occurrence of events are computed at a certain point of time with successive probabilities multiplied by any earlier computed probabilities to get a final estimate. The survival probability at any particular time is calculated as the number of subjects surviving divided by the number of subjects at risk. Subjects who have died, dropped out, or have been censored from the study are not counted as at risk.

The term “labeling” as used herein refers to a process of distinguishing a compound, structure, protein, peptide, antibody, cell or cell component by introducing a traceable constituent. Common traceable constituents include, but are not limited to, a fluorescent antibody, a fluorophore, a dye or a fluorescent dye, a stain or a fluorescent stain, a marker, a fluorescent marker, a chemical stain, a differential stain, a differential label, and a radioisotope.

The terms “marker” or “cell surface marker” are used interchangeably herein to refer to an antigenic determinant or epitope found on the surface of a specific type of cell. Cell surface markers can facilitate the characterization of a cell type, its identification, and eventually its isolation. Cell sorting techniques are based on cellular biomarkers where a cell surface marker(s) may be used for either positive selection or negative selection, i.e., for inclusion or exclusion, from a cell population.

The terms “mixed lymphocyte tumor reaction” or “MLTR” are used interchangeably herein to refer to a reaction similar to a mixed lymphocyte reaction but rather than using allogeneic lymphocytes to stimulate a response, allogeneic tumor cells are used instead. The MLTR method comprises contacting tumor cells being tested for immunogenic potential with mixed lymphocytes from peripheral blood mononuclear cells, followed by measuring one or more of cellular proliferation of the lymphocytes, cellular subset differentiation of the lymphocytes, cytokine release profile of the lymphocytes, and tumor cell death.

The term “modified” or “modulated” as used herein with respect to immune response to tumor cells refers to changing the form or character of the immune response to the tumor cells via one or more recombinant DNA techniques such that the immune cells are able to recognize and kill tumor cells.

The term “myeloid suppressor cells” or “myeloid-derived suppressor cells” as used herein refers to the heterogeneous population of cells characterized by myeloid origin, immature state, and ability to potentially suppress T cell responses. These cells regulate immune responses and tissue repair in healthy individuals and the population rapidly expands during inflammation.

The term “open reading frame” as used herein refers to a sequence of nucleotides in a DNA molecule that has the potential to encode a peptide or protein: it starts with a start triplet (ATG), is followed by a string of triplets each of which encodes an amino acid, and ends with a stop triplet (TAA, TAG or TGA).

The phrase “operably linked” refers (1) to a first sequence(s) or domain being positioned sufficiently proximal to a second sequence(s) or domain so that the first sequence(s) or domain can exert influence over the second sequence(s) or domain or a region under control of that second sequence or domain; and (2) to a functional linkage between a promoter and a second sequence, wherein the promoter sequence initiates and mediates transcription of the DNA sequence corresponding to the second sequence. Generally, operably linked means that the nucleic acid sequences being linked are contiguous and, where necessary to join two protein coding regions, are in the same reading frame. According to some embodiments, the phrase “operatively linked” refers to a linkage in which two or more protein domains or polypeptides are ligated or combined via recombinant DNA technology or chemical reaction such that each protein domain or polypeptide of the resulting fusion protein retains its original function.

The term “overall survival” (OS) as used herein refers to the length of time from either the date of diagnosis or the start of treatment for a disease, such as cancer, that patients diagnosed with the disease are still alive.

The term “parenteral” and its other grammatical forms as used herein refers to administration of a substance occurring in the body other than by the mouth or alimentary canal. For example, the term “parenteral” as used herein refers to introduction into the body by way of an injection (i.e., administration by injection), including, for example, subcutaneously (i.e., an injection beneath the skin), intramuscularly (i.e., an injection into a muscle); intravenously (i.e., an injection into a vein), intrathecally (i.e., an injection into the space around the spinal cord or under the arachnoid membrane of the brain), intrasternal injection, or infusion techniques.

The terms “peripheral blood mononuclear cells” or “PBMCs” are used interchangeably herein to refer to blood cells having a single round nucleus such as, for example, a lymphocyte or a monocyte.

The term “pharmaceutical composition” as used herein refers to a composition that is employed to prevent, reduce in intensity, cure or otherwise treat a target condition, syndrome, disorder or disease.

The term “pharmaceutically acceptable carrier” as used herein refers to any substantially non-toxic carrier conventionally useable for administration of pharmaceuticals in which the isolated polypeptide of the present invention will remain stable and bioavailable. The pharmaceutically acceptable carrier must be of sufficiently high purity and of sufficiently low toxicity to render it suitable for administration to the mammal being treated. It further should maintain the stability and bioavailability of an active agent. The pharmaceutically acceptable carrier can be liquid or solid and is selected, with the planned manner of administration in mind, to provide for the desired bulk, consistency, etc., when combined with an active agent and other components of a given composition.

The term “pharmaceutically acceptable salt” as used herein refers to those salts which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response and the like and are commensurate with a reasonable benefit/risk ratio. When used in medicine the salts should be pharmaceutically acceptable, but non-pharmaceutically acceptable salts may conveniently be used to prepare pharmaceutically acceptable salts thereof. Such salts include, but are not limited to, those prepared from the following acids: hydrochloric, hydrobromic, sul-

phuric, nitric, phosphoric, maleic, acetic, salicylic, p-toluene sulphonic, tartaric, citric, methane sulphonic, formic, malonic, succinic, naphthalene-2-sulphonic, and benzene sulphonic. Also, such salts may be prepared as alkaline metal or alkaline earth salts, such as sodium, potassium or calcium salts of the carboxylic acid group. By "pharmaceutically acceptable salt" is meant those salts which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response and the like and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well-known in the art. For example, P. H. Stahl, et al. describe pharmaceutically acceptable salts in detail in "Handbook of Pharmaceutical Salts: Properties, Selection, and Use" (Wiley VCH, Zurich, Switzerland: 2002). The salts may be prepared in situ during the final isolation and purification of the compounds described within the present invention or separately by reacting a free base function with a suitable organic acid. Representative acid addition salts include, but are not limited to, acetate, adipate, alginate, citrate, aspartate, benzoate, benzenesulfonate, bisulfate, butyrate, camphorate, camphorsulfonate, digluconate, glycerophosphate, hemisulfate, heptanoate, hexanoate, fumarate, hydrochloride, hydrobromide, hydroiodide, 2-hydroxyethanesulfonate (isethionate), lactate, maleate, methanesulfonate, nicotinate, 2-naphthalene-sulfonate, oxalate, pamoate, pectinate, persulfate, 3-phenylpropionate, picrate, pivalate, propionate, succinate, tartrate, thiocyanate, phosphate, glutamate, bicarbonate, p-toluene-sulfonate and undecanoate. Also, the basic nitrogen-containing groups may be quaternized with such agents as lower alkyl halides such as methyl, ethyl, propyl, and butyl chlorides, bromides and iodides; dialkyl sulfates like dimethyl, diethyl, dibutyl and diamyl sulfates; long chain halides such as decyl, lauryl, myristyl and stearyl chlorides, bromides and iodides; arylalkyl halides like benzyl and phenethyl bromides and others. Water or oil-soluble or dispersible products are thereby obtained. Examples of acids which may be employed to form pharmaceutically acceptable acid addition salts include such inorganic acids as hydrochloric acid, hydrobromic acid, sulphuric acid and phosphoric acid and such organic acids as oxalic acid, maleic acid, succinic acid and citric acid. Basic addition salts may be prepared in situ during the final isolation and purification of compounds described within the invention by reacting a carboxylic acid-containing moiety with a suitable base such as the hydroxide, carbonate or bicarbonate of a pharmaceutically acceptable metal cation or with ammonia or an organic primary, secondary or tertiary amine. Pharmaceutically acceptable salts include, but are not limited to, cations based on alkali metals or alkaline earth metals such as lithium, sodium, potassium, calcium, magnesium and aluminum salts and the like and nontoxic quaternary ammonia and amine cations including ammonium, tetramethylammonium, tetraethylammonium, methylamine, dimethylamine, trimethylamine, triethylamine, diethylamine, ethylamine and the like. Other representative organic amines useful for the formation of base addition salts include ethylenediamine, ethanolamine, diethanolamine, piperidine, piperazine and the like. Pharmaceutically acceptable salts also may be obtained using standard procedures well known in the art, for example by reacting a sufficiently basic compound such as an amine with a suitable acid affording a physiologically acceptable anion. Alkali metal (for example, sodium, potassium or lithium) or alkaline earth metal (for example calcium or magnesium) salts of carboxylic acids may also be made.

The terms "polypeptide", "peptide" and "protein" are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The essential nature of such analogues of naturally occurring amino acids is that, when incorporated into a protein, that protein is specifically reactive to antibodies elicited to the same protein but consisting entirely of naturally occurring amino acids.

The terms "polypeptide", "peptide" and "protein" also are inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulfation, gamma-carboxylation of glutamic acid residues, hydroxylation, and ADP-ribosylation. It will be appreciated, as is well known and as noted above, that polypeptides may not be entirely linear. For instance, polypeptides may be branched as a result of ubiquitination, and they may be circular, with or without branching, generally as a result of posttranslational events, including natural processing event and events brought about by human manipulation which do not occur naturally. Circular, branched and branched circular polypeptides may be synthesized by non-translation natural process and by entirely synthetic methods, as well. According to some embodiments, the peptide is of any length or size.

The terms "protein domain" and "domain" are used interchangeably to refer to a portion of a protein that has its own tertiary structure. Large proteins are generally composed of several domains connected to one another via flexible regions of polypeptide chain.

The following terms are used herein to describe the sequence relationships between two or more nucleic acids or polynucleotides: (a) "reference sequence", (b) "comparison window", (c) "sequence identity", (d) "percentage of sequence identity", and (e) "substantial identity." (a) The term "reference sequence" refers to a sequence used as a basis for sequence comparison. A reference sequence may be a subset or the entirety of a specified sequence; for example, as a segment of a full-length cDNA or gene sequence, or the complete cDNA or gene sequence. (b) The term "comparison window" refers to a contiguous and specified segment of a polynucleotide sequence, wherein the polynucleotide sequence may be compared to a reference sequence and wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. Generally, the comparison window is at least 20 contiguous nucleotides in length, and optionally can be at least 30 contiguous nucleotides in length, at least 40 contiguous nucleotides in length, at least 50 contiguous nucleotides in length, at least 100 contiguous nucleotides in length, or longer. Those of skill in the art understand that to avoid a high similarity to a reference sequence due to inclusion of gaps in the polynucleotide sequence, a gap penalty typically is introduced and is subtracted from the number of matches. Methods of alignment of sequences for comparison are well-known in the art. Optimal alignment of sequences for comparison may be conducted by the local homology algorithm of Smith and Waterman, *Adv. Appl. Math.* 2:482 (1981); by the homology alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.* 48:443 (1970); by the search for similarity method of Pearson and Lipman, *Proc. Natl. Acad. Sci.* 85:2444 (1988); by computerized implementations of these algorithms, including, but not limited to: CLUSTAL in the PC/Gene program by Intelli-

genetics, Mountain View, Calif; GAP, BESTFIT, BLAST, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group (GCG), 575 Science Dr., Madison, Wis., USA; the CLUSTAL program is well described by Higgins and Sharp, *Gene* 73:237-244 (1988); Higgins and Sharp, *CABIOS* 5:151-153 (1989); Corpet, et al., *Nucleic Acids Research* 16:10881-90 (1988); Huang, et al., *Computer Applications in the Biosciences*, 8:155-65 (1992), and Pearson, et al., *Methods in Molecular Biology*, 24:307-331 (1994). The BLAST family of programs, which can be used for database similarity searches, includes: BLASTN for nucleotide query sequences against nucleotide database sequences; BLASTX for nucleotide query sequences against protein database sequences; BLASTP for protein query sequences against protein database sequences; TBLASTN for protein query sequences against nucleotide database sequences; and TBLASTX for nucleotide query sequences against nucleotide database sequences. See, *Current Protocols in Molecular Biology*, Chapter 19, Ausubel, et al., Eds., Greene Publishing and Wiley-Interscience, New York (1995). Unless otherwise stated, sequence identity/similarity values provided herein refer to the value obtained using the BLAST 2.0 suite of programs using default parameters. Altschul et al., *Nucleic Acids Res.* 25:3389-3402 (1997). Software for performing BLAST analyses is publicly available, e.g., through the National Center for Biotechnology-Information. This algorithm involves first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold (Altschul et al., supra). These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits then are extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T, and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a word length (W) of 11, an expectation (E) of 10, a cutoff of 100, M=5, N=-4, and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a word length (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff & Henikoff (1989) *Proc. Natl. Acad. Sci. USA* 89:10915). In addition to calculating percent sequence identity, the BLAST algorithm also performs a statistical analysis of the similarity between two sequences (see, e.g., Karlin & Altschul, *Proc. Natl. Acad. Sci. USA* 90:5873-5787 (1993)). One measure of similarity provided by the BLAST algorithm is the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. BLAST searches assume that proteins may be

homopolymeric tracts, short-period repeats, or regions enriched in one or more amino acids. Such low-complexity regions may be aligned between unrelated proteins even though other regions of the protein are entirely dissimilar. A number of low-complexity filter programs may be employed to reduce such low-complexity alignments. For example, the SEG (Wooten and Federhen, *Comput. Chem.*, 17:149-163 (1993)) and XNU (Claverie and States, *Comput. Chem.*, 17:191-201 (1993)) low-complexity filters may be employed alone or in combination. (c) The term "sequence identity" or "identity" in the context of two nucleic acid or polypeptide sequences is used herein to refer to the residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window. When percentage of sequence identity is used in reference to proteins it is recognized that residue positions that are not identical often differ by conservative amino acid substitutions, i.e., where amino acid residues are substituted for other amino acid residues with similar chemical properties (e.g. charge or hydrophobicity) and therefore do not change the functional properties of the molecule. Where sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Sequences that differ by such conservative substitutions are said to have "sequence similarity" or "similarity." Means for making this adjustment are well-known to those of skill in the art. Typically this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated, e.g., according to the algorithm of Meyers and Miller, *Computer Applic. Biol. Sci.*, 4:11-17 (1988) e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, Calif, USA). (d) The term "percentage of sequence identity" is used herein mean the value determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity. (e) The term "substantial identity" of polynucleotide sequences means that a polynucleotide comprises a sequence that has at least 70% sequence identity, at least 80% sequence identity, at least 90% sequence identity and at least 95% sequence identity, compared to a reference sequence using one of the alignment programs described using standard parameters. One of skill will recognize that these values may be adjusted appropriately to determine corresponding identity of proteins encoded by two nucleotide sequences by taking into account codon degeneracy, amino acid similarity, reading frame positioning and the like. Substantial identity of amino acid sequences for these purposes normally means sequence identity of at least 60%, or at least 70%, at least 80%, at least 90%, or at least 95%. Another indication that nucleotide sequences are substantially identical is if two molecules hybridize to each other under stringent condi-

tions. However, nucleic acids that do not hybridize to each other under stringent conditions are still substantially identical if the polypeptides that they encode are substantially identical. This may occur, e.g., when a copy of a nucleic acid is created using the maximum codon degeneracy permitted by the genetic code. One indication that two nucleic acid sequences are substantially identical is that the polypeptide that the first nucleic acid encodes is immunologically cross reactive with the polypeptide encoded by the second nucleic acid. Mutations may also be made to the nucleotide sequences of the present proteins by reference to the genetic code, including taking into account codon degeneracy.

The term "prime" (or "priming") as used herein refers to the process of increasing sensitivity to. When used in an immunological sense it refers to a process whereby a specific antigen is presented to naïve lymphocytes causing them to differentiate.

The term "progression free survival" or "PFS" as used herein refers to length of time during and after the treatment of a disease, such as cancer, that a patient lives with the disease but it does not get worse. In a clinical trial, measuring the progression free survival is one way to determine how well a new treatment works.

The term "recurrence" as used herein with respect to cancer refers to a cancer that has recurred (come back), usually after a period of time during which the cancer could not be detected. The cancer may come back to the same place as the original (primary) tumor or to another place in the body.

The term "relapse-free survival (RFS)" as used herein refers to the length of time after primary treatment for a cancer during which the patient survives without any signs or symptoms of that cancer. Also called disease-free survival (DFS) and progression free survival (PFS).

The term "response rate" as used herein refers to the percentage of patients whose cancer shrinks or disappears after treatment.

The term "resistant cancer" as used herein refers to a cancer that does not respond to a treatment at the beginning of such treatment or sometime during such treatment.

The term "reporter gene" ("reporter") or "assay marker" refers to a gene and/or peptide that can be detected, or easily identified and measured. The expression of the reporter may be measured at either the RNA level, or at the protein level. The gene product, which may be detected in an experimental assay protocol, includes, but is not limited to, marker enzymes, antigens, amino acid sequence markers, cellular phenotypic markers, nucleic acid sequence markers, and the like. Researchers may attach a reporter gene to another gene of interest in cell culture, bacteria, animals, or plants. For example, some reporters are selectable markers, or confer characteristics upon on organisms expressing them allowing the organism to be easily identified and assayed. To introduce a reporter gene into an organism, researchers may place the reporter gene and the gene of interest in the same DNA construct to be inserted into the cell or organism. For bacteria or eukaryotic cells in culture, this may be in the form of a plasmid. Commonly used reporter genes may include, but are not limited to, fluorescent proteins, luciferase, beta-galactosidase, and selectable markers, such as chloramphenicol and kanomycin.

The term "stimulate" in any of its grammatical forms as used herein refers to inducing activation or increasing activity.

As used herein, the terms "subject" or "individual" or "patient" are used interchangeably to refer to a member of an animal species of mammalian origin, including humans.

The phrase "subject in need thereof" as used herein refers to a patient that (i) will be administered an immunogenic composition according to the described invention, (ii) is receiving an immunogenic composition according to the described invention; or (iii) has received an immunogenic composition according to the described invention, unless the context and usage of the phrase indicates otherwise.

The term "therapeutic agent" as used herein refers to a drug, molecule, nucleic acid, protein, metabolite, composition or other substance that provides a therapeutic effect. The term "active" as used herein refers to the ingredient, component or constituent of the compositions of the described invention responsible for the intended therapeutic effect. The terms "therapeutic agent" and "active agent" are used interchangeably herein. The term "therapeutic component" as used herein refers to a therapeutically effective dosage (i.e., dose and frequency of administration) that eliminates, reduces, or prevents the progression of a particular disease manifestation in a percentage of a population. An example of a commonly used therapeutic component is the ED₅₀ which describes the dose in a particular dosage that is therapeutically effective for a particular disease manifestation in 50% of a population.

The terms "therapeutic amount", "therapeutically effective amount", an "amount effective", or "pharmaceutically effective amount" of an active agent is used interchangeably to refer to an amount that is sufficient to provide the intended benefit of treatment. However, dosage levels are based on a variety of factors, including the type of injury, the age, weight, sex, medical condition of the patient, the severity of the condition, the route of administration, and the particular active agent employed. Thus the dosage regimen may vary widely, but can be determined routinely by a physician using standard methods. Additionally, the terms "therapeutic amount", "therapeutically effective amounts" and "pharmaceutically effective amounts" include prophylactic or preventative amounts of the compositions of the described invention. In prophylactic or preventative applications of the described invention, pharmaceutical compositions or medicaments are administered to a patient susceptible to, or otherwise at risk of, a disease, disorder or condition in an amount sufficient to eliminate or reduce the risk, lessen the severity, or delay the onset of the disease, disorder or condition, including biochemical, histologic and/or behavioral symptoms of the disease, disorder or condition, its complications, and intermediate pathological phenotypes presenting during development of the disease, disorder or condition. It is generally preferred that a maximum dose be used, that is, the highest safe dose according to some medical judgment. The terms "dose" and "dosage" are used interchangeably herein.

The term "therapeutic effect" as used herein refers to a consequence of treatment, the results of which are judged to be desirable and beneficial. A therapeutic effect can include, directly or indirectly, the arrest, reduction, or elimination of a disease manifestation. A therapeutic effect can also include, directly or indirectly, the arrest reduction or elimination of the progression of a disease manifestation.

For any therapeutic agent described herein the therapeutically effective amount may be initially determined from preliminary in vitro studies and/or animal models. A therapeutically effective dose may also be determined from human data. The applied dose may be adjusted based on the relative bioavailability and potency of the administered compound. Adjusting the dose to achieve maximal efficacy

based on the methods described above and other well-known methods is within the capabilities of the ordinarily skilled artisan.

General principles for determining therapeutic effectiveness, which may be found in Chapter 1 of Goodman and Gilman's *The Pharmacological Basis of Therapeutics*, 10th Edition, McGraw-Hill (New York) (2001), incorporated herein by reference, are summarized below.

Pharmacokinetic principles provide a basis for modifying a dosage regimen to obtain a desired degree of therapeutic efficacy with a minimum of unacceptable adverse effects. In situations where the drug's plasma concentration can be measured and related to the therapeutic window, additional guidance for dosage modification can be obtained.

Drug products are considered to be pharmaceutical equivalents if they contain the same active ingredients and are identical in strength or concentration, dosage form, and route of administration. Two pharmaceutically equivalent drug products are considered to be bioequivalent when the rates and extents of bioavailability of the active ingredient in the two products are not significantly different under suitable test conditions.

The term "therapeutic window" refers to a concentration range that provides therapeutic efficacy without unacceptable toxicity. Following administration of a dose of a drug, its effects usually show a characteristic temporal pattern. A lag period is present before the drug concentration exceeds the minimum effective concentration ("MEC") for the desired effect. Following onset of the response, the intensity of the effect increases as the drug continues to be absorbed and distributed. This reaches a peak, after which drug elimination results in a decline in the effect's intensity that disappears when the drug concentration falls back below the MEC. Accordingly, the duration of a drug's action is determined by the time period over which concentrations exceed the MEC. The therapeutic goal is to obtain and maintain concentrations within the therapeutic window for the desired response with a minimum of toxicity. Drug response below the MEC for the desired effect will be subtherapeutic, whereas for an adverse effect, the probability of toxicity will increase above the MEC. Increasing or decreasing drug dosage shifts the response curve up or down the intensity scale and is used to modulate the drug's effect. Increasing the dose also prolongs a drug's duration of action but at the risk of increasing the likelihood of adverse effects. Accordingly, unless the drug is nontoxic, increasing the dose is not a useful strategy for extending a drug's duration of action.

Instead, another dose of drug should be given to maintain concentrations within the therapeutic window. In general, the lower limit of the therapeutic range of a drug appears to be approximately equal to the drug concentration that produces about half of the greatest possible therapeutic effect, and the upper limit of the therapeutic range is such that no more than about 5% to about 10% of patients will experience a toxic effect. These figures can be highly variable, and some patients may benefit greatly from drug concentrations that exceed the therapeutic range, while others may suffer significant toxicity at much lower values. The therapeutic goal is to maintain steady-state drug levels within the therapeutic window. For most drugs, the actual concentrations associated with this desired range are not and need not be known, and it is sufficient to understand that efficacy and toxicity are generally concentration-dependent, and how drug dosage and frequency of administration affect the drug level. For a small number of drugs where there is a small (two- to three-fold) difference between concentrations resulting in

efficacy and toxicity, a plasma-concentration range associated with effective therapy has been defined.

In this case, a target level strategy is reasonable, wherein a desired target steady-state concentration of the drug (usually in plasma) associated with efficacy and minimal toxicity is chosen, and a dosage is computed that is expected to achieve this value. Drug concentrations subsequently are measured and dosage is adjusted if necessary to approximate the target more closely.

In most clinical situations, drugs are administered in a series of repetitive doses or as a continuous infusion to maintain a steady-state concentration of drug associated with the therapeutic window. To maintain the chosen steady-state or target concentration ("maintenance dose"), the rate of drug administration is adjusted such that the rate of input equals the rate of loss. If the clinician chooses the desired concentration of drug in plasma and knows the clearance and bioavailability for that drug in a particular patient, the appropriate dose and dosing interval can be calculated.

As used herein the term "treating" includes abrogating, substantially inhibiting, slowing or reversing the progression of a condition, substantially ameliorating clinical symptoms of a condition, or substantially preventing the appearance of clinical symptoms of a condition. Treating further refers to accomplishing one or more of the following: (a) reducing the severity of the disorder; (b) limiting development of symptoms characteristic of the disorder(s) being treated; (c) limiting worsening of symptoms characteristic of the disorder(s) being treated; (d) limiting recurrence of the disorder(s) in patients that have previously had the disorder(s); and (e) limiting recurrence of symptoms in patients that were previously asymptomatic for the disorder(s).

The term "vaccinated" as used herein refers to being treated with a vaccine.

The term "vaccination" as used herein refers to treatment with a vaccine.

The term "vaccine" as used herein refers a substance or group of substances meant to cause the immune system to respond to a tumor or to microorganisms, or help the body recognize and destroy cancer cells or microorganisms. The term vaccine also refers to an artificial stimulus used to stimulate a robust immune response against that exposure (e.g. infectious agent, cancer cell).

The term "vaccine therapy" as used herein refers to a type of treatment that uses a substance or group of substances to stimulate the immune system to destroy a tumor or infectious microorganisms.

Allogeneic Vaccine

Vaccine proteins can induce immune responses that find use in the described invention. According to one aspect, the described invention comprises a tumor-type specific allogeneic tumor vaccine for the treatment of cancer. According to some embodiments, the cancer is prostate cancer. According to some embodiments, the vaccine comprises an allogeneic cancer cell line that is genetically modified by two or more immunomodulatory molecules. According to some embodiments, the tumor cell provides a broad array of tumor specific antigens, most of which are of unknown nature. According to some embodiments, the immunomodulatory molecules genetically engineered or added to the cells are selected from a group for their ability to either initiate or sustain an anti-tumor immune response, or alternatively for their ability to abrogate pre-existing immunosuppression characteristically present in cancer patients, or a combination of all three. According to some embodiments, combinations of immunomodulatory molecules are evaluated and selected by a human mixed lymphocyte tumor cell reaction.

According to some embodiments, the allogeneic vaccine composition is administered to a subject diagnosed with cancer in combination with an agent that inhibits immunosuppressive molecules produced by tumor cells.

According to some embodiments, the allogeneic vaccine further comprises one or more checkpoint inhibitors that are sufficient to prevent premature termination of an effective immune response once such an immune response is initiated.

According to some embodiments, a subject (i.e. a subject diagnosed with cancer) is treated by checkpoint inhibitor therapy prior to or concurrently with administration of the allogeneic vaccine composition. In certain embodiments, the cancer is a melanoma.

Checkpoint Blockade/Blockage of Tumor Immunosuppression

Some human tumors can be eliminated by a patient's immune system. For example, administration of a monoclonal antibody targeted to an immune "checkpoint" molecule can lead to complete response and tumor remission. A mode of action of such antibodies is through inhibition of an immune regulatory molecule that the tumors have co-opted as protection from an anti-tumor immune response. By inhibiting these "checkpoint" molecules (e.g., with an antagonistic antibody), a patient's CD8+ T cells may be allowed to proliferate and destroy tumor cells.

For example, administration of a monoclonal antibody targeted to by way of example, without limitation, CTLA-4 or PD-1 can lead to a complete response and tumor remission. The mode of action of such antibodies is through inhibition of CTLA-4 or PD-1 that the tumors have co-opted as protection from an anti-tumor immune response. By inhibiting these "checkpoint" molecules (e.g., with an antagonistic antibody), a patient's CD8+ T cells may be allowed to proliferate and destroy tumor cells.

Thus, the allogeneic vaccine compositions provided herein can be used in combination with one or more blocking antibodies targeted to an immune "checkpoint" molecule. For instance, in some embodiments, the allogeneic vaccine compositions provided herein can be used in combination with one or more blocking antibodies targeted to a molecule such as CTLA-4 or PD-1. For example, the allogeneic vaccine compositions provided herein may be used in combination with an agent that blocks, reduces and/or inhibits PD-1 and PD-L1 or PD-L2 and/or the binding of PD-1 with PD-L1 or PD-L2 (by way of non-limiting example, one or more of nivolumab (ONO-4538/BMS-936558, MDX1106, OPDIVO, BRISTOL MYERS SQUIBB), pembrolizumab (KEYTRUDA, Merck), pidilizumab (CT-011, CURE TECH), MK-3475 (MERCK), BMS 936559 (BRISTOL MYERS SQUIBB), MPDL3280A (ROCHE)). In an embodiment, the allogeneic vaccine compositions provided herein may be used in combination with an agent that blocks, reduces and/or inhibits the activity of CTLA-4 and/or the binding of CTLA-4 with one or more receptors (e.g. CD80, CD86, AP2M1, SHP-2, and PPP2R5A). For instance, in some embodiments, the immune-modulating agent is an antibody such as, by way of non-limitation, ipilimumab (MDX-010, MDX-101, Yervoy, BMS) and/or tremelimumab (Pfizer). Blocking antibodies against these molecules can be obtained from, for example, Bristol Myers Squibb (New York, N.Y.), Merck (Kenilworth, N.J.), MedImmune (Gaithersburg, Md.), and Pfizer (New York, N.Y.).

Further, the allogeneic immune compositions provided herein can be used in combination with one or more blocking antibodies targeted to an immune "checkpoint" molecule

such as for example, BTLA, HVEM, TIM3, GALS, LAG3, VISTA, KIR, 2B4, CD160 (also referred to as BY55), CGEN-15049, CHK 1 and CHK2 kinases, A2aR, CEACAM (e.g., CEACAM-1, CEACAM-3 and/or CEACAM-5), GITR, GITRL, galectin-9, CD244, CD160, TIGIT, SIRPα, ICOS, CD172a, and TMIGD2 and various B-7 family ligands (including, but are not limited to, B7-1, B7-2, B7-DC, B7-H1, B7-H2, B7-H3, B7-H4, B7-H5, B7-H6 and B7-H7).

According to some embodiments, the allogeneic vaccine is adapted for rapid in vitro evaluation using human peripheral blood mononuclear cells from healthy subjects and cancer patients to examine inter-individual variability as well as normal to patient differences, thus avoiding animal experimentation.

According to some embodiments, the described invention comprises an allogeneic tumor cell vaccine for an active immunotherapy that can be universally administered to all patients with a particular type of cancer. According to some embodiments, the allogeneic vaccine comprises a genetically modified allogeneic tumor-type specific cell, or a membrane lysate derived from modified allogeneic tumor-type specific cells, formulated in a pharmaceutically acceptable carrier. According to some embodiments, the modified allogeneic tumor-type specific cells are derived from previously established cell lines.

According to some embodiments, the allogeneic vaccine is adapted to treat patients with minimal residual disease and a functional immune system. For example, according to some embodiments, the allogeneic vaccine is adapted to treat a patient with minimal residual disease obtained shortly after a primary lesion is surgically removed. According to some embodiments, the allogeneic vaccine is adapted for subcutaneous administration of the vaccine. According to some embodiments, the dose and schedule for administering the allogeneic vaccine are determined by using immunologic responses to the vaccine as a guide for eventual enhancement of overall survival.

According to some embodiments, the allogeneic vaccine is adapted to provide clinical benefit in the short term by the induction of strong anti-allogeneic vaccine responses, and, in the long term, to provide a long lived and cross reactive response to the endogenous unmodified host tumor. According to some embodiments, the immune response against the allogeneic tumor cell vaccine comprises a heteroclitic cross reaction between a peptide native to the tumor cell line and a peptide native to the tumor cells of a patient receiving the vaccine (See, e.g., FIG. 1). According to some embodiments, the heteroclitic cross reaction enhances immunogenicity via enhanced binding of a T cell receptor with a tumor cell peptide-MHC complex that normally provides a non-immunogenic surface. According to some embodiments, the allogeneic tumor cell vaccine comprises peptides altered relative to tumor cells of a subject with cancer, where the altered peptides provide an immunogenic surface that results in a heteroclitic cross-reaction to the non-immunogenic peptide of tumor cells from the subject with cancer. According to some embodiments, the heteroclitic recognition and alloreactive antigen recognition of the tumor cell vaccine provides a broad array of antigens useful to elicit an immune response against the tumor cells of a patient receiving the vaccine. According to some embodiments, the allogeneic vaccine is adapted to provide a clinical benefit, e.g., in the form of progression free survival, relapse-free survival, or overall survival. According to some embodiments, the allogeneic vaccine is effective to provide heteroclitic immunization induced tumor immunity (Dyall R., et al., Heteroclitic

Immunization Induces Tumor Immunity, *J. Exp. Med.*, Vol. 188, No. 9, Nov. 2, 1998, incorporated by reference herein in its entirety).

According to some embodiments, the allogeneic vaccine is derived from tumor cell lines genetically modified to comprise recombinant immunomodulatory signals that are expressed in therapeutic amounts. According to some embodiments, the allogeneic vaccine is derived from a uniform starting material, such as a tumor cell line, wherein multiple discrete biologics are expressed in the starting material in either soluble or membrane bound form. According to some embodiments, expression and activity of the soluble and membrane bound forms are confirmed, *in vitro*, by flow cytometry and mixed lymphocyte tumor assays using peripheral blood mononuclear cells, respectively. According to some embodiments, expression and activity of the soluble and membrane bound forms are confirmed, *in vitro*, by flow cytometry and mixed lymphocyte tumor assays using peripheral blood mononuclear cells of the vaccinated cancer patient against the allogeneic tumor cells used to immunize.

According to some embodiments, the allogeneic vaccine comprises genetically modified immunomodulatory molecules each encoding a membrane bound or secreted signaling molecule. According to some embodiments, each membrane bound immunomodulatory molecule is adapted to deliver a therapeutic amount in sub-pharmacologic doses that is active in a spatially and temporally restricted manner to provide signaling predominantly at the time and place of antigen presentation. According to some embodiments, the membrane bound immunomodulatory molecules are adapted to decrease the probability of systemic side effects. According to some embodiments, the secreted immunomodulatory molecules are adapted to deliver local, not systemic, signals.

According to some aspects, the allogeneic vaccine comprises genetic material that is effective to genetically introduce one or more immunomodulatory molecules into a tumor cell line. According to some embodiments, the genetic material can be introduced by viral transduction techniques and isolated by positive selection for the genetically introduced immune modulator. For example, according to some embodiments, the positive selection of the genetically introduced immune modulator molecule comprises selection using antibodies. According to some embodiments, the immunomodulatory molecules are diverse and complementary with respect to impact on key immune cell subsets such as dendritic cell, lymphocyte sub-populations (e.g. T cells, Natural Killer cells, and T-regulatory cells). According to some embodiments, the allogeneic vaccine comprises a variety of immunomodulatory molecules directed to a variety of immunomodulatory pathways on various immune cell subsets, wherein not all pathways will equally contribute to immunogenic response in individual cancer patients. According to some embodiments, the immunomodulatory molecules genetically introduced into a tumor cell line are stably expressed.

Tumor Antigen Specificity

Immunologic antigenic specificity may arise from one or more of the amino acid sequence of the antigen, from the degree of expression of that antigen by the tumor cell, from post-translational modification of the antigen, and the like.

Immunologic antigen specificity to a certain type of cancer cell may also arise from one or more of a particular fingerprint of a plurality of tumor antigens, from the fact that a particular antigen, while expressed by a wide variety of tumor cells, has particular use in immunotherapy against a

smaller number of tumor types, from the fact that a particular collection of MHC class I presentable and MHC class II presentable epitopes exist on a particular polypeptide or polypeptide fragment, and by omitting one or more peptides that may provoke immunotolerance. The skilled artisan can locate the relevant nucleic acid and polypeptide sequences, e.g., on the U.S. government's web site, at ncbi.nlm.nih.

According to some embodiments, tumor antigen specificity of the present invention may be determined by the parental tumor cell line that is selected for modification with immune modulators.

Parent Cell Lines

According to some embodiments, tumor cell line variants may be derived from established cell lines from either public sources (e.g. NIH, DCTD Tumor Repository operated by Charles River Laboratories Inc.) or commercial sources (e.g. ATCC® (American Type Culture Collection), Sigma Aldrich, Thermo Fischer Scientific, Genescript, DSM2). According to some embodiments, new cell lines can be established *de novo* from tumor cells taken from the tumor of a cancer patient.

According to some embodiments, cancer tissues, cancer cells, cells infected with a cancer-causing agent, other pre-neoplastic cells, and cell lines of human origin can be used. In some embodiments, a cancer cell can be from an established tumor cell line such as, without limitation, an established non-small cell lung carcinoma (NSCLC), bladder cancer, melanoma, ovarian cancer, renal cell carcinoma, prostate carcinoma, sarcoma, breast carcinoma, squamous cell carcinoma, head and neck carcinoma, hepatocellular carcinoma, pancreatic carcinoma, or colon carcinoma cell line.

According to some embodiments, the established cell lines comprise the LNCaP clone FGC (ATCC® CRL-1740), which itself is derived from a metastatic prostate cancer that had migrated to a lymph node. According to some embodiments, the established cell lines comprise the PC-3 (ATCC® CRL-1435) cell line, which itself is derived from metastatic prostate cancer that migrated to bone. According to some embodiments, the tumor cell line variants are derived from one or more of the following ATCC® cell lines: VCaP (ATCC® CRL-2876); MDA PCa 2b (ATCC® CRL-2422); or DU 145 (ATCC® HTB-81).

According to some embodiments, the established cell lines comprise the SK-MEL-2 clone (ATCC® HTB-68), which itself is derived from metastasis on skin of thigh.

According to some embodiments, the established cell lines comprise one or more of mammary carcinoma cell lines designated COO-G, DU4475, ELL-G, HIG-G, MCF7, MDA-MB-436, MX-1, SW-613, and VAN-G. According to some embodiments, the established cell lines comprise one or more of alveolar soft part sarcoma cell lines designated ASPS, and ASPS-1. According to some embodiments, the established cell lines comprise one or more lung cell lines designated LX-1, COS-G, H-MESO-1, H-MESO-1A, NCI-H23, and NCI-H460. According to some embodiments, the established cell lines comprise one or more colon cancer cell lines designated CX-5, GOB-G, HCC-2998, HCT-15, KLOG, KM20L2, MRI-H-194, LOVO I, LOVO II, and MRI-H-250. According to some embodiments, the established cell lines comprise one or more melanoma cell lines designated NIS-G, TRI-G, WIL-G, MRI-H-121B, MRI-H-187, MRI-H-221, and MRI-H-255. According to some embodiments, the established cell lines comprise one or more cervical cancer cell lines designated MRI-H-177, MRI-H-186, MRI-H-196, and MRI-H-215. According to some embodiments, the established cell lines comprise one or more kidney

cancer cell lines designated MRI-H-121 and MRI-H-166. According to some embodiments, the established cell lines comprise one or more endometrium cancer cell lines designated MRI-H-147 and MRI-H-220. According to some embodiments, the established cell lines comprise one or more ovarian cancer cell lines designated MRI-H-258, MRI-H-273, MRI-H-1834, and SWA-G. According to some embodiments, the established cell lines comprise one or more sarcoma cell lines designated HS-1, OGL-G, and DEL-G. According to some embodiments, the established cell lines comprise the epidermoid cell line designated DEAC-1. According to some embodiments, the established cell line comprises the glioblastoma cell line designated SF 295. According to some embodiments, the established cell line comprises the prostate cancer cell line designated CWR-22. According to some embodiments, the established cell line comprises the Burkitt's lymphoma cell line designated DAU.

According to some embodiments, exemplary established cell lines comprise one or more of the following cell lines:

Designation	Tissue of Origin	Histologic Type
786-0	Kidney	Renal Cell Carcinoma
A2780	Ovary	Adenocarcinoma
A498	Kidney	Renal Cell Carcinoma
A549	Lung	Non-small Cell
A704	Kidney	Renal Cell Carcinoma
ACHN	Kidney	Renal Cell Carcinoma
ASPS-1	Lymph Node	Alveolar Soft Part Sarcoma
BT-549	Breast	Adenocarcinoma
CAKI-1	Kidney	Renal Cell Carcinoma
CCRF-CEM	Lymph	Leukemia
CCRF-SB	Lymph	Leukemia
CHA-59	Bone	Osteosarcoma
COLO 205	Colon	Adenocarcinoma
DMS-114	Lung	Small Cell
DU-145	Prostate	Carcinoma
EK VX	Lung	Adenocarcinoma
HCC-2998	Colon	Adenocarcinoma
HCT-15	Colon	Carcinoma
HCT-116	Colon	Adenocarcinoma
HOP-18	Lung	Large Cell Carcinoma
HOP-62	Lung	Adenocarcinoma
HL-60	Ascites	Pro-myelocytic Leukemia
H-MESO-1		Mesothelioma
HS 578T	Breast	Adenocarcinoma
HS 913T	Lung	Mixed Cell
HT-29	Colon	Adenocarcinoma
IGR-OV1	Ovary	Adenocarcinoma
KM-12	Colon	Adenocarcinoma
KM 20L2	Colon	Adenocarcinoma
K-562	Lymph	Leukemia
LOVO	Colon	Adenocarcinoma
LOX IMVI	Lymph Node Metastasis	Amelanotic Melanoma
LXFL 529	Lung	Large Cell Carcinoma
NCI-H1299	Lung	Adenocarcinoma
NCI-H2887	Lung	Adenocarcinoma
NCI-H3122	Lung	Adenocarcinoma
NCI-H322M	Lung	Adenocarcinoma
NCI-H3255	Lung	Adenocarcinoma
NCI-H358M	Lung	Bronchioalveolar Carcinoma
NCI-H460	Lung	Large Cell
NCI-H522	Lung	Adenocarcinoma
NCI-H69	Lung	Small Cell Carcinoma
NCI-H82	Lung	Small Cell Carcinoma
NCI-H838	Lung	Adenocarcinoma
NCI/ADR-RES	Ovary	Adenocarcinoma
OVCAR-3	Ovary	Adenocarcinoma
OVCAR-4	Ovary	Adenocarcinoma
OVCAR-5	Ovary	Adenocarcinoma
OVCAR-8	Ovary	Adenocarcinoma
PC-3	Prostate	Carcinoma
PC-3/M	Prostate	Carcinoma
RPMI-7951	Skin	Melanoma

-continued

Designation	Tissue of Origin	Histologic Type
RPMI-8226	Lymph	Leukemia
RXF 393	Kidney	Renal Cell Carcinoma
RXF 631	Kidney	Renal Cell Carcinoma
TK-10	Kidney	Renal Cell Carcinoma
UACC-62	Skin	Melanoma
UACC-257	Skin	Melanoma
UCSD 242L	Skin	Melanoma
UCSD 354K	Skin	Melanoma
UO-31	Kidney	Renal Cell Carcinoma
U-251	CNS	Glioblastoma
WIDR	Colon	Adenocarcinoma
XF 498	CNS	Glioblastoma

According to some embodiments, the choice of the parental cell line from which the tumor cell line variant may be derived affects the specificity of the allogeneic vaccine. For example, the use of a tumor cell line variant derived from metastatic prostate cancer that migrated to the bone of a patient may result in an allogeneic vaccine that elicits an immune response specific for metastatic prostate cancer in the bone of a patient.

According to some embodiments, the tumor cell line variants may be derived from a parental cell that comprises a universal cancer specific antigen. For example, the use of a parental tumor cell line variant derived from metastatic prostate cancer that migrated to the bone of a patient may result in an allogeneic vaccine that elicits an immune response against all prostate cancer cells.

According to some embodiments, the tumor cell line variants are derived from patient derived cells derived from various cancers. According to some embodiments, fresh tissue surgically removed from a tumor is enzymatically digested by type IV collagenase, followed by collection of disaggregated cells. According to some embodiments, disaggregated cells may then be grown in vitro in growth media with 10% fetal bovine serum on an extracellular matrix substrate, such as collagen or fibronectin, to promote attachment. According to some embodiments, adherent cells may then be passaged until the immortal cancer cells outgrow the non-cancerous fibroblast cells.

For example, according to some embodiments, the tumor cell line variants may be derived from a solid tumor comprising tumor cells, including cancer stem cells, a metastatic cancer comprising metastatic tumor cells, comprising cancer stem cells, or a non-metastatic cancer. According to some embodiments, the cancer may originate in the bladder, blood, bone, bone marrow, brain, breast, colon, esophagus, duodenum, small intestine, large intestine, colon, rectum, anus, gum, head, kidney, liver, lung, nasopharynx, neck, ovary, prostate, skin, stomach, testis, tongue, or uterus. According to some embodiments, the cancer may be of a histological type, e.g., a cancer that begins in the skin or tissues that line or cover internal organs (carcinoma); a cancer that begins in bone or in the soft tissue of the body including cartilage, fat, muscle, blood vessels, and fibrous tissue (sarcoma); a cancer that starts in blood-forming tissue (leukemia); a cancer that begins in cells of the immune system (lymphoma); a cancer that arises in plasma cells (myeloma), or a brain/spinal cord cancer.

Examples of carcinomas include, without limitation, giant and spindle cell carcinoma; small cell carcinoma; papillary carcinoma; squamous cell carcinoma; lymphoepithelial carcinoma; basal cell carcinoma; pilomatrix carcinoma; transitional cell carcinoma; papillary transitional cell carcinoma; an adenocarcinoma; a gastrinoma, a cholangiocarcinoma; a

hepatocellular carcinoma; a combined hepatocellular carcinoma and cholangiocarcinoma; a trabecular adenocarcinoma; an adenoid cystic carcinoma; an adenocarcinoma in adenomatous polyp; an adenocarcinoma, familial polyposis coli; a solid carcinoma; a carcinoid tumor; a bronchioloalveolar adenocarcinoma; a papillary adenocarcinoma; a chromophobe carcinoma; an acidophil carcinoma; an oxyphilic adenocarcinoma; a basophil carcinoma; a clear cell adenocarcinoma; a granular cell carcinoma; a follicular adenocarcinoma; a non-encapsulating sclerosing carcinoma; adrenal cortical carcinoma; an endometroid carcinoma; a skin appendage carcinoma; an apocrine adenocarcinoma; a sebaceous adenocarcinoma; a ceruminous adenocarcinoma; a mucoepidermoid carcinoma; a cystadenocarcinoma; a papillary cystadenocarcinoma; a papillary serous cystadenocarcinoma; a mucinous cystadenocarcinoma; a mucinous adenocarcinoma; a signet ring cell carcinoma; an infiltrating duct carcinoma; a medullary carcinoma; a lobular carcinoma; an inflammatory carcinoma; paget's disease, a mammary acinar cell carcinoma; an adenosquamous carcinoma; an adenocarcinoma w/squamous metaplasia; a sertoli cell carcinoma; embryonal carcinoma; choriocarcinoma.

Examples of sarcomas include, without limitation, gliomangiosarcoma; sarcoma; fibrosarcoma; myxosarcoma; liposarcoma; leiomyosarcoma; rhabdomyosarcoma; embryonal rhabdomyosarcoma; alveolar rhabdomyosarcoma; stromal sarcoma; carcinosarcoma; synovial sarcoma; hemangiosarcoma; kaposi's sarcoma; lymphangiosarcoma; osteosarcoma; juxtacortical osteosarcoma; chondrosarcoma; mesenchymal chondrosarcoma; giant cell tumor of bone; ewing's sarcoma; odontogenic tumor, malignant; ameloblastic odontosarcoma; ameloblastoma, malignant; ameloblastic fibrosarcoma; myeloid sarcoma; mast cell sarcoma.

Examples of leukemias include, without limitation, leukemia; lymphoid leukemia; plasma cell leukemia; erythroleukemia; lymphosarcoma cell leukemia; myeloid leukemia; basophilic leukemia; eosinophilic leukemia; monocytic leukemia; mast cell leukemia; megakaryoblastic leukemia; and hairy cell leukemia.

Examples of lymphomas and myelomas include, without limitation, malignant lymphoma; hodgkin's disease; hodgkin's; paragranuloma; malignant lymphoma, small lymphocytic; malignant lymphoma, large cell, diffuse; malignant lymphoma, follicular; mycosis fungoides; other specified non-hodgkin's lymphomas; malignant melanoma; amelanotic melanoma; superficial spreading melanoma; malignant melanoma in giant pigmented nevus; epithelioid cell melanoma; multiple myeloma.

Examples of brain/spinal cord cancers include, without limitation, pinealoma, malignant; chordoma; glioma, malignant; ependymoma; astrocytoma; protoplasmic astrocytoma; fibrillary astrocytoma; astroblastoma; glioblastoma; oligodendroglioma; oligodendroblastoma; primitive neuroectodermal; cerebellar sarcoma; ganglioneuroblastoma; neuroblastoma; retinoblastoma; olfactory neurogenic tumor; meningioma, malignant; neurofibrosarcoma; neurilemmoma, malignant.

Examples of other cancers include, without limitation, a thymoma; an ovarian stromal tumor; a thecoma; a granulosa cell tumor; an androblastoma; a leydig cell tumor; a lipid cell tumor; a paraganglioma; an extra-mammary paraganglioma; a pheochromocytoma; blue nevus, malignant; fibrous histiocytoma, malignant; mixed tumor, malignant; mullerian mixed tumor; nephroblastoma; hepatoblastoma; mesenchymoma, malignant; brenner tumor, malignant; phyllodes tumor, malignant; mesothelioma, malignant; dysgerminoma; teratoma, malignant; struma ovarii, malignant; meso-

nephroma, malignant; hemangioendothelioma, malignant; hemangiopericytoma, malignant; chondroblastoma, malignant; granular cell tumor, malignant; malignant histiocytosis; immunoproliferative small intestinal disease.

For any given tumor type, several tumor cell lines may be commercially available. According to some embodiments, pooling of several of these cells lines, either as a mixture of whole cells or by making a membrane preparation out of the mixture of whole cells, may provide an array of cell surface tumor antigens for that tumor type.

Selection of Immune Modulators

According to some embodiments, the tumor cell line variants may be engineered to express two or more recombinant sequences of DNA and protein that are then presented on the tumor cell and are functional.

IgG Heavy Chain Constant and Variable Region

Immunoglobulins (Ig) are glycoproteins produced by immune cells. Antibodies are serum proteins, the molecules of which possess small areas of their surface that are complementary to small chemical groupings on their targets. These complementary regions (referred to as complementary determining regions (CDRs), or antibody combining sites, or antigen binding sites) of which there are at least two per antibody molecule, and in some types of antibody molecules ten, eight, or in some species as many as 12, may react with their corresponding complementary region on the antigen (the antigenic determinant or epitope) to link several molecules of multivalent antigen together to form a lattice. Immunoglobulins play a critical role in an immune response by binding to particular antigens, such as those exhibited by bacteria or viruses. According to some embodiments, the binding of immunoglobulins to antigens may target them for destruction by the subject's immune cells.

The basic structural unit of a whole antibody molecule consists of four polypeptide chains, two identical light (L) chains (each containing about 220 amino acids) and two identical heavy (H) chains (each usually containing about 440 amino acids). The two heavy chains and two light chains are held together by a combination of noncovalent and covalent (disulfide) bonds. The molecule is composed of two identical halves, each with an identical antigen-binding site composed of the N-terminal region of a light chain and the N-terminal region of a heavy chain. Both light and heavy chains usually cooperate to form the antigen binding surface.

In mammals, there are five classes of antibodies, IgA, IgD, IgE, IgG, and IgM, each with its own class of heavy chain— α (for IgA), δ (for IgD), ϵ (for IgE), γ (for IgG) and μ (for IgM). In addition, there are four subclasses of IgG immunoglobulins (IgG1, IgG2, IgG3, IgG4) having $\gamma 1$, $\gamma 2$, $\gamma 3$, and $\gamma 4$ heavy chains respectively. In its secreted form, IgM is a pentamer composed of five four-chain units, giving it a total of 10 antigen binding sites. Each pentamer contains one copy of a J chain, which is covalently inserted between two adjacent tail regions.

Diverse libraries of immunoglobulin heavy (VH) and light (Vk and V λ) chain variable genes from peripheral blood lymphocytes also can be amplified by polymerase chain reaction (PCR) amplification. Genes encoding single polypeptide chains in which the heavy and light chain variable domains are linked by a polypeptide spacer can be made by randomly combining heavy and light chain V-genes using PCR.

According to some embodiments, the tumor cell line variants may be engineered to express an IgG1 heavy chain constant region. In nature, the Ig gamma-1 (IgG-1) chain C region is a protein encoded by the IGHG1 gene in humans.

According to some embodiments, a tumor cell line variant may express a membrane bound form IgG-1 chain C protein of SEQ ID NO: 1. According to some embodiments, a tumor cell line variant may express a secreted form of IgG-1 chain C of SEQ ID NO: 2. According to some embodiments, a tumor cell line variant may express a secreted form of IgG-1 chain C of SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to one or more of proteins with an amino acid sequence SEQ ID NO: 1, SEQ ID NO: 2, and SEQ ID NO: 3.

According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33,

SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to one or more proteins with amino acid sequence SEQ ID NO: 12, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 45, and SEQ ID NO: 46.

According to some embodiments, the tumor cell line variant may be engineered to express an IgG protein that is capable of binding to tumor cell specific antigens. For example, the tumor cell line variant may be engineered to express an IgG protein capable of binding to a prostate cancer specific antigen; e.g., the extracellular region of prostate-specific membrane antigen (PSMA) (See Chang, S., Overview of Prostate-Specific Membrane Antigen, Reviews in Urology, Vol. 6 Suppl. 10, S13 (2004)). According to some embodiments, the tumor cell line variant may be engineered to express an IgG protein that is capable of binding to immune cell specific antigens. For example, the tumor cell line variant may be engineered to express an IgG protein capable of binding to T cell markers, e.g., CD3, CD4, or CD8. According to another example, the tumor cell line variant may be engineered to express an IgG protein capable of binding to dendritic cell markers, e.g. CD11c or CD123.

According to some embodiments, the tumor cell line variants may be engineered to express an IgG3 heavy chain constant region. In nature, the IgG3 heavy chain constant region comprises CH1-hinge-CH2-CH3 domains, and is encoded by the IGHG3 gene in humans; the IGHG3 gene comprises structural polymorphisms comprising different hinge lengths. According to some embodiments, a tumor cell line variant may express an IgG-3 heavy chain constant region of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may express a derivative of SEQ ID NO: 4 with amino acids 1-76 missing. According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 with amino acids 1-76 missing.

According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 with amino acids 77-98 replaced with amino acids QMQGVNCTVSS. According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising an E213Q variant (SEQ ID NO: 16). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising a P221L variant (SEQ ID NO: 17). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising an E224Q variant (SEQ ID NO: 18). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising a Y226F variant (SEQ ID NO: 19). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising a D242N variant (SEQ ID NO: 20). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising a N245D variant (SEQ ID NO: 21). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising a T269A variant (SEQ ID NO: 22). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising a S314N variant (SEQ ID NO: 23). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising a deleted S314 (SEQ ID NO: 24). According to some embodiments, a tumor cell line variant may express the derivative of SEQ ID NO: 4 comprising F366Y variant (SEQ ID NO: 25).

According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 4. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 4.

According to some embodiments, a tumor cell line variant may be engineered to express one or more IgG heavy chain variable regions. According to some embodiments, a tumor cell line variant may be engineered to express a lambda/kappa light chain constant and/or light chain variable region. According to some embodiments, the hinge region of IgG binds to the FcγR receptors on immune cells. According to some embodiments, the IgG is effective to activate the FcγR and enhance presentation of antigens (e.g. PSA associated with prostate cancer cells).

According to some embodiments, a tumor cell line variant may be engineered to express an intact monoclonal or

polyclonal antibody on the cell surface of the tumor cell. According to some embodiments, the intact monoclonal or polyclonal antibody may be designed to deliver a molecule that elicits an immunogenic response. For example, according to some embodiments, the intact monoclonal antibody may be designed to bind to DNA to deliver CpG motifs to immune cells.

According to some embodiments, the immunostimulatory activity of bacterial DNA may be mimicked by engineering an immunomodulator to deliver unmethylated CpG motifs to immune cells. For example, according to some embodiments, the IgG may be engineered to bind to biotin, which is then capable of delivering biotinylated CpG to cells of the immune system. According to some embodiments, CpG motifs may be bound directly or indirectly to the surface of the tumor cells of the tumor cell vaccine to prevent systemic effects. According to some embodiments, CpG motifs may be conjugated to one or more antigens presented on the surface of tumor cells from the tumor cell line variant. According to some embodiments, the CpG is a class A CpG. According to some embodiments, the CpG is a class B CpG. According to some embodiments, the CpG is a class C CpG. According to some embodiments, the CpG is a CpG 30-mer of the sequence 5' EEAACCGTATCGGCGATATCGGT-TEEEEEG 3'. As used herein with respect to CpG motifs, "E" is a G-phosphorothioate and this linkage refers to the 3' end of the nucleotide (i.e. the phosphorothioate bond substitutes a sulfur atom for a non-bridging oxygen in the nucleotide backbone). According to some embodiments, the CpG is a biotinylated 30-mer of the sequence 5'-biotin-EEAACCGTATCGGCGATATCGGT-TEEEEEG-3'. According to some embodiments, the CpG is a CpG 30-mer of the sequence 5' EEAACCGTATGCGGCATATCGGT-TEEEEEG 3'. According to some embodiments, the CpG is a biotinylated CpG 30-mer of the sequence 5'-biotin-EEAACCGTATGCGGCATATCGGT-TEEEEEG-3'.

According to some embodiments, the IgG may be engineered as a hybrid of one or more IgG subclasses. For example, according to some embodiments, the IgG comprises sequences from IgG1 and IgG3. According to some embodiments, the IgG may be engineered to have an affinity for biotin. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 45. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 45. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 45. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 45. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 45. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 45. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 45. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 45. According to some embodi-

ments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 45.

According to some embodiments, the IgG comprises one or more mutations relative to wild type IgG that enhance affinity for FcγR. According to some embodiments, a tumor cell line variant may comprise one or more proteins of SEQ ID NO: 45 with one or more of mutations T323A and E325A. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the proteins of one or more of SEQ ID NO: 41, SEQ ID NO: 30, and SEQ ID NO: 43.

CD40L

The ligand of CD40, known as CD154 or CD40L, is a type II transmembrane protein, with a variable molecular weight between 32 and 39 kDa because of post-translation modifications (Elgueta R et al., Molecular mechanism and function of CD40/CD40L engagement in the immune system. *Immunological reviews*. 2009; 229(1):10.1111/j.1600-065X.2009.00782.x. doi:10.1111/j.1600-065X.2009.00782.x, citing van Kooten C et al., *J. Leukoc Biol*. 2000 January; 67(1):2-17.). A soluble form of CD40L has been reported that has activities similar to the transmembrane form (Id. citing Graf D et al., *Eur J Immunol*. 1995 June; 25(6):1749-54; Mazzei G J et al., *J Biol Chem*. 1995 Mar. 31; 270(13):7025-8.).

In nature, CD40L is a member of the TNF superfamily and is characterized by a sandwich extracellular structure that is composed of a β-sheet, α-helix loop, and a β-sheet, which allows for the trimerization of CD40L (Id. citing Karpusas M et al., *Structure*. 1995 Oct. 15; 3(10):1031-9). CD40L is expressed primarily by activated T cells, as well as activated B cells and platelets; under inflammatory conditions it is also induced on monocytic cells, natural killer cells, mast cells, and basophils (Id. citing Carbone E et al., *J Exp Med*. 1997 Jun. 16; 185(12):2053-60). The wide

spread expression of the costimulatory pair of CD40L and CD40 indicates the pivotal roles they play in different cellular immune processes.

CD40L has three binding partners: CD40, α5β1 integrin and αIIbβ3 integrin. CD40L acts as a costimulatory molecule and is particularly important on a subset of T cells called T follicular helper cells (TFH cells), where it promotes B cell maturation and function by engaging CD40 on the B cell surface facilitating cell-cell communication. A defect in the CD40L gene results in an inability to undergo immunoglobulin class switching and is associated with hyper-IgM syndrome. Absence of CD40L also stops the formation of germinal centers thereby prohibiting antibody affinity maturation, an important process in the adaptive immune system.

CD40 has been found to be expressed on APCs, while its ligand, CD40L, has been found on activated T cells. CD40 has been found to play a critical role in the humoral immune response, and has been identified as enabling APCs to activate T cells. Several pathologies have been associated with the CD40/CD40L pathway including lupus and atherosclerosis, but anti-CD40L antibodies have been limited to clinical applications of thrombotic complications from CD40 expression on activated platelets (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)).

CD40 has also been found on several types of cancer, including solid tumors and hematologic malignancies. Signaling through CD40 in hematological cancer may mediate growth or regression, while CD40 signaling in solid tumors is only tumoricidal. These characteristics are found even in SCID mouse models, and therefore are likely due to TNF death domain signaling. There is also evidence of immune modulation, for example blockade of the CD40/CD40L pathway mitigates the protective effect of GM-CSF secreting melanoma vaccines (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)).

Tumor cell vaccines expressing CD40L have proved useful in cancer models. For example, ligation of CD40 with CD40L or anti-CD40 antibodies has shown synergy with GM-CSF, IFN-γ, IL-2, and CTLA-4 blockade. Furthermore, anti-CD40 antibodies have been reported to have anti-tumor activity in a pre-clinical mouse model (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)).

According to some embodiments of the disclosed invention, the tumor cell line variant may be engineered to express the cleavable CD40L peptide of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant

may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 6. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 6.

According to some embodiments, the tumor cell line variant may be engineered to express the non-cleavable membrane bound CD40L peptide of SEQ ID NO: 7 on the membrane surface of the tumor cell. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 7. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 7.

Tumor Necrosis Factor Alpha

Tumor necrosis factor (TNF; tumor necrosis factor alpha (TNF α); cachexin, cachectin) is a cytokine, primarily produced by activated macrophages and lymphocytes, which is involved in systemic inflammation. It is also one of the cytokines involved in the acute phase of an immunogenic response. TNF may be produced by other cell types such as, for example, CD4+ lymphocytes, NK cells, neutrophils, mast cells, eosinophils, and neurons.

In its primary role as a regulator of immune cells, TNF is capable of inducing fever, apoptotic cell death, cachexia, inflammation, and inhibition of tumorigenesis; of inhibiting viral replication; and of initiating a response to sepsis via IL-1 and IL-6 producing cells. Dysregulated TNF production has been associated with a wide array of human diseases, including Alzheimer's disease, major depression, psoriasis, and inflammatory bowel disease (IBD). TNF can be produced ectopically in the setting of malignancy and parallels parathyroid hormone both in causing secondary hypercalcemia and in the cancers with which excessive production is associated.

TNF comprises a 26 kDa membrane bound form and 17 kDa soluble cytokine form. The soluble form of TNF is derived from proteolytic cleavage of the membrane bound form by TNF-alpha converting enzyme (TACE) (Grell M. et al., *The Transmembrane Form of Tumor Necrosis Factor Is the Prime Activating Ligand of the 80 kDa Tumor Necrosis Factor Receptor*, Cell, Vol. 83, 793-802). TACE is a matrix metalloprotease that recognizes a cleavage site in the extra-

cellular domain of full-length TNF (Rieger, R., Chimeric form of tumor necrosis factor-alpha has enhanced surface expression and antitumor activity, *Cancer Gene Therapy*, 2009, 16, 53-64). Deletion of the cleavage site on TNF results in enhanced membrane stability of TNF (Id.).

TNF has antiproliferative and cytotoxic effects on cells, is known to reduce tumor blood flow and tumor vascular damage, and is able to modulate immune response by stimulating macrophage and NK cell activity. However, the use of TNF as a therapeutic itself has been limited by dose-dependent hypotension and capillary leak that can cause a sepsis-like syndrome. For that reason, it must be delivered in a manner that limits systemic effects. TNF has been added to standard chemotherapy agents to improve response rates. Other approaches to administering TNF include injection of adenovirus altered to express TNF in gastrointestinal malignancies. A tumor vascular-targeted TNF compound has also been developed (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). Recombinant TNF has been used as an immunostimulant under the name tasonermin, while HUMIRA® is an antibody to TNF, useful for the treatment of inflammatory diseases (e.g. psoriasis and rheumatoid arthritis). In recognition of this role, molecules such as antibodies have been designed to interfere with TNF activity. However, such therapies pose the risk of initiating a cytokine storm caused by the inappropriate systemic release of cytokines, resulting in a positive feedback loop of white blood cell activation/cytokine release that potentially can be fatal.

According to some embodiments, a tumor cell line variant may express the membrane bound form of TNF on the membrane of the tumor cell. For example, according to some embodiments, the cell line variants comprise the peptide of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 8. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 8.

According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF. For example, according to some embodiments, the tumor cell line variant may comprise the TNF protein of SEQ ID NO: 8 with one or more of amino acids VRSSRTPSDKP deleted (see e.g. SEQ ID NO: 26).

According to some embodiments, a tumor cell line variant may express a soluble form of TNF. For example, according to some embodiments, the tumor cell line variant may express the TNF protein of SEQ ID NO: 8 with part or the entire transmembrane region removed. For example, according to some embodiments, the tumor cell line variant may comprise a derivative TNF protein of SEQ ID NO: 8 with one or more of amino acids F, S, F, L, I, V, A, G, A, T, T, L, F, C, L, L, H, F, G, V, I deleted (see e.g. SEQ ID NO: 27).

According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound chimeric form of CD40L and TNF. For example, according to some embodiments, the ligand binding portion of a TNF molecule may be fused with the transmembrane and proximal extracellular domains of CD40L, such that the TNF lacks a defined TNF alpha cleaving enzyme (TACE) site. According to some embodiments, the intracellular, transmembrane, and partial extracellular portions CD40L may be fused with the extracellular region of TNF distal to the TACE cleavage site. According to some embodiments, the chimeric form of CD40L/TNF may comprise the CD40L sequence of SEQ ID NO: 9 and the TNF sequence of SEQ ID NO: 10. According to some embodiments, the CD40L/TNF sequences are operably linked via a linking peptide between 1 and 30 amino acids in length. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 60% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 70% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 80% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 90% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 95% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 96% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 97% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 98% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 99% to the proteins of SEQ ID NO: 9 and SEQ ID NO: 10.

According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 60% to the protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 70% to the protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 80% to the protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 90% to the

protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 95% to the protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 96% to the protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 97% to the protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 98% to the protein of SEQ ID NO: 11. According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound form of TNF with a sequence identity of at least 99% to the protein of SEQ ID NO: 11.

According to some embodiments, a tumor cell line variant may express a non-cleavable membrane bound chimeric form of CD40L and TNF. For example, according to some embodiments, the ligand portion of a TNF molecule may be fused with extracellular portions of CD40L, wherein CD40L comprises an extracellular portion that is non-cleavable and the TNF lacks a defined TACE site (e.g. cleavage site between amino acids 76 and 77). According to some embodiments, some or all of a CD40L peptide sequence is fused with the extracellular region of a TNF peptide sequence distal to the TACE cleavage site. According to some embodiments, the chimeric form of CD40L/TNF may comprise the sequence of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 60% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 70% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 80% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 90% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 95% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 96% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 97% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 98% to the protein of SEQ ID NO: 31. According to some embodiments, a tumor cell line variant may comprise a fusion protein with a sequence identity of at least 99% to the protein of SEQ ID NO: 31.

Granulocyte-Macrophage Colony Stimulating Factor (GM-CSF)

Granulocyte-macrophage colony-stimulating factor (GM-CSF; colony stimulating factor 2; CSF2) is found in monocytes/macrophages and activated T cells, and can act as a growth factor to stimulate and recruit dendritic cells. GM-CSF is a monomeric glycoprotein secreted by cells of the immune system, as well as endothelial cells and fibroblasts. Human GM-CSF is a 144 amino acid protein comprising a

17 amino acid signal peptide that can be cleaved to produce a mature 127 amino acid protein. Biological activity of GM-CSF occurs via binding to heteromeric cell surface receptors that are expressed on monocytes, macrophages, granulocytes, lymphocytes, endothelial cells and alveolar epithelial cells. The GM-CSF receptor (GM-CSFR) typically has a low expression (e.g. 20-200/cell), but has a high affinity (Shi Y et al., Granulocyte-macrophage colony-stimulating factor (GM-CSF) and T-cell responses: what we do and don't know, *Cell Research* (2006) 16: 126-133).

In some mouse models, vaccination with syngeneic mouse melanoma cells that secrete GM-CSF stimulates a more potent and long-lasting antitumor immunity than vaccines produced by other cytokines. Melanoma patients treated with soluble GM-CSF as an adjuvant therapy displayed an increase in disease free survival compared to controls. GM-CSF has been used as an immune adjuvant in various ways, including, without limitation, systemic and topical application of soluble GM-CSF, GM-CSF fusion proteins, transfection of tumor cells with GM-CSF and injection of GM-CSF DNA. Recombinant GM-CSF has been used an adjuvant for various peptide, protein, and viral vaccines, and has been shown to be an effective adjuvant in patients with melanoma, breast, and ovarian cancer. A fusion protein comprising GM-CSF has also been shown to enhance immunogenicity of an antigen. GM-CSF has been tested for use in a gene therapy approach where allogeneic or autologous GM-CSF expressing cells are used as a vaccine (Kaufman and Wolchok eds., *General Principles of Tumor Immunotherapy*, Chpt 5, 67-121 (2007)). Such vaccines have had varying degrees of effectiveness among several different cancer types.

According to some embodiments, a tumor cell line variant may express the GM-CSF peptide of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 13. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 13.

According to some embodiments, a tumor cell line variant may comprise one or more proteins comprising a fusion between GM-CSF and HLA-I to enable membrane expression. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 42 or

SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 42 or SEQ ID NO: 5. Fms-Like Tyrosine Kinase-3 Ligand (Flt-3L)

The human Flt3L protein is a membrane bound hematopoietic four helical bundle cytokine encoded by the FLT3LG gene. Flt3L acts as a growth factor that stimulates proliferation and differentiation of various blood cell progenitors, and is crucial for production and development of dendritic cells. Mice that lack Flt3L have low levels of dendritic cells, while Flt3L administered to mice or humans results in very high levels of dendritic cells (Shortman et al., Steady-state and inflammatory dendritic-cell development, *Nature Reviews Immunology*, Vol. 7. 19-30 (2007)).

According to some embodiments, a tumor cell line variant expresses the Flt3L peptide of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 14. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 14.

According to some embodiments, a tumor cell line variant comprises a soluble form of Flt3L. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 60% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 70% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 80% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 90% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 95% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 96% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 97% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 98% to the protein of SEQ ID NO: 44. According to some embodiments, a tumor cell line variant may comprise one or more proteins with a sequence identity of at least 99% to the protein of SEQ ID NO: 44.

Vectors and Host Cells

The described invention provides nucleic acid constructs that encode two or more immune modulators that can be expressed in prokaryotic and eukaryotic cells. For example, the described invention provides expression vectors (e.g., DNA- or RNA-based vectors) containing nucleotide sequences that encode two or more immune modulators. In addition, the described invention provides methods for making the vectors described herein, as well as methods for introducing the vectors into appropriate host cells for expression of the encoded polypeptides. In general, the methods provided herein include constructing nucleic acid sequences encoding two or more immune modulators, and cloning the sequences into an expression vector. The expression vector can be introduced into host cells or incorporated into virus particles, either of which can be administered to a subject to, for example, treat cancer.

cDNA or DNA sequences encoding two or more immune modulators can be obtained (and, if desired, modified) using conventional DNA cloning and mutagenesis methods, DNA amplification methods, and/or synthetic methods. In general, a sequence encoding two or more immune modulators can be inserted into a cloning vector for genetic modification and replication purposes prior to expression. Each coding sequence can be operably linked to a regulatory element, such as a promoter, for purposes of expressing the encoded protein in suitable host cells in vitro and in vivo.

Expression vectors can be introduced into host cells for producing secreted immune modulators. There are a variety of techniques available for introducing nucleic acids into viable cells. Techniques suitable for the transfer of nucleic acid into mammalian cells in vitro include the use of liposomes, electroporation, microinjection, cell fusion, polymer-based systems, DEAE-dextran, viral transduction, the calcium phosphate precipitation method, etc. For in vivo gene transfer, a number of techniques and reagents may also be used, including liposomes; and natural polymer-based delivery vehicles, such as chitosan and gelatin; viral vectors

are also suitable for in vivo transduction. In some situations it is desirable to provide a targeting agent, such as an antibody or ligand specific for a cell surface membrane protein. Where liposomes are employed, proteins which bind to a cell surface membrane protein associated with endocytosis may be used for targeting and/or to facilitate uptake, e.g., capsid proteins or fragments thereof tropic for a particular cell type, antibodies for proteins which undergo internalization in cycling, proteins that target intracellular localization and enhance intracellular half-life. The technique of receptor-mediated endocytosis is described, for example, by Wu et al., *J. Biol. Chem.* 262, 4429-4432 (1987); and Wagner et al., *Proc. Natl. Acad. Sci. USA* 87, 3410-3414 (1990).

Where appropriate, gene delivery agents such as, e.g., integration sequences can also be employed. Numerous integration sequences are known in the art (see, e.g., Nunes-Duby et al., *Nucleic Acids Res.* 26:391-406, 1998; Sadowski, *J. Bacteriol.*, 165:341-357, 1986; Bestor, *Cell*, 122 (3):322-325, 2005; Plasterk et al., *TIG* 15:326-332, 1999; Kootstra et al., *Ann. Rev. Pharm. Toxicol.*, 43:413-439, 2003). These include recombinases and transposases. Examples include Cre (Sternberg and Hamilton, *J. Mol. Biol.*, 150:467-486, 1981), lambda (Nash, *Nature*, 247, 543-545, 1974), FIp (Broach, et al., *Cell*, 29:227-234, 1982), R (Matsuzaki, et al., *J. Bacteriology*, 172:610-618, 1990), cpC31 (see, e.g., Groth et al., *J. Mol. Biol.* 335:667-678, 2004), sleeping beauty, transposases of the mariner family (Plasterk et al., *supra*), and components for integrating viruses such as AAV, retroviruses, and antiviruses having components that provide for virus integration such as the LTR sequences of retroviruses or lentivirus and the ITR sequences of AAV (Kootstra et al., *Ann. Rev. Pharm. Toxicol.*, 43:413-439, 2003).

Cells may be cultured in vitro or genetically engineered, for example. Host cells can be obtained from normal or affected subjects, including healthy humans, cancer patients, private laboratory deposits, public culture collections such as the American Type Culture Collection, or from commercial suppliers.

Cells that can be used for production and secretion of two or more immune modulators in vivo include, without limitation, epithelial cells, endothelial cells, keratinocytes, fibroblasts, muscle cells, hepatocytes; blood cells such as T lymphocytes, B lymphocytes, monocytes, macrophages, neutrophils, eosinophils, megakaryocytes, or granulocytes, various stem or progenitor cells, such as hematopoietic stem or progenitor cells (e.g., as obtained from bone marrow), umbilical cord blood, peripheral blood, fetal liver, etc., and tumor cells (e.g., human tumor cells). The choice of cell type depends on the type of tumor or infectious disease being treated or prevented, and can be determined by one of skill in the art.

Different host cells have characteristic and specific mechanisms for post-translational processing and modification of proteins. A host cell may be chosen which modifies and processes the expressed gene products in a specific fashion similar to the way the recipient processes its heat shock proteins (hsps).

In some embodiments, an expression construct as provided herein can be introduced into an antigenic cell. As used herein, antigenic cells can include preneoplastic cells that are infected with a cancer-causing infectious agent, such as a virus, but that are not yet neoplastic, or antigenic cells that have been exposed to a mutagen or cancer-causing agent, such as a DNA-damaging agent or radiation, for example. Other cells that can be used are preneoplastic cells

that are in transition from a normal to a neoplastic form as characterized by morphology or physiological or biochemical function.

Typically, the cancer cells and preneoplastic cells used in the methods provided herein are of mammalian origin. In some embodiments, cancer cells (e.g., human tumor cells) can be used in the methods described herein. Cell lines derived from a preneoplastic lesion, cancer tissue, or cancer cells also can be used. Cancer tissues, cancer cells, cells infected with a cancer-causing agent, other preneoplastic cells, and cell lines of human origin can be used. In some embodiments, a cancer cell can be from an established tumor cell line such as, without limitation, an established non-small cell lung carcinoma (NSCLC), bladder cancer, melanoma, ovarian cancer, renal cell carcinoma, prostate carcinoma, sarcoma, breast carcinoma, squamous cell carcinoma, head and neck carcinoma, hepatocellular carcinoma, pancreatic carcinoma, or colon carcinoma cell line.

Parent cell lines are described supra.

Further, in some embodiments, the allogeneic tumor cell vaccines provide for an adjuvant effect that further allows the immune system of a patient, when used in the various methods described herein, to be activated against a disease of interest.

Both prokaryotic and eukaryotic vectors can be used for expression of the two or more immune modulators in the methods provided herein. Prokaryotic vectors include constructs based on *E. coli* sequences (see, e.g., Makrides, Microbiol Rev 1996, 60:512-538). Non-limiting examples of regulatory regions that can be used for expression in *E. coli* include lac, trp, lpp, phoA, recA, tac, T3, T7 and lambda P_L. Non-limiting examples of prokaryotic expression vectors may include the Agt vector series such as lambda.gtl1 (Huynh et al., in "DNA Cloning Techniques, Vol. I: A Practical Approach," 1984, (D. Glover, ed.), pp. 49-78, IRL Press, Oxford), and the pET vector series (Studier et al., Methods Enzymol 1990, 185:60-89).

A variety of regulatory regions can be used for expression of the allogeneic tumor vaccines in mammalian host cells. For example, the SV40 early and late promoters, the cytomegalovirus (CMV) immediate early promoter, and the Rous sarcoma virus long terminal repeat (RSV-LTR) promoter can be used. Inducible promoters that may be useful in mammalian cells include, without limitation, promoters associated with the metallothionein II gene, mouse mammary tumor virus glucocorticoid responsive long terminal repeats (MMTV-LTR), the n-interferon gene, and the hsp70 gene (see, Williams et al., Cancer Res 1989, 49:2735-42; and Taylor et al., Mol Cell Biol 1990, 10:165-75). Heat shock promoters or stress promoters also may be advantageous for driving expression of the fusion proteins in recombinant host cells.

Animal regulatory regions that exhibit tissue specificity and have been utilized in transgenic animals also can be used in tumor cells of a particular tissue type: the elastase I gene control region that is active in pancreatic acinar cells (Swift et al., Cell 1984, 38:639-646; Ornitz et al., Cold Spring Harbor Symp Quant Biol 1986, 50:399-409; and MacDonald, Hepatology 1987, 7:425-515); the insulin gene control region that is active in pancreatic beta cells (Hanahan, Nature 1985, 315:115-122), the immunoglobulin gene control region that is active in lymphoid cells (Grosschedl et al., Cell 1984, 38:647-658; Adames et al., Nature 1985, 318:533-538; and Alexander et al., Mol Cell Biol 1987, 7:1436-1444), the mouse mammary tumor virus control region that is active in testicular, breast, lymphoid and mast cells (Leder et al., Cell 1986, 45:485-495), the albumin gene control

region that is active in liver (Pinkert et al., Genes Devel, 1987, 1:268-276), the alpha-fetoprotein gene control region that is active in liver (Krumlauf et al., Mol Cell Biol 1985, 5:1639-1648; and Hammer et al., Science 1987, 235:53-58); the alpha 1-antitrypsin gene control region that is active in liver (Kelsey et al., Genes Devel 1987, 1:161-171), the beta-globin gene control region that is active in myeloid cells (Mogam et al., Nature 1985, 315:338-340; and Kollias et al., Cell 1986, 46:89-94); the myelin basic protein gene control region that is active in oligodendrocyte cells in the brain (Readhead et al., Cell 1987, 48:703-712); the myosin light chain-2 gene control region that is active in skeletal muscle (Sani, Nature 1985, 314:283-286), and the gonadotropic releasing hormone gene control region that is active in the hypothalamus (Mason et al., Science 1986, 234:1372-1378).

An expression vector also can include transcription enhancer elements, such as those found in SV40 virus, Hepatitis B virus, cytomegalovirus, immunoglobulin genes, metallothionein, and beta-actin (see, Bittner et al., Meth Enzymol 1987, 153:516-544; and Gorman, Curr Op Biotechnol 1990, 1:36-47). In addition, an expression vector can contain sequences that permit maintenance and replication of the vector in more than one type of host cell, or integration of the vector into the host chromosome. Such sequences include, without limitation, to replication origins, autonomously replicating sequences (ARS), centromere DNA, and telomere DNA.

In addition, an expression vector can contain one or more selectable or screenable marker genes for initially isolating, identifying, or tracking host cells that contain DNA encoding the immunogenic proteins as described herein. For long term, high yield production of gp96-Ig and T cell costimulatory fusion proteins, stable expression in mammalian cells can be useful. A number of selection systems can be used for mammalian cells. For example, the Herpes simplex virus thymidine kinase (Wigler et al., Cell 1977, 11:223), hypoxanthine-guanine phosphoribosyltransferase (Szybalski and Szybalski, Proc Natl Acad Sci USA 1962, 48:2026), and adenine phosphoribosyltransferase (Lowy et al., Cell 1980, 22:817) genes can be employed in tk⁻, hgp^r, or apr^r cells, respectively. In addition, antimetabolite resistance can be used as the basis of selection for dihydrofolate reductase (dhfr), which confers resistance to methotrexate (Wigler et al., Proc Natl Acad Sci USA 1980, 77:3567; O'Hare et al., Proc Natl Acad Sci USA 1981, 78:1527); gpt, which confers resistance to mycophenolic acid (Mulligan and Berg, Proc Natl Acad Sci USA 1981, 78:2072); neomycin phosphotransferase (neo), which confers resistance to the aminoglycoside G-418 (Colberre-Garapin et al., J Mol Biol 1981, 150:1); and hygromycin phosphotransferase (hyg), which confers resistance to hygromycin (Santerre et al., Gene 1984, 30:147). Other selectable markers such as histidinol and ZeocinTM also can be used.

A number of viral-based expression systems also can be used with mammalian cells to produce the allogeneic tumor cell vaccines. Vectors using DNA virus backbones have been derived from simian virus 40 (SV40) (Hamer et al., Cell 1979, 17:725), adenovirus (Van Doren et al., Mol Cell Biol 1984, 4:1653), adeno-associated virus (McLaughlin et al., J Virol 1988, 62:1963), and bovine papillomas virus (Zinn et al., Proc Natl Acad Sci USA 1982, 79:4897). When an adenovirus is used as an expression vector, the donor DNA sequence may be ligated to an adenovirus transcription/translation control complex, e.g., the late promoter and tripartite leader sequence. This fusion gene may then be inserted in the adenovirus genome by in vitro or in vivo

recombination. Insertion in a non-essential region of the viral genome (e.g., region E1 or E3) can result in a recombinant virus that is viable and capable of expressing heterologous products in infected hosts. (See, e.g., Logan and Shenk, *Proc Natl Acad Sci USA* 1984, 81:3655-3659).

Bovine papillomavirus (BPV) can infect many higher vertebrates, including man, and its DNA replicates as an episome. A number of shuttle vectors have been developed for recombinant gene expression, which exist as stable, multicopy (20-300 copies/cell) extrachromosomal elements in mammalian cells. Typically, these vectors contain a segment of BPV DNA (the entire genome or a 69% transforming fragment), a promoter with a broad host range, a polyadenylation signal, splice signals, a selectable marker, and "poisonless" plasmid sequences that allow the vector to be propagated in *E. coli*. Following construction and amplification in bacteria, the expression gene constructs are transfected into cultured mammalian cells by, for example, calcium phosphate coprecipitation. For those host cells that do not manifest a transformed phenotype, selection of transformants is achieved by use of a dominant selectable marker, such as histidinol and G418 resistance.

Alternatively, the vaccinia 7.5K promoter can be used. (See, e.g., Mackett et al., *Proc Natl Acad Sci USA* 1982, 79:7415-7419; Mackett et al., *J Virol* 1984, 49:857-864; and Panicali et al., *Proc Natl Acad Sci USA* 1982, 79:4927-4931.) In cases where a human host cell is used, vectors based on the Epstein-Barr virus (EBV) origin (OriP) and EBV nuclear antigen 1 (EBNA-1; a trans-acting replication factor) can be used. Such vectors can be used with a broad range of human host cells, e.g., EBO-pCD (Spickofsky et al., *DNA Prot Eng Tech* 1990, 2:14-18); pDR2 and λ mdaDR2 (available from Clontech Laboratories).

Allogeneic tumor cell vaccines also can be made with retrovirus-based expression systems. Retroviruses, such as Moloney murine leukemia virus, can be used since most of the viral gene sequence can be removed and replaced with exogenous coding sequence while the missing viral functions can be supplied in trans. In contrast to transfection, retroviruses can efficiently infect and transfer genes to a wide range of cell types including, for example, primary hematopoietic cells. Moreover, the host range for infection by a retroviral vector can be manipulated by the choice of envelope used for vector packaging.

For example, a retroviral vector can comprise a 5' long terminal repeat (LTR), a 3' LTR, a packaging signal, a bacterial origin of replication, and a selectable marker. The gp96-Ig fusion protein coding sequence, for example, can be inserted into a position between the 5' LTR and 3' LTR, such that transcription from the 5' LTR promoter transcribes the cloned DNA. The 5' LTR contains a promoter (e.g., an LTR promoter), an R region, a U5 region, and a primer binding site, in that order. Nucleotide sequences of these LTR elements are well known in the art. A heterologous promoter as well as multiple drug selection markers also can be included in the expression vector to facilitate selection of infected cells. See, McLaughlin et al., *Prog Nucleic Acid Res Mol Biol* 1990, 38:91-135; Morgenstern et al., *Nucleic Acid Res* 1990, 18:3587-3596; Choulaka et al., *J Virol* 1996, 70:1792-1798; Boesen et al., *Biotherapy* 1994, 6:291-302; Salmons and Gunzberg, *Human Gene Ther* 1993, 4:129-141; and Grossman and Wilson, *Curr Opin Genet Devel* 1993, 3:110-114.

Any of the cloning and expression vectors described herein may be synthesized and assembled from known DNA sequences using techniques that are known in the art. The regulatory regions and enhancer elements can be of a variety

of origins, both natural and synthetic. Some vectors and host cells may be obtained commercially. Non-limiting examples of useful vectors are described in Appendix 5 of *Current Protocols in Molecular Biology*, 1988, ed. Ausubel et al., Greene Publish. Assoc. & Wiley Interscience, which is incorporated herein by reference; and the catalogs of commercial suppliers such as Clontech Laboratories, Stratagene Inc., and Invitrogen, Inc.

Recombinant Immunomodulators

According to some embodiments, two or more immune modulators may be cloned into plasmid constructs for transfection (via, e.g., lipids, calcium phosphate, cationic polymers, DEAE-dextran, activated dendrimers, magnetic beads, electroporation, biolistic technology, microinjection, laserfection/optoinjection) or transduction (via, e.g., retrovirus, lentivirus, adenovirus, adeno-associated virus) into cells of tumor cell lines. According to some embodiments, recombinant DNA encoding each immune modulator protein may be cloned into a lentiviral vector plasmid for integration into the genome of cells of tumor cell lines. According to some embodiments, recombinant DNA encoding the immune modulator protein may be cloned into a plasmid DNA construct encoding a selectable trait, such as an antibiotic resistance gene. According to some embodiments, recombinant DNA encoding the immune modulator protein may be cloned into a plasmid construct that is adapted to stably express each recombinant protein in the cells of the tumor cell line. According to some embodiments, the transfected or transduced tumor cells may be clonally expanded to achieve a cell line variant with a homogenous site of integration of the recombinant DNA encoding each immune modulator protein into the genome of the cells of the tumor cell line.

Lentiviral Constructs

According to some embodiments, the DNA sequences coding for immune modulator proteins may be cloned into a lentiviral vector for transduction into mammalian cells. According to some embodiments, the lentiviral system may comprise a lentiviral transfer plasmid encoding the two or more immune modulator sequences, packaging plasmids encoding the GAG, POL, TAT, and REV sequences, and an envelope plasmid encoding the ENV sequences. According to some embodiments, the lentiviral transfer plasmid uses a viral LTR promoter for gene expression. According to some embodiments, the lentiviral transfer plasmid uses a hybrid promoter, or other specialized promoter. According to some embodiments, the promoter of the lentiviral transfer plasmid is selected to express the two or more immune modulator sequences at a desired level relative to other immunomodulatory sequences. According to some embodiments, the relative level is measured on the level of transcription as mRNA transcripts. According to some embodiments, the relative level is measured on the level of translation as protein expression.

Multicistronic Plasmid Constructs

According to some embodiments, one or more immune modulator sequence may be cloned in a multicistronic vector for co-expression of one immune modulator with a second immune modulator or other recombinant sequence. According to some embodiments, an immune modulator sequence may be cloned into a plasmid comprising an IRES element to promote translation of two or more proteins from a single transcript. According to some embodiments, one or more immune modulator sequences is cloned into a multicistronic vector comprising sequences for a self cleaving 2A peptide to produce two or more immune modulator proteins from a single transcript.

Genetic Introduction of Immune Modulators

According to some embodiments, plasmid constructs comprising the recombinant immune modulator sequences may be transfected or transduced into tumor cell lines.

Lentiviral System

According to some embodiments, the lentiviral system may be employed where the transfer vector with immune modulator sequences, an envelope vector, and a packaging vector are each transfected into host cells for virus production. According to some embodiments, the lentiviral vectors may be transfected into 293T cells by any of calcium phosphate precipitation transfection, lipid based transfection, or electroporation, and incubated overnight. For embodiments where the immune modulator sequence may be accompanied by a fluorescence reporter, inspection of the 293T cells for fluorescence may be checked after overnight incubation. The culture medium of the 293T cells comprising virus particles may be harvested 2 or 3 times every 8-12 hours and centrifuged to sediment detached cells and debris. The culture medium may then be used directly, frozen or concentrated as needed.

Tumor cell lines may be grown to a confluency of about 70% under standard tissue culture conditions. The cells may then be treated with hexadimethrine bromide (to enhance transduction of cells) and lentiviral particles comprising recombinant constructs in fresh media, and incubated for 18-20 hours followed by a media change.

Lipid Based Transfection

According to some embodiments, cells of tumor cell lines may be transfected with immune modulator sequences using a lipid based transfection method. According to some embodiments, established lipid based transfection reagents, such as LIPOFECTAMINE™, may be used. Tumor cell lines may be grown to about 70-90% confluence in a tissue culture vessel. Appropriate amounts of LIPOFECTAMINE™ and plasmid construct comprising the immune modulator sequences may be separately diluted in tissue culture media and briefly incubated at room temperature. The diluted LIPOFECTAMINE™ and plasmid constructs in media may be mixed together and incubated briefly at room temperature. The plasmid LIPOFECTAMINE™ mixture may then be added to the cells of the tumor cell lines in the tissue culture vessel and incubated for 1-3 days under standard tissue culture conditions.

Selection of Expressing Clones

According to some embodiments, tumor cells of the tumor cell line that have been transfected with immune modulator sequences may be selected for various levels of expression.

According to some embodiments, the immune modulator sequences may be accompanied by antibiotic resistance genes, which may be used to select for clones with stable integration of the recombinant DNA encoding the immune modulator sequences. According to some embodiments, the immune modulator sequences may be cloned into a plasmid construct comprising antibiotic resistance, such as the Neomycin/Kanamycin resistance gene. Transfected cells are treated with antibiotics according to the manufacturer's protocol for 1-2 weeks or more with daily media changes. At some point during antibiotic treatment, there is massive tumor cell death of all cells that have not stably integrated the antibiotic resistance gene, leaving behind small colonies of stably expressing clones. Each of the stably expressing clones may be picked, cultured in a separate tissue culture container, and tested for levels of immune modulator expression by any established method, such as western blot, flow cytometry, and fluorescence microscopy.

According to some embodiments, transfected tumor cells may be selected for high expression of the immune modulators by fluorescence activated cell sorting (FACS). According to some embodiments, immune modulator sequences may be accompanied by one or more fluorescent proteins (e.g. GFP), which can be used to quantify expression of immune modulator. For example, a bicistronic plasmid comprising an immune modulator sequence connected to a GFP sequence via IRES sequence would result in both an immune modulator and GFP protein translated from the same transcript. Thus, the GFP expression level would act as a proxy for the expression level of immune modulator. Single cell suspensions of immune modulator/GFP transfected tumor cells could be selected for the desired level of expression by FACS based on the fluorescence intensity. Any fluorescent protein may be used in this regard. For example, any of the following recombinant fluorescent proteins may be used: EBFP, ECFP, EGFP, YFP, mHoneydew, mBanana, mOrange, tdTomato, mTangerine, mStrawberry, mCherry, mGrape, mRaspberry, mGrape2, mPlum.

Alternatively, the expression of the recombinant immune modulator may be directly observed by fluorescent antibodies specific to each immune modulator or specific to a tag engineered onto each immune modulator. For example, according to some embodiments the extracellular region of an immune modulator sequence may be fused with a FLAG tag or HA tag. Anti-FLAG or anti-HA antibodies may be used, along with a fluorophore attached to the primary antibody or a secondary antibody) to detect the expression of the immune modulator on the surface of the transfected tumor cells. Tumor cells expressing the desired level of immune modulator may be selected by FACS sorting and cultured separately.

Testing of Clones for Immunogenic Potential
Mixed Lymphocyte Tumor Cell Reactivity

According to some embodiments, the genetically introduced immunomodulators may be assessed for their immunogenic potential by a mixed lymphocyte tumor cell reaction (MLTR). The MLTR assay comprises incubating mixed lymphocytes with tumor cell line variants (or controls) for several days to allow the tumor cells of the tumor cell line variant to elicit an immune response from the mixed lymphocytes in vitro. This method may provide a rapid in vitro method to assess mixed lymphocyte responses (such as cellular proliferation of lymphocytes, cellular subset differentiation of lymphocytes, cytokine release profile of lymphocytes, and tumor cell death) to tumor cells or lysates. This approach may enable comprehensive monitoring of cellular, humoral, or both, immunity responses to phenotypically modified transfected tumor cells using human peripheral blood mononuclear cells. The MLTR also may provide an alternative to murine tumor survival studies, and may result in selection of optimal tumor cell line variants for anti-tumor response. A similar assay has been described by Hunter T B et al., (2007) Scandanavian J. Immunology 65, 479-486, which is incorporated herein by reference in its entirety.

According to some embodiments, tumor cell line variants may be tested for immunogenic potential by contacting transfected tumor cells with mixed lymphocytes from peripheral blood mononuclear cells, followed by measuring cellular proliferation, cellular subset differentiation, cytokine release profile, and tumor cell lysate.

According to some embodiments, mixed lymphocytes may be obtained from peripheral blood mononuclear cells isolated by a FICOLL-PAQUE® gradient. Briefly, anticoagulant-treated blood may be diluted in the range of 1:2 to

1:4 with PBS/EDTA to reduce aggregation of erythrocytes. The diluted blood may then be layered above a FICOLL-PAQUE® solution in a centrifuge tube, without mixing. The layered blood/FICOLL-PAQUE® may be centrifuged for 40 minutes at 400×g between 18° and 20° C., without the use of the centrifuge brake, resulting in the formation of blood fractions comprising, from top to bottom, a first fraction comprising blood plasma; a second fraction comprising mononuclear cells; a third fraction comprising FICOLL-PAQUE® media; and a fourth fraction comprising granulocytes and erythrocytes.

The fractions may be further processed to isolate specific fraction components. For example, to further process mononuclear cells, the second fraction comprising mononuclear cells may be carefully removed from the FICOLL-PAQUE® gradient using a Pasteur pipet. Alternatively, the second fraction may be removed directly by puncturing the tube with a needle and directly withdrawing the second fraction. The second fraction may then be washed and centrifuged at 300×g, 18° and 20° C., three times with PBS/EDTA, discarding the supernatant after each round.

According to some embodiments, tumor cell line variants may be co-cultured with the PBMCs comprising lymphocytes for seven days to allow for direct evaluation of activation of anti-tumor response in the presence of immune modulators from the tumor cell line variants.

According to some embodiments, one parameter used for measuring activation of lymphocytes may be cellular proliferation. According to some embodiments, proliferation may be detected by ³H-thymidine incorporation. Briefly, approximately 5×10³ tumor cell line variant cells may be co-cultured with approximately 1×10⁶ mixed lymphocytes in round bottomed 96-well plates. After three days of culture, cells may be pulsed with 1 μCi of ³H-thymidine for 18 hours. The cells may then be harvested onto filter mats, and ³H-thymidine incorporation may be measured using a scintillation counter. Proliferation of tumor cell line variants compared to non-transfected tumor cell controls may be measured. An increase, a decrease, or no change in proliferation relative to controls, are possible outcomes.

According to some embodiments, another parameter for measuring activation of lymphocytes may be the cytokine release profile. For example, the number of responsive T cells in the mixed lymphocyte population may be quantified by enzyme linked immunospot (ELISpot) analysis of IFN-gamma and/or IL-2 production by PBMCs. Briefly, PBMCs comprising mixed lymphocytes and a tumor cell line variant may be co-cultured between 3 and 7 days. Co-cultured cells may then be harvested and incubated on ELISpot plates pre-coated with anti-IFN-gamma and/or anti-IL-2 antibodies. After 20 hours, cells may be removed by washing 2 times in distilled water and two times in washing buffer. ELISpot plates may then be contacted with biotinylated anti-IFN-gamma and/or anti-IL-2 antibodies and streptavidin alkaline phosphatase in blocking buffer for 1-2 hours. After washing, plates may be contacted with alkaline phosphatase substrate until dark spot emerge. Plates may then be washed in tap water and air dried. Spots are then quantified manually or by plate reader and compared to non-transfected tumor cell line control group.

According to some embodiments, another parameter for measuring activation of lymphocytes may be by quantifying cellular subset differentiation. For example, the differentiation of CD45+/CD3+ T-lymphocytes to CD45+/CD3+/CD4+ helper T-lymphocytes, CD45+/CD3+/CD8+ cyto-

toxic T-lymphocytes, and CD45+/CD3+/CD25+ activated T-lymphocytes may be quantified by flow cytometry analysis.

According to some embodiments, another parameter for measuring activation of lymphocytes may be by quantifying tumor cell cytotoxicity. Cytotoxicity of tumor cells may be measured by any number of established methods. For example, according to some embodiments, an LDH-Cytotoxicity colorimetric assay kit (BioVision Cat. #K311-400) may be used to measure cytotoxicity of tumor cells by testing for lactate dehydrogenase (LDH) released from damaged cells into the growth media. Briefly, 100 μl of media from each of the control group (comprising untransfected tumor cells), the experimental group (comprising immune modulator transfected tumor cells), and media alone may be pipetted into the wells of a 96 well plate. 100 μl of the LDH reaction mixture, comprising dye solution and catalyst solution, may then be added to the wells of the 96 well plate and incubated for 30 minutes at room temperature. Then the samples may be measured for light absorbance at 490-500 nm using a microtiter plate reader.

Sequentially Add New Plasmid Constructs to the Clones

According to some embodiments, tumor cell line variants that express one or more immune modulator sequences are transfected with additional immune modulators for stable expression in a sequential manner. By sequentially adding recombinant immune modulators in successive fashion, cells of a tumor cell line variant may be created that express several immune modulators simultaneously. According to some embodiments, a tumor cell line variant may be created that expresses two immune modulators simultaneously. According to some embodiments, a tumor cell line variant may be created that expresses three immune modulators simultaneously. According to some embodiments, a tumor cell line variant may be created that expresses four immune modulators simultaneously. According to some embodiments, a tumor cell line variant may be created that expresses five immune modulators simultaneously.

Variably Expressing Clones

According to one aspect of the disclosed invention, multiple recombinant immune modulator peptides may be expressed in a single clonally derived tumor cell line variant. According to some embodiments, the amount (or level) of each individual immune modulator expressed in each cell is the same as the level of expression of all other immune modulator peptides. According to some embodiments, however, the level of each individual immune modulator expressed in each cell is different from the level of expression of the other immune modulators expressed in the cell. According to some embodiments, clonally derived tumor cell line variants that express the same complement of immune modulators stably express those immune modulators in varying amounts relative to each other.

The relative amount of recombinant immune modulator expressed within each clonally derived tumor cell line variant, and between tumor cell line variants, can be measured on the level of transcription or translation. For example, the relative amount of recombinant immune modulator can be quantified by western blot, RT-PCR, flow cytometry, immunofluorescence, and northern blot, among others.

According to some embodiments, the differences in the amount of expressed immune modulators relative to one another may be a result of random integration into more or less transcriptionally active regions of the genome of the tumor cell line variant. According to some embodiments, the relative differences in the amount of expressed immune

modulator may be achieved by elements engineered into the transfected or transduced DNA used to create the tumor cell line variant.

For example, according to some embodiments, the level of expression of the immune modulator proteins may be achieved on the transcriptional level by engineering stronger or weaker gene promoter sequences to control expression of the immune modulator gene. According to some embodiments, one or more of the following promoters may be used to control expression of immune modulators: simian virus 40 early promoter (SV40), cytomegalovirus immediate-early promoter (CMV), human Ubiquitin C promoter (UBC), human elongation factor 1 α promoter (EF1A), mouse phosphoglycerate kinase 1 promoter (PGK), and chicken (3-Actin promoter coupled with CMV early enhancer (CAGG).

According to some embodiments, the level of expression of the immune modulator proteins may be achieved on the translational level by engineering stronger or weaker Kozak consensus sequences around the start codon of the immune modulator transcript. According to some embodiments, the following nucleotide sequences may be provided to control immune modulator translation: GCCGCC(A/G)CCAUGG (SEQ ID NO: 15). According to some embodiments, a sequence that is at least 60% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 70% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 80% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 90% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 95% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 96% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 97% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 98% identical to SEQ ID NO: 15 may be provided to control immune modulator translation. According to some embodiments, a sequence that is at least 99% identical to SEQ ID NO: 15 may be provided to control immune modulator translation.

Therapeutic Compositions

According to another aspect of the disclosed invention, an immunogenic composition may comprise an amount of a tumor cell line variant comprising two or more genes encoding human immune modulators. According to some embodiments, clones of tumor cell line variants that maximally express the human immune modulators are identified and selected. According to some embodiments, expression of the human immune modulators by populations of the tumor cell line variants is determined by flow cytometry. According to some embodiments, flow cytometry is used to gate on the maximally expressing population(s) of tumor cell line variants.

According to some embodiments, the immunogenic amount may be effective to stimulate an anti-tumor immune response to one or more tumor specific antigens. According to some embodiments, the immunogenic amount may be titrated to provide both safety and efficacy.

According to some embodiments, the immunogenic composition comprises a pharmaceutically acceptable carrier.

According to some embodiments, the immunogenic composition further comprises an adjuvant.

According to some embodiments, the tumor cell line variant may comprise tumor cells derived from an established cell line. According to some embodiments, the tumor cell line variant comprises tumor cells derived from a patient with cancer, wherein the tumor cells are derived from a solid tumor.

According to some embodiments, the tumor cell line variant comprises an immunogenic amount of a disrupted tumor cell line variant. Examples of methods for physical disruption include, without limitation, sonication, cavitation, dehydration, ion depletion, or by toxicity from exposure to one or more salts.

According to some embodiments, the immunogenic amount of the immunogenic composition may comprise at least 1×10^3 whole or disrupted tumor cell line variant cells. According to some embodiments, the amount of the immunogenic composition may comprise at least 1×10^4 whole or disrupted tumor cell line variant cells. According to some embodiments, the amount of the immunogenic composition may comprise at least 1×10^5 whole or disrupted tumor cell line variant cells. According to some embodiments, the amount of the immunogenic composition may comprise at least 1×10^6 whole or disrupted tumor cell line variant cells. According to some embodiments, the amount of the immunogenic composition may comprise at least 1×10^7 whole or disrupted tumor cell line variant cells. According to some embodiments, the amount of the immunogenic composition may comprise at least 1×10^8 whole or disrupted tumor cell line variant cells. According to some embodiments, the amount of the immunogenic composition may comprise at least 1×10^9 whole or disrupted tumor cell line variant cells. According to some embodiments, the immunogenic amount may be a therapeutic amount.

According to some embodiments, the immunogenic amount may be effective (1) to stimulate an effective immune response comprising one or more of cytotoxic T cells, natural killer cells, antibodies, APCs, T cells, B cells, and dendritic cells; and (2) to improve a clinical outcome parameter selected from one or more of progression-free survival, disease-free survival, time to progression, time to distant metastasis, and overall survival of the subject, when compared to a suitable control.

According to some embodiments, the immunogenic composition may be administered once per week, twice per week, once every two weeks, once every three weeks, once every four weeks, once per month, once every two months, once every three months, once every four months, once every five months, once every six months, once every seven months, once every eight months, once every nine months, once every ten months, once every eleven months, or once a year. According to some embodiments, administration occurs in one day or over 2 days, 3 days, 4, days, 5 days, 6 days, 7 days, 8, days, 9 days, 10 days, 11 days, 12 days, 13 days, 14 days, 15 days, 16 days, 17 days, 18 days, 19 days, 20 days, or more. According to some embodiments, administration may involve two or more administrations on the same day.

Combination Therapies

According to some embodiments, the disclosure provides methods that further comprise administering an additional agent to a subject. In some embodiments, the invention pertains to co-administration and/or co-formulation.

In some embodiments, administration of the immunogenic composition acts synergistically when co-administered with another agent and is administered at doses that are lower than the doses commonly employed when such agents are used as monotherapy.

In some embodiments, inclusive of, without limitation, cancer applications, the present invention pertains to chemotherapeutic agents as additional agents. Examples of chemotherapeutic agents include, but are not limited to, alkylating agents such as thiopeta and CYTOXAN cyclophosphamide; alkyl sulfonates such as busulfan, improsulfan and piposulfan; aziridines such as benzodopa, carboquone, meturedopa, and uredopa; ethylenimines and methylamelamines including altretamine, triethylenemelamine, trietylenephosphoramide, triethylenethiophosphoramide and trimethylolomelamine; acetogenins (e.g., bullatacin and bullatacinone); a camptothecin (including the synthetic analogue topotecan); bryostatin; cally statin; CC-1065 (including its adozelesin, carzelesin and bizelesin synthetic analogues); cryptophycins (e.g., cryptophycin 1 and cryptophycin 8); dolastatin; duocarmycin (including the synthetic analogues, KW-2189 and CB 1-TM1); eleutherobin; pancratistatin; a sarcodictyin; spongistatin; nitrogen mustards such as chlorambucil, chlornaphazine, cholophosphamide, estramustine, ifosfamide, mechlorethamine, mechlorethamine oxide hydrochloride, melphalan, novembichin, phenesterine, prednimustine, trofosfamide, uracil mustard; nitrosureas such as carmustine, chlorozotocin, fotemustine, lomustine, nimustine, and ranimustine; antibiotics such as the enediyne antibiotics (e.g., calicheamicin, especially calicheamicin gamma11 and calicheamicin omega11; dynemicin, including dynemicin A; bisphosphonates, such as clodronate; an esperamicin; as well as neocarzinostatin chromophore and related chromoprotein enediyne antibiotic chromophores), aclacinomycin, actinomycin, anthramycin, azaserine, bleomycins, cactinomycin, carabacin, caminomycin, carzinophilin, chromomycinis, dactinomycin, daunorubicin, detorubicin, 6-diazo-5-oxo-L-norleucine, ADRIAMYCIN doxorubicin (including morpholino-doxorubicin, cyanomorpholino-doxorubicin, 2-pyrrolino-doxorubicin and deoxy doxorubicin), epirubicin, esorubicin, idarubicin, marcellomycin, mitomycins such as mitomycin C, mycophenolic acid, nogalamycin, olivomycins, peplomycin, potfiromycin, puromycin, quelamycin, rodorubicin, streptonigrin, streptozocin, tubercidin, ubenimex, zinostatin, zorubicin; anti-metabolites such as methotrexate and 5-fluorouracil (5-FU); folic acid analogues such as denopterin, methotrexate, pteropterin, trimetrexate; purine analogs such as fludarabine, 6-mercaptopurine, thiamiprine, thioguanine; pyrimidine analogs such as ancitabine, azacitidine, 6-azauridine, carmofur, cytarabine, dideoxyuridine, doxifluridine, enocitabine, floxuridine; androgens such as calusterone, dromostanolone propionate, epitioestanol, mepitioestane, testolactone; anti-adrenals such as minoglutethimide, mitotane, trilostane; folic acid replenisher such as frolinic acid; aceglutone; aldophosphamide glycoside; aminolevulinic acid; eniluracil; amsacrine; bestrabucil; bisantrene; edatraxate; demecolcine; diaziquone; elformithine; elliptinium acetate; an epothilone; etoglucid; gallium nitrate; hydroxyurea; lentinan; lonidainine; maytansinoids such as maytansine and ansamitocins; mitoguazone; mitoxantrone; mopidanmol; nitraerine; pentostatin; phenamet; pirarubicin; losoxantrone; podophyllinic acid; 2-ethylhydrazide; procarbazine; PSK polysaccharide complex; razoxane; rhizoxin; sizofuran; spirogermanium; tenuazonic acid; triaziquone; 2,2',2"-trichlorotriethylamine; trichothecenes (e.g., T-2 toxin, verracurin A, roridin A and

anguidine); urethan; vindesine; dacarbazine; mannomustine; mitobronitol; mitolactol; pipobroman; gacytosine; arabinoside ("Ara-C"); cyclophosphamide; thiopeta; taxoids, e.g., TAXOL paclitaxel (Bristol-Myers Squibb Oncology, Princeton, N.J.), ABRAXANE Cremophor-free, albumin-engineered nanoparticle formulation of paclitaxel, and TAXOTERE doxetaxel; chloranbucil; GEMZAR gemcitabine; 6-thioguanine; mercaptopurine; methotrexate; platinum analogs such as cisplatin, oxaliplatin and carboplatin; vinblastine; platinum; etoposide (VP-16); ifosfamide; mitoxantrone; vincristine; NAVELBINE. vinorelbine; novantrone; teniposide; edatrexate; daunomycin; aminopterin; xeloda; ibandronate; irinotecan (Camptosar, CPT-11) (including the treatment regimen of irinotecan with 5-FU and leucovorin); topoisomerase inhibitor RFS 2000; difluoromethylornithine (DMFO); retinoids such as retinoic acid; capecitabine; combretastatin; leucovorin (LV); oxaliplatin, including the oxaliplatin treatment regimen (FOLFOX); lapatinib (TYK-ERB); inhibitors of PKC- α , Raf, H-Ras, EGFR (e.g., erlotinib (Tarceva)) and VEGF-A that reduce cell proliferation and pharmaceutically acceptable salts, acids or derivatives of any of the above. In addition, the methods of treatment can further include the use of radiation.

Other additional agents are described elsewhere herein, including the blocking antibodies targeted to an immune "checkpoint" molecule.

According to some embodiments, the treatment regimen may comprise a standard anti-tumor therapy (such as surgery, radiation therapy, a targeted therapy that precisely identifies and attacks cancer cells, a hormone therapy, or a combination thereof). According to some embodiments, the standard anti-tumor therapy is effective to treat the tumor while preserving any existing anti-tumor immune response. According to some embodiments, the immunogenic composition is not applied after chemotherapy. According to some embodiments, the immunogenic composition is applied after low-dose chemotherapy.

According to some embodiments, the immunogenic composition comprises two or more clonally derived tumor cell line variants. According to some embodiments, the two or more tumor cell line variants comprise the same complement of recombinant immune modulators. According to some embodiments, the two or more tumor cell line variants comprise different panels of recombinant immune modulators.

According to some embodiments, the tumor cell line variants are treated with an agent that prevents cell division prior to administration to a subject. According to some embodiments, the tumor cell line variants are irradiated. According to some embodiments, the tumor cell line variants are treated with a chemical agent that prevents proliferation.

According to some embodiments, the tumor cell line variants may be administered parenterally. According to some embodiments, the tumor cell line variants may be administered locally into a surgical excision cavity. According to some embodiments, the tumor cell variants may be administered by intradermal injection. According to some embodiments, the tumor cell line variants may be administered by subcutaneous injection. According to some embodiments, the tumor cell line variants may be administered by intramuscular injection.

Methods of Treatment

Tumor cell line variants as provided herein can be incorporated into a composition for administration to a subject (e.g., a research animal or a mammal, such as a human, having a clinical condition such as cancer or an infection).

For example, an allogeneic tumor cell vaccine comprising a tumor cell line variant comprising two or more stably expressed recombinant membrane bound immunomodulatory molecules selected from IgG1, CD40L, TNF-alpha, and Flt-3L peptides; and stably expressed recombinant soluble GM-CSF peptides; and a pharmaceutically acceptable carrier; can be administered to a subject for the treatment of cancer. In another example, an allogeneic tumor cell vaccine comprising a tumor-type specific cell line variant is used to deliver a broad array of tumor antigens in the context of immunomodulatory signals sufficient to elicit a potent anti-tumor response as reflected in improved progression free survival, overall survival, or both relative to placebo controls, wherein the immunomodulatory signals are comprised of two or more stably expressed recombinant membrane bound immunomodulatory molecules selected from membrane expressed IgG1, CD40L, TNF-alpha, as well as membrane and soluble forms of GM-CSF, and Flt-3L.

Thus, the described invention provides methods for treating clinical conditions such as cancer with the allogeneic tumor vaccines provided herein.

In various embodiments, the described invention pertains to cancers and/or tumors; for example, the treatment or prevention of cancers and/or tumors. The phrase "cancers or tumors" refers to an uncontrolled growth of cells and/or abnormal increased cell survival and/or inhibition of apoptosis which interferes with the normal functioning of the bodily organs and systems. Included are benign and malignant cancers, polyps, hyperplasia, as well as dormant tumors or micrometastases. Also included are cells having abnormal proliferation that is not impeded by the immune system (e.g. virus infected cells). The cancer may be a primary cancer or a metastatic cancer. The primary cancer may be an area of cancer cells at an originating site that becomes clinically detectable, and may be a primary tumor. In contrast, the metastatic cancer may be the spread of a disease from one organ or part to another non-adjacent organ or part. The metastatic cancer may be caused by a cancer cell that acquires the ability to penetrate and infiltrate surrounding normal tissues in a local area, forming a new tumor, which may be a local metastasis. The cancer may also be caused by a cancer cell that acquires the ability to penetrate the walls of lymphatic and/or blood vessels, after which the cancer cell is able to circulate through the bloodstream (thereby being a circulating tumor cell) to other sites and tissues in the body. The cancer may be due to a process such as lymphatic or hematogeneous spread. The cancer may also be caused by a tumor cell that comes to rest at another site, re-penetrates through the vessel or walls, continues to multiply, and eventually forms another clinically detectable tumor. The cancer may be this new tumor, which may be a metastatic (or secondary) tumor.

The cancer may be caused by tumor cells that have metastasized, which may be a secondary or metastatic tumor. The cells of the tumor may be like those in the original tumor. As an example, if a breast cancer or colon cancer metastasizes to the liver, the secondary tumor, while present in the liver, is made up of abnormal breast or colon cells, not of abnormal liver cells. The tumor in the liver may thus be a metastatic breast cancer or a metastatic colon cancer, not liver cancer.

Illustrative cancers that may be treated include, but are not limited to, carcinomas, e.g. various subtypes, including, for example, adenocarcinoma, basal cell carcinoma, squamous cell carcinoma, and transitional cell carcinoma), sarcomas (including, for example, bone and soft tissue), leukemias (including, for example, acute myeloid, acute lymphoblas-

tic, chronic myeloid, chronic lymphocytic, and hairy cell), lymphomas and myelomas (including, for example, Hodgkin and non-Hodgkin lymphomas, light chain, non-secretory, MGUS, and plasmacytomas), and central nervous system cancers (including, for example, brain (e.g. gliomas (e.g. astrocytoma, oligodendroglioma, and ependymoma), meningioma, pituitary adenoma, and neuromas, and spinal cord tumors (e.g. meningiomas and neurofibroma).

In certain embodiments, cancers/tumors that may be treated are those where the standard of care is no longer chemotherapy, since chemotherapy is known to interfere with immune responses, which are expected to occur during a successful vaccination protocol. Exemplary tumor types include tumor types treated with hormonal therapies such as prostate and breast cancers (e.g. Abiraterone® for prostate cancer and Tamoxifen® for breast cancer), tumor types treated with targeted therapies like antibodies (e.g. Rituxan® for B cell malignancies, Herceptin® for breast cancer), tumor types treated with kinase inhibitors such as GLEEVEC™ for chronic myelogenous leukemia and tumor types treated with other immune system sparing or enhancing modalities, such as checkpoint inhibitors, oncolytic viruses and CAR-T cells

Representative cancers and/or tumors of the present invention are described herein. The described invention also provides compositions containing an allogeneic tumor cell vaccine comprising a tumor cell line variant comprising two or more stably expressed recombinant membrane bound immunomodulatory molecules selected from IgG1, CD40L, TNF-alpha, and Flt-3L peptides; and stably expressed recombinant soluble GM-CSF peptides; and a pharmaceutically acceptable carrier, as described herein, in combination with a physiologically and pharmaceutically acceptable carrier. The physiologically and pharmaceutically acceptable carrier can include any of the well-known components useful for immunization. The carrier can facilitate or enhance an immune response to an antigen administered in a vaccine. The cell formulations can contain buffers to maintain a preferred pH range, salts or other components that present an antigen to an individual in a composition that stimulates an immune response to the antigen. The physiologically acceptable carrier also can contain one or more adjuvants that enhance the immune response to an antigen. Pharmaceutically acceptable carriers include, for example, pharmaceutically acceptable solvents, suspending agents, or any other pharmacologically inert vehicles for delivering compounds to a subject. Pharmaceutically acceptable carriers can be liquid or solid, and can be selected with the planned manner of administration in mind so as to provide for the desired bulk, consistency, and other pertinent transport and chemical properties, when combined with one or more therapeutic compounds and any other components of a given pharmaceutical composition. Typical pharmaceutically acceptable carriers include, without limitation: water, saline solution, binding agents (e.g., polyvinylpyrrolidone or hydroxypropyl methylcellulose); fillers (e.g., lactose or dextrose and other sugars, gelatin, or calcium sulfate), lubricants (e.g., starch, polyethylene glycol, or sodium acetate), disintegrates (e.g., starch or sodium starch glycolate), and wetting agents (e.g., sodium lauryl sulfate). Compositions can be formulated for subcutaneous, intramuscular, or intradermal administration, or in any manner acceptable for immunization.

An "adjuvant" refers to a substance which, when added to an immunogenic agent such as a tumor cell expressing secreted vaccine protein, nonspecifically enhances or potentiates an immune response to the agent in the recipient host

upon exposure to the mixture. Adjuvants can include, for example, oil-in-water emulsions, water-in oil emulsions, alum (aluminum salts), liposomes and microparticles, such as, polystyrene, starch, polyphosphazene and polylactide/polyglycosides.

Adjuvants can also include, for example, squalene mixtures (SAF-I), muramyl peptide, saponin derivatives, mycobacterium cell wall preparations, monophosphoryl lipid A, mycolic acid derivatives, nonionic block copolymer surfactants, QUIL-A®, cholera toxin B subunit, polyphosphazene and derivatives, and immunostimulating complexes (ISCs) such as those described by Takahashi et al., Nature 1990, 344:873-875. For veterinary use and for production of antibodies in animals, mitogenic components of Freund's adjuvant (both complete and incomplete) can be used. In humans, Incomplete Freund's Adjuvant (IFA) is a useful adjuvant. Various appropriate adjuvants are well known in the art (see, for example, Warren and Chedid, CRC Critical Reviews in Immunology 1988, 8:83; and Allison and Byars, in Vaccines: New Approaches to Immunological Problems, 1992, Ellis, ed., Butterworth-Heinemann, Boston). Additional adjuvants include, for example, bacille Calmett-Guerin (BCG), DETOX (containing cell wall skeleton of *Mycobacterium phlei* (CWS) and monophosphoryl lipid A from *Salmonella minnesota* (MPL)), and the like (see, for example, Hoover et al., J Clin Oncol 1993, 11:390; and Woodlock et al., J Immunother 1999, 22:251-259).

In some embodiments, an allogeneic tumor cell vaccine can be administered to a subject one or more times (e.g., once, twice, two to four times, three to five times, five to eight times, six to ten times, eight to 12 times, or more than 12 times). An allogeneic tumor cell vaccine as provided herein can be administered one or more times per day, one or more times per week, every other week, one or more times per month, once every two to three months, once every three to six months, or once every six to 12 months. An allogeneic tumor cell vaccine can be administered over any suitable period of time, such as a period from about 1 day to about 12 months. In some embodiments, for example, the period of administration can be from about 1 day to 90 days; from about 1 day to 60 days; from about 1 day to 30 days; from about 1 day to 20 days; from about 1 day to 10 days; from about 1 day to 7 days. In some embodiments, the period of administration can be from about 1 week to 50 weeks; from about 1 week to 50 weeks; from about 1 week to 40 weeks; from about 1 week to 30 weeks; from about 1 week to 24 weeks; from about 1 week to 20 weeks; from about 1 week to 16 weeks; from about 1 week to 12 weeks; from about 1 week to 8 weeks; from about 1 week to 4 weeks; from about 1 week to 3 weeks; from about 1 week to 2 weeks; from about 2 weeks to 3 weeks; from about 2 weeks to 4 weeks; from about 2 weeks to 6 weeks; from about 2 weeks to 8 weeks; from about 3 weeks to 8 weeks; from about 3 weeks to 12 weeks; or from about 4 weeks to 20 weeks.

In some embodiments, after an initial dose (sometimes referred to as a "priming" dose) of an allogeneic tumor cell vaccine has been administered and a maximal antigen-specific immune response has been achieved, one or more boosting doses can be administered. For example, a boosting dose can be administered about 10 to 30 days, about 15 to 35 days, about 20 to 40 days, about 25 to 45 days, or about 30 to 50 days after a priming dose.

In some embodiments, the methods provided herein can be used for controlling solid tumor growth and/or metastasis. The methods can include administering an effective

amount of an allogeneic tumor cell vaccine as described herein to a subject in need thereof.

The vectors and methods provided herein can be useful for stimulating an immune response against a tumor. Such immune response is useful in treating or alleviating a sign or symptom associated with the tumor. A practitioner will appreciate that the methods described herein are to be used in concomitance with continuous clinical evaluations by a skilled practitioner (physician or veterinarian) to determine subsequent therapy. Such evaluations will aid and inform in evaluating whether to increase, reduce, or continue a particular treatment dose, mode of administration, etc.

The methods provided herein can thus be used to treat a tumor, including, for example, a cancer. The methods can be used, for example, to inhibit the growth of a tumor by preventing further tumor growth, by slowing tumor growth, or by causing tumor regression. Thus, the methods can be used, for example, to treat a cancer. It will be understood that the subject to which a compound is administered need not suffer from a specific traumatic state. Indeed, the allogeneic tumor cell vaccine described herein may be administered prophylactically, prior to development of symptoms (e.g., a patient in remission from cancer).

Anti-tumor and anti-cancer effects include, without limitation, modulation of tumor growth (e.g., tumor growth delay), tumor size, or metastasis, the reduction of toxicity and side effects associated with a particular anti-cancer agent, the amelioration or minimization of the clinical impairment or symptoms of cancer, extending the survival of the subject beyond that which would otherwise be expected in the absence of such treatment, and the prevention of tumor growth in an animal lacking tumor formation prior to administration, i.e., prophylactic administration.

Therapeutically effective amounts can be determined by, for example, starting at relatively low amounts and using step-wise increments with concurrent evaluation of beneficial effects. The methods provided herein thus can be used alone or in combination with other well-known tumor therapies, to treat a patient having a tumor. One skilled in the art will readily understand advantageous uses of the allogeneic tumor cell vaccines and methods provided herein, for example, in prolonging the life expectancy of a cancer patient and/or improving the quality of life of a cancer patient (e.g., a lung cancer patient).

Subjects

The methods described herein are intended for use with any subject that may experience the benefits of these methods. Thus, "subjects," "patients," and "individuals" (used interchangeably) include humans as well as non-human subjects, particularly domesticated animals.

In some embodiments, the subject and/or animal is a mammal, e.g., a human, mouse, rat, guinea pig, dog, cat, horse, cow, pig, rabbit, sheep, or non-human primate, such as a monkey, chimpanzee, or baboon. In other embodiments, the subject and/or animal is a non-mammal, such, for example, a zebrafish. In some embodiments, the subject and/or animal may comprise fluorescently-tagged cells (with e.g. GFP). In some embodiments, the subject and/or animal is a transgenic animal comprising a fluorescent cell.

In some embodiments, the subject and/or animal is a human. In some embodiments, the human is a pediatric human. In other embodiments, the human is an adult human. In other embodiments, the human is a geriatric human. In other embodiments, the human may be referred to as a patient.

In certain embodiments, the human has an age in a range of from about 0 months to about 6 months old, from about

6 to about 12 months old, from about 6 to about 18 months old, from about 18 to about 36 months old, from about 1 to about 5 years old, from about 5 to about 10 years old, from about 10 to about 15 years old, from about 15 to about 20 years old, from about 20 to about 25 years old, from about 25 to about 30 years old, from about 30 to about 35 years old, from about 35 to about 40 years old, from about 40 to about 45 years old, from about 45 to about 50 years old, from about 50 to about 55 years old, from about 55 to about 60 years old, from about 60 to about 65 years old, from about 65 to about 70 years old, from about 70 to about 75 years old, from about 75 to about 80 years old, from about 80 to about 85 years old, from about 85 to about 90 years old, from about 90 to about 95 years old or from about 95 to about 100 years old.

In other embodiments, the subject is a non-human animal, and therefore the invention pertains to veterinary use. In a specific embodiment, the non-human animal is a household pet. In another specific embodiment, the non-human animal is a livestock animal. In certain embodiments, the subject is a human cancer patient that cannot receive chemotherapy, e.g. the patient is unresponsive to chemotherapy or too ill to have a suitable therapeutic window for chemotherapy (e.g. experiencing too many dose- or regimen-limiting side effects). In certain embodiments, the subject is a human cancer patient having advanced and/or metastatic disease.

Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the invention. The upper and lower limits of these smaller ranges which may independently be included in the smaller ranges is also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either both of those included limits are also included in the invention.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present invention, the preferred methods and materials are now described. All publications mentioned herein are incorporated herein by reference to disclose and described the methods and/or materials in connection with which the publications are cited.

It must be noted that as used herein and in the appended claims, the singular forms "a", "an", and "the" include plural references unless the context clearly dictates otherwise. All technical and scientific terms used herein have the same meaning.

The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application and each is incorporated by reference in its entirety. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed.

EXAMPLES

The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure

and description of how to make and use the present invention, and are not intended to limit the scope of what the inventors regard as their invention nor are they intended to represent that the experiments below are all or the only experiments performed. Efforts have been made to ensure accuracy with respect to numbers used (e.g. amounts, temperature, etc.) but some experimental errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, molecular weight is weight average molecular weight, temperature is in degrees Centigrade, and pressure is at or near atmospheric.

Example 1

Examples 2-5 make use of, but are not limited to, the methods described hereinbelow.

Western Blotting

Briefly, cells are lysed with cold lysis buffer and centrifuged to pellet cellular debris. Protein concentration of the supernatant is determined by a protein quantification assay (e.g., Bradford Protein Assay, Bio-Rad Laboratories). The lysate supernatant is then combined with an equal volume of 2× SDS sample buffer and boiled at 100° C. for 5 minutes. Equal amounts of protein in sample buffer are loaded into the wells of an SDS-PAGE gel along with molecular weight marker and electrophoresed for 1-2 hours at 100 V. Proteins are then transferred to a nitrocellulose or PVDF membrane. The membrane is then blocked for 1 hour at room temperature using 5% non-fat dry milk in TBST blocking buffer. The membrane is then incubated with a 1:500 dilution of primary antibody in 5% non-fat dry milk in TBST blocking buffer, followed by three washes in 20 mM Tris (tris(hydroxymethyl)aminomethane), Ph 7.5; 150 mM NaCl, 0.1% Tween 20® (Polysorbate 20) (TBST) for 5 minutes. The membrane is then incubated with conjugated secondary antibody at a 1:2000 dilution in 5% non-fat dry milk in TBST blocking buffer for 1 hour at room temperature, followed by three washes in TBST for 5 minutes each. Images of the blot are obtained using dark room development techniques for chemiluminescence detection, or using image scanning techniques for colorimetric or fluorescent detection.

Real Time PCR

Real-time PCR techniques may be performed as described to analyze expression level of mRNAs (Zhao Y. et al., Biochemical and Biophysical Research Communications 360 (2007) 205-211). Briefly, total RNA is extracted from cells using the Qiagen kit (Valencia CA), followed by first strand cDNA synthesis using random hexamer primers (Fermentas, Hanover MD). Real-time PCR is performed on each sample using the Mx3000p Quantitative PCR system (Stratagene, La Jolla, CA), for 40 cycles using validated gene specific RT-PCR primer sets for each gene of interest. Relative expression level of each transcript is corrected for that of the house keeping gene beta-actin as an internal control.

Immunofluorescence

Briefly, adherent tumor cell line variant cells are fixed with 4% formaldehyde diluted in warm PBS for 15 minutes at room temperature. The fixative is aspirated and the cells washed three times with PBS for 5 minutes each. Cells are blocked in a 5% BSA blocking buffer for 60 minutes at room temperature. Blocking buffer is then aspirated and a solution of primary antibody (e.g. 1:100 dilution) is incubated with the cells overnight at 4° C. Cells are then rinsed three times with PBS for 5 minutes each, and subsequently incubated with a solution of fluorochrome conjugated secondary antibody (e.g. 1:1000 dilution) for 1-2 hours at room tempera-

ture. Cells are then washed three times with PBS for 5 minutes each and visualized by fluorescence microscopy. Flow Cytometry

Flow Cytometry analysis may be performed as described (Zhao Y. et al., Exp. Cell Res., 312, 2454 (2006)). Briefly, tumor cell line variant cells that are either treated with trypsin/EDTA or left untreated are collected by centrifugation and re-suspended in PBS. The cells are fixed in 4% formaldehyde for 10 minutes at 37° C. For extracellular staining with antibodies, cells are not permeabilized. For intracellular staining, cells are permeabilized by adding ice-cold 100% methanol to pre-chilled cells to a final concentration of 90% methanol and incubated on ice for 30 minutes. Cells are immunostained by first resuspending cells in incubation buffer and adding dilutions of primary antibody. Cells are incubated with primary antibody for 1 hour at room temperature, followed by three washes with incubation buffer. Cells are then resuspended in incubation buffer with dilutions of conjugated secondary antibody for 30 minutes at room temperature, followed by three washes in incubation buffer. Stained cells are then analyzed by flow cytometry.

Enzyme-Linked Immunosorbent Assay (ELISA)

Briefly, a capture antibody, specific for a protein of interest, is coated onto the wells of a microplate. Samples, including a standard containing protein of interest, control specimens, and unknowns, are pipetted into wells of the microplate, where the protein antigen binds to the capture antibody. After washing 4 times, a detection antibody is added to the wells for one hour, binding to the immobilized protein captured during the first incubation. After removal of excess detection antibody and washing 4 times, a horse radish peroxidase (HRP) conjugate (secondary antibody or streptavidin) is added for 30 minutes to bind to the detection antibody. After washing 4 more times to remove the excess HRP conjugate, a substrate solution is added for 30 minutes in the dark to be converted by the enzyme to a detectable form (color signal). A stop solution is added to each well of the microplate and evaluated within 30 minutes of stopping the reaction. Intensity of the colored product may be directly proportional to the concentration of antigen present in the original specimen.

Human Mixed Lymphocyte Tumor Reaction (MLTR) Testing

The mixed lymphocyte tumor reaction (MLTR) is an all human, in vitro assay, designed to optimize lead candidates. In the MLTR, optimization is achieved through the qualitative and quantitative assessment of human peripheral blood mononuclear cell (PBMC) responses to engineered allogeneic tumor cells. The MLTR assay permits assessment of proliferation and differentiation by flow cytometry and mass cytometry (CyTOF). Cytotoxicity, can be measured by lactate dehydrogenase (LDH) release assay, and the cytokine profile can be measured by LUMINEX® multiplex assay. In certain embodiments, allogeneic cell pools expressing a single immunomodulatory protein are used in the MLTR. In other embodiments, allogeneic cell pools expressing multiple immunomodulatory proteins are used in the MLTR.

The basic MLTR one day procedure is carried out as follows:

Frozen human PBMC are thawed. Cells are then washed in dPBS. PMBC cells are resuspended at 2.5×10^6 cell per ml stock in X-VIVO™ serum free media (Invitrogen). The cells are characterized by flow cytometry to document and confirm the phenotypic nature of the cell population.

The MLTR set up in the following fashion:

2.5×10^5 cell PBMC (100 μ l of stock)

0.5×10^5 allogeneic cells (100 μ l of stock), when used

Positive control 50 μ l of a 6 $\times$ stock (anti-CD28/CD3)

Total volume 300 μ l in a 96-well flat bottom—total volume of a 96-well is 300 μ l.

Incubate for 1 day.

100 μ l is removed for cytokine analysis by LUMINEX® multiplex assay

CyTOF is conducted on the cellular component.

CyTOF has been previously described, for example in Bendall et al. (Science, Vol. 332, 6 May 2011) and Bendall and Nolan (Nature Biotechnology, Vol. 30 No. 7, July 2012), both of which are incorporated by reference in their entireties herein. Human markers employed in CyTOF staining are shown below in Table 1.

TABLE 1

Human Markers for CyTOF Staining			
Marker	Clone	Metal	
HLA-DR	L243	89Y	
CD3	UCHT1	115In	
CD27	O323	141Pr	
CD19	HIB 19	142Nd	
CD134/OX40	Ber-ACT35	143Nd	
* Granzyme B	GB11	144Nd	
CD258/LIGHT	115520	145Nd	
CD8A	RPA t8	146Nd	
CD45RO	UCHL1	147Sm	
CD226/DNAM-1	11A8	149Sm	
CD194/CCR4	L291H4	150Nd	
PD1 (CD279)	EH12.2H7	151Eu	
CD170	1A5	152Sm	
CD69	FN50	153Eu	
CD70	113-16	154Sm	
CD4	RPA T4	155Gd	
CD8b	SID18BEE	156Gd	
IL-17R	W15177A	158Gd	
* CTLA-4 CD152	L3D10	159Tb	
CD278/ICOS	C398.4A	160Gd	
* AHR	FF3399	161Dy	
CD56	NCAM16.2	162Dy	
CD195/CCR5	J418F1	163Dy	
* Ki67	8D5	164Dy	
* FoxP3	Use Ebio	165Ho	
CD40	5C3	166Er	
* Helios	22F6	168Er	
* PU.1	puph13	169Tm	
* RORgt	1181A	170Er	
CD127/IL-7R	40131	171Yb	
CD38	HIT2	172Yb	
CD25	M-A251	173Yb	
CD86	IT2.2	174Yb	
* T-bet	4B10	175Lu	
* Perforin	dG9	176Yb	

* denotes intracellular targets while all others are cell surface targets

LUMINEX® Multiplex Assay

The LUMINEX® xMAP technology (formerly LabMAP, FlowMetrix) uses digital signal processing capable of classifying polystyrene beads (microspheres) dyed with distinct proportions of red and near-infrared fluorophores. These proportions define 'spectral addresses' for each bead population. As a result, up to one hundred different detection reactions can be carried out simultaneously on the various bead populations in very small sample volumes (Earley et al. Report from a Workshop on Multianalyte Microsphere Arrays. Cytometry 2002; 50:239-242; Oliver et al. Clin Chem 1998; 44(9):2057-2060; Eishal and McCoy, Methods 38(4): 317-323, April 2006, all of which are incorporated by reference in their entireties herein). The LUMINEX® Multiplex Assay is commercially available and is described at Thermofisher's official website, incorporated by reference in

its entirety herein. *Mitomycin C Preparation of Allogeneic Cells for MLTR Assay of Greater than 1 Day Duration.*

Mitomycin C is prepared from dry powder (2 mg per vial) using 400 μ l of DMSO (500 $\times$ stock=5 mg/ml), dissolved completely and aliquoted into 25 μ l volumes, and stored at -80C. 20 μ l of a single aliquot is used in 10 ml warmed C5 to yield 10 μ g/ml final working solution. The solution is filter sterilized.

The solution can be used on resuspended cells or adherent cells in flasks.

Cells are incubated at 37C for 30 minutes in the dark, then washed in warm C5 cell culture media (RPMI, supplemented with non-essential amino acids, glutamine, antibiotics and 5% Fetal calf serum) three times. Cells are resuspended in 1 ml X-VIVOTM, counted and final concentration adjusted to 1 $\times$ 10⁶/ml stock solution in X-VIVOTM (serum free media, Lonza).

Example 2

The described invention provides an approach for restoring immunologic balance in, for example, treating cancer, by targeting multiple immunomodulators with a single cellular platform. This approach enables the simultaneous modulation of multiple signals, and affords a spatially and temporally restricted method of modulating the immune response, an important feature that differentiates this methodology from traditional approaches using systemic administration of biologic agents to act on a single immunomodulatory pathway at a time.

According to one aspect of the disclosed invention, a tumor-type specific cell line variant expressing five or more recombinant peptides may be generated for use as a tumor cell vaccine to treat that cancer type. For example, a tumor cell line may be selected for modification and lentiviral transfection of recombinant immunomodulator sequences may be used to stably integrate immune modulators into the cell genome. Example 3 below describes 7 lentiviral vectors (vector 1, vector 2, vector 3, vector 4, vector 5, vector 6 and vector 7) that may be used to stably integrate immune modulators into the cell genome.

According to some embodiments, two recombinant immunomodulator proteins may be transfected simultaneously, followed by transfections of two more recombinant immunomodulator proteins simultaneously, followed by transfection of a single recombinant immunomodulator protein to achieve the total of five recombinant peptides for use as a tumor cell vaccine. According to some embodiments, two recombinant peptides may be transfected simultaneously, followed by transfection of a single recombinant peptide, followed by transfection of a single recombinant peptide to achieve the total of five recombinant peptides for use as a tumor cell vaccine. According to some embodiments, a single recombinant peptide is transfected, followed by transfection of two recombinant peptides simultaneously, followed by transfection of two recombinant peptides simultaneously to achieve a total of five recombinant peptide for use as a tumor cell vaccine.

According to one embodiment of the disclosed invention, combinations of allogeneic cell pools, each expressing a single immunomodulatory protein, are used to model what a single cell expressing multiple immunomodulatory proteins might do (e.g. additivity, synergy, interference).

According to one aspect of the disclosed invention, a tumor cell line variant expressing one, two, three, four, five or more recombinant peptides may be generated for use as

a tumor cell vaccine to treat skin cancer. For example, the SK-MEL2 human melanoma cell line (ATCC[®] HTB-68) may be selected for modification, and lentiviral transfection of recombinant immune modulator sequences may be used to stably integrate immune modulators into the cell genome.

According to one aspect of the disclosed invention, a tumor cell line variant expressing one, two, three, four, five or more recombinant peptides may be generated for use as a tumor cell vaccine to treat a prostate cancer. For example, the DU-145 human prostate carcinoma cell line may be selected for modification, and lentiviral transfection of recombinant immune modulator sequences may be used to stably integrate immune modulators into the cell genome. According to some embodiments, two recombinant immunomodulator proteins may be transfected simultaneously, followed by transfections of two more recombinant immunomodulator proteins simultaneously, followed by transfection of a single recombinant immunomodulator protein to achieve the total of five recombinant peptides for use as a tumor cell vaccine. According to some embodiments, two recombinant peptides may be transfected simultaneously, followed by transfection of a single recombinant peptide, followed by transfection of a single recombinant peptide, followed by transfection of a single recombinant peptide to achieve the total of five recombinant peptides for use as a tumor cell vaccine. According to some embodiments, a single recombinant peptide is transfected, followed by transfection of two recombinant peptides simultaneously, followed by transfection of two recombinant peptides simultaneously to achieve a total of five recombinant peptide for use as a tumor cell vaccine.

According to another aspect of the present invention, two or more tumor cell line variants expressing one or more recombinant peptides may be generated for use as a tumor cell vaccine to treat a prostate cancer. For example, the DU-145 and PC-3 human prostate carcinoma cell line may be selected for modification, and lentiviral transfection of recombinant immune modulator sequences may be used to stably integrate immune modulators into the cell genome. CD40L Immunomodulator

The CD40L immune modulator cDNA sequence may be cloned into the lentiviral transfer plasmid construct pLenti-puro (Addgene Cat. No. 39481) driven by a CMV promoter with puromycin selectable marker. The CD40L immune modulator cDNA sequence may be engineered to be non-cleavable, which ultimately keeps the translated CD40L protein in a membrane bound state (e.g. SEQ ID NO: 7).

Each of the lentiviral transfer plasmid, packaging plasmid, and envelope plasmid may be transfected into log phase growth 293T cells using LIPOFECTAMINETM 2000 (ThermoFisher Cat. No. 11668027). Briefly, cells are seeded at 70% to 90% confluence. On the day of transfection, 12 μ l of LIPOFECTAMINETM reagent is diluted in 150 μ l of serum free cell media. 5 μ g of DNA for transfection is also diluted in 150 μ l of serum free media. The diluted DNA is then added to the diluted LIPOFECTAMINETM and incubated for 5 minutes at room temperature. The total volume of the mixture is then added dropwise to the media of the seeded 293T cells while swirling. Cells are then incubated for one to three days at 37 degrees.

The 293T cell culture medium comprising virus particle is harvested 3 times every 8-12 hours and centrifuged to pellet detached cells and debris. The culture medium containing virus particles is used directly to infect the DU-145 cell line.

The DU-145 cell line is cultured in Eagle's Minimum Essential Medium (EMEM) with 10% fetal bovine serum to a confluency of about 70%. Hexadimethrine bromide

(Sigma-Aldrich Cat No. H9268) is then combined with media containing virus particles to make a final concentration of 8 g/mL Hexadimethrine bromide. Culture media of the DU-145 cells is aspirated and replaced with media containing virus particles and 8 g/mL Hexadimethrine bromide. DU-145 cells are cultured for 18-20 hours followed by media change.

Infected DU-145 cells are then grown in media containing 1 g/mL Puromycin (ThermoFisher Cat. No. A1113802) until cell die off begins after about a week. Multiple surviving colonies of transfected cells are picked for expansion and tested for CD40L expression by Western blot. The Western blot is probed with mouse monoclonal anti-HA primary antibodies (Abcam Cat. No. ab18181) and goat anti-mouse HRP (Abcam Cat. No. ab205719) secondary antibodies to quantify the relative amounts of recombinant CD40L expressed in each clonal line. The highest stably expressing DU-145 line is labeled DU145-Gen1 and selected for further manipulation.

TNF-alpha GM-CSF

The DU145-Gen1 cells transfected to express CD40L are further transfected with a bi-cistronic lentiviral vector comprising TNF-alpha and GM-CSF sequences. Each of TNF-alpha cDNA and GM-CSF cDNA is first cloned into the pEF1α-IRES bicistronic mammalian expression vector (Clontech Cat. No. 631970) under the control of the human elongation factor 1 alpha (EF1α) promoter. A variant of TNF-alpha that cannot be cleaved by TACE is used so that the translated protein remains in membrane bound form. The TNF-alpha sequence is provided with a FLAG tag sequence on the extracellular region of TNF-alpha for easy detection of translated protein. The FLAG tag peptide sequence is DYKDDDDK (SEQ ID NO: 29). GM-CSF sequences capable of forming soluble GM-CSF are used. The entirety of the pEF1 promoter, TNF-alpha sequences, IRES sequences, and GM-CSF sequences is then cloned into the pLenti-puro (Addgene Cat. No. 39481) lentiviral vector (the original CMV promoter from the vector is removed during this process). Packaging plasmid psPAX2 (AddGene Cat. No. 12260) and envelope plasmid pLTR-RD114A (AddGene Cat. No. 17576) are also selected.

Each of the lentiviral transfer plasmid, packaging plasmid, and envelope plasmid is transfected into log phase growth 293T cells using LIPOFECTAMINE™ 2000 (ThermoFisher Cat. No. 11668027). Briefly, cells are seeded at 70% to 90% confluence. On the day of transfection, 12 µl of LIPOFECTAMINE™ reagent is diluted in 150 µl of serum free cell media. 5 µg of DNA for transfection is also diluted in 150 µl of serum free media. The diluted DNA is then added to the diluted LIPOFECTAMINE™ and incubated for 5 minutes at room temperature. The total volume of the mixture is then added dropwise to the media of the seeded 293T cells while swirling. Cells are then incubated for one to three days at 37 degrees.

The 293T cell culture medium comprising virus particle is harvested 3 times every 8-12 hours and centrifuged to pellet detached cells and debris. The culture medium containing virus particles is used directly to infect the DU145-Gen1 cell line.

The DU145-Gen1 cell line is cultured to a confluency of about 70%. Hexadimethrine bromide (Sigma-Aldrich Cat No. H9268) is then combined with media containing virus particles to make a final concentration of 8 g/mL Hexadimethrine bromide. Culture media of the DU145-Gen1 cells is aspirated and replaced with media containing virus particles and 8 g/mL Hexadimethrine bromide. DU145-Gen1 cells are cultured for 18-20 hours followed by media change.

The transduced DU145-Gen1 cells are then selected for clones that stably express the recombinant immune modulators. The selection process is performed by fluorescence activated cell sorting using the FLAG tag on the TNF-alpha to identify cells that have integrated the immune modulators. Live cells are probed with mouse monoclonal anti-FLAG antibody (Sigma Aldrich F3040) and rabbit anti-mouse FITC conjugated secondary antibody (Sigma Aldrich ASB3701170) in PBS with blocking buffer. The highest expressing cells are sorted, isolated, and cultured for further processing. After sorting based on the presence of the FLAG tag, expression of soluble GM-CSF is confirmed by Western blot. Concentrated media of sorted cultured cells is resolved by SDS-PAGE and probed by Western blot with mouse anti-GM-CSF antibody (ThermoFisher Cat. No. 3092) and goat anti-mouse HRP conjugated secondary antibody. Cell lysate may also be resolved by SDS-PAGE and probed for FLAG tag to verify the presence of TNF. Cell cultures that express high levels of recombinant GM-CSF and TNF-alpha are designated DU145-Gen2 and selected for further processing.

Flt-3L

The DU145-Gen2 cells transfected to express CD40L, GM-CSF, and TNF are further transfected with a lentiviral vector comprising Flt-3L immune modulator sequences. The Flt-3L cDNA is cloned into a pEF1α-IRES bicistronic mammalian expression vector (Clontech Cat. No. 631970), along with GFP protein sequences to be used as a marker for integration and expression. The sequence of Flt-3L is translated into a membrane bound peptide, while the GFP remains cytoplasmic. The entirety of the pEF1 promoter, Flt-3L sequences, IRES sequences, and GFP sequences is then cloned into the pLenti-puro (Addgene Cat. No. 39481) lentiviral vector (the original CMV promoter from the vector is removed during this process). Packaging plasmid psPAX2 (AddGene Cat. No. 12260) and envelope plasmid pLTR-RD114A (AddGene Cat. No. 17576) are also selected.

Each of the lentiviral transfer plasmid, packaging plasmid, and envelope plasmid is transfected into log phase growth 293T cells using LIPOFECTAMINE™ 2000 (ThermoFisher Cat. No. 11668027). Briefly, cells are seeded at 70% to 90% confluence. On the day of transfection, 12 µl of LIPOFECTAMINE™ reagent is diluted in 150 µl of serum free cell media. 5 µg of DNA for transfection is also diluted in 150 µl of serum free media. The diluted DNA is then added to the diluted LIPOFECTAMINE™ and incubated for 5 minutes at room temperature. The total volume of the mixture is then added dropwise to the media of the seeded 293T cells while swirling. Cells are then incubated for one to three days at 37 degrees.

The 293T cell culture medium comprising virus particle is harvested 3 times every 8-12 hours and centrifuged to pellet detached cells and debris. The culture medium containing virus particles is used directly to infect the DU145-Gen2 cell line.

The DU145-Gen2 cell line is cultured to a confluency of about 70%. Hexadimethrine bromide (Sigma-Aldrich Cat No. H9268) is then combined with media containing virus particles to make a final concentration of 8 g/mL Hexadimethrine bromide. Culture media of the DU145-Gen2 cells is aspirated and replaced with media containing virus particles and 8 g/mL Hexadimethrine bromide. DU145-Gen2 cells are cultured for 18-20 hours followed by media change.

The DU145-Gen2 cells are then selected for cells stably expressing the Flt-3L sequences using the GFP marker. The selection process is performed by fluorescence activated cell sorting (FACS) using the GFP marker to identify cells that

have integrated the immune modulator. The highest expressing cells are sorted, isolated, and cultured for further processing. After sorting based on the presence of the GFP marker, the expression of Flt-3L is confirmed by Western blot. Cultured cell lysates are resolved by SDS-PAGE and probed by Western blot with rabbit polyclonal anti-Flt-3L antibody (AbCam Cat. No. ab9688) and goat anti-rabbit HRP conjugated secondary antibody (AbCam Cat. No. ab205718). Cell cultures that express high levels of recombinant Flt-3L are designated DU145-Gen3 and are selected for further processing.

IgG Heavy and Light Chains

The DU145-Gen3 cells transfected to express CD40L, GM-CSF, TNF-alpha, and Flt-3L are further transfected with a lentiviral vector comprising IgG 1 (SEQ ID NO: 1), a membrane bound IgG1 heavy chain fragment. The IgG1 heavy chain cDNA is cloned into pEF1α-IRES bicistronic mammalian expression vector (Clontech Cat. No. 631970), along with RFP protein sequences to be used as a marker for integration and expression. The sequence of IgG1 heavy chain is translated into a membrane bound peptide, while the RFP remains cytoplasmic. The entirety of the pEF1 promoter, IgG1 heavy chain sequence, IRES sequence, and RFP sequence is then cloned into the pLenti-puro (Addgene Cat. No. 39481) lentiviral vector (the original CMV promoter from the vector is removed during this process). Packaging plasmid psPAX2 (AddGene Cat. No. 12260) and envelope plasmid pLTR-RD 114A (AddGene Cat. No. 17576) are also selected.

Each of the lentiviral transfer plasmid, packaging plasmid, and envelope plasmid is transfected into log phase growth 293T cells using LIPOFECTAMINE™ 2000 (ThermoFisher Cat. No. 11668027). Briefly, cells are seeded at 70% to 90% confluence. On the day of transfection, 12 µl of LIPOFECTAMINE™ reagent is diluted in 150 µl of serum free cell media. 5 µg of DNA for transfection is also diluted in 150 µl of serum free media. The diluted DNA is then added to the diluted LIPOFECTAMINE™ and incubated for 5 minutes at room temperature. The total volume of the mixture is then added dropwise to the media of the seeded 293T cells while swirling. Cells are then incubated for one to three days at 37 degrees.

The 293T cell culture medium comprising virus particle is harvested 3 times every 8-12 hours and centrifuged to pellet detached cells and debris. The culture medium containing virus particles is used directly to infect the DU145-Gen3 cell line.

The DU145-Gen3 cell line is cultured to a confluency of about 70%. Hexadimethrine bromide (Sigma-Aldrich Cat No. H9268) is then combined with media containing virus particles to make a final concentration of 8 g/mL Hexadimethrine bromide. Culture media of the DU145-Gen2 cells is aspirated and replaced with media containing virus particles and 8 µg/mL Hexadimethrine bromide. DU145-Gen3 cells are cultured for 18-20 hours followed by media change.

The DU145-Gen3 cells are then selected for cells stably expressing the IgG1 heavy chain sequences using the RFP marker. The selection process is performed by fluorescence activated cell sorting (FACS) using the RFP marker to

identify cells that have integrated the immune modulator. The highest expressing cells are sorted, isolated, and cultured for further processing. After sorting based on the presence of the RFP marker, the expression of IgG1 heavy chain is confirmed by Western blot. Cell cultures that express high levels of recombinant IgG1 heavy chain are designated DU145-Gen4 and are selected for further processing.

The DU145-Gen4 tumor cell line transfected to express CD40L, GM-CSF, TNF, Flt-3L, and IgG1 heavy chain is characterized by RT-PCR, immunofluorescence, and Western blotting to confirm all recombinant immune modulators are expressed by the cells and are in the right location (e.g. on the membrane of the cell).

Human Mixed Lymphocyte Tumor Reaction (MLTR) Testing

The DU145-Gen4 cells are tested for their immunomodulatory potential by primary and secondary MLTR assay against each of the other generations (i.e. DU145-Gen2 and DU145-Gen3) of modified cells and unmodified DU145 cells.

Peripheral blood mononuclear cells (PBMCs) are obtained from the peripheral blood of healthy individuals and from prostate cancer patients, and the blood cells separated using a FICOLL-PAQUE® gradient. Anticoagulant-treated blood is diluted in the range of 1:2 to 1:4 with PBS/EDTA to reduce aggregation of erythrocytes. The diluted blood is then layered above a FICOLL-PAQUE® solution in a centrifuge tube, without mixing. The layered blood/FICOLL-PAQUE® is centrifuged for 40 minutes at 400×g between 18° and 20° C., without the use of the centrifuge brake, resulting in the formation of blood fractions. The fraction comprising mononuclear cells is selected for further processing.

Each of the cells from the transfected tumor cell line variants and from parental tumor cell line DU-145 (control) is co-cultured with PBMCs for seven days under standard tissue culture conditions, followed by evaluation for immune cell proliferation, immune cell differentiation, measured by flow cytometry and CyTOF, cytokine release profile, and cytotoxicity, measured by LDH release assay.

Example 3

A schematic of the core lentiviral vectors employed in the experiments described herein is shown in FIG. 2A and a schematic of the encoded proteins is shown in FIG. 2B. The promoter is human elongation factor 1 alpha (EF1α) promoter and the internal ribosomal entry sequence (IRES) is derived from encephalomyocarditis virus (EMCV). The core vectors are described in detail hereinbelow as follows:

Vector 1. Immunomodulator:
scFv-anti-biotin-G3hinge-mIgG1 (to Generate Surface IgG)

A schematic of the organization of vector 1, used for the immunomodulator scFv-anti-biotin-G3hinge-mIgG1 is shown in FIG. 3A. The nucleotide sequence of vector 1 (SEQ ID NO. 47) is shown in FIG. 3B. Table 2, below, shows the vector component name, the corresponding nucleotide position in SEQ ID NO. 47, the full name of the component and a description.

TABLE 2

Component Name	Nucleotide Position	Full Name	Description
RSV promoter	1-229	Rous sarcoma virus (RSV) enhancer/promoter	Allows Tat-independent production of viral mRNA.
Δ 5' LTR	230-410	HIV-1 truncated 5' LTR	Permits viral packaging and reverse transcription of the viral mRNA.
Ψ	521-565	HIV-1 psi packaging signal	Allows viral packaging.
RRET	1075-1308	HIV-1 Rev response element	Permits Rev-dependent nuclear export of unspliced viral mRNA.
cPPT	1803-1920	Central polypurine tract	Facilitates the nuclear import of HIV-1 cDNA through a central DNA flap.
EF1A	1959-3137	EF1A	Component entered by user
Kozak	3162-3167	Kozak	Component entered by user
{ORF1}	3168-5005	{ORF1}	Component entered by user
WPRES	5044-5641	Woodchuck hepatitis virus posttranscriptional regulatory element	Facilitates effective transcription termination at the 3'TLTR.T
Δ U3/3' LTR	5723-5957	HIV-1 truncated 3' LTR	Allows viral packaging but self-inactivates the 5'LTR for biosafety purposes. The element also contains apolyadenylation signal for transcription termination and polyadenylation of mRNA in transduced cells.
SV40 early pA	6030-6164	SV40 early polyadenation signal	Allows transcription termination and polyadenylation of mRNA.
Ampicillin	7118-7978	Ampicillin resistance gene	Allows selection of the plasmid in <i>E. coli</i> .
pUC ori	8149-8737	pUC origin of replication	Permits high-copy replication and maintenance in <i>E. coli</i> .

When vector 1 is employed, anti-IgG is used for flow detection. A biotin+fluorescent labelled oligodeoxynucleotides (ODN) is used as a secondary detection method.

The following is a description of the immunomodulator scFv-anti-biotin-G3hinge-IgG1-Tm.

Type:

Immunoglobulin

Annotation:

H7 heavy chain leader

Anti-biotin Variable Heavy chain (VH) allows for loading biotin labeled CpG

Inter-domain disulfide linkage VH44 (G->C) and VL100 (G->C)

IgG3 hinge to enhance FcyR interaction

Linkage is standard

IgG1 (CH2-CH3-Tm-Cyt) used for interaction with FcyR/FcRn and membrane anchoring

T233A mutation to enhance FcRn and FcyR interaction

The sequences are shown as follows:

H7 heavy chain leader

MEFGLSWVFLVALFRGVQC

anti-biotin murine vH with inserted Cys for inter-domain linkage

QVKLQESGPG LVAPSQSLSI TCTVSGFSLT AYGVVDWVRQP

PGKCLEWLGV IWGGGRNTYN SGLMSRLSIR KDNSKSQVFL

TMNSLQTDIT AKYYCVKHTN WDGGFAYWGQ GTTVTVSS

linker

GGGGSGGGGS GGGGS

Light Chain Variable (human lambda variable)

-continued

(SEQ ID NO. 57)

GSPGQSVSIS CSGSSSNIGN NYVYWYQHLP GTAPKLLIYS

DTKRPSGVDP RISGSKSGTS ASLAISGLQS EDEADYYCAS

WDDSLDGPVF GCGTKLTVL

IgG3 hinge for greater accessibility to FcyR

(SEQ ID NO. 58)

LKTPGLDTHTCPR CPEPKSCDTP PPCPRCPEPK SCDTPPPCPR

CPEPKSCDTP PPCPRCP

IgG1 CH2, CH3 Tm and cytoplasmic tail (T256A)

(SEQ ID NO. 59)

LLGGPSVFLF PPKPKDTLMI SRAPEVTCVV VDVSHEDPEV

45 KFNWYVDGVE VHNATKPRE EQYNSTYRVV SVLTVLHQDW

LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP

SRDELTKNQV SLTCLVKGFY PSDIAVEWES NGQPENNYKT

50 TPPVLDSGGS FFLYSKLTVD KSRWQQGNVF SCSVMHEALH

NHYTQKSLSL SPELQLEESC AEAQDGELDG LWTITITIFIT

LFLLSVCYSA TVTFFVKWVI FSSVVDLKQT IIPDYRNMIG

55 QGA*

The following shows the sequence of scFv-anti-biotin-G3hinge-IgG1-Tm (598 ORF1)

(SEQ ID NO. 60)

MEFGLSWVFLVALFRGVQCQVKLQESGPGVLVAPSQSLSIITCTVSGFSLT

60 AYGVVDWVRQPPGKCLEWLGVWGGGRNTYNSGLMSRLSIRKDNSKSQVF

LTMSLQTDITAKYYCVKHTNWDGGFAYWGQGTTVTVSSGGGGSGGGGS

GGGGSGSPGQSVSISCSGSSSNIGNNYVYWYQHLPGTAPKLLIYSDTKR

65 PSVPDRISGSKSGTSASLAISGLQSEDEADYYCASWDDSLDGPVFGCG

-continued

TKLTVLLKTPLGDTTHTCPRCPEPKSCDTPPPCPRCPEPKSCDTPPPCP
 RCPEPKSCDTPPPCPRCPAPELLGGPSVFLFPPKPKDTLMISRPEVTC
 VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREQYNSTYRVVSVLTVLH
 QDWLNGKEYCKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELT
 KNQVSLTCLVKGFPYSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYS
 KLTVDKSRWQQGNVFSQVMHEALHNHYTQKSLSLSPQLLEESCAEAQ
 DGELDGLWTTITITITLFLLSVCYSATVTFKVKWIFSSVVDLKQTIIP
 DYRNMIQQGA*

5

10

15

Vector 2. Immunomodulator: Full Anti-Biotin
 G3hinge-mIgG1 (Using Heavy Chain/Ires/Light
 Chain)

A schematic of the organization of vector 2, used for the
 immunomodulator full anti-biotin—G3hinge-mIgG1 is
 shown in FIG. 4A. Vector 2 is bicistronic. The nucleotide
 sequence of vector 2 (SEQ ID NO. 48) is shown in FIG. 4B.
 Table 3, below, shows the vector component name, the
 corresponding nucleotide position in SEQ ID NO. 48, the
 full name of the component and a description.

20

TABLE 3

Component Name	Nucleotide Position	Full Name	Description
RSV promoter	1-229	Rous sarcoma virus (RSV) enhancer/promoter	Allows Tat-independent production of viral mRNA.
Δ5' LTR	230-410	HIV-1 truncated 5' LTR	Permits viral packaging and reverse transcription of the viral mRNA.
Ψ	521-56g	HIV-1 psi packaging signal	Allows viral packaging.
RRET	1075-1308	HIV-1 Rev response element	Permits Rev-dependent nuclear export of unspliced viral mRNA.
cPPT	1803-1920	Central polypurine tract	Facilitates the nuclear import of HIV-1 cDNA through a central DNA flap.
EF1A	1950-3128	EF1A	Component entered by user
Kozak	3153-3158	Kozak	Component entered by user
{ORF1}	3159-5342	{ORF2A}	Component entered by user
WPRES	6703-7300	Woodchuck hepatitis virus posttranscriptional regulatory element	Facilitates effective transcription termination at the 3'TLTR.T
ΔU3/3' LTR	7382-7616	HIV-1 truncated 3' LTR	Allows viral packaging but self-inactivates the 5'LTR for biosafety purposes. The element also contains apolyadenylation signal for transcription termination and polyadenylation of mRNA in transduced cells.
SV40 early pA	7689-7823	SV40 early polyadenation signal	Allows transcription termination and polyadenylation of mRNA.
Ampicillin	8777-9637	Ampicillin resistance gene	Allows selection of the plasmid in <i>E. coli</i> .
pUC ori	9808-10396	pUC origin of replication	Permits high-copy replication and maintenance in <i>E. coli</i> .

When vector 2 is employed, anti-IgG is used for flow detection. Biotin+fluorescent labelled ODN is used as a secondary detection method.

The following is a description of the immunomodulator full anti-biotin—G3hinge-mIgG1 (using heavy chain/ires/light chain).

Type:

Membrane anchored Immunoglobulin

Annotation:

H7 heavy chain leader

IgG3 hinge to enhance FcγR interaction

T233A mutation to enhance FcγR and FcγR interaction
 Anti-biotin Variable H allows for loading biotin labeled
 CpG

CH1 (generic)

LC Variable (human lambda variable)

LC Constant Region 1 from Lambda

Interdomain disulfide linkage VH44 (G->C) and VL100 (G->C) (ref)

Linkage is standard

IgG1 (CH2-CH3-Tm-Cyt) for interaction with FcγR/
 FcγR and membrane anchoring

65

91

L1 light chain leader (modified for improved IRES expression)
MATDMRVPAQLLGLLLLWLSGARC (SEQ ID NO. 61)
The sequences are shown as follows:

H7 heavy chain leader (SEQ ID NO. 61)
MEFGLSWVFLVALFRGVQC
anti-biotin vH (murine) (SEQ ID NO. 62)
QVKLQESGPG LVAPSQSLSI TCTVSGFSLT AYGVVDWVRQP
PGKGLEWLGW IWGGGRNTYN SGLMSRLSIR KDNSKSQVFL
TMNSLQTDdT AKYYCVKHTN WDGGFAYWGQ GTTVTVSS
CH1 (generic) (SEQ ID NO. 63)
PSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV
HTFPAVLQSS GLYSLSSVVT VPSSSLGTQT YICNVNHKPS
NTKVDKKVE
IgG3 hinge for greater accessibility to FcγR (SEQ ID NO. 64)
LKTP LGDTTHTCPR CPEPKSCDTP PPCPRCPEPK SCDTPPPCPR
CPEPKSCDTP PPCPRCP
IgG1 CH2, CH3 Tm and cytoplasmic tail (T256A) (SEQ ID NO. 65)
APELLGGPSVFLF PPKPKDTLMI SRAPEVTCVV VDVSHEDPEV
KFNWYVDGVE VHNATKPRE EQNSTYRVV SVLTVLHQDW
LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP
SRDELTKNQV SLTCLVKGFY PSDIAVEWES NGQPENNYKT
TPPVLDSDGS FFLYSKLTVD KSRWQQGNVF SCSVMHEALH
NHYTQKSLSL SPELQLEESC AEAQDGELDG LWTITITIPIT
LFLLSVCYSA TVTFFKVKWI FSSVVDLKQT IIPDYRNMIG
QGA*
Summary (578 ORF2a) (SEQ ID NO. 66)
MEFGLSWVFLVALFRGVQCQVKLQESGPGLVAPSQSLSICTVSGFSLTA
YGVVDWVRQPPGKGLEWLGVIWGGGRNTYNSGLMSRLSIRKDNSKSQVFLT
MNSLQTDdTAKYYCVKHTNWDGGFAYWGQGTTVTVSSPSVFPPLAPSSKST
SGGTAALGCLVKDYFPEPVTVS WNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNHKPSNTKVDKKVELKTPLGDTTHTCPRCPEPK

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SCDTPPPCPRCPEPKSCDTPPPCPRCPEPKSCDTPPPCPRCPAPELLGGP
SVFLFPPKPKDTLMISRAPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAK
TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK
AKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPVLDSGSGFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNYHTQ
KSLSLSPQLLEESCAEAQDGELDLWTITITIFITLFLLSVCYSATVTFF
KVKWIFSSVVDLKQTIIPDYRNMIGQGA*
(SEQ ID NO. 67)
IRES
L1 Signal (modified to be IRES compatible) (SEQ ID NO. 68)
MATDMR VPAQLLGLLLLWLSGARC
LC Variable (human lambda variable) (SEQ ID NO. 69)
GSPGQSVSIS CSGSSSNIGN NYVYWYQHLP GTAPKLLIYS
DTKRPSGVPD RISGSKSGTS ASLAISGLOS EDEADYYCAS
WDDSLDGPVF GGGTKLTVL
LC Constant Region 1 from Lambda (irrelevant) (SEQ ID NO. 70)
GQPKANPTVT LFPSSSEELQ ANKATLVCLI SDFYPGAVTV
AWKADGSPVK AGVETTKPSK QSNNKYAASS YLSLTPEQWK
SHRSYSCQVT HEGSTVEKTV APTECS*
Summary (229 ORF2b) (SEQ ID NO. 71)
MATDMRVPAQLLGLLLLWLSGARCSPGQSVSISCSGSSSNIGNNYVYWY
QHLPGTAPKLLIYSDTKRPSGVPDRISGSKSGTSASLAISGLQSEADY
YCASWDDSLDGPVFGGGTKLTVLGQPKANPTVTLFPSSSEELQANKATLV
CLISDFYPGAVTVAWKADGSPVKAGVETTKPSKQSNNKYAASSYLSLTPE
QWKSHRSYSCQVTHEGSTVEKTVAPTECS*

Vector 3. Immunomodulator: sGM-CSF/ires/mFLT3L

A schematic of the organization of vector 3, used for the immunomodulator sGM-CSF/ires/mFLT3L is shown in FIG. 5A. Vector 3 is bicistronic. The nucleotide sequence of vector 3 (SEQ ID NO. 49) is shown in FIG. 5B. Table 4, below, shows the vector component name, the corresponding nucleotide position in SEQ ID NO. 49, the full name of the component and a description.

TABLE 4

Component Name	Nucleotide Position	Full Name	Description
RSV promoter	1-229	Rous sarcoma virus (RSV) enhancer/promoter	Allows Tat-independent production of viral mRNA.
Δ5' LTR	230-410	HIV-1 truncated 5' LTR	Permits viral packaging and reverse transcription of the viral mRNA.
Ψ	521-565	HIV-1 psi packaging signal	Allows viral packaging.
RRE	1075-1308	HIV-1 Rev response element	Permits Rev-dependent nuclear export of unspliced viral mRNA.
cPPT	1803-1920	Central polypurine tract	Facilitates the nuclear import of HIV-1 cDNA through a central DNA flap.

TABLE 4-continued

Component Name	Nucleotide Position	Full Name	Description
EF1A	1950-3128	EF1A	Component entered by user
Kozak	3153-3158	Kozak	Component entered by user
{ORF3A_wSPACER}	3159-4040	{ORF3A_wSPACER}	Component entered by user
IRES	4065-4652	IRES	Component entered by user
{ORF3B}	4653-5392	{ORF3B}	Component entered by user
WPRES	5422-6019	Woodchuck hepatitis virus posttranscriptional regulatory element	Facilitates effective transcription termination at the 3' LTR.
AU3/3' LTR	6101-6335	HIV-1 truncated 3' LTR	Allows viral packaging but self-inactivates the 5'LTR for biosafety purposes. The element also contains apolyadenylation signal for transcription termination and polyadenylation of mRNA in transduced cells.
SV40 early pA	6408-6542	SV40 early polyadenation signal	Allows transcription termination and polyadenylation of mRNA.
Ampicillin	7496-8356	Ampicillin resistance gene	Allows selection of the plasmid in <i>E. coli</i> .
pUC ori	8527-9115	pUC origin of replication	Permits high-copy replication and maintenance in <i>E. coli</i> .

When vector 3 is employed, anti-FLT3L is used for flow detection. The highest surface FLT3L expressor will have the highest secreted GM-CSF expression.

The following is a description of the immunomodulator sGM-CSF/ires/mFLT3L.

Type:

cytokine, growth and differentiation factor

Annotation:

wild-type sequence

The sequences are shown as follows:

GM-CSF signal sequence

MWLQSLLLLG TVACSI

wild type GM-CSF sequence

APA RSPSPSTQPW EHVNAIQEAR RLLNLSRDTA AEMNETVEVI

SEMFDLQEP T CLQTRLELYK QGLRGLSLTKL KGPLTMMASH

YKQHCPTPE TSCATQIITF ESFKENLKDF LL VIPFDCWE

PVQE*

IRES

FLT3L signal (modified to be IRES friendly)

MATVLAPAWSP TTYLLLLLLL SGLS

FLT3L

GTQDC SFQHSPISS FAVKIRELSD YLLQDYPVTV ASNQDEELC

GGLWRLVLAQ RWMERLKTVA GSKMQGLLER VNTEIHFVTK

CAFQPPPSCL RFVQTNISRL LQETSEQLVA LKPWITRQNF

SRCLQLQCQP DSSTLPPWSP PRPLEATAPT APQPPLLLLLL

LLPVGLLLLLL AAWCLHWQRT RRRTPRPGEQ VPPVPSPQDL

LLVEH*

-continued

Summary (144 ORF3a)

(SEQ ID NO. 77)

MWLQSLLLLGTVACSIAPARSPSPSTQPEWHVNAIQEARLLNLSRDT

30

AAEMNETVEVISEMFDLQEP T CLQTRLELYKQGLRGLSLTKLKGPLTMMASH

SHYQHCPTPETSCATQIITFESFKENLKDFLLVIPFDCWEPVQE*

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Summary (236 ORF3b)

(SEQ ID NO. 78)

MATVLAPAWSP TTYLLLLLLL SGLSGTQDCSFQHSPISSDFAVKIREL

40

SDYLLQDYPVTVASNLQDEELCGGLWRLVLAQRWMERLKTAVAGSKMQGL

LERVNTEIHFVTKCAFQPPPSCLRFVQTNISRLQLQETSEQLVALKPWIT

45

RQNF SRCLQLQCQPDSSTLPPWSPRPLEATAPTAPQPPLLLLLLLPVG

LLLLAAWCLHWQRT RRRTPRPGEQ VPPVPSPQDL LVEH*

50

Vector 4. Immunomodulator: sFLT3L/ires/(FLT3 signal-GM-CSF-Tm)

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A schematic of the organization of vector 4, used for the immunomodulator sFLT3L/ires/(FLT3 signal-GM-CSF-Tm) is shown in FIG. 6A. Vector 4 is bicistronic. The nucleotide sequence of vector 4 (SEQ ID NO. 50) is shown in FIG. 6B. Table 5, below, shows the vector component name, the corresponding nucleotide position in SEQ ID NO. 50, the full name of the component and a description.

TABLE 5

Component Name	Nucleotide Position	Full Name	Description
RSV promoter	1-229	Rous sarcoma virus (RSV) enhancer/promoter	Allows Tat-independent production of viral mRNA.
Δ5' LTR	230-410	HIV-1 truncated 5' LTR	Permits viral packaging and reverse transcription of the viral mRNA.
Ψ	521-565	HIV-1 psi packaging signal	Allows viral packaging.
RRE	1075-1308	HIV-1 Rev response element	Permits Rev-dependent nuclear export of unspliced viral mRNA.
cPPT	1803-1920	Central polypurine tract	Facilitates the nuclear import of HIV-1 cDNA through a central DNA flap.
EF1A	1950-3128	EF1A	Component entered by user
Kozak	3153-3158	Kozak	Component entered by user
{ORF4A_wSPACER}	3159-4157	{ORF4A_wSPACER}	Component entered by user
IRES	4182-4769	IRES	Component entered by user
{ORF4B}	4770-5557	{ORF4B}	Component entered by user
WPRE	5587-6184	Woodchuck hepatitis virus posttranscriptional regulatory element	Facilitates effective transcription termination at the 3' LTR.
ΔU3/3' LTR	6266-6500	HIV-1 truncated 3' LTR	Allows viral packaging but self-inactivates the 5'LTR for biosafety purposes. The element also contains apolyadenylation signal for transcription termination and polyadenylation of mRNA in transduced cells.
SV40 early pA	6573-6707	SV40 early polyadenation signal	Allows transcription termination and polyadenylation of mRNA.
Ampicillin	7661-8521	Ampicillin resistance gene	Allows selection of the plasmid in <i>E. coli</i> .
pUC ori	8692-9280	pUC origin of replication	Permits high-copy replication and maintenance in <i>E. coli</i> .

When vector 4 is employed, anti-GM-CSF is used for flow detection. The highest surface GMCSF expressor will have highest secreted FLT3L expression.

The following is a description of the immunomodulator sFLT3L/ires/(FLT3 signal-GM-CSF-Tm)

Type:

cytokine, growth and differentiation factor

Annotation:

wild-type sequence

The sequences are shown as follows:

wild type FLT3L sequence with transmembrane deleted

(SEQ ID NO. 79)

MTVLAPAWSP TTYLLLLLLL SSGLSGTQDC SFQHSPISSD

FAVKIRELSD YLLQDYPVTV ASNLQDEELC GGLWRLVLAQ

RWMERLKTVA GSKMQGLLER VNTEIHVFTK CAFQPPPSCL

RFVQTNISRL LQETSEQLVA LKPWITRQNF SRCLELQCQP

DSSTLPPPWS PRPLEATAPT APQ*

(SEQ ID NO. 80)

IRES

FLT3L signal (modified to be IRES friendly)

(SEQ ID NO. 81)

MATVLAPAWSP TTYLLLLLLL SSGLS

wild type GM-CSF sequence (minus native signal)

(SEQ ID NO. 82)

APA RSPSPSTQPW EHVNAIQEAR RLLNLSRDTA AEMNETVEVI

SEMFDLQEP T CLQTRLELYK QGLRGS LTKL KGPLTMASH

YKQHCPTTPE TSCATQIITF ESFKENLKDF LLVIPFDCWE

PVQE

-continued

CD8alpha transmembrane and cytoplasmic domain
(SEQ ID NO. 83)

PTTTP APRPPTPAPTASQPLSLRP EACRPAAGGA VHTRGLDFAC

DIYIWAPLAG TCGVLLLSLVITLYCNHRNR RRVCKCPRPV

VKSGDKPSLS ARYV*

40 Summary (183 ORF4a)

(SEQ ID NO. 84)

MTVLAPAWSP TTYLLLLLLL SSGLSGTQDCSFQHSPISSDFAVKIRELS

DYLLQDYPVTVASNLQDEELCGGLWRLVLAQRWMERLKTAVGSKMQGLL

45 ERVNTIEHFVTKCAFQPPPSCLRFVQTNISRL LQETSEQLVALKPWITR

QNF SRCLELQCQPDSTLPPPWSRPLEATAPTAPQ*

Summary for CYAGEN (253 ORF4b)

(SEQ ID NO. 85)

50 MATVLAPAWSP TTYLLLLLLL SSGLSAPARSPSPSTQWEHVNAIQEAR

RLLNLSRDTA AEMNETVEVISEMFDLQEP T CLQTRLELYKQGLRGS LTK

LKGPLTMASHYKQHCPTT PETSCATQIITFESFKENLKDFLLVIPFDC

55 WEPVQEPTTTTAPRPPTPAPTASQPLSLRPEACRPAAGGAVHTRGLDF

ACDIYIWAPLAGTCGVLLLSLVITLYCNHRNR RRVCKCPRPVVKSGDKP

SLSARYV*

Vector 5. Immunomodulator: mCD40L

A schematic of the organization of vector 5, used for the immunomodulator mCD40L is shown in FIG. 7A. Vector 5 is monocistronic. The nucleotide sequence of vector 5 (SEQ ID NO. 51) is shown in FIG. 7B. Table 6, below, shows the vector component name, the corresponding nucleotide position in SEQ ID NO. 51, the full name of the component and a description.

TABLE 6

Component Name	Nucleotide Position	Full Name	Description
RSV promoter	1-229	Rous sarcoma virus (RSV) enhancer/promoter	Allows Tat-independent production of viral mRNA.
Δ 5' LTR	230-410	HIV-1 truncated 5' LTR	Permits viral packaging and reverse transcription of the viral mRNA.
Ψ	521-565	HIV-1 psi packaging signal	Allows viral packaging.
RRE	1075-1308	HIV-1 Rev response element	Permits Rev-dependent nuclear export of unspliced viral mRNA.
cPPT	1803-1920	Central polypurine tract	Facilitates the nuclear import of HIV-1 cDNA through a central DNA flap.
EF1A	1959-3137	EF1A	Component entered by user
Kozak	3162-3167	Kozak	Component entered by user
{ORF5}	3168-3991	{ORF5}	Component entered by user
WPPE	4030-4627	Woodchuck hepatitis virus posttranscriptional regulatory element	Facilitates effective transcription termination at the 3' LTR.
Δ U3/3' LTR	4709-4943	HIV-1 truncated 3' LTR	Allows viral packaging but self-inactivates the 5'LTR for biosafety purposes. The element also contains apolyadenylation signal for transcription termination and polyadenylation of mRNA in transduced cells.
SV40 early pA	5016-5150	SV40 early polyadenation signal	Allows transcription termination and polyadenylation of mRNA.
Ampicillin	6104-6964	Ampicillin resistance gene	Allows selection of the plasmid in <i>E. coli</i> .
pUC ori	7135-7723	pUC origin of replication	Permits high-copy replication and maintenance in <i>E. coli</i> .

When Vector 5 is employed, anti-CD40L is used for flow detection.

The following is a description of the immunomodulator mCD40L.

Type:

TNF type II transmembrane protein

Annotation:

Mutations (UNDERLINED) introduced to make a non-cleavable version

The sequences are shown as follows:

Modified sequence to stop cleavage

(SEQ ID NO. 86)
MIETYNQTSP RSAATGLPIS MKIFMYLLTV FLITQMIGSA

LFAVYLHRRLL DKIEDERNLH EDFVFMKTIQ RCNTGERSLS

LLNCEEIKSQ FEGFVKDIML NKEETKKENS FEMPRCEEDS

QIAAHVISEA SSKTTSVLQW AEKGYYTMSN NLVTLENGKQ

LTVKRQGLYY IYAQVTFCSN REASSQAPFI ASLCLKSPGR

30

-continued

FERILLRAAN THSSAKPCGQ QSIHLGGVFE LQPGASVFN

VTDPSQVSHG TGFTSFGLLK L*

35

Summary (261 ORF5)

(SEQ ID NO. 87)

MIETYNQTSPRSAATGLPISMKIFMYLLTVFLITQMIGSALFAVYLHRR

LDKIEDERNLHEDFVFMKTIQRCNTGERSLSLLNCEEIKSQFEGFVKDI

40

MLNKEETKKENSFEMPRGEEDSQIAAHVISEASSKTTSVLQWAEKGYTT

MSNNLVTLENGKQLTVKRQGLYYIYAQVTFCSNREASSQAPFIASLCLK

SPGRFERILLRAANTHSSAKPCGQSQSIHLGGVFELQPGASVFNVTDPS

45

QVSHGTGFTSFGLLKL*

A schematic of the organization of vector 6, used for the immunomodulator mTNF α is shown in FIG. 8A. Vector 6 is monocistronic. The nucleotide sequence of vector 6 (SEQ TD NO. 52) is shown in FIG. 8B. Table 7, below, shows the vector component name, the corresponding nucleotide position in SEQ TD NO. 52, the full name of the component and a description.

TABLE 7

Component Name	Nucleotide Position	Full Name	Description
RSV promoter	1-229	Rous sarcoma virus (RSV) enhancer/promoter	Allows Tat-independent production of viral mRNA.
Δ 5' LTR	230-410	HIV-1 truncated 5' LTR	Permits viral packaging and reverse transcription of the viral mRNA.
Ψ	521-565	HIV-1 psi packaging signal	Allows viral packaging.
RRE	1075-1308	HIV-1 Rev response element	Permits Rev-dependent nuclear export of unspliced viral mRNA.
cPPT	1803-1920	Central polypurine tract	Facilitates the nuclear import of HIV-1 cDNA through a central DNA flap.

TABLE 7-continued

Component Name	Nucleotide Position	Full Name	Description
EF1A	1959-3137	EF1A	Component entered by user
Kozak	3162-3167	Kozak	Component entered by user
{ORF6}	3168-3871	{ORF6}	Component entered by user
WPPE	3910-4507	Woodchuck hepatitis virus posttranscriptional regulatory element	Facilitates effective transcription termination at the 3' LTR.
ΔU3/3' LTR	4859-4823	HIV-1 truncated 3' LTR	Allows viral packaging but self-inactivates the 5'LTR for biosafety purposes. The element also contains apolyadenylation signal for transcription termination and polyadenylation of mRNA in transduced cells.
SV40 early pA	4896-5030	SV40 early polyadenation signal	Allows transcription termination and polyadenylation of mRNA.
Ampicillin	5984-6844	Ampicillin resistance gene	Allows selection of the plasmid in <i>E. coli</i> .
pUC ori	7015-7603	pUC origin of replication	Permits high-copy replication and maintenance in <i>E. coli</i> .

When vector 6 is employed, anti-TNF α is used for flow detection.

The following is a description of the immunomodulator mTNF α .

Type:

TNF type II transmembrane protein

Annotation:

Mutations introduced to make a non-cleavable version, as described below in SEQ ID NO. 88

The sequences are shown as follows:

Modified to stop cleavage

(SEQ ID NO. 88)

MSTESMIRDV ELAEEALPKK TGGPQGSRRCLFLSLFSLIVAGATTLFC

VAGATTLFCL LHFVGIVGQR EEFPRDLSLI SPLAQVAHVANPQAEGLQWLNRRAN

.....VA HVVANPQAEGLQWLNRRAN ALLANGVELR

DNQLVVPSEG LYLIYSQVLF KGQGCPSHVL LLTHTISR

VSYQTKVNLL SAIKSPCQRE TPEGAEAKPW YEPIYLGGVF

QLEKGDRLSA EINRPDYLDFAESGQVYFGI IAL*

-continued

Summary (221 ORF6)

(SEQ ID NO. 89)

MSTESMIRDV ELAEEALPKK TGGPQGSRRCLFLSLFSLIVAGATTLFC

LLHFVGIVGQREEFPRDLSLI SPLAQVAHVANPQAEGLQWLNRRAN

ALLANGVELRDNQLVVPSEG LYLIYSQVLF KGQGCPSHVL LLTHTISR

AVSYQTKVNLL SAIKSPCQRE TPEGAEAKPW YEPIYLGGVF QLEKGDRL

SAEINRPDYLDFAESGQVYFGI IAL*

Vector 7. Immunomodulator: mRANKL/ires/FLT3 signal-V5-scFV anti-biotin-Tm

A schematic of the organization of vector 7, used for the immunomodulator mRANKL/ires/FLT3 signal-V5-scFV anti-biotin-Tm is shown in FIG. 9A. The nucleotide sequence of vector 7 (SEQ ID NO. 53) is shown in FIG. 9B. Table 8, below, shows the vector component name, the corresponding nucleotide position in SEQ ID NO. 53, the full name of the component and a description.

TABLE 8

Component Name	Nucleotide Position	Full Name	Description
RSV promoter	1-229	Rous sarcoma virus (RSV) enhancer/promoter	Allows Tat-independent production of viral mRNA.
Δ5' LTR	230-410	HIV-1 truncated 5' LTR	Permits viral packaging and reverse transcription of the viral mRNA.
Ψ	521-565	HIV-1 psi packaging signal	Allows viral packaging.
RRE	1075-1308	HIV-1 Rev response element	Permits Rev-dependent nuclear export of unspliced viral mRNA.
cPPT	1803-1920	Central polypurine tract	Facilitates the nuclear import of HIV-1 cDNA through a central DNA flap.
EF1A	1950-3128	EF1A	Component entered by user
Kozak	3153-3158	Kozak	Component entered by user
{ORF7_wSPACER}	3159-4091	{ORF7_wSPACER}	Component entered by user
IRE	4116-4703	IRE	Component entered by user
{ORF7B}	4704-5878	{ORF7B}	Component entered by user
WPPE	5908-6505	Woodchuck hepatitis virus posttranscriptional regulatory element	Facilitates effective transcription termination at the 3' LTR.

TABLE 8-continued

Component Name	Nucleotide Position	Full Name	Description
AU3/3' LTR	6587-3821	HIV-1 truncated 3' LTR	Allows viral packaging but self-inactivates the 5'LTR for biosafety purposes. The element also contains apolyadenylation signal for transcription termination and polyadenylation of mRNA in transduced cells.
SV40 early pA	6894-7028	SV40 early polyadenation signal	Allows transcription termination and polyadenylation of mRNA.
Ampicillin	7982-8842	Ampicillin resistance gene	Allows selection of the plasmid in <i>E. coli</i> .
pUC ori	9013-9601	pUC origin of replication	Permits high-copy replication and maintenance in <i>E. coli</i> .

When vector 7 is employed, anti-RANKL is used for flow detection. Anti-V5 mAb is used as a secondary detection method.

The following is a description of the immunomodulator mRANKL/ires/FLT3 signal-V5-scFV anti-biotin-Tm.

Type:

TNF type II transmembrane protein

Annotation:

wild-type sequence

The sequences are shown as follows:

wild-type

(SEQ ID NO. 90)
MDPNRISEDG THCIYRILRL HENADFQDTT LESQDTKLIP
DSCRRIKQAF QGAVQKELQH IVGSQHIAE KAMVDGSLWLD
LAKRSKLEAQ PFAHLTINAT DIPSGSHKVS LSSWYHDRGW
AKISNMTFSN GKLVNQDGF YYLYANICFR HHETSGDLAT
EYLQLMVYVT KTSIKIPSSH TLMKGGSTKY WSGNSEFHFY
SINVGGFFKL RSGEEISIEV SNPSLLDPDQ DATYFGAFKV
RDID*"

(SEQ ID NO. 91)
IRES

(SEQ ID NO. 92)
FLT3L signal (modified to be IRES friendly)
MATVLAPAWSP TTYLLLLLLL SSGLS

(SEQ ID NO. 93)
Linker
GGGS

(SEQ ID NO. 94)
V5 epitope tag for flow detection
GKPIPNNLLGLDST

(SEQ ID NO. 93)
Linker
GGGS

(SEQ ID NO. 95)
anti-biotin murine vH with inserted Cys for intralinkage
QVKLQESGPG LVAPSQSLSI TCTVSGFSLT AYGVWDVRQP
PGKLEWLGW IWGGGRTNYN SGLMSRLSIR KDNSKSVFL
TMNSLQTDAT AKYYCVKHTN WGGGFAYWQG GTTVTVSS

-continued

linker
20 GGGGSGGGGS GGGGS (SEQ ID NO. 96)
LC Variable (human lambda variable) (SEQ ID NO. 97)
GSPGQSVSIS CSGSSNIGN NYVYVYQHLP GTAPKLLIYS
25 DTKRPSGVPD RISGSKSGTS ASLAISGLQS EDEADYYCAS
WDDSLDGPVF GCGTKLTVL
CD8alpha transmembrane and cytoplasmic domain (SEQ ID NO. 98)
30 PTTTP APRPPTPAPTIASQPLSLRP EACRPAAGGA VHTRGLDFAC
DIYIWAPLAG TCGVLLLSLVITLYCNHRNR RRVCKCPRPV
VKSGDKPSLS ARYV*
35 Summary (244 ORF7a) (SEQ ID NO. 99)
MDPNRISEDGTHCIYRILRLHENADFQDTTLESQDTKLIPDSCRRIKQA
FQGAVQKELQHIVGSQHIAEKAMVDGSLWDLAKRSKLEAQPFPAHLTIN
40 ATDIPSGSHKVSLSWYHDRGWAKISNMTFSNGKLVNQDGFYYLYANI
CFRHETSGDLATEYLMVYVTKTSIKIPSSHTLMKGGSTKYWSGNSE
FHFYSINVGGFFKLRSAGEEISIEVSNPSLLDPDQDATYFGAFKVRDID*"
45 Summary (381aa ORF7b) (SEQ ID NO. 100)
MATVLAPAWSPTTYLLLLLLLSSGLSGGGGSGKPIPNNLLGLDSTGGGG
SQVKLQESGPGLVAPSQSLSICTVSGFSLTAYGVWDVRQPPGKCLEWL
GVIWGGGRTNYSGLMSRLSIRKDNSKSVFLTMNSLQTDATKYCVK
50 HTNWDGGFAYWQGTITVTVSSGGGSGGGGSGGGGSGSPGQSVSISCSG
SSSNIGNNYVYVYQHLPGTAPKLLIYSDTKRPSGVPDRISGSKSGTSAS
LAISGLQSEADYYCASWDDSLDGPVFGCGTKLTVLPTTTPAPRPPTP
55 APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLL
LSLVITLYCNHRNRNRVCKCPRPVVKSGDKPSLSARYV*

60 According to one embodiment, a tumor cell line is selected for modification, and vector 2 is used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 3 is used to stably integrate immune modulators into the cell genome.

65 According to one embodiment, a tumor cell line is selected for modification, and vector 4 is used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 5 is used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 6 is used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2 and vector 3 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2 and vector 4 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2 and vector 5 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 3 and vector 4 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 3 and vector 5 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 3 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3 and vector 4 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3 and vector 5 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 3, vector 4 and vector 5 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 3, vector 4 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3, vector 4 and vector 5 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3, vector 4 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3, vector 5 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 3, vector 4, vector 5 and vector 6 are used to stably integrate immune modulators into the cell genome.

According to one embodiment, a tumor cell line is selected for modification, and vector 2, vector 3, vector 4, vector 5 and vector 6 are used to stably integrate immune modulators into the cell genome.

Example 4

Experiments were carried out to demonstrate that the immunomodulators described herein, expressed on the tumor cell line SK-MEL2 differentially impact the proliferation and differentiation of human PBMC. FIG. 10 is a schematic that shows the general experimental design. The following allogeneic cell lines are tested:

SK-MEL (Parental line) ("SK")

SK modified with Vector 2 only ("2")

SK modified with Vector 3 only ("3")

SK modified with Vector 4 only ("4")

SK modified with Vector 6 only ("6")

SK modified with Vector 3 and Vector 4 ("3-4")

SK modified with Vector 3, Vector 4 and Vector 5 ("3-4-5")

SK modified with Vector 3, Vector 5 and Vector 6 ("3-5-6")

Functional characterization of the allogeneic cell lines was performed using a primary MLTR assay, as described herein. The MLTR assay was set up with 250,000 freshly thawed PBMC and 50,000 of select engineered allogeneic cell lines. The following outputs were measured: 1) Proliferation is measured by flow on CFSE labeled PMBC; 2) Differentiation is measured by CyTOF on unlabeled PMBC; 3) Cytokine profiling is performed by LUMINEX®. Flow Cytometry Data

The experiments described herein detect hPBMC activation from direct allorecognition of allogeneic cells versus a pan-T-cell activation using anti-CD3 and anti-CD28 mAbs. It was found that hPBMC activation via direct allorecognition of allogeneic cells displays a fundamentally different response compared to pan-T-cell activation with anti-CD3/CD28 treatment. Three key observations were made with regard to this differential hPBMC activation: 1) that ~10% of hPBMC proliferate in response to incubation with allogeneic cells compared to ~50% with anti-CD3/CD28 treatment; 2) that hPBMC proliferate through more cell divisions in response to activation with allogeneic cells compared to activation with anti-CD3/CD28 treatment; 3) that hPBMC take on a more highly varied morphology as measured by side scatter as compared to the more uniform cell morphology when hPBMC are stimulated with anti-CD3/CD28 treatment.

FIG. 11 is a panel of graphs that show the results of flow cytometry experiments. Forward (FSC) and side scatter (SSC) plots for size and granularity are shown. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF-alpha; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3, 4 and 5; and 3-4-6 is a combination of 3, 4 and 6. Cell lines 6, 3-4-5 and 3-4-6 display a larger and more granular phenotype likely owing to the presence of receptors for TNF-alpha and CD40L on cells of epithelial origin.

FIG. 12 is a panel of graphs that show representative flow cytometry stains for the indicated engineered surface markers; GM-CSF, FLT3L, TNF-a and CD40L. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a

non-cleavable form of CD40L; 6, a non-cleavable form of TNF-alpha; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3,4 and 5; and 3-4-6 is a combination of 3, 4 and 6.

FIG. 13 is a panel of graphs that show representative flow cytometry stains for the indicated engineered surface markers; GM-CSF, FLT3L, TNF-a and CD40L. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT-3L; 4, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF-a; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3,4 and 5; and 3-4-6 is a combination of 3,4 and 6.

CyTOF Data

CyTOF mass cytometry single-cell phenotype analysis of hPBMc response to SK melanoma cells with modification by expression of immunomodulatory factors is shown in FIG. 14A and FIG. 14B. The SK melanoma cell line and hPBMCs were cultured for 24 hours. Cells were harvested from cultures and stained with a 32-marker CyTOF antibody panel to detect multiple immune cell subsets as well as cell-surface and intracellular phenotyping markers. CyTOF mass cytometry data was generated on a Helios instrument. The data were normalized for signal using equilibration beads. Cell staining data were analyzed using Cytobank—a cloud computing suite for CyTOF data analysis that includes cell gating functions and an array of data visualization methods.

The data shown in FIG. 14A and FIG. 14B were plotted using viSNE, which is a dimensional reduction method that converts multidimensional staining signals from single cells into plots for visualization. FIG. 14A shows viSNE density contour plots of CyTOF staining data showing relative changes in immune cell subset abundance and phenotype. FIG. 14B shows single-cell phenotype analysis. viSNE density contour plots were generated by viSNE from ungated total PBMCs that were cultured with SK melanoma cells or modified SK melanoma cells. The plots illustrate relative changes in cell density for hPBMc immune cell subsets. The inserted viSNE plot identifies the immune cell subsets that are found within the clusters of the viSNE density plots. The arrows in the density contour plots point to the obvious changes in immune cell subsets between hPBMCs, SK cells, and the modified SK cells.

FIGS. 15A-FIG. 15D show CyTOF monocyte cluster analysis of hPBMc indicating changes in the activation markers CD40 (FIG. 15A), CD86 (FIG. 15B), CD69 (FIG. 15C) and CD25 (FIG. 15D) expression following 1 day stimulation with the indicated genetically modified SK lines at a 1:5 cell ratio. FIG. 15E shows CyTOF monocyte cluster analysis of hPBMc indicating relative median expression levels of monocyte markers CD40 and CD86. FIG. 15E shows CyTOF monocyte cluster analysis of hPBMc indicating relative median expression levels of CD4 T cell markers CD69 and CD25.

Cytokine Data

LUMINEX® multiplex cytokine profiling of human PBMC responses to SK parent line and genetically modified SK lines is shown in FIG. 16. SK cells or the indicated modified cell lines were cultured for 24 hours with human PBMCs at a 1:5 cell ratio. Control cultures included SK cells alone, hPBMCs alone, and hPBMCs stimulated with a mixture of anti-CD3 and anti-CD28 antibodies (1 pg/ml final concentration). Supernatants were screened for cytokine levels using a multiplexed LUMINEX® bead array assay to detect IL-1a, IL-1b, IL-1ra, IL-2, IL-4, IL-6, IL-8, IL-10,

IL-12p40, IL-12p70, IL-13, IL-17A, IL-23, TNFa, IFNg, G-CSF, GM-CSF, MIP1b, MCP-1, Rantes, Tweak, and TREM-1. Those cytokines found to be specifically induced by the SK parent line and modified SK lines are shown in the plots. Symbols indicate cytokine levels in pg/ml as estimated from a standard curve using recombinant cytokines. Absence of symbols indicates the cytokine was not detected. SK lines are represented by a number code; SK, unmodified parent line; 3, secreted GM-CSF and membrane expressed FLT-3L; 3, secreted FLT3L and membrane expressed GM-CSF; 5, a non-cleavable form of CD40L; 6, a non-cleavable form of TNF-alpha; 3-4 is a combination of 3 and 4; 3-4-5 is a combination of 3,4 and 5; and 3-4-6 is a combination of 3,4 and 6.

The described study provides a proof of concept that the complex combinatorial space of immunomodulators can be rapidly and efficiently assessed using an all human in vitro MLTR assay.

“Allorecognition” is a term used to define immunological recognition of histoincompatible antigens between genetically disparate individuals within the same species. “Direct allorecognition” is a mechanism by which recipient T cells recognize determinants on MHC-molecule-peptide complexes displayed on the surface of transplanted cells without the requirement for antigen processing by recipient APCs. The direct allorecognition response is detected early in the course of the MLTR assay (up to 1 day duration) during which time no antigen processing by host APC is required.

“Indirect allorecognition” refers to recognition of processed antigens of allogeneic cell origin presented by self-HLA on the surface of host antigen presenting cells. The indirect allorecognition response can be detected later in the course of the MLTR assay (greater than 3 days duration) during which time antigen processing by host APC has occurred.

Approximately 10% of peripheral blood T cells bear a TCR capable of allorecognition of the allogeneic tumor type specific cells used for vaccination. This is called “direct allorecognition” and occurs early in the course of events post vaccination. Direct allorecognition targets a T cell mediated immune response against the allogeneic cells resulting in their death and release of tumor type specific neoantigens (and shared normal antigens. These tumor neoantigens (and normal antigens) are taken up by host antigen presenting cells, processed and presented in the context of host HLA. This “indirect allorecognition” occurs late in the course of events post vaccination. The TCRs activated during indirect allorecognition are different from those involved earlier during direct allorecognition, but both processes occur in a local environment after exposure to allogeneic cells. The presence of genetically introduced immunomodulators on the allogeneic cells altered the allorecognition response in qualitative and quantitative ways.

Epitope spreading is a process of expanding an immune response to include distinct but closely related T cell epitopes. This is generally described as a maturation of the immune response. The differential maturation of the immune response against tumor neoantigens versus self-antigens is driven by the fact that tolerance mechanisms are in place to differentially protect against immune responses against self-antigens. While self-tolerance can be broken, it is more difficult than the response against a tumor neoantigen.

Without being limited by theory, since all tumors of a given type share many antigens, the T-cell mediated response initially driven by indirect allorecognition of the immune response will cross react against the host tumor of

the same type. According to some embodiments, since the tumor microenvironment may provide an insurmountable negative immunomodulatory hurdle, this approach may best be used in combination with checkpoint inhibitors in the setting of minimal residual disease after a debulking therapy (e.g. surgery, radiation or oncolytic viruses).

While the present invention has been described with reference to the specific embodiments thereof it should be

understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the invention. In addition, many modifications may be made to adopt a particular situation, material, composition of matter, process, process step or steps, to the objective spirit and scope of the present invention. All such modifications are intended to be within the scope of the claims appended hereto.

 SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 104

<210> SEQ ID NO 1

<211> LENGTH: 399

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: membrane IgG-1 heavy chain

<400> SEQUENCE: 1

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
1           5           10           15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
          20           25           30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
          35           40           45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
          50           55           60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
          65           70           75           80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
          85           90           95

Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys
          100          105          110

Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro
          115          120          125

Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys
          130          135          140

Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp
          145          150          155          160

Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu
          165          170          175

Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu
          180          185          190

His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn
          195          200          205

Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly
          210          215          220

Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu
          225          230          235          240

Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr
          245          250          255

Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn
          260          265          270

Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe
          275          280          285

Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn
          290          295          300

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-continued

Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
305 310 315 320

Gln Lys Ser Leu Ser Leu Ser Pro Glu Leu Gln Leu Glu Glu Ser Cys
325 330 335

Ala Glu Ala Gln Asp Gly Glu Leu Asp Gly Leu Trp Thr Thr Ile Thr
340 345 350

Ile Phe Ile Thr Leu Phe Leu Leu Ser Val Cys Tyr Ser Ala Thr Val
355 360 365

Thr Phe Phe Lys Val Lys Trp Ile Phe Ser Ser Val Val Asp Leu Lys
370 375 380

Gln Thr Ile Ile Pro Asp Tyr Arg Asn Met Ile Gly Gln Gly Ala
385 390 395

<210> SEQ ID NO 2

<211> LENGTH: 330

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: secreted IgG-1 heavy chain

<400> SEQUENCE: 2

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
1 5 10 15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
20 25 30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
35 40 45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
50 55 60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
65 70 75 80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
85 90 95

Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys
100 105 110

Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro
115 120 125

Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys
130 135 140

Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp
145 150 155 160

Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu
165 170 175

Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu
180 185 190

His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn
195 200 205

Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly
210 215 220

Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu
225 230 235 240

Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr
245 250 255

Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn
260 265 270

-continued

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Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe
   275                               280               285

Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn
   290                               295               300

Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
   305                               310               315               320

Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
           325               330

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<210> SEQ ID NO 3
<211> LENGTH: 295
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: fragment secreted IgG-1 heavy chain

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<400> SEQUENCE: 3

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
 1           5           10           15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20           25           30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35           40           45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50           55           60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
 65           70           75           80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
 85           90           95

Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys
100          105          110

Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro
115          120          125

Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys
130          135          140

Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp
145          150          155          160

Tyr Val Asp Gly Val Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu
165          170          175

Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
180          185          190

Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys
195          200          205

Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp
210          215          220

Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
225          230          235          240

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser
245          250          255

Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
260          265          270

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
275          280          285

Leu Ser Leu Ser Pro Gly Lys
290          295

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-continued

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<210> SEQ ID NO 4
<211> LENGTH: 377
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG-3 heavy chain constant region

<400> SEQUENCE: 4
Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
1      5      10      15
Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
20     25     30
Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
35     40     45
Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
50     55     60
Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
65     70     75     80
Tyr Thr Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
85     90     95
Arg Val Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro
100    105    110
Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg
115    120    125
Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys
130    135    140
Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro
145    150    155    160
Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
165    170    175
Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
180    185    190
Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Lys Trp Tyr
195    200    205
Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
210    215    220
Gln Tyr Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Leu His
225    230    235    240
Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
245    250    255
Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln
260    265    270
Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met
275    280    285
Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
290    295    300
Ser Asp Ile Ala Val Glu Trp Glu Ser Ser Gly Gln Pro Glu Asn Asn
305    310    315    320
Tyr Asn Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu
325    330    335
Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Ile
340    345    350
Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn Arg Phe Thr Gln
355    360    365

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-continued

Lys Ser Leu Ser Leu Ser Pro Gly Lys
370 375

<210> SEQ ID NO 5
<211> LENGTH: 184
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mGM-CSF HLA derived TM and SHORTENED
cytoplasmic domain

<400> SEQUENCE: 5

Met Trp Leu Gln Ser Leu Leu Leu Leu Gly Thr Val Ala Cys Ser Ile
1 5 10 15

Ser Ala Pro Ala Arg Ser Pro Ser Pro Ser Thr Gln Pro Trp Glu His
20 25 30

Val Asn Ala Ile Gln Glu Ala Arg Arg Leu Leu Asn Leu Ser Arg Asp
35 40 45

Thr Ala Ala Glu Met Asn Glu Thr Val Glu Val Ile Ser Glu Met Phe
50 55 60

Asp Leu Gln Glu Pro Thr Cys Leu Gln Thr Arg Leu Glu Leu Tyr Lys
65 70 75 80

Gln Gly Leu Arg Gly Ser Leu Thr Lys Leu Lys Gly Pro Leu Thr Met
85 90 95

Met Ala Ser His Tyr Lys Gln His Cys Pro Pro Thr Pro Glu Thr Ser
100 105 110

Cys Ala Thr Gln Ile Ile Thr Phe Glu Ser Phe Lys Glu Asn Leu Lys
115 120 125

Asp Phe Leu Leu Val Ile Pro Phe Asp Cys Trp Glu Pro Val Gln Glu
130 135 140

Glu Leu Ser Ser Gln Pro Thr Ile Pro Ile Val Gly Ile Ile Ala Gly
145 150 155 160

Leu Val Leu Leu Gly Ala Val Ile Thr Gly Ala Val Val Ala Ala Val
165 170 175

Met Trp Arg Arg Lys Ser Ser Asp
180

<210> SEQ ID NO 6
<211> LENGTH: 261
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD40L

<400> SEQUENCE: 6

Met Ile Glu Thr Tyr Asn Gln Thr Ser Pro Arg Ser Ala Ala Thr Gly
1 5 10 15

Leu Pro Ile Ser Met Lys Ile Phe Met Tyr Leu Leu Thr Val Phe Leu
20 25 30

Ile Thr Gln Met Ile Gly Ser Ala Leu Phe Ala Val Tyr Leu His Arg
35 40 45

Arg Leu Asp Lys Ile Glu Asp Glu Arg Asn Leu His Glu Asp Phe Val
50 55 60

Phe Met Lys Thr Ile Gln Arg Cys Asn Thr Gly Glu Arg Ser Leu Ser
65 70 75 80

Leu Leu Asn Cys Glu Glu Ile Lys Ser Gln Phe Glu Gly Phe Val Lys
85 90 95

-continued

Asp Ile Met Leu Asn Lys Glu Glu Thr Lys Lys Glu Asn Ser Phe Glu
 100 105 110
 Met Gln Lys Gly Asp Gln Asn Pro Gln Ile Ala Ala His Val Ile Ser
 115 120 125
 Glu Ala Ser Ser Lys Thr Thr Ser Val Leu Gln Trp Ala Glu Lys Gly
 130 135 140
 Tyr Tyr Thr Met Ser Asn Asn Leu Val Thr Leu Glu Asn Gly Lys Gln
 145 150 155 160
 Leu Thr Val Lys Arg Gln Gly Leu Tyr Tyr Ile Tyr Ala Gln Val Thr
 165 170 175
 Phe Cys Ser Asn Arg Glu Ala Ser Ser Gln Ala Pro Phe Ile Ala Ser
 180 185 190
 Leu Cys Leu Lys Ser Pro Gly Arg Phe Glu Arg Ile Leu Leu Arg Ala
 195 200 205
 Ala Asn Thr His Ser Ser Ala Lys Pro Cys Gly Gln Gln Ser Ile His
 210 215 220
 Leu Gly Gly Val Phe Glu Leu Gln Pro Gly Ala Ser Val Phe Val Asn
 225 230 235 240
 Val Thr Asp Pro Ser Gln Val Ser His Gly Thr Gly Phe Thr Ser Phe
 245 250 255
 Gly Leu Leu Lys Leu
 260

<210> SEQ ID NO 7
 <211> LENGTH: 261
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: non-cleavable CD40L

<400> SEQUENCE: 7

Met Ile Glu Thr Tyr Asn Gln Thr Ser Pro Arg Ser Ala Ala Thr Gly
 1 5 10 15
 Leu Pro Ile Ser Met Lys Ile Phe Met Tyr Leu Leu Thr Val Phe Leu
 20 25 30
 Ile Thr Gln Met Ile Gly Ser Ala Leu Phe Ala Val Tyr Leu His Arg
 35 40 45
 Arg Leu Asp Lys Ile Glu Asp Glu Arg Asn Leu His Glu Asp Phe Val
 50 55 60
 Phe Met Lys Thr Ile Gln Arg Cys Asn Thr Gly Glu Arg Ser Leu Ser
 65 70 75 80
 Leu Leu Asn Cys Glu Glu Ile Lys Ser Gln Phe Glu Gly Phe Val Lys
 85 90 95
 Asp Ile Met Leu Asn Lys Glu Glu Thr Lys Lys Glu Asn Ser Phe Glu
 100 105 110
 Met Pro Arg Gly Glu Glu Asp Ser Gln Ile Ala Ala His Val Ile Ser
 115 120 125
 Glu Ala Ser Ser Lys Thr Thr Ser Val Leu Gln Trp Ala Glu Lys Gly
 130 135 140
 Tyr Tyr Thr Met Ser Asn Asn Leu Val Thr Leu Glu Asn Gly Lys Gln
 145 150 155 160
 Leu Thr Val Lys Arg Gln Gly Leu Tyr Tyr Ile Tyr Ala Gln Val Thr
 165 170 175
 Phe Cys Ser Asn Arg Glu Ala Ser Ser Gln Ala Pro Phe Ile Ala Ser
 180 185 190

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Leu Cys Leu Lys Ser Pro Gly Arg Phe Glu Arg Ile Leu Leu Arg Ala
 195                200                205

Ala Asn Thr His Ser Ser Ala Lys Pro Cys Gly Gln Gln Ser Ile His
 210                215                220

Leu Gly Gly Val Phe Glu Leu Gln Pro Gly Ala Ser Val Phe Val Asn
 225                230                235                240

Val Thr Asp Pro Ser Gln Val Ser His Gly Thr Gly Phe Thr Ser Phe
      245                250                255

Gly Leu Leu Lys Leu
      260

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<210> SEQ ID NO 8
<211> LENGTH: 233
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TNF

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<400> SEQUENCE: 8

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Met Ser Thr Glu Ser Met Ile Arg Asp Val Glu Leu Ala Glu Glu Ala
 1          5          10          15

Leu Pro Lys Lys Thr Gly Gly Pro Gln Gly Ser Arg Arg Cys Leu Phe
      20          25          30

Leu Ser Leu Phe Ser Phe Leu Ile Val Ala Gly Ala Thr Thr Leu Phe
      35          40          45

Cys Leu Leu His Phe Gly Val Ile Gly Pro Gln Arg Glu Glu Phe Pro
      50          55          60

Arg Asp Leu Ser Leu Ile Ser Pro Leu Ala Gln Ala Val Arg Ser Ser
      65          70          75          80

Ser Arg Thr Pro Ser Asp Lys Pro Val Ala His Val Val Ala Asn Pro
      85          90          95

Gln Ala Glu Gly Gln Leu Gln Trp Leu Asn Arg Arg Ala Asn Ala Leu
      100         105         110

Leu Ala Asn Gly Val Glu Leu Arg Asp Asn Gln Leu Val Val Pro Ser
      115         120         125

Glu Gly Leu Tyr Leu Ile Tyr Ser Gln Val Leu Phe Lys Gly Gln Gly
      130         135         140

Cys Pro Ser Thr His Val Leu Leu Thr His Thr Ile Ser Arg Ile Ala
      145         150         155         160

Val Ser Tyr Gln Thr Lys Val Asn Leu Leu Ser Ala Ile Lys Ser Pro
      165         170         175

Cys Gln Arg Glu Thr Pro Glu Gly Ala Glu Ala Lys Pro Trp Tyr Glu
      180         185         190

Pro Ile Tyr Leu Gly Gly Val Phe Gln Leu Glu Lys Gly Asp Arg Leu
      195         200         205

Ser Ala Glu Ile Asn Arg Pro Asp Tyr Leu Asp Phe Ala Glu Ser Gly
      210         215         220

Gln Val Tyr Phe Gly Ile Ile Ala Leu
      225         230

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<210> SEQ ID NO 9
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD40L piece

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<400> SEQUENCE: 9

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Met Ile Glu Thr Tyr Asn Gln Thr Ser Pro Arg Ser Ala Ala Thr Gly
1           5           10          15
Leu Pro Ile Ser Met Lys Ile Phe Met Tyr Leu Leu Thr Val Phe Leu
          20           25           30
Ile Thr Gln Met Ile Gly Ser Ala Leu Phe Ala Val Tyr Leu His Arg
          35           40           45
Arg Leu Asp Lys Ile Glu Asp Glu Arg Asn Leu His Glu Asp Phe Val
          50           55           60
Phe Met Lys Thr Ile Gln Arg Cys Asn Thr Gly Glu Arg Ser Leu Ser
65           70           75           80
Leu Leu Asn Cys Glu Glu Ile Lys Ser Gln Phe Glu Gly Phe Val Lys
          85           90           95
Asp Ile Met Leu Asn Lys Glu Glu Thr Lys Lys Glu
          100          105

```

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<210> SEQ ID NO 10
<211> LENGTH: 141
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TNF piece

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<400> SEQUENCE: 10

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Val Ala Asn Pro Gln Ala Glu Gly Gln Leu Gln Trp Leu Asn Arg Arg
1           5           10          15
Ala Asn Ala Leu Leu Ala Asn Gly Val Glu Leu Arg Asp Asn Gln Leu
          20           25           30
Val Val Pro Ser Glu Gly Leu Tyr Leu Ile Tyr Ser Gln Val Leu Phe
          35           40           45
Lys Gly Gln Gly Cys Pro Ser Thr His Val Leu Leu Thr His Thr Ile
          50           55           60
Ser Arg Ile Ala Val Ser Tyr Gln Thr Lys Val Asn Leu Leu Ser Ala
65           70           75           80
Ile Lys Ser Pro Cys Gln Arg Glu Thr Pro Glu Gly Ala Glu Ala Lys
          85           90           95
Pro Trp Tyr Glu Pro Ile Tyr Leu Gly Gly Val Phe Gln Leu Glu Lys
          100          105          110
Gly Asp Arg Leu Ser Ala Glu Ile Asn Arg Pro Asp Tyr Leu Asp Phe
          115          120          125
Ala Glu Ser Gly Gln Val Tyr Phe Gly Ile Ile Ala Leu
          130          135          140

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<210> SEQ ID NO 11
<211> LENGTH: 233
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mTNF-a

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<400> SEQUENCE: 11

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```

Met Ser Thr Glu Ser Met Ile Arg Asp Val Glu Leu Ala Glu Glu Ala
1           5           10          15
Leu Pro Lys Lys Thr Gly Gly Pro Gln Gly Ser Arg Arg Cys Leu Phe
          20           25           30
Leu Ser Leu Phe Ser Phe Leu Ile Val Ala Gly Ala Thr Thr Leu Phe
          35           40           45
Cys Leu Leu His Phe Gly Val Ile Gly Pro Gln Arg Glu Glu Phe Pro

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50	55	60
Arg Asp Leu Ser Leu Ile Ser Pro Leu Ala Gln Ala Ser Gly Ser Gly		
65	70	75 80
Ser Gly Ser Gly Ser Gly Glu Pro Val Ala His Val Val Ala Asn Pro		
	85	90 95
Gln Ala Glu Gly Gln Leu Gln Trp Leu Asn Arg Arg Ala Asn Ala Leu		
	100	105 110
Leu Ala Asn Gly Val Glu Leu Arg Asp Asn Gln Leu Val Val Pro Ser		
	115	120 125
Glu Gly Leu Tyr Leu Ile Tyr Ser Gln Val Leu Phe Lys Gly Gln Gly		
	130	135 140
Cys Pro Ser Thr His Val Leu Leu Thr His Thr Ile Ser Arg Ile Ala		
	145	150 155 160
Val Ser Tyr Gln Thr Lys Val Asn Leu Leu Ser Ala Ile Lys Ser Pro		
	165	170 175
Cys Gln Arg Glu Thr Pro Glu Gly Ala Glu Ala Lys Pro Trp Tyr Glu		
	180	185 190
Pro Ile Tyr Leu Gly Gly Val Phe Gln Leu Glu Lys Gly Asp Arg Leu		
	195	200 205
Ser Ala Glu Ile Asn Arg Pro Asp Tyr Leu Asp Phe Ala Glu Ser Gly		
	210	215 220
Gln Val Tyr Phe Gly Ile Ile Ala Leu		
	225	230

<210> SEQ ID NO 12
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: HC Constant Region 3 from IgG1

<400> SEQUENCE: 12

Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp		
1	5	10 15
Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe		
	20	25 30
Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu		
	35	40 45
Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe		
	50	55 60
Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly		
	65	70 75 80
Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr		
	85	90 95
Thr Gln Lys Ser Leu Ser Leu Ser Pro Glu Leu Gln Leu Glu Glu Ser		
	100	105 110

Cys

<210> SEQ ID NO 13
 <211> LENGTH: 144
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: native GM-CSF

<400> SEQUENCE: 13

Met Trp Leu Gln Ser Leu Leu Leu Leu Gly Thr Val Ala Cys Ser Ile

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1	5	10	15
Ser Ala Pro	Ala Arg Ser Pro	Ser Pro Ser Thr Gln Pro	Trp Glu His
	20	25	30
Val Asn Ala	Ile Gln Glu Ala Arg	Arg Leu Leu Asn Leu Ser	Arg Asp
	35	40	45
Thr Ala Ala	Glu Met Asn Glu Thr Val	Glu Val Ile Ser Glu Met	Phe
	50	55	60
Asp Leu Gln	Glu Pro Thr Cys Leu Gln Thr	Arg Leu Glu Leu Tyr Lys	
65	70	75	80
Gln Gly Leu	Arg Gly Ser Leu Thr Lys	Leu Lys Gly Pro Leu Thr	Met
	85	90	95
Met Ala Ser	His Tyr Lys Gln His Cys Pro	Pro Thr Pro Glu Thr Ser	
	100	105	110
Cys Ala Thr	Gln Ile Ile Thr Phe Glu Ser	Phe Lys Glu Asn Leu Lys	
	115	120	125
Asp Phe Leu	Leu Val Ile Pro Phe Asp Cys Trp	Glu Pro Val Gln Glu	
	130	135	140

<210> SEQ ID NO 14
 <211> LENGTH: 235
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Flt3L

<400> SEQUENCE: 14

Met Thr Val	Leu Ala Pro Ala Trp Ser	Pro Thr Thr Tyr Leu Leu Leu
1	5	10 15
Leu Leu Leu	Leu Ser Ser Gly Leu Ser Gly Thr	Gln Asp Cys Ser Phe
	20	25 30
Gln His Ser	Pro Ile Ser Ser Asp Phe Ala Val	Lys Ile Arg Glu Leu
	35	40 45
Ser Asp Tyr	Leu Leu Gln Asp Tyr Pro Val Thr	Val Ala Ser Asn Leu
	50	55 60
Gln Asp Glu	Glu Leu Cys Gly Gly Leu Trp Arg	Leu Val Leu Ala Gln
65	70	75 80
Arg Trp Met	Glu Arg Leu Lys Thr Val Ala Gly	Ser Lys Met Gln Gly
	85	90 95
Leu Leu Glu	Arg Val Asn Thr Glu Ile His Phe	Val Thr Lys Cys Ala
	100	105 110
Phe Gln Pro	Pro Pro Ser Cys Leu Arg Phe Val	Gln Thr Asn Ile Ser
	115	120 125
Arg Leu Leu	Gln Glu Thr Ser Glu Gln Leu Val	Ala Leu Lys Pro Trp
	130	135 140
Ile Thr Arg	Gln Asn Phe Ser Arg Cys Leu Glu	Leu Gln Cys Gln Pro
145	150	155 160
Asp Ser Ser	Thr Leu Pro Pro Pro Trp Ser Pro	Arg Pro Leu Glu Ala
	165	170 175
Thr Ala Pro	Thr Ala Pro Gln Pro Pro Leu Leu	Leu Leu Leu Leu
	180	185 190
Pro Val Gly	Leu Leu Leu Leu Ala Ala Ala Trp	Cys Leu His Trp Gln
	195	200 205
Arg Thr Arg	Arg Arg Thr Pro Arg Pro Gly Glu	Gln Val Pro Pro Val
	210	215 220

Pro Ser Pro Gln Asp Leu Leu Leu Val Glu His

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225 230 235

<210> SEQ ID NO 15
 <211> LENGTH: 13
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Kozak sequence
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (7)..(7)
 <223> OTHER INFORMATION: n is A or G

<400> SEQUENCE: 15

gccgccncca ugg

<210> SEQ ID NO 16
 <211> LENGTH: 377
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG-3 heavy chain constant region E213Q

<400> SEQUENCE: 16

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
 1 5 10 15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20 25 30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35 40 45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50 55 60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
 65 70 75 80

Tyr Thr Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
 85 90 95

Arg Val Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro
 100 105 110

Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg
 115 120 125

Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys
 130 135 140

Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro
 145 150 155 160

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
 165 170 175

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
 180 185 190

Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Lys Trp Tyr
 195 200 205

Val Asp Gly Val Gln Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
 210 215 220

Gln Tyr Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Leu His
 225 230 235 240

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
 245 250 255

Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln
 260 265 270

13

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Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met
   275                               280                               285

Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
   290                               295                               300

Ser Asp Ile Ala Val Glu Trp Glu Ser Ser Gly Gln Pro Glu Asn Asn
   305                               310                               315                               320

Tyr Asn Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu
                               325                               330                               335

Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Ile
   340                               345                               350

Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn Arg Phe Thr Gln
   355                               360                               365

Lys Ser Leu Ser Leu Ser Pro Gly Lys
   370                               375

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<210> SEQ ID NO 17
<211> LENGTH: 377
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG-3 heavy chain constant region P221L

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<400> SEQUENCE: 17

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
 1                               5                               10                               15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
   20                               25                               30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
   35                               40                               45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
   50                               55                               60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
   65                               70                               75                               80

Tyr Thr Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
   85                               90                               95

Arg Val Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro
   100                               105                               110

Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg
   115                               120                               125

Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys
   130                               135                               140

Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro
   145                               150                               155                               160

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
   165                               170                               175

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
   180                               185                               190

Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Lys Trp Tyr
   195                               200                               205

Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Leu Arg Glu Glu
   210                               215                               220

Gln Tyr Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Leu His
   225                               230                               235                               240

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
   245                               250                               255

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Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln
			260					265					270		
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met
		275					280					285			
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
		290				295					300				
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Ser	Gly	Gln	Pro	Glu	Asn	Asn
305					310					315					320
Tyr	Asn	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
				325					330					335	
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Ile
			340					345					350		
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	Arg	Phe	Thr	Gln
		355					360					365			
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys							
		370				375									

<210> SEQ ID NO 18
 <211> LENGTH: 377
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG-3 heavy chain constant region E224Q

<400> SEQUENCE: 18

Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg
1				5					10					15	
Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr
			20					25					30		
Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser
		35					40				45				
Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser
		50				55					60				
Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr
65					70					75					80
Tyr	Thr	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys
			85					90						95	
Arg	Val	Glu	Leu	Lys	Thr	Pro	Leu	Gly	Asp	Thr	Thr	His	Thr	Cys	Pro
			100					105					110		
Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg
		115					120					125			
Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys
		130				135						140			
Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro
145					150					155					160
Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys
				165					170					175	
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val
			180					185					190		
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Lys	Trp	Tyr
		195					200					205			
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Gln
		210				215					220				
Gln	Tyr	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
225					230					235					240

-continued

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
245 250 255

Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln
260 265 270

Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met
275 280 285

Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
290 295 300

Ser Asp Ile Ala Val Glu Trp Glu Ser Ser Gly Gln Pro Glu Asn Asn
305 310 315 320

Tyr Asn Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu
325 330 335

Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Ile
340 345 350

Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn Arg Phe Thr Gln
355 360 365

Lys Ser Leu Ser Leu Ser Pro Gly Lys
370 375

<210> SEQ ID NO 19
 <211> LENGTH: 377
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG-3 heavy chain constant region Y226F

<400> SEQUENCE: 19

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
1 5 10 15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
20 25 30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
35 40 45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
50 55 60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
65 70 75 80

Tyr Thr Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
85 90 95

Arg Val Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro
100 105 110

Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg
115 120 125

Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys
130 135 140

Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro
145 150 155 160

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
165 170 175

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
180 185 190

Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Lys Trp Tyr
195 200 205

Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
210 215 220

-continued

Gln Phe Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Leu His
 225 230 235 240

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
 245 250 255

Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln
 260 265 270

Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met
 275 280 285

Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
 290 295 300

Ser Asp Ile Ala Val Glu Trp Glu Ser Ser Gly Gln Pro Glu Asn Asn
 305 310 315 320

Tyr Asn Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu
 325 330 335

Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Ile
 340 345 350

Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn Arg Phe Thr Gln
 355 360 365

Lys Ser Leu Ser Leu Ser Pro Gly Lys
 370 375

<210> SEQ ID NO 20
 <211> LENGTH: 377
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG-3 heavy chain constant region D242N

<400> SEQUENCE: 20

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
 1 5 10 15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20 25 30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35 40 45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50 55 60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
 65 70 75 80

Tyr Thr Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
 85 90 95

Arg Val Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro
 100 105 110

Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg
 115 120 125

Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys
 130 135 140

Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro
 145 150 155 160

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
 165 170 175

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
 180 185 190

Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Lys Trp Tyr
 195 200 205

-continued

Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
210						215					220				
Gln	Tyr	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
225					230					235					240
Gln	Asn	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
			245						250					255	
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln
			260					265					270		
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met
		275					280					285			
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
	290					295					300				
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Ser	Gly	Gln	Pro	Glu	Asn	Asn
305					310					315					320
Tyr	Asn	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
				325					330					335	
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Ile
			340					345					350		
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	Arg	Phe	Thr	Gln
		355					360					365			
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys							
	370					375									

<210> SEQ ID NO 21
 <211> LENGTH: 377
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG-3 heavy chain constant region N245D

<400> SEQUENCE: 21

Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg
1				5					10					15	
Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr
			20					25					30		
Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser
		35					40				45				
Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser
	50					55					60				
Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr
65					70					75				80	
Tyr	Thr	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys
			85					90						95	
Arg	Val	Glu	Leu	Lys	Thr	Pro	Leu	Gly	Asp	Thr	Thr	His	Thr	Cys	Pro
			100					105					110		
Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg
		115					120					125			
Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys
	130					135						140			
Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro
145					150					155					160
Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys
				165					170					175	
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val
				180				185						190	

-continued

Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Lys	Trp	Tyr
	195						200					205			
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
	210					215					220				
Gln	Tyr	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
	225				230					235					240
Gln	Asp	Trp	Leu	Asp	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
			245						250					255	
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln
			260					265					270		
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met
		275					280					285			
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
	290					295					300				
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Ser	Gly	Gln	Pro	Glu	Asn	Asn
	305				310					315					320
Tyr	Asn	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
			325						330					335	
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Ile
			340				345						350		
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	Arg	Phe	Thr	Gln
		355					360					365			
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys							
	370					375									

<210> SEQ ID NO 22

<211> LENGTH: 377

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: IgG-3 heavy chain constant region T269A

<400> SEQUENCE: 22

Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg
1				5					10					15	
Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr
			20					25					30		
Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser
		35					40				45				
Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser
	50					55					60				
Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr
	65				70				75					80	
Tyr	Thr	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys
			85					90						95	
Arg	Val	Glu	Leu	Lys	Thr	Pro	Leu	Gly	Asp	Thr	Thr	His	Thr	Cys	Pro
			100					105					110		
Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg
		115					120					125			
Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys
	130					135						140			
Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro
	145				150					155					160
Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys
			165					170							175

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Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	
			180					185					190			
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Lys	Trp	Tyr	
		195					200				205					
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	
	210					215					220					
Gln	Tyr	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	
225					230					235					240	
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	
			245						250					255		
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	
			260					265					270			
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	
		275					280					285				
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	
	290					295					300					
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Ser	Gly	Gln	Pro	Glu	Asn	Asn	
305					310					315					320	
Tyr	Asn	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	
				325					330					335		
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Ile	
			340				345						350			
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	Arg	Phe	Thr	Gln	
		355					360					365				
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys								
	370					375										

<210> SEQ ID NO 23
 <211> LENGTH: 377
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG-3 heavy chain constant region S314N

<400> SEQUENCE: 23

Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	
1				5					10					15		
Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	
			20					25					30			
Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	
		35					40				45					
Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	
	50					55					60					
Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	
65					70				75					80		
Tyr	Thr	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	
			85					90						95		
Arg	Val	Glu	Leu	Lys	Thr	Pro	Leu	Gly	Asp	Thr	Thr	His	Thr	Cys	Pro	
		100						105					110			
Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	
		115					120					125				
Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	
	130					135						140				
Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro	
145					150					155					160	

Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	
				165					170						175	
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	
			180					185						190		
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Lys	Trp	Tyr	
		195					200					205				
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	
	210					215					220					
Gln	Tyr	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	
225					230					235					240	
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	
			245						250					255		
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln	
			260					265						270		
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	
		275					280					285				
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	
	290					295					300					
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	
305					310					315					320	
Tyr	Asn	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	
				325					330					335		
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Ile	
			340					345					350			
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	Arg	Phe	Thr	Gln	
		355					360					365				
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys								
	370					375										
<210> SEQ ID NO 24																
<211> LENGTH: 376																
<212> TYPE: PRT																
<213> ORGANISM: Artificial Sequence																
<220> FEATURE:																
<223> OTHER INFORMATION: IgG-3 heavy chain constant region S314 del																
<400> SEQUENCE: 24																
Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	
1				5					10					15		
Ser	Thr	Ser	Gly	Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	

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Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro
145          150          155          160

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
          165          170          175

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
          180          185          190

Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Lys Trp Tyr
          195          200          205

Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
          210          215          220

Gln Tyr Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Leu His
225          230          235          240

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
          245          250          255

Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln
          260          265          270

Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met
          275          280          285

Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
          290          295          300

Ser Asp Ile Ala Val Glu Trp Glu Ser Gly Gln Pro Glu Asn Asn Tyr
305          310          315          320

Asn Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr
          325          330          335

Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Ile Phe
          340          345          350

Ser Cys Ser Val Met His Glu Ala Leu His Asn Arg Phe Thr Gln Lys
          355          360          365

Ser Leu Ser Leu Ser Pro Gly Lys
          370          375

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<210> SEQ ID NO 25
<211> LENGTH: 377
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG-3 heavy chain constant region F366Y

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<400> SEQUENCE: 25

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
1          5          10          15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
          20          25          30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
          35          40          45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
          50          55          60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
65          70          75          80

Tyr Thr Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
          85          90          95

Arg Val Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro
          100          105          110

Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg
          115          120          125

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Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys
130						135					140				
Pro	Glu	Pro	Lys	Ser	Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro
145					150					155					160
Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys
				165					170					175	
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val
			180					185					190		
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Lys	Trp	Tyr
		195					200					205			
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
	210					215					220				
Gln	Tyr	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
225					230					235					240
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
			245						250					255	
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln
			260					265					270		
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met
		275					280					285			
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
	290					295					300				
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Ser	Gly	Gln	Pro	Glu	Asn	Asn
305					310					315					320
Tyr	Asn	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
			325						330					335	
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Ile
			340					345					350		
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	Arg	Tyr	Thr	Gln
		355					360					365			
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys							
	370					375									

<210> SEQ ID NO 26

<211> LENGTH: 221

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: TNF - VRSSSRTPSDKP del

<400> SEQUENCE: 26

Met	Ser	Thr	Glu	Ser	Met	Ile	Arg	Asp	Val	Glu	Leu	Ala	Glu	Glu	Ala
1				5					10					15	
Leu	Pro	Lys	Lys	Thr	Gly	Gly	Pro	Gln	Gly	Ser	Arg	Arg	Cys	Leu	Phe
		20						25					30		
Leu	Ser	Leu	Phe	Ser	Phe	Leu	Ile	Val	Ala	Gly	Ala	Thr	Thr	Leu	Phe
		35				40						45			
Cys	Leu	Leu	His	Phe	Gly	Val	Ile	Gly	Pro	Gln	Arg	Glu	Glu	Phe	Pro
	50					55					60				
Arg	Asp	Leu	Ser	Leu	Ile	Ser	Pro	Leu	Ala	Gln	Ala	Val	Ala	His	Val
65					70					75					80
Val	Ala	Asn	Pro	Gln	Ala	Glu	Gly	Gln	Leu	Gln	Trp	Leu	Asn	Arg	Arg
			85					90					95		
Ala	Asn	Ala	Leu	Leu	Ala	Asn	Gly	Val	Glu	Leu	Arg	Asp	Asn	Gln	Leu
			100					105						110	

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Val	Val	Pro	Ser	Glu	Gly	Leu	Tyr	Leu	Ile	Tyr	Ser	Gln	Val	Leu	Phe
		115					120					125			
Lys	Gly	Gln	Gly	Cys	Pro	Ser	Thr	His	Val	Leu	Leu	Thr	His	Thr	Ile
	130					135					140				
Ser	Arg	Ile	Ala	Val	Ser	Tyr	Gln	Thr	Lys	Val	Asn	Leu	Leu	Ser	Ala
145					150					155					160
Ile	Lys	Ser	Pro	Cys	Gln	Arg	Glu	Thr	Pro	Glu	Gly	Ala	Glu	Ala	Lys
			165					170						175	
Pro	Trp	Tyr	Glu	Pro	Ile	Tyr	Leu	Gly	Gly	Val	Phe	Gln	Leu	Glu	Lys
			180					185					190		
Gly	Asp	Arg	Leu	Ser	Ala	Glu	Ile	Asn	Arg	Pro	Asp	Tyr	Leu	Asp	Phe
	195						200					205			
Ala	Glu	Ser	Gly	Gln	Val	Tyr	Phe	Gly	Ile	Ile	Ala	Leu			
	210					215					220				

<210> SEQ ID NO 27
 <211> LENGTH: 212
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: TNF - FSFLIVAGATTLCFLHFGVI del

<400> SEQUENCE: 27

Met	Ser	Thr	Glu	Ser	Met	Ile	Arg	Asp	Val	Glu	Leu	Ala	Glu	Glu	Ala
1				5					10					15	
Leu	Pro	Lys	Lys	Thr	Gly	Gly	Pro	Gln	Gly	Ser	Arg	Arg	Cys	Leu	Phe
			20					25					30		
Leu	Ser	Leu	Gly	Pro	Gln	Arg	Glu	Glu	Phe	Pro	Arg	Asp	Leu	Ser	Leu
		35					40					45			
Ile	Ser	Pro	Leu	Ala	Gln	Ala	Val	Arg	Ser	Ser	Ser	Arg	Thr	Pro	Ser
	50					55						60			
Asp	Lys	Pro	Val	Ala	His	Val	Val	Ala	Asn	Pro	Gln	Ala	Glu	Gly	Gln
65					70					75					80
Leu	Gln	Trp	Leu	Asn	Arg	Arg	Ala	Asn	Ala	Leu	Leu	Ala	Asn	Gly	Val
				85					90					95	
Glu	Leu	Arg	Asp	Asn	Gln	Leu	Val	Val	Pro	Ser	Glu	Gly	Leu	Tyr	Leu
			100					105					110		
Ile	Tyr	Ser	Gln	Val	Leu	Phe	Lys	Gly	Gln	Gly	Cys	Pro	Ser	Thr	His
		115						120				125			
Val	Leu	Leu	Thr	His	Thr	Ile	Ser	Arg	Ile	Ala	Val	Ser	Tyr	Gln	Thr
	130						135				140				
Lys	Val	Asn	Leu	Leu	Ser	Ala	Ile	Lys	Ser	Pro	Cys	Gln	Arg	Glu	Thr
145					150					155					160
Pro	Glu	Gly	Ala	Glu	Ala	Lys	Pro	Trp	Tyr	Glu	Pro	Ile	Tyr	Leu	Gly
			165						170					175	
Gly	Val	Phe	Gln	Leu	Glu	Lys	Gly	Asp	Arg	Leu	Ser	Ala	Glu	Ile	Asn
			180					185					190		
Arg	Pro	Asp	Tyr	Leu	Asp	Phe	Ala	Glu	Ser	Gly	Gln	Val	Tyr	Phe	Gly
		195					200					205			
Ile	Ile	Ala	Leu												
	210														

<210> SEQ ID NO 28
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

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<223> OTHER INFORMATION: HA tag seq

<400> SEQUENCE: 28

Tyr Pro Tyr Asp Val Pro Asp Tyr Ala
1 5

<210> SEQ ID NO 29

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: FLAG TAG

<400> SEQUENCE: 29

Asp Tyr Lys Asp Asp Asp Lys
1 5

<210> SEQ ID NO 30

<211> LENGTH: 583

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: mIgG Heavy Chain IgG 1/3 hybrid anti-biotin
heavy chain - E325A mutant

<400> SEQUENCE: 30

Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Phe Arg Gly
1 5 10 15

Val Gln Cys Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala
20 25 30

Pro Ser Gln Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu
35 40 45

Thr Ala Tyr Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Gly Leu
50 55 60

Glu Trp Leu Gly Val Ile Trp Gly Gly Gly Arg Thr Asn Tyr Asn Ser
65 70 75 80

Gly Leu Met Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln
85 90 95

Val Phe Leu Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr
100 105 110

Tyr Cys Val Lys His Thr Asn Trp Asp Gly Gly Phe Ala Tyr Trp Gly
115 120 125

Gln Gly Thr Thr Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser
130 135 140

Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala
145 150 155 160

Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val
165 170 175

Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala
180 185 190

Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val
195 200 205

Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His
210 215 220

Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Leu Lys Thr Pro
225 230 235 240

Leu Gly Asp Thr Thr His Thr Cys Pro Arg Cys Pro Glu Pro Lys Ser
245 250 255

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Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys
      260      265      270

Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys Asp
      275      280      285

Thr Pro Pro Pro Cys Pro Arg Cys Pro Ala Pro Glu Leu Leu Gly Gly
      290      295      300

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
305      310      315      320

Ser Arg Thr Pro Ala Val Thr Cys Val Val Val Asp Val Ser His Glu
      325      330      335

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
      340      345      350

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
      355      360      365

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
      370      375      380

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
385      390      395      400

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
      405      410      415

Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu
      420      425      430

Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
      435      440      445

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
      450      455      460

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
465      470      475      480

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
      485      490      495

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
      500      505      510

Glu Leu Gln Leu Glu Glu Ser Cys Ala Glu Ala Gln Asp Gly Glu Leu
      515      520      525

Asp Gly Leu Trp Thr Thr Ile Thr Ile Phe Ile Thr Leu Phe Leu Leu
      530      535      540

Ser Val Cys Tyr Ser Ala Thr Val Thr Phe Phe Lys Val Lys Trp Ile
545      550      555      560

Phe Ser Ser Val Val Asp Leu Lys Gln Thr Ile Ile Pro Asp Tyr Arg
      565      570      575

Asn Met Ile Gly Gln Gly Ala
      580

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<210> SEQ ID NO 31
<211> LENGTH: 223
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD40L-TNFα fusion protein

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<400> SEQUENCE: 31

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Met Ile Glu Thr Tyr Asn Gln Thr Ser Pro Arg Ser Ala Ala Thr Gly
1           5           10          15

Leu Pro Ile Ser Met Lys Ile Phe Met Tyr Leu Leu Thr Val Phe Leu
20          25          30

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Ile Thr Gln Met Ile Gly Ser Ala Leu Phe Ala Val Tyr Leu His Arg
    35              40              45

Arg Leu Asp Lys Ile Glu Asp Glu Arg Asn Leu His Glu Asp Phe Val
    50              55              60

Phe Met Lys Thr Ile Gln Arg Cys Asn Thr Gly Glu Arg Pro Val Ala
    65              70              75              80

His Val Val Ala Asn Pro Gln Ala Glu Gly Gln Leu Gln Trp Leu Asn
    85              90              95

Arg Arg Ala Asn Ala Leu Leu Ala Asn Gly Val Glu Leu Arg Asp Asn
    100             105             110

Gln Leu Val Val Pro Ser Glu Gly Leu Tyr Leu Ile Tyr Ser Gln Val
    115             120             125

Leu Phe Lys Gly Gln Gly Cys Pro Ser Thr His Val Leu Leu Thr His
    130             135             140

Thr Ile Ser Arg Ile Ala Val Ser Tyr Gln Thr Lys Val Asn Leu Leu
    145             150             155             160

Ser Ala Ile Lys Ser Pro Cys Gln Arg Glu Thr Pro Glu Gly Ala Glu
    165             170             175

Ala Lys Pro Trp Tyr Glu Pro Ile Tyr Leu Gly Gly Val Phe Gln Leu
    180             185             190

Glu Lys Gly Asp Arg Leu Ser Ala Glu Ile Asn Arg Pro Asp Tyr Leu
    195             200             205

Asp Phe Ala Glu Ser Gly Gln Val Tyr Phe Gly Ile Ile Ala Leu
    210             215             220

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<210> SEQ ID NO 32
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: HC Signal

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<400> SEQUENCE: 32

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Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Phe Arg Gly
 1          5              10              15

Val Gln Cys

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<210> SEQ ID NO 33
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: HC Variable anti-biotin

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<400> SEQUENCE: 33

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Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala Pro Ser Gln
 1          5              10              15

Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ala Tyr
    20              25              30

Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
    35              40              45

Gly Val Ile Trp Gly Gly Gly Arg Thr Asn Tyr Asn Ser Gly Leu Met
    50              55              60

Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln Val Phe Leu
    65              70              75              80

Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr Tyr Cys Val
    85              90              95

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-continued

Lys His Thr Asn Trp Asp Gly Gly Phe Ala Tyr Trp Gly Gln Gly Thr
 100 105 110

Thr Val Thr Val Ser Ser
 115

<210> SEQ ID NO 34
 <211> LENGTH: 99
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: HC Constant Region 1 from IgG1

<400> SEQUENCE: 34

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
 1 5 10 15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20 25 30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35 40 45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50 55 60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
 65 70 75 80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
 85 90 95

Lys Val Glu

<210> SEQ ID NO 35
 <211> LENGTH: 61
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: HC Hinge Region from IgG3

<400> SEQUENCE: 35

Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro Arg Cys Pro
 1 5 10 15

Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu
 20 25 30

Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro
 35 40 45

Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro
 50 55 60

<210> SEQ ID NO 36
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: HC Constant Region 2 from IgG1

<400> SEQUENCE: 36

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
 1 5 10 15

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
 20 25 30

Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr
 35 40 45

Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
 50 55 60

-continued

Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
65 70 75 80

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
85 90 95

Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys
100 105 110

<210> SEQ ID NO 37

<211> LENGTH: 63

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: HC Transmembrane and Cytoplasmic region from IgG1

<400> SEQUENCE: 37

Ala Glu Ala Gln Asp Gly Glu Leu Asp Gly Leu Trp Thr Thr Ile Thr
1 5 10 15

Ile Phe Ile Thr Leu Phe Leu Leu Ser Val Cys Tyr Ser Ala Thr Val
20 25 30

Thr Phe Phe Lys Val Lys Trp Ile Phe Ser Ser Val Val Asp Leu Lys
35 40 45

Gln Thr Ile Ile Pro Asp Tyr Arg Asn Met Ile Gly Gln Gly Ala
50 55 60

<210> SEQ ID NO 38

<211> LENGTH: 22

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: LC Signal

<400> SEQUENCE: 38

Met Lys Tyr Leu Leu Pro Thr Ala Ala Ala Gly Leu Leu Leu Ala
1 5 10 15

Ala Gln Pro Ala Met Ala
20

<210> SEQ ID NO 39

<211> LENGTH: 99

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: LC Variable

<400> SEQUENCE: 39

Gly Ser Pro Gly Gln Ser Val Ser Ile Ser Cys Ser Gly Ser Ser Ser
1 5 10 15

Asn Ile Gly Asn Asn Tyr Val Tyr Trp Tyr Gln His Leu Pro Gly Thr
20 25 30

Ala Pro Lys Leu Leu Ile Tyr Ser Asp Thr Lys Arg Pro Ser Gly Val
35 40 45

Pro Asp Arg Ile Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala
50 55 60

Ile Ser Gly Leu Gln Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ser
65 70 75 80

Trp Asp Asp Ser Leu Asp Gly Pro Val Phe Gly Gly Gly Thr Lys Leu
85 90 95

Thr Val Leu

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<210> SEQ ID NO 40
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: LC Constant Region 1

<400> SEQUENCE: 40

Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser Ser
 1 5 10 15
 Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp
 20 25 30
 Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser Pro
 35 40 45
 Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn Asn
 50 55 60
 Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys
 65 70 75 80
 Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val
 85 90 95
 Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
 100 105

<210> SEQ ID NO 41
 <211> LENGTH: 583
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: mIgG Heavy Chain IgG 1/3 hybrid anti-biotin
 heavy chain - T323A mutant

<400> SEQUENCE: 41

Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Phe Arg Gly
 1 5 10 15
 Val Gln Cys Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala
 20 25 30
 Pro Ser Gln Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu
 35 40 45
 Thr Ala Tyr Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Gly Leu
 50 55 60
 Glu Trp Leu Gly Val Ile Trp Gly Gly Gly Arg Thr Asn Tyr Asn Ser
 65 70 75 80
 Gly Leu Met Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln
 85 90 95
 Val Phe Leu Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr
 100 105 110
 Tyr Cys Val Lys His Thr Asn Trp Asp Gly Gly Phe Ala Tyr Trp Gly
 115 120 125
 Gln Gly Thr Thr Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser
 130 135 140
 Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala
 145 150 155 160
 Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val
 165 170 175
 Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala
 180 185 190
 Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val

-continued

195	200	205
Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His 210 215 220		
Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Leu Lys Thr Pro 225 230 235 240		
Leu Gly Asp Thr Thr His Thr Cys Pro Arg Cys Pro Glu Pro Lys Ser 245 250 255		
Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys 260 265 270		
Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys Asp 275 280 285		
Thr Pro Pro Pro Cys Pro Arg Cys Pro Ala Pro Glu Leu Leu Gly Gly 290 295 300		
Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile 305 310 315 320		
Ser Arg Ala Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu 325 330 335		
Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His 340 345 350		
Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg 355 360 365		
Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys 370 375 380		
Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu 385 390 395 400		
Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr 405 410 415		
Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu 420 425 430		
Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp 435 440 445		
Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val 450 455 460		
Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp 465 470 475 480		
Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His 485 490 495		
Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro 500 505 510		
Glu Leu Gln Leu Glu Glu Ser Cys Ala Glu Ala Gln Asp Gly Glu Leu 515 520 525		
Asp Gly Leu Trp Thr Thr Ile Thr Ile Phe Ile Thr Leu Phe Leu Leu 530 535 540		
Ser Val Cys Tyr Ser Ala Thr Val Thr Phe Phe Lys Val Lys Trp Ile 545 550 555 560		
Phe Ser Ser Val Val Asp Leu Lys Gln Thr Ile Ile Pro Asp Tyr Arg 565 570 575		
Asn Met Ile Gly Gln Gly Ala 580		

<210> SEQ ID NO 42

<211> LENGTH: 211

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: GM-CSF - HLA-I fusion peptide

<400> SEQUENCE: 42

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Met Trp Leu Gln Ser Leu Leu Leu Leu Gly Thr Val Ala Cys Ser Ile
1           5           10           15
Ser Ala Pro Ala Arg Ser Pro Ser Pro Ser Thr Gln Pro Trp Glu His
          20           25           30
Val Asn Ala Ile Gln Glu Ala Arg Arg Leu Leu Asn Leu Ser Arg Asp
          35           40           45
Thr Ala Ala Glu Met Asn Glu Thr Val Glu Val Ile Ser Glu Met Phe
50           55           60
Asp Leu Gln Glu Pro Thr Cys Leu Gln Thr Arg Leu Glu Leu Tyr Lys
65           70           75           80
Gln Gly Leu Arg Gly Ser Leu Thr Lys Leu Lys Gly Pro Leu Thr Met
          85           90           95
Met Ala Ser His Tyr Lys Gln His Cys Pro Pro Thr Pro Glu Thr Ser
          100          105          110
Cys Ala Thr Gln Ile Ile Thr Phe Glu Ser Phe Lys Glu Asn Leu Lys
          115          120          125
Asp Phe Leu Leu Val Ile Pro Phe Asp Cys Trp Glu Pro Val Gln Glu
          130          135          140
Glu Leu Ser Ser Gln Pro Thr Ile Pro Ile Val Gly Ile Ile Ala Gly
          145          150          155          160
Leu Val Leu Leu Gly Ala Val Ile Thr Gly Ala Val Val Ala Ala Val
          165          170          175
Met Trp Arg Arg Lys Ser Ser Asp Arg Lys Gly Gly Ser Tyr Thr Gln
          180          185          190
Ala Ala Ser Ser Asp Ser Ala Gln Gly Ser Asp Val Ser Leu Thr Ala
          195          200          205
Cys Lys Val
          210

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<210> SEQ ID NO 43

<211> LENGTH: 583

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: mIgG Heavy Chain IgG 1/3 hybrid anti-biotin
heavy chain - E325A, T323A mutant

<400> SEQUENCE: 43

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Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Phe Arg Gly
1           5           10           15
Val Gln Cys Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala
          20           25           30
Pro Ser Gln Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu
          35           40           45
Thr Ala Tyr Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Gly Leu
50           55           60
Glu Trp Leu Gly Val Ile Trp Gly Gly Gly Arg Thr Asn Tyr Asn Ser
65           70           75           80
Gly Leu Met Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln
          85           90           95
Val Phe Leu Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr
          100          105          110

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Tyr	Cys	Val	Lys	His	Thr	Asn	Trp	Asp	Gly	Gly	Phe	Ala	Tyr	Trp	Gly
115						120						125			
Gln	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser
130						135						140			
Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala
145						150						155		160	
Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val
				165						170				175	
Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala
				180				185						190	
Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val
195						200						205			
Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His
210						215						220			
Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Leu	Lys	Thr	Pro
225						230				235				240	
Leu	Gly	Asp	Thr	Thr	His	Thr	Cys	Pro	Arg	Cys	Pro	Glu	Pro	Lys	Ser
				245						250				255	
Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys
				260				265						270	
Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp
275						280						285			
Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly
290						295						300			
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile
305						310				315				320	
Ser	Arg	Ala	Pro	Ala	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu
				325				330						335	
Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His
		340						345				350			
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg
355						360						365			
Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys
370						375						380			
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu
385						390				395				400	
Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr
				405				410						415	
Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu
		420						425				430			
Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp
435						440						445			
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val
450						455						460			
Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp
465						470				475				480	
Lys	Ser	Arg	Trp	Gln	Gln	G									

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530	535	540
Ser Val Cys Tyr Ser Ala Thr Val Thr Phe Phe Lys Val Lys Trp Ile		
545	550	555 560
Phe Ser Ser Val Val Asp Leu Lys Gln Thr Ile Ile Pro Asp Tyr Arg		
	565	570 575
Asn Met Ile Gly Gln Gly Ala		
	580	

<210> SEQ ID NO 44
 <211> LENGTH: 183
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Soluble Flt3-L

<400> SEQUENCE: 44

Met Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu Leu		
1	5	10 15
Leu Leu Leu Leu Ser Ser Gly Leu Ser Gly Thr Gln Asp Cys Ser Phe		
	20	25 30
Gln His Ser Pro Ile Ser Ser Asp Phe Ala Val Lys Ile Arg Glu Leu		
	35	40 45
Ser Asp Tyr Leu Leu Gln Asp Tyr Pro Val Thr Val Ala Ser Asn Leu		
	50	55 60
Gln Asp Glu Glu Leu Cys Gly Gly Leu Trp Arg Leu Val Leu Ala Gln		
	65	70 75 80
Arg Trp Met Glu Arg Leu Lys Thr Val Ala Gly Ser Lys Met Gln Gly		
	85	90 95
Leu Leu Glu Arg Val Asn Thr Glu Ile His Phe Val Thr Lys Cys Ala		
	100	105 110
Phe Gln Pro Pro Pro Ser Cys Leu Arg Phe Val Gln Thr Asn Ile Ser		
	115	120 125
Arg Leu Leu Gln Glu Thr Ser Glu Gln Leu Val Ala Leu Lys Pro Trp		
	130	135 140
Ile Thr Arg Gln Asn Phe Ser Arg Cys Leu Glu Leu Gln Cys Gln Pro		
	145	150 155 160
Asp Ser Ser Thr Leu Pro Pro Pro Trp Ser Pro Arg Pro Leu Glu Ala		
	165	170 175
Thr Ala Pro Thr Ala Pro Gln		
	180	

<210> SEQ ID NO 45
 <211> LENGTH: 583
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: mIgG Heavy Chain IgG 1/3 hybrid anti-biotin heavy chain

<400> SEQUENCE: 45

Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Phe Arg Gly		
1	5	10 15
Val Gln Cys Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala		
	20	25 30
Pro Ser Gln Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu		
	35	40 45
Thr Ala Tyr Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Gly Leu		
	50	55 60

-continued

Glu	Trp	Leu	Gly	Val	Ile	Trp	Gly	Gly	Gly	Arg	Thr	Asn	Tyr	Asn	Ser		
65					70					75					80		
Gly	Leu	Met	Ser	Arg	Leu	Ser	Ile	Arg	Lys	Asp	Asn	Ser	Lys	Ser	Gln		
				85					90					95			
Val	Phe	Leu	Thr	Met	Asn	Ser	Leu	Gln	Thr	Asp	Asp	Thr	Ala	Lys	Tyr		
				100				105					110				
Tyr	Cys	Val	Lys	His	Thr	Asn	Trp	Asp	Gly	Gly	Phe	Ala	Tyr	Trp	Gly		
		115					120					125					
Gln	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser		
		130				135					140						
Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala		
145					150					155					160		
Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val		
				165					170					175			
Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala		
		180						185					190				
Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val		
		195				200						205					
Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His		
		210				215					220						
Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Leu	Lys	Thr	Pro		
225					230					235				240			
Leu	Gly	Asp	Thr	Thr	His	Thr	Cys	Pro	Arg	Cys	Pro	Glu	Pro	Lys	Ser		
				245					250					255			
Cys	Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys		
			260					265					270				
Asp	Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro	Glu	Pro	Lys	Ser	Cys	Asp		
		275					280					285					
Thr	Pro	Pro	Pro	Cys	Pro	Arg	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly		
		290				295					300						
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile		
305					310					315				320			
Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu		
			325					330					335				
Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His		
			340					345					350				
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg		
		355					360					365					
Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys		
		370				375					380						
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu		
385					390					395				400			
Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr		
				405					410					415			
Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu		
			420					425					430				
Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp		
		435					440					445					
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val		
		450				455					460						
Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp		
465					470					475				480			

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Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
 485 490 495
 Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 500 505 510
 Glu Leu Gln Leu Glu Glu Ser Cys Ala Glu Ala Gln Asp Gly Glu Leu
 515 520 525
 Asp Gly Leu Trp Thr Thr Ile Thr Ile Phe Ile Thr Leu Phe Leu Leu
 530 535 540
 Ser Val Cys Tyr Ser Ala Thr Val Thr Phe Phe Lys Val Lys Trp Ile
 545 550 555 560
 Phe Ser Ser Val Val Asp Leu Lys Gln Thr Ile Ile Pro Asp Tyr Arg
 565 570 575
 Asn Met Ile Gly Gln Gly Ala
 580

<210> SEQ ID NO 46
 <211> LENGTH: 227
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG Light Chain

<400> SEQUENCE: 46

Met Lys Tyr Leu Leu Pro Thr Ala Ala Ala Gly Leu Leu Leu Leu Ala
 1 5 10 15
 Ala Gln Pro Ala Met Ala Gly Ser Pro Gly Gln Ser Val Ser Ile Ser
 20 25 30
 Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn Tyr Val Tyr Trp Tyr
 35 40 45
 Gln His Leu Pro Gly Thr Ala Pro Lys Leu Leu Ile Tyr Ser Asp Thr
 50 55 60
 Lys Arg Pro Ser Gly Val Pro Asp Arg Ile Ser Gly Ser Lys Ser Gly
 65 70 75 80
 Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln Ser Glu Asp Glu Ala
 85 90 95
 Asp Tyr Tyr Cys Ala Ser Trp Asp Asp Ser Leu Asp Gly Pro Val Phe
 100 105 110
 Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln Pro Lys Ala Asn Pro
 115 120 125
 Thr Val Thr Leu Phe Pro Pro Ser Ser Glu Glu Leu Gln Ala Asn Lys
 130 135 140
 Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr Pro Gly Ala Val Thr
 145 150 155 160
 Val Ala Trp Lys Ala Asp Gly Ser Pro Val Lys Ala Gly Val Glu Thr
 165 170 175
 Thr Lys Pro Ser Lys Gln Ser Asn Asn Lys Tyr Ala Ala Ser Ser Tyr
 180 185 190
 Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His Arg Ser Tyr Ser Cys
 195 200 205
 Gln Val Thr His Glu Gly Ser Thr Val Glu Lys Thr Val Ala Pro Thr
 210 215 220
 Glu Cys Ser
 225

<210> SEQ ID NO 47
 <211> LENGTH: 9203

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<212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Vector 1

<400> SEQUENCE: 47

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aatgtagtct tatgcaatac tctttagtgc ttgcaacatg gtaacgatga gttagcaaca    60
tgcccttaca ggagagaaaa agcaccgtgc atgccgattg gtggaagtaa ggtggtacga    120
tcgtgcctta ttaggaaagg aacagacggg tctgacatgg attggacgaa ccactgaatt    180
gccgcattgc agagatatgt tatttaagtg cctagctcga tacataaacg ggtctctctg    240
gtagaccag atctgagcct gggagctctc tggctaacta gggaaccac tgcttaagcc    300
tcaataaagc ttgccttgag tgcttcaagt agtgtgtgcc cgtctgttgt gtgactctgg    360
taactagaga tccctcagac ccttttagtc agtgtggaaa atctctagca gtggcgcccg    420
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cttcctttct cgcacgcttc gccggtttc cccgtcaagc tctaaatcgg gggctccctt	7560
tagggttccg atttagtgct ttacggcacc tcgaccccaa aaaacttgat tagggtagtg	7620
gttcacgtag tgggccatcg cctgataga cggtttttcg ccctttgacg ttggagtcca	7680
cgttctttaa tagtggaact ttgttccaaa ctggaacaac actcaacct atctcggctc	7740
attcttttga tttataaggg attttgccga ttccggccta ttggttaaaa aatgagctga	7800
tttaacaaaa atttaacgcg aattttaaca aaatattaac gcttacaatt taggtggcac	7860
ttttcgggga aatgtgcgcg gaaccttat ttgtttattt ttctaaatac attcaaatat	7920
gtatccgctc atgagacaat aacctgata aatgcttcaa taatattgaa aaaggaagag	7980
tatgagtatt caacatttcc gtgtcgccct tattcccttt ttgctggcat ttgccttcc	8040
tgtttttgct caccagaaa cgtcgttgaa agtaaaagat gctgaagatc agttgggtgc	8100
acgagtgggt tacatcgaac tggatctcaa cagcggtaag atccttgaga gttttcgccc	8160
cgaagaacgt tttccaatga tgagcacttt taaagttctg ctatgtggcg cggtattatc	8220
ccgtattgac gccgggcaag agcaactcgg tcgccgcata cactattctc agaatgactt	8280
ggttgagtac tcaccagtc cagaaaagca tcttacggat ggcatgacag taagagaatt	8340
atgcagtgtc gccataacca tgagtataa cactcgggcc aacttacttc tgacaacgat	8400
cggaggaccg aaggagctaa ccgctttttt gcacaacatg ggggatcatg taactcgct	8460
tgatcgttgg gaaccggagc tgaatgaag cataccaaac gacgagcgtg acaccagat	8520

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gcctgtagca atggcaacaa cgttgcgcaa actattaact ggcgaaactac ttactctagc 8580
ttcccggcaa caattaatag actggatgga ggcggataaa gttgcaggac cacttctgcg 8640
ctcgccctt ccggtcggtt ggtttattgc tgataaatct ggagccggtg agcgtgggtc 8700
tcgcggtatc attgcagcac tggggccaga tggtaagccc tccggtatcg tagttatcta 8760
cacgacgggg agtcaggcaa ctatggatga acgaaataga cagatcgctg agataggtgc 8820
ctcactgatt aagcattggt aactgtcaga ccaagtttac tcatatatac tttagattga 8880
tttaaaactt catttttaat ttaaaaggat ctaggatgaag atcctttttg ataattctcat 8940
gacaaaaatc ccttaacgtg agttttcgtt ccaactgagcg tcagaccccg tagaaaagat 9000
caaaggatct tcttgagatc ctttttttct gcgcgtaatc tgctgcttgc aaacaaaaaa 9060
accaccgcta ccagcggttg tttgtttgcc ggatcaagag ctaccaactc tttttccgaa 9120
ggtaactggc ttcagcagag cgcagatacc aaatactgtt cttctagtgt agccgtagtt 9180
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gttaccggat aaggcgcagc ggtcgggctg aacggggggt tcgtgcacac agcccagctt 9360
ggagcgaaac acctacaccg aactgagata cctacagcgt gagctatgag aaagcgccac 9420
gcttcccga gagagaaagg cggacaggta tccgtaagc ggcagggtcg gaacaggaga 9480
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gttctttcct gcgttatccc ctgattctgt ggataaccgt attaccgctt ttgagtgagc 9720
tgataccgct cgccgcagcc gaacgaccga gcgcagcgag tcagtgcgag aggaagcgga 9780
agagcgccca atacgcaaac cgctctccc cgcggttg cagattcatt aatgcagctg 9840
gcacgacagg tttccgact ggaaagcggg cagtgcgagc aacgcaatta atgtgagtta 9900
gctcactcat taggcacccc aggccttaca ctttatgctt ccggctcgta tgttgtgtgg 9960
aattgtgagc ggataacaat ttcacacagg aaacagctat gaccatgatt acgccaagcg 10020
cgcaattaac cctcactaaa gggaacaaaa gctggagctg caagctt 10067

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<210> SEQ ID NO 54
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: L1 light chain leader

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<400> SEQUENCE: 54

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Met Ala Thr Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu
1           5           10           15

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Leu Trp Leu Ser Gly Ala Arg Cys
20

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<210> SEQ ID NO 55
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: anti-biotin murine vH with inserted Cys for
inter-domain linkage

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<400> SEQUENCE: 55

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-continued

Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala Pro Ser Gln
 1 5 10 15
 Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ala Tyr
 20 25 30
 Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Cys Leu Glu Trp Leu
 35 40 45
 Gly Val Ile Trp Gly Gly Gly Arg Thr Asn Tyr Asn Ser Gly Leu Met
 50 55 60
 Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln Val Phe Leu
 65 70 75 80
 Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr Tyr Cys Val
 85 90 95
 Lys His Thr Asn Trp Asp Gly Gly Phe Ala Tyr Trp Gly Gln Gly Thr
 100 105 110
 Thr Val Thr Val Ser Ser
 115

<210> SEQ ID NO 56
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: linker

<400> SEQUENCE: 56

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 1 5 10 15

<210> SEQ ID NO 57
 <211> LENGTH: 99
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Light Chain Variable (human lambda variable)

<400> SEQUENCE: 57

Gly Ser Pro Gly Gln Ser Val Ser Ile Ser Cys Ser Gly Ser Ser Ser
 1 5 10 15
 Asn Ile Gly Asn Asn Tyr Val Tyr Trp Tyr Gln His Leu Pro Gly Thr
 20 25 30
 Ala Pro Lys Leu Leu Ile Tyr Ser Asp Thr Lys Arg Pro Ser Gly Val
 35 40 45
 Pro Asp Arg Ile Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala
 50 55 60
 Ile Ser Gly Leu Gln Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ser
 65 70 75 80
 Trp Asp Asp Ser Leu Asp Gly Pro Val Phe Gly Cys Gly Thr Lys Leu
 85 90 95
 Thr Val Leu

<210> SEQ ID NO 58
 <211> LENGTH: 61
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG3 hinge for greater accessibility to FcyR

<400> SEQUENCE: 58

Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro Arg Cys Pro

-continued

1	5	10	15
Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu	20	25	30
Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro	35	40	45
Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro	50	55	60

<210> SEQ ID NO 59
 <211> LENGTH: 283
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG1 CH2, CH3 Tm and cytoplasmic tail (T256A)

<400> SEQUENCE: 59

Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp	1	5	10	15
Thr Leu Met Ile Ser Arg Ala Pro Glu Val Thr Cys Val Val Val Asp	20	25	30	
Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly	35	40	45	
Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn	50	55	60	
Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp	65	70	75	80
Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro	85	90	95	
Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu	100	105	110	
Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn	115	120	125	
Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile	130	135	140	
Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr	145	150	155	160
Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys	165	170	175	
Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys	180	185	190	
Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu	195	200	205	
Ser Leu Ser Pro Glu Leu Gln Leu Glu Glu Ser Cys Ala Glu Ala Gln	210	215	220	
Asp Gly Glu Leu Asp Gly Leu Trp Thr Thr Ile Thr Ile Phe Ile Thr	225	230	235	240
Leu Phe Leu Leu Ser Val Cys Tyr Ser Ala Thr Val Thr Phe Phe Lys	245	250	255	
Val Lys Trp Ile Phe Ser Ser Val Val Asp Leu Lys Gln Thr Ile Ile	260	265	270	
Pro Asp Tyr Arg Asn Met Ile Gly Gln Gly Ala	275	280		

<210> SEQ ID NO 60

<211> LENGTH: 598

<212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: scFv-anti-biotin-G3hinge-IgG1-Tm (598 ORF1)

<400> SEQUENCE: 60

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Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Phe Arg Gly
 1              5              10              15

Val Gln Cys Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala
 20              25              30

Pro Ser Gln Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu
 35              40              45

Thr Ala Tyr Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Cys Leu
 50              55              60

Glu Trp Leu Gly Val Ile Trp Gly Gly Gly Arg Thr Asn Tyr Asn Ser
 65              70              75              80

Gly Leu Met Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln
 85              90              95

Val Phe Leu Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr
 100             105             110

Tyr Cys Val Lys His Thr Asn Trp Asp Gly Gly Phe Ala Tyr Trp Gly
 115             120             125

Gln Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly
 130             135             140

Gly Gly Ser Gly Gly Gly Gly Ser Gly Ser Pro Gly Gln Ser Val Ser
 145             150             155             160

Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn Tyr Val Tyr
 165             170             175

Trp Tyr Gln His Leu Pro Gly Thr Ala Pro Lys Leu Leu Ile Tyr Ser
 180             185             190

Asp Thr Lys Arg Pro Ser Gly Val Pro Asp Arg Ile Ser Gly Ser Lys
 195             200             205

Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln Ser Glu Asp
 210             215             220

Glu Ala Asp Tyr Tyr Cys Ala Ser Trp Asp Asp Ser Leu Asp Gly Pro
 225             230             235             240

Val Phe Gly Cys Gly Thr Lys Leu Thr Val Leu Leu Lys Thr Pro Leu
 245             250             255

Gly Asp Thr Thr His Thr Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys
 260             265             270

Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys Asp
 275             280             285

Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr
 290             295             300

Pro Pro Pro Cys Pro Arg Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro
 305             310             315             320

Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
 325             330             335

Arg Ala Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp
 340             345             350

Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
 355             360             365

Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val
 370             375             380

Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu

```

385				390				395				400			
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys
405				410				415							
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr
420				425				430							
Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr
435				440				445							
Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu
450				455				460							
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu
465				470				475				480			
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys
485				490				495							
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu
500				505				510							
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Glu
515				520				525							
Leu	Gln	Leu	Glu	Glu	Ser	Cys	Ala	Glu	Ala	Gln	Asp	Gly	Glu	Leu	Asp
530				535				540							
Gly	Leu	Trp	Thr	Thr	Ile	Thr	Ile	Phe	Ile	Thr	Leu	Phe	Leu	Leu	Ser
545				550				555				560			
Val	Cys	Tyr	Ser	Ala	Thr	Val	Thr	Phe	Phe	Lys	Val	Lys	Trp	Ile	Phe
565				570				575							
Ser	Ser	Val	Val	Asp	Leu	Lys	Gln	Thr	Ile	Ile	Pro	Asp	Tyr	Arg	Asn
580				585				590							
Met	Ile	Gly	Gln	Gly	Ala										
595															

<400> SEQUENCE: 61

Val Gln Cys

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<210> SEQ ID NO 62
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: anti-biotin vH
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<400> SEQUENCE: 62

Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln Val Phe Leu

-continued

65	70	75	80
Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr Tyr Cys Val			
	85	90	95
Lys His Thr Asn Trp Asp Gly Gly Phe Ala Tyr Trp Gly Gln Gly Thr			
	100	105	110
Thr Val Thr Val Ser Ser			
	115		

<210> SEQ ID NO 63
 <211> LENGTH: 94
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CH1

<400> SEQUENCE: 63

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly			
1	5	10	15
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val			
	20	25	30
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe			
	35	40	45
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val			
	50	55	60
Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val			
65	70	75	80
Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu			
	85	90	

<210> SEQ ID NO 64
 <211> LENGTH: 61
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG3 hinge

<400> SEQUENCE: 64

Leu Lys Thr Pro Leu Gly Asp Thr Thr His Thr Cys Pro Arg Cys Pro			
1	5	10	15
Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu			
	20	25	30
Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro Glu Pro			
	35	40	45
Lys Ser Cys Asp Thr Pro Pro Pro Cys Pro Arg Cys Pro			
	50	55	60

<210> SEQ ID NO 65
 <211> LENGTH: 286
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG1 CH2, CH3 Tm and cytoplasmic tail

<400> SEQUENCE: 65

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys			
1	5	10	15
Pro Lys Asp Thr Leu Met Ile Ser Arg Ala Pro Glu Val Thr Cys Val			
	20	25	30
Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr			
	35	40	45

-continued

Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
50						55					60				
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
65					70					75				80	
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
			85						90					95	
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln
			100					105						110	
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu
		115					120					125			
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
		130				135					140				
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn
145					150					155				160	
Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
				165					170					175	
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val
			180					185					190		
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
		195					200					205			
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Glu	Leu	Gln	Leu	Glu	Glu	Ser	Cys	Ala
		210				215					220				
Glu	Ala	Gln	Asp	Gly	Glu	Leu	Asp	Gly	Leu	Trp	Thr	Thr	Ile	Thr	Ile
225					230					235					240
Phe	Ile	Thr	Leu	Phe	Leu	Leu	Ser	Val	Cys	Tyr	Ser	Ala	Thr	Val	Thr
			245						250					255	
Phe	Phe	Lys	Val	Lys	Trp	Ile	Phe	Ser	Ser	Val	Val	Asp	Leu	Lys	Gln
			260					265					270		
Thr	Ile	Ile	Pro	Asp	Tyr	Arg	Asn	Met	Ile	Gly	Gln	Gly	Ala		
		275					280					285			

<210> SEQ ID NO 66

<211> LENGTH: 578

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Summary (578 ORF2a)

<400> SEQUENCE: 66

Met	Glu	Phe	Gly	Leu	Ser	Trp	Val	Phe	Leu	Val	Ala	Leu	Phe	Arg	Gly
1				5					10					15	
Val	Gln	Cys	Gln	Val	Lys	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Ala
		20				25							30		
Pro	Ser	Gln	Ser	Leu	Ser	Ile	Thr	Cys	Thr	Val	Ser	Gly	Phe	Ser	Leu
		35				40						45			
Thr	Ala	Tyr	Gly	Val	Asp	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Gly	Leu
		50				55					60				
Glu	Trp	Leu	Gly	Val	Ile	Trp	Gly	Gly	Gly	Arg	Thr	Asn	Tyr	Asn	Ser
65					70					75				80	
Gly	Leu	Met	Ser	Arg	Leu	Ser	Ile	Arg	Lys	Asp	Asn	Ser	Lys	Ser	Gln
			85						90					95	
Val	Phe	Leu	Thr	Met	Asn	Ser	Leu	Gln	Thr	Asp	Asp	Thr	Ala	Lys	Tyr
			100					105					110		
Tyr	Cys	Val	Lys	His	Thr	Asn	Trp	Asp	Gly	Gly	Phe	Ala	Tyr	Trp	Gly
		115					120					125			

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Gln Gly Thr Thr Val Thr Val Ser Ser Pro Ser Val Phe Pro Leu Ala															
130						135				140					
Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu															
145				150					155						160
Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly															
			165					170						175	
Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser															
			180					185						190	
Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu															
			195				200				205				
Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr															
			210			215				220					
Lys Val Asp Lys Lys Val Glu Leu Lys Thr Pro Leu Gly Asp Thr Thr															
225				230					235						240
His Thr Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro															
			245					250						255	
Pro Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro															
			260				265						270		
Cys Pro Arg Cys Pro Glu Pro Lys Ser Cys Asp Thr Pro Pro Pro Cys															
			275			280				285					
Pro Arg Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu															
			290			295			300						
Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Ala Pro Glu															
305				310					315						320
Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys															
			325				330							335	
Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys															
			340			345						350			
Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu															
			355			360						365			
Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys															
			370			375			380						
Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys															
385				390					395						400
Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser															
			405				410							415	
Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys															
			420				425					430			
Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln															
			435			440			445						
Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly															
			450			455			460						
Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln															
465				470				475							480
Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn															
			485				490							495	
His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Glu Leu Gln Leu Glu															
			500				505						510		
Glu Ser Cys Ala Glu Ala Gln Asp Gly Glu Leu Asp Gly Leu Trp Thr															
			515				520					525			
Thr Ile Thr Ile Phe Ile Thr Leu Phe Leu Leu Ser Val Cys Tyr Ser															
			530			535						540			

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Ala Thr Val Thr Phe Phe Lys Val Lys Trp Ile Phe Ser Ser Val Val
545 550 555 560

Asp Leu Lys Gln Thr Ile Ile Pro Asp Tyr Arg Asn Met Ile Gly Gln
565 570 575

Gly Ala

<210> SEQ ID NO 67
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 67

Ile Arg Glu Ser
1

<210> SEQ ID NO 68
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: L1 Signal

<400> SEQUENCE: 68

Met Ala Thr Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu
1 5 10 15

Leu Trp Leu Ser Gly Ala Arg Cys
20

<210> SEQ ID NO 69
<211> LENGTH: 99
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LC Variable

<400> SEQUENCE: 69

Gly Ser Pro Gly Gln Ser Val Ser Ile Ser Cys Ser Gly Ser Ser Ser
1 5 10 15

Asn Ile Gly Asn Asn Tyr Val Tyr Trp Tyr Gln His Leu Pro Gly Thr
20 25 30

Ala Pro Lys Leu Leu Ile Tyr Ser Asp Thr Lys Arg Pro Ser Gly Val
35 40 45

Pro Asp Arg Ile Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala
50 55 60

Ile Ser Gly Leu Gln Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ser
65 70 75 80

Trp Asp Asp Ser Leu Asp Gly Pro Val Phe Gly Gly Gly Thr Lys Leu
85 90 95

Thr Val Leu

<210> SEQ ID NO 70
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LC Constant Region 1

<400> SEQUENCE: 70

Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser Ser
1 5 10 15

-continued

Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp
 20 25 30
 Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser Pro
 35 40 45
 Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn Asn
 50 55 60
 Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys
 65 70 75 80
 Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val
 85 90 95
 Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
 100 105

<210> SEQ ID NO 71
 <211> LENGTH: 229
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Summary (229 ORF2b)

<400> SEQUENCE: 71

Met Ala Thr Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu
 1 5 10 15
 Leu Trp Leu Ser Gly Ala Arg Cys Gly Ser Pro Gly Gln Ser Val Ser
 20 25 30
 Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn Tyr Val Tyr
 35 40 45
 Trp Tyr Gln His Leu Pro Gly Thr Ala Pro Lys Leu Leu Ile Tyr Ser
 50 55 60
 Asp Thr Lys Arg Pro Ser Gly Val Pro Asp Arg Ile Ser Gly Ser Lys
 65 70 75 80
 Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln Ser Glu Asp
 85 90 95
 Glu Ala Asp Tyr Tyr Cys Ala Ser Trp Asp Asp Ser Leu Asp Gly Pro
 100 105 110
 Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln Pro Lys Ala
 115 120 125
 Asn Pro Thr Val Thr Leu Phe Pro Pro Ser Ser Glu Glu Leu Gln Ala
 130 135 140
 Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr Pro Gly Ala
 145 150 155 160
 Val Thr Val Ala Trp Lys Ala Asp Gly Ser Pro Val Lys Ala Gly Val
 165 170 175
 Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn Asn Lys Tyr Ala Ala Ser
 180 185 190
 Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His Arg Ser Tyr
 195 200 205
 Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu Lys Thr Val Ala
 210 215 220
 Pro Thr Glu Cys Ser
 225

<210> SEQ ID NO 72
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: GM-CSF signal sequence

<400> SEQUENCE: 72

Met Trp Leu Gln Ser Leu Leu Leu Leu Gly Thr Val Ala Cys Ser Ile
1 5 10 15

Ser

<210> SEQ ID NO 73
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild type GM-CSF sequence

<400> SEQUENCE: 73

Ala Pro Ala Arg Ser Pro Ser Pro Ser Thr Gln Pro Trp Glu His Val
1 5 10 15

Asn Ala Ile Gln Glu Ala Arg Arg Leu Leu Asn Leu Ser Arg Asp Thr
20 25 30

Ala Ala Glu Met Asn Glu Thr Val Glu Val Ile Ser Glu Met Phe Asp
35 40 45

Leu Gln Glu Pro Thr Cys Leu Gln Thr Arg Leu Glu Leu Tyr Lys Gln
50 55 60

Gly Leu Arg Gly Ser Leu Thr Lys Leu Lys Gly Pro Leu Thr Met Met
65 70 75 80

Ala Ser His Tyr Lys Gln His Cys Pro Pro Thr Pro Glu Thr Ser Cys
85 90 95

Ala Thr Gln Ile Ile Thr Phe Glu Ser Phe Lys Glu Asn Leu Lys Asp
100 105 110

Phe Leu Leu Val Ile Pro Phe Asp Cys Trp Glu Pro Val Gln Glu
115 120 125

<210> SEQ ID NO 74
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 74

Ile Arg Glu Ser
1

<210> SEQ ID NO 75
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FLT3L signal

<400> SEQUENCE: 75

Met Ala Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu
1 5 10 15

Leu Leu Leu Leu Leu Ser Ser Gly Leu Ser
20 25

<210> SEQ ID NO 76
<211> LENGTH: 210
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: FLT3L

<400> SEQUENCE: 76

Gly Thr Gln Asp Cys Ser Phe Gln His Ser Pro Ile Ser Ser Asp Phe
 1 5 10 15
 Ala Val Lys Ile Arg Glu Leu Ser Asp Tyr Leu Leu Gln Asp Tyr Pro
 20 25 30
 Val Thr Val Ala Ser Asn Leu Gln Asp Glu Glu Leu Cys Gly Gly Leu
 35 40 45
 Trp Arg Leu Val Leu Ala Gln Arg Trp Met Glu Arg Leu Lys Thr Val
 50 55 60
 Ala Gly Ser Lys Met Gln Gly Leu Leu Glu Arg Val Asn Thr Glu Ile
 65 70 75 80
 His Phe Val Thr Lys Cys Ala Phe Gln Pro Pro Pro Ser Cys Leu Arg
 85 90 95
 Phe Val Gln Thr Asn Ile Ser Arg Leu Leu Gln Glu Thr Ser Glu Gln
 100 105 110
 Leu Val Ala Leu Lys Pro Trp Ile Thr Arg Gln Asn Phe Ser Arg Cys
 115 120 125
 Leu Glu Leu Gln Cys Gln Pro Asp Ser Ser Thr Leu Pro Pro Pro Trp
 130 135 140
 Ser Pro Arg Pro Leu Glu Ala Thr Ala Pro Thr Ala Pro Gln Pro Pro
 145 150 155 160
 Leu Leu Leu Leu Leu Leu Leu Pro Val Gly Leu Leu Leu Leu Ala Ala
 165 170 175
 Ala Trp Cys Leu His Trp Gln Arg Thr Arg Arg Arg Thr Pro Arg Pro
 180 185 190
 Gly Glu Gln Val Pro Pro Val Pro Ser Pro Gln Asp Leu Leu Leu Val
 195 200 205
 Glu His
 210

<210> SEQ ID NO 77

<211> LENGTH: 144

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Summary (144 ORF3a)

<400> SEQUENCE: 77

Met Trp Leu Gln Ser Leu Leu Leu Leu Gly Thr Val Ala Cys Ser Ile
 1 5 10 15
 Ser Ala Pro Ala Arg Ser Pro Ser Pro Ser Thr Gln Pro Trp Glu His
 20 25 30
 Val Asn Ala Ile Gln Glu Ala Arg Arg Leu Leu Asn Leu Ser Arg Asp
 35 40 45
 Thr Ala Ala Glu Met Asn Glu Thr Val Glu Val Ile Ser Glu Met Phe
 50 55 60
 Asp Leu Gln Glu Pro Thr Cys Leu Gln Thr Arg Leu Glu Leu Tyr Lys
 65 70 75 80
 Gln Gly Leu Arg Gly Ser Leu Thr Lys Leu Lys Gly Pro Leu Thr Met
 85 90 95
 Met Ala Ser His Tyr Lys Gln His Cys Pro Pro Thr Pro Glu Thr Ser
 100 105 110
 Cys Ala Thr Gln Ile Ile Thr Phe Glu Ser Phe Lys Glu Asn Leu Lys
 115 120 125

-continued

Asp Phe Leu Leu Val Ile Pro Phe Asp Cys Trp Glu Pro Val Gln Glu
 130 135 140

<210> SEQ ID NO 78
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Summary (236 ORF3b)

<400> SEQUENCE: 78

Met Ala Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu
 1 5 10 15
 Leu Leu Leu Leu Leu Ser Ser Gly Leu Ser Gly Thr Gln Asp Cys Ser
 20 25 30
 Phe Gln His Ser Pro Ile Ser Ser Asp Phe Ala Val Lys Ile Arg Glu
 35 40 45
 Leu Ser Asp Tyr Leu Leu Gln Asp Tyr Pro Val Thr Val Ala Ser Asn
 50 55 60
 Leu Gln Asp Glu Glu Leu Cys Gly Gly Leu Trp Arg Leu Val Leu Ala
 65 70 75 80
 Gln Arg Trp Met Glu Arg Leu Lys Thr Val Ala Gly Ser Lys Met Gln
 85 90 95
 Gly Leu Leu Glu Arg Val Asn Thr Glu Ile His Phe Val Thr Lys Cys
 100 105 110
 Ala Phe Gln Pro Pro Pro Ser Cys Leu Arg Phe Val Gln Thr Asn Ile
 115 120 125
 Ser Arg Leu Leu Gln Glu Thr Ser Glu Gln Leu Val Ala Leu Lys Pro
 130 135 140
 Trp Ile Thr Arg Gln Asn Phe Ser Arg Cys Leu Glu Leu Gln Cys Gln
 145 150 155 160
 Pro Asp Ser Ser Thr Leu Pro Pro Pro Trp Ser Pro Arg Pro Leu Glu
 165 170 175
 Ala Thr Ala Pro Thr Ala Pro Gln Pro Pro Leu Leu Leu Leu Leu
 180 185 190
 Leu Pro Val Gly Leu Leu Leu Leu Ala Ala Ala Trp Cys Leu His Trp
 195 200 205
 Gln Arg Thr Arg Arg Arg Thr Pro Arg Pro Gly Glu Gln Val Pro Pro
 210 215 220
 Val Pro Ser Pro Gln Asp Leu Leu Leu Val Glu His
 225 230 235

<210> SEQ ID NO 79
 <211> LENGTH: 183
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: wild type FLT3L sequence with transmembrane
 deleted

<400> SEQUENCE: 79

Met Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu Leu
 1 5 10 15
 Leu Leu Leu Leu Ser Ser Gly Leu Ser Gly Thr Gln Asp Cys Ser Phe
 20 25 30
 Gln His Ser Pro Ile Ser Ser Asp Phe Ala Val Lys Ile Arg Glu Leu
 35 40 45

-continued

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Ser Asp Tyr Leu Leu Gln Asp Tyr Pro Val Thr Val Ala Ser Asn Leu
 50          55          60

Gln Asp Glu Glu Leu Cys Gly Gly Leu Trp Arg Leu Val Leu Ala Gln
 65          70          75          80

Arg Trp Met Glu Arg Leu Lys Thr Val Ala Gly Ser Lys Met Gln Gly
          85          90          95

Leu Leu Glu Arg Val Asn Thr Glu Ile His Phe Val Thr Lys Cys Ala
 100          105          110

Phe Gln Pro Pro Pro Ser Cys Leu Arg Phe Val Gln Thr Asn Ile Ser
 115          120          125

Arg Leu Leu Gln Glu Thr Ser Glu Gln Leu Val Ala Leu Lys Pro Trp
 130          135          140

Ile Thr Arg Gln Asn Phe Ser Arg Cys Leu Glu Leu Gln Cys Gln Pro
 145          150          155          160

Asp Ser Ser Thr Leu Pro Pro Pro Trp Ser Pro Arg Pro Leu Glu Ala
          165          170          175

Thr Ala Pro Thr Ala Pro Gln
          180

```

```

<210> SEQ ID NO 80
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 80

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Ile Arg Glu Ser
1

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<210> SEQ ID NO 81
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FLT3L signal

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<400> SEQUENCE: 81

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Met Ala Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu
 1          5          10          15

Leu Leu Leu Leu Leu Ser Ser Gly Leu Ser
          20          25

```

```

<210> SEQ ID NO 82
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild type GM-CSF sequence

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<400> SEQUENCE: 82

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Ala Pro Ala Arg Ser Pro Ser Pro Ser Thr Gln Pro Trp Glu His Val
 1          5          10          15

Asn Ala Ile Gln Glu Ala Arg Arg Leu Leu Asn Leu Ser Arg Asp Thr
          20          25          30

Ala Ala Glu Met Asn Glu Thr Val Glu Val Ile Ser Glu Met Phe Asp
          35          40          45

Leu Gln Glu Pro Thr Cys Leu Gln Thr Arg Leu Glu Leu Tyr Lys Gln
          50          55          60

Gly Leu Arg Gly Ser Leu Thr Lys Leu Lys Gly Pro Leu Thr Met Met

```

-continued

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65              70              75              80
Ala Ser His Tyr Lys Gln His Cys Pro Pro Thr Pro Glu Thr Ser Cys
              85              90              95
Ala Thr Gln Ile Ile Thr Phe Glu Ser Phe Lys Glu Asn Leu Lys Asp
              100             105             110
Phe Leu Leu Val Ile Pro Phe Asp Cys Trp Glu Pro Val Gln Glu
              115             120             125

<210> SEQ ID NO 83
<211> LENGTH: 99
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD8alpha transmembrane and cytoplasmic domain

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```

<400> SEQUENCE: 83

Pro Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile
1              5              10              15
Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala
              20              25              30
Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr
              35              40              45
Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu
              50              55              60
Val Ile Thr Leu Tyr Cys Asn His Arg Asn Arg Arg Arg Val Cys Lys
65              70              75              80
Cys Pro Arg Pro Val Val Lys Ser Gly Asp Lys Pro Ser Leu Ser Ala
              85              90              95

Arg Tyr Val

```

```

<210> SEQ ID NO 84
<211> LENGTH: 183
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Summary (183 ORF4a)

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<400> SEQUENCE: 84

Met Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu Leu
1              5              10              15
Leu Leu Leu Leu Ser Ser Gly Leu Ser Gly Thr Gln Asp Cys Ser Phe
              20              25              30
Gln His Ser Pro Ile Ser Ser Asp Phe Ala Val Lys Ile Arg Glu Leu
              35              40              45
Ser Asp Tyr Leu Leu Gln Asp Tyr Pro Val Thr Val Ala Ser Asn Leu
              50              55              60
Gln Asp Glu Glu Leu Cys Gly Gly Leu Trp Arg Leu Val Leu Ala Gln
65              70              75              80
Arg Trp Met Glu Arg Leu Lys Thr Val Ala Gly Ser Lys Met Gln Gly
              85              90              95
Leu Leu Glu Arg Val Asn Thr Glu Ile His Phe Val Thr Lys Cys Ala
              100             105             110
Phe Gln Pro Pro Pro Ser Cys Leu Arg Phe Val Gln Thr Asn Ile Ser
              115             120             125
Arg Leu Leu Gln Glu Thr Ser Glu Gln Leu Val Ala Leu Lys Pro Trp
              130             135             140
Ile Thr Arg Gln Asn Phe Ser Arg Cys Leu Glu Leu Gln Cys Gln Pro

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-continued

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145              150              155              160
Asp Ser Ser Thr Leu Pro Pro Pro Trp Ser Pro Arg Pro Leu Glu Ala
              165              170              175

Thr Ala Pro Thr Ala Pro Gln
      180

<210> SEQ ID NO 85
<211> LENGTH: 252
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Summary for CYAGEN (253 ORF4b)

<400> SEQUENCE: 85

Met Ala Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu
1              5              10              15

Leu Leu Leu Leu Leu Ser Ser Gly Leu Ser Ala Pro Ala Arg Ser Pro
      20              25              30

Ser Pro Ser Thr Gln Pro Trp Glu His Val Asn Ala Ile Gln Glu Ala
      35              40              45

Arg Arg Leu Leu Asn Leu Ser Arg Asp Thr Ala Ala Glu Met Asn Glu
      50              55              60

Thr Val Glu Val Ile Ser Glu Met Phe Asp Leu Gln Glu Pro Thr Cys
65              70              75              80

Leu Gln Thr Arg Leu Glu Leu Tyr Lys Gln Gly Leu Arg Gly Ser Leu
      85              90              95

Thr Lys Leu Lys Gly Pro Leu Thr Met Met Ala Ser His Tyr Lys Gln
      100             105             110

His Cys Pro Pro Thr Pro Glu Thr Ser Cys Ala Thr Gln Ile Ile Thr
      115             120             125

Phe Glu Ser Phe Lys Glu Asn Leu Lys Asp Phe Leu Leu Val Ile Pro
      130             135             140

Phe Asp Cys Trp Glu Pro Val Gln Glu Pro Thr Thr Thr Pro Ala Pro
      145             150             155             160

Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu
      165             170             175

Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg
      180             185             190

Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly
      195             200             205

Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Asn
      210             215             220

His Arg Asn Arg Arg Arg Val Cys Lys Cys Pro Arg Pro Val Val Lys
      225             230             235             240

Ser Gly Asp Lys Pro Ser Leu Ser Ala Arg Tyr Val
      245             250

```

```

<210> SEQ ID NO 86
<211> LENGTH: 261
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mCD40L modified to stop cleavage

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<400> SEQUENCE: 86

Met Ile Glu Thr Tyr Asn Gln Thr Ser Pro Arg Ser Ala Ala Thr Gly
1              5              10              15

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-continued

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Leu Pro Ile Ser Met Lys Ile Phe Met Tyr Leu Leu Thr Val Phe Leu
      20                25                30

Ile Thr Gln Met Ile Gly Ser Ala Leu Phe Ala Val Tyr Leu His Arg
      35                40                45

Arg Leu Asp Lys Ile Glu Asp Glu Arg Asn Leu His Glu Asp Phe Val
      50                55                60

Phe Met Lys Thr Ile Gln Arg Cys Asn Thr Gly Glu Arg Ser Leu Ser
      65                70                75                80

Leu Leu Asn Cys Glu Glu Ile Lys Ser Gln Phe Glu Gly Phe Val Lys
      85                90                95

Asp Ile Met Leu Asn Lys Glu Glu Thr Lys Lys Glu Asn Ser Phe Glu
      100               105               110

Met Pro Arg Gly Glu Glu Asp Ser Gln Ile Ala Ala His Val Ile Ser
      115               120               125

Glu Ala Ser Ser Lys Thr Thr Ser Val Leu Gln Trp Ala Glu Lys Gly
      130               135               140

Tyr Tyr Thr Met Ser Asn Asn Leu Val Thr Leu Glu Asn Gly Lys Gln
      145               150               155               160

Leu Thr Val Lys Arg Gln Gly Leu Tyr Tyr Ile Tyr Ala Gln Val Thr
      165               170               175

Phe Cys Ser Asn Arg Glu Ala Ser Ser Gln Ala Pro Phe Ile Ala Ser
      180               185               190

Leu Cys Leu Lys Ser Pro Gly Arg Phe Glu Arg Ile Leu Leu Arg Ala
      195               200               205

Ala Asn Thr His Ser Ser Ala Lys Pro Cys Gly Gln Gln Ser Ile His
      210               215               220

Leu Gly Gly Val Phe Glu Leu Gln Pro Gly Ala Ser Val Phe Val Asn
      225               230               235               240

Val Thr Asp Pro Ser Gln Val Ser His Gly Thr Gly Phe Thr Ser Phe
      245               250               255

Gly Leu Leu Lys Leu
      260

```

<210> SEQ ID NO 87

<211> LENGTH: 261

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Summary (261 ORF5)

<400> SEQUENCE: 87

```

Met Ile Glu Thr Tyr Asn Gln Thr Ser Pro Arg Ser Ala Ala Thr Gly
1      5      10      15

Leu Pro Ile Ser Met Lys Ile Phe Met Tyr Leu Leu Thr Val Phe Leu
      20                25                30

Ile Thr Gln Met Ile Gly Ser Ala Leu Phe Ala Val Tyr Leu His Arg
      35                40                45

Arg Leu Asp Lys Ile Glu Asp Glu Arg Asn Leu His Glu Asp Phe Val
      50                55                60

Phe Met Lys Thr Ile Gln Arg Cys Asn Thr Gly Glu Arg Ser Leu Ser
      65                70                75                80

Leu Leu Asn Cys Glu Glu Ile Lys Ser Gln Phe Glu Gly Phe Val Lys
      85                90                95

Asp Ile Met Leu Asn Lys Glu Glu Thr Lys Lys Glu Asn Ser Phe Glu
      100               105               110

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-continued

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Met  Pro  Arg  Gly  Glu  Glu  Asp  Ser  Gln  Ile  Ala  Ala  His  Val  Ile  Ser
      115                      120                      125

Glu  Ala  Ser  Ser  Lys  Thr  Thr  Ser  Val  Leu  Gln  Trp  Ala  Glu  Lys  Gly
      130                      135                      140

Tyr  Tyr  Thr  Met  Ser  Asn  Asn  Leu  Val  Thr  Leu  Glu  Asn  Gly  Lys  Gln
      145                      150                      155                      160

Leu  Thr  Val  Lys  Arg  Gln  Gly  Leu  Tyr  Tyr  Ile  Tyr  Ala  Gln  Val  Thr
      165                      170                      175

Phe  Cys  Ser  Asn  Arg  Glu  Ala  Ser  Ser  Gln  Ala  Pro  Phe  Ile  Ala  Ser
      180                      185                      190

Leu  Cys  Leu  Lys  Ser  Pro  Gly  Arg  Phe  Glu  Arg  Ile  Leu  Leu  Arg  Ala
      195                      200                      205

Ala  Asn  Thr  His  Ser  Ser  Ala  Lys  Pro  Cys  Gly  Gln  Gln  Ser  Ile  His
      210                      215                      220

Leu  Gly  Gly  Val  Phe  Glu  Leu  Gln  Pro  Gly  Ala  Ser  Val  Phe  Val  Asn
      225                      230                      235                      240

Val  Thr  Asp  Pro  Ser  Gln  Val  Ser  His  Gly  Thr  Gly  Phe  Thr  Ser  Phe
      245                      250                      255

Gly  Leu  Leu  Lys  Leu
      260

```

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<210> SEQ ID NO 88
<211> LENGTH: 221
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mTNFalpha modified to stop cleavage

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<400> SEQUENCE: 88

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Met  Ser  Thr  Glu  Ser  Met  Ile  Arg  Asp  Val  Glu  Leu  Ala  Glu  Glu  Ala
 1      5      10      15

Leu  Pro  Lys  Lys  Thr  Gly  Gly  Pro  Gln  Gly  Ser  Arg  Arg  Cys  Leu  Phe
      20      25      30

Leu  Ser  Leu  Phe  Ser  Phe  Leu  Ile  Val  Ala  Gly  Ala  Thr  Thr  Leu  Phe
      35      40      45

Cys  Leu  Leu  His  Phe  Gly  Val  Ile  Gly  Pro  Gln  Arg  Glu  Glu  Phe  Pro
      50      55      60

Arg  Asp  Leu  Ser  Leu  Ile  Ser  Pro  Leu  Ala  Gln  Ala  Val  Ala  His  Val
      65      70      75      80

Val  Ala  Asn  Pro  Gln  Ala  Glu  Gly  Gln  Leu  Gln  Trp  Leu  Asn  Arg  Arg
      85      90      95

Ala  Asn  Ala  Leu  Leu  Ala  Asn  Gly  Val  Glu  Leu  Arg  Asp  Asn  Gln  Leu
      100     105     110

Val  Val  Pro  Ser  Glu  Gly  Leu  Tyr  Leu  Ile  Tyr  Ser  Gln  Val  Leu  Phe
      115     120     125

Lys  Gly  Gln  Gly  Cys  Pro  Ser  Thr  His  Val  Leu  Leu  Thr  His  Thr  Ile
      130     135     140

Ser  Arg  Ile  Ala  Val  Ser  Tyr  Gln  Thr  Lys  Val  Asn  Leu  Leu  Ser  Ala
      145     150     155     160

Ile  Lys  Ser  Pro  Cys  Gln  Arg  Glu  Thr  Pro  Glu  Gly  Ala  Glu  Ala  Lys
      165     170     175

Pro  Trp  Tyr  Glu  Pro  Ile  Tyr  Leu  Gly  Gly  Val  Phe  Gln  Leu  Glu  Lys
      180     185     190

Gly  Asp  Arg  Leu  Ser  Ala  Glu  Ile  Asn  Arg  Pro  Asp  Tyr  Leu  Asp  Phe
      195     200     205

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-continued

Ala Glu Ser Gly Gln Val Tyr Phe Gly Ile Ile Ala Leu
210 215 220

<210> SEQ ID NO 89
 <211> LENGTH: 221
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Summary (221 ORF6)

<400> SEQUENCE: 89

Met Ser Thr Glu Ser Met Ile Arg Asp Val Glu Leu Ala Glu Glu Ala
1 5 10 15
 Leu Pro Lys Lys Thr Gly Gly Pro Gln Gly Ser Arg Arg Cys Leu Phe
20 25 30
 Leu Ser Leu Phe Ser Phe Leu Ile Val Ala Gly Ala Thr Thr Leu Phe
35 40 45
 Cys Leu Leu His Phe Gly Val Ile Gly Pro Gln Arg Glu Glu Phe Pro
50 55 60
 Arg Asp Leu Ser Leu Ile Ser Pro Leu Ala Gln Ala Val Ala His Val
65 70 75 80
 Val Ala Asn Pro Gln Ala Glu Gly Gln Leu Gln Trp Leu Asn Arg Arg
85 90 95
 Ala Asn Ala Leu Leu Ala Asn Gly Val Glu Leu Arg Asp Asn Gln Leu
100 105 110
 Val Val Pro Ser Glu Gly Leu Tyr Leu Ile Tyr Ser Gln Val Leu Phe
115 120 125
 Lys Gly Gln Gly Cys Pro Ser Thr His Val Leu Leu Thr His Thr Ile
130 135 140
 Ser Arg Ile Ala Val Ser Tyr Gln Thr Lys Val Asn Leu Leu Ser Ala
145 150 155 160
 Ile Lys Ser Pro Cys Gln Arg Glu Thr Pro Glu Gly Ala Glu Ala Lys
165 170 175
 Pro Trp Tyr Glu Pro Ile Tyr Leu Gly Gly Val Phe Gln Leu Glu Lys
180 185 190
 Gly Asp Arg Leu Ser Ala Glu Ile Asn Arg Pro Asp Tyr Leu Asp Phe
195 200 205
 Ala Glu Ser Gly Gln Val Tyr Phe Gly Ile Ile Ala Leu
210 215 220

<210> SEQ ID NO 90
 <211> LENGTH: 244
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: wild-type

<400> SEQUENCE: 90

Met Asp Pro Asn Arg Ile Ser Glu Asp Gly Thr His Cys Ile Tyr Arg
1 5 10 15
 Ile Leu Arg Leu His Glu Asn Ala Asp Phe Gln Asp Thr Thr Leu Glu
20 25 30
 Ser Gln Asp Thr Lys Leu Ile Pro Asp Ser Cys Arg Arg Ile Lys Gln
35 40 45
 Ala Phe Gln Gly Ala Val Gln Lys Glu Leu Gln His Ile Val Gly Ser
50 55 60
 Gln His Ile Arg Ala Glu Lys Ala Met Val Asp Gly Ser Trp Leu Asp
65 70 75 80

-continued

Leu Ala Lys Arg Ser Lys Leu Glu Ala Gln Pro Phe Ala His Leu Thr
 85 90 95
 Ile Asn Ala Thr Asp Ile Pro Ser Gly Ser His Lys Val Ser Leu Ser
 100 105 110
 Ser Trp Tyr His Asp Arg Gly Trp Ala Lys Ile Ser Asn Met Thr Phe
 115 120 125
 Ser Asn Gly Lys Leu Ile Val Asn Gln Asp Gly Phe Tyr Tyr Leu Tyr
 130 135 140
 Ala Asn Ile Cys Phe Arg His His Glu Thr Ser Gly Asp Leu Ala Thr
 145 150 155 160
 Glu Tyr Leu Gln Leu Met Val Tyr Val Thr Lys Thr Ser Ile Lys Ile
 165 170 175
 Pro Ser Ser His Thr Leu Met Lys Gly Gly Ser Thr Lys Tyr Trp Ser
 180 185 190
 Gly Asn Ser Glu Phe His Phe Tyr Ser Ile Asn Val Gly Gly Phe Phe
 195 200 205
 Lys Leu Arg Ser Gly Glu Glu Ile Ser Ile Glu Val Ser Asn Pro Ser
 210 215 220
 Leu Leu Asp Pro Asp Gln Asp Ala Thr Tyr Phe Gly Ala Phe Lys Val
 225 230 235 240
 Arg Asp Ile Asp

<210> SEQ ID NO 91
 <211> LENGTH: 4
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 91

Ile Arg Glu Ser
 1

<210> SEQ ID NO 92
 <211> LENGTH: 26
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FLT3L signal

<400> SEQUENCE: 92

Met Ala Thr Val Leu Ala Pro Ala Trp Ser Pro Thr Thr Tyr Leu Leu
 1 5 10 15
 Leu Leu Leu Leu Leu Ser Ser Gly Leu Ser
 20 25

<210> SEQ ID NO 93
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Linker

<400> SEQUENCE: 93

Gly Gly Gly Gly Ser
 1 5

<210> SEQ ID NO 94
 <211> LENGTH: 14
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: V5 epitope tag for flow detection

<400> SEQUENCE: 94

Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr
1           5           10

<210> SEQ ID NO 95
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: anti-biotin murine vH with inserted Cys for
        intralinkage

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<400> SEQUENCE: 95

Gln Val Lys Leu Gln Glu Ser Gly Pro Gly Leu Val Ala Pro Ser Gln
1           5           10           15

Ser Leu Ser Ile Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ala Tyr
        20           25           30

Gly Val Asp Trp Val Arg Gln Pro Pro Gly Lys Cys Leu Glu Trp Leu
        35           40           45

Gly Val Ile Trp Gly Gly Gly Arg Thr Asn Tyr Asn Ser Gly Leu Met
        50           55           60

Ser Arg Leu Ser Ile Arg Lys Asp Asn Ser Lys Ser Gln Val Phe Leu
        65           70           75           80

Thr Met Asn Ser Leu Gln Thr Asp Asp Thr Ala Lys Tyr Tyr Cys Val
        85           90           95

Lys His Thr Asn Trp Asp Gly Gly Phe Ala Tyr Trp Gly Gln Gly Thr
        100          105          110

Thr Val Thr Val Ser Ser
        115

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<210> SEQ ID NO 96
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

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<400> SEQUENCE: 96

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
1           5           10           15

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<210> SEQ ID NO 97
<211> LENGTH: 99
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LC Variable

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<400> SEQUENCE: 97

Gly Ser Pro Gly Gln Ser Val Ser Ile Ser Cys Ser Gly Ser Ser Ser
1           5           10           15

Asn Ile Gly Asn Asn Tyr Val Tyr Trp Tyr Gln His Leu Pro Gly Thr
        20           25           30

Ala Pro Lys Leu Leu Ile Tyr Ser Asp Thr Lys Arg Pro Ser Gly Val
        35           40           45

Pro Asp Arg Ile Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala
        50           55           60

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Ile Ser Gly Leu Gln Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ser
65 70 75 80

Trp Asp Asp Ser Leu Asp Gly Pro Val Phe Gly Cys Gly Thr Lys Leu
85 90 95

Thr Val Leu

<210> SEQ ID NO 98

<211> LENGTH: 99

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: CD8alpha transmembrane and cytoplasmic domain

<400> SEQUENCE: 98

Pro Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile
1 5 10 15

Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala
20 25 30

Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr
35 40 45

Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Ser Leu
50 55 60

Val Ile Thr Leu Tyr Cys Asn His Arg Asn Arg Arg Arg Val Cys Lys
65 70 75 80

Cys Pro Arg Pro Val Val Lys Ser Gly Asp Lys Pro Ser Leu Ser Ala
85 90 95

Arg Tyr Val

<210> SEQ ID NO 99

<211> LENGTH: 244

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Summary (244 ORF7a)

<400> SEQUENCE: 99

Met Asp Pro Asn Arg Ile Ser Glu Asp Gly Thr His Cys Ile Tyr Arg
1 5 10 15

Ile Leu Arg Leu His Glu Asn Ala Asp Phe Gln Asp Thr Thr Leu Glu
20 25 30

Ser Gln Asp Thr Lys Leu Ile Pro Asp Ser Cys Arg Arg Ile Lys Gln
35 40 45

Ala Phe Gln Gly Ala Val Gln Lys Glu Leu Gln His Ile Val Gly Ser
50 55 60

Gln His Ile Arg Ala Glu Lys Ala Met Val Asp Gly Ser Trp Leu Asp
65 70 75 80

Leu Ala Lys Arg Ser Lys Leu Glu Ala Gln Pro Phe Ala His Leu Thr
85 90 95

Ile Asn Ala Thr Asp Ile Pro Ser Gly Ser His Lys Val Ser Leu Ser
100 105 110

Ser Trp Tyr His Asp Arg Gly Trp Ala Lys Ile Ser Asn Met Thr Phe
115 120 125

Ser Asn Gly Lys Leu Ile Val Asn Gln Asp Gly Phe Tyr Tyr Leu Tyr
130 135 140

Ala Asn Ile Cys Phe Arg His His Glu Thr Ser Gly Asp Leu Ala Thr
145 150 155 160

Glu Tyr Leu Gln Leu Met Val Tyr Val Thr Lys Thr Ser Ile Lys Ile

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165					170					175					
Pro	Ser	Ser	His	Thr	Leu	Met	Lys	Gly	Gly	Ser	Thr	Lys	Tyr	Trp	Ser
			180					185					190		
Gly	Asn	Ser	Glu	Phe	His	Phe	Tyr	Ser	Ile	Asn	Val	Gly	Gly	Phe	Phe
		195					200					205			
Lys	Leu	Arg	Ser	Gly	Glu	Glu	Ile	Ser	Ile	Glu	Val	Ser	Asn	Pro	Ser
		210					215					220			
Leu	Leu	Asp	Pro	Asp	Gln	Asp	Ala	Thr	Tyr	Phe	Gly	Ala	Phe	Lys	Val
		225					230					235			240
Arg Asp Ile Asp															
<210> SEQ ID NO 100															
<211> LENGTH: 381															
<212> TYPE: PRT															
<213> ORGANISM: Artificial Sequence															
<220> FEATURE:															
<223> OTHER INFORMATION: Summary (381aa ORF7b)															
<400> SEQUENCE: 100															
Met	Ala	Thr	Val	Leu	Ala	Pro	Ala	Trp	Ser	Pro	Thr	Thr	Tyr	Leu	Leu
1				5					10					15	
Leu	Leu	Leu	Leu	Leu	Ser	Ser	Gly	Leu	Ser	Gly	Gly	Gly	Gly	Ser	Gly
			20					25					30		
Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser	Thr	Gly	Gly	Gly
		35					40					45			
Gly	Ser	Gln	Val	Lys	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Ala	Pro
		50					55					60			
Ser	Gln	Ser	Leu	Ser	Ile	Thr	Cys	Thr	Val	Ser	Gly	Phe	Ser	Leu	Thr
		65					70					75			80
Ala	Tyr	Gly	Val	Asp	Trp	Val	Arg	Gln	Pro	Pro	Gly	Lys	Cys	Leu	Glu
			85					90						95	
Trp	Leu	Gly	Val	Ile	Trp	Gly	Gly	Gly	Arg	Thr	Asn	Tyr	Asn	Ser	Gly
		100					105						110		
Leu	Met	Ser	Arg	Leu	Ser	Ile	Arg	Lys	Asp	Asn	Ser	Lys	Ser	Gln	Val
		115					120					125			
Phe	Leu	Thr	Met	Asn	Ser	Leu	Gln	Thr	Asp	Asp	Thr	Ala	Lys	Tyr	Tyr
		130					135					140			
Cys	Val	Lys	His	Thr	Asn	Trp	Asp	Gly	Gly	Phe	Ala	Tyr	Trp	Gly	Gln
		145					150					155			160
Gly	Thr	Thr	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
			165					170					175		
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Ser	Pro	Gly	Gln	Ser	Val	Ser	Ile
		180					185					190			
Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Asn	Asn	Tyr	Val	Tyr	Trp
		195					200					205			
Tyr	Gln	His	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu	Leu	Ile	Tyr	Ser	Asp
		210					215					220			
Thr	Lys	Arg	Pro	Ser	Gly	Val	Pro	Asp	Arg	Ile	Ser	Gly	Ser	Lys	Ser
		225					230					235			240
Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Ser	Gly	Leu	Gln	Ser	Glu	Asp	Glu
			245					250					255		
Ala	Asp	Tyr	Tyr	Cys	Ala	Ser	Trp	Asp	Asp	Ser	Leu	Asp	Gly	Pro	Val
		260					265					270			
Phe	Gly	Cys	Gly	Thr	Lys	Leu	Thr	Val	Leu	Pro	Thr	Thr	Thr	Pro	Ala
		275					280					285			

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Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser
 290 295 300

Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr
 305 310 315 320

Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala
 325 330 335

Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys
 340 345 350

Asn His Arg Asn Arg Arg Arg Val Cys Lys Cys Pro Arg Pro Val Val
 355 360 365

Lys Ser Gly Asp Lys Pro Ser Leu Ser Ala Arg Tyr Val
 370 375 380

<210> SEQ ID NO 101
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

<400> SEQUENCE: 101

Gln Met Gln Gly Val Asn Cys Thr Val Ser Ser
 1 5 10

<210> SEQ ID NO 102
 <211> LENGTH: 30
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CpG
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(2)
 <223> OTHER INFORMATION: n is a G-phosphorothioate
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (25)..(29)
 <223> OTHER INFORMATION: n is a G-phosphorothioate

<400> SEQUENCE: 102

nnaaccgtat cggcgatatc ggttnnnnng

30

<210> SEQ ID NO 103
 <211> LENGTH: 30
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CpG
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(2)
 <223> OTHER INFORMATION: n is a G-phosphorothioate
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (25)..(29)
 <223> OTHER INFORMATION: n is a G-phosphorothioate

<400> SEQUENCE: 103

nnaaccgtat gcggcatatc ggttnnnnng

30

<210> SEQ ID NO 104
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic peptide

-continued

<400> SEQUENCE: 104

Val Arg Ser Ser Ser Arg Thr Pro Ser Asp Lys Pro
 1 5 10

What is claimed is:

1. An allogeneic tumor cell vaccine used to elicit and stimulate an immune response against a melanoma or prostate cancer comprising:

- (1) a clonally-derived variant of a melanoma tumor cell line or a prostate tumor cell line comprising:
 - (a) two or more stably expressed recombinant membrane-bound immunomodulator molecules selected from the group consisting of an IgG1 peptide, a CD40L peptide, a TNF-alpha peptide, and an Flt-3L peptide; and
 - (b) a stably expressed recombinant soluble GM-CSF peptide; and
 - (2) a pharmaceutically acceptable carrier;

wherein the clonally-derived variant of the melanoma tumor cell line or the prostate tumor cell line is proliferation-incompetent;

wherein the clonally-derived variant of the melanoma tumor cell line or the prostate tumor cell line is effective to elicit stimulation of the immune response to a subject's melanoma or prostate cancer, respectively, comprising activation of one or more of T cells, B cells, and dendritic cells as measured by one or more parameters selected from cellular proliferation, cellular subset differentiation, cytokine release profile, and tumor cell lysis that improves overall survival of the subject when compared to a control comprising a non-transfected melanoma tumor cell line or a prostate tumor cell line; wherein the clonally-derived variant of the melanoma tumor cell line or the prostate tumor cell line expresses the soluble GM-CSF peptide comprising an amino acid sequence as set forth in SEQ ID NO: 13 and two or more of:

- (a) the membrane-bound IgG1 immunomodulator peptide comprising an amino acid sequence as set forth in SEQ ID NO: 45;

(b) the membrane-bound CD40L immunomodulator peptide comprising an amino acid sequence as set forth in SEQ ID NO: 7;

(c) the membrane-bound form of TNF-alpha immunomodulator peptide comprising an amino acid sequence as set forth in SEQ ID NO: 11; and

(d) the membrane-bound form of Flt-3L immunomodulator peptide comprising an amino acid sequence as set forth in SEQ ID NO: 14;

wherein the melanoma tumor cell line is SK-MEL2, NIS-G, TRI-G, WIL-G, MRI-H-121B, MRI-H-187, MRI-H-221, or MRI-H-255; and

wherein the prostate tumor cell line is CWR-22, DU145, PC-3/M, PC-3, VCaP, MDA PCa 2b, or LNCaP clone FGC.

2. The allogeneic tumor cell vaccine of claim 1, wherein the clonally-derived variant of the melanoma tumor cell line or prostate tumor cell line

comprises a membrane-bound chimeric fusion protein comprising a fusion of the CD40L immunomodulator peptide and the TNF-alpha immunomodulator peptide.

3. The allogeneic tumor cell vaccine of claim 2, wherein the membrane-bound chimeric fusion protein comprises the CD40L immunomodulator peptide as set forth in SEQ ID NO: 9 and the TNF-alpha immunomodulator peptide as set forth in SEQ ID NO: 10.

4. The allogeneic tumor cell vaccine of claim 1, wherein the stably expressed recombinant membrane-bound immunomodulator molecules in (a) include the membrane-bound form of TNF-alpha immunomodulator peptide with an amino acid sequence as set forth in SEQ ID NO: 11.

5. The allogeneic tumor cell vaccine of claim 1, wherein the subject includes a member of an animal species of mammalian origin.

* * * * *